exploring cell types with the allen institute

Rachel Hostetler, Irika Sinha, Jeremy Miller

11 November 2026

allen institute/

allen institute

agenda/morning

9:00-9:30 | check-in / breakfast

9:30-9:45 | welcome, logistics, and overview of allen institute (Rachel)

9:45-10:30 | introduction to transcriptomics for cell typing (Jeremy)

10:30-10:45 | break

10:45-12:00 | cell type data sets on allen brain map

- overview of mouse data sets – Rachel
- cross-primate healthy brain atlases – Jeremy
- cell typing in aged humans, SEA-AD and beyond - Irika

12:00-1:00 | lunch

github.com/AllenInstitute/SPOCTT/tree/main/2026_Additional_Presentations

agenda/afternoon

12:00-1:00 | lunch

1:00-2:15 | walkthrough of allen brain map tools (Rachel / Jeremy)

- walkthrough of allen brain map
- Brain Knowledge Platform: ABC Atlas, MapMyCells, Genetics Tools Atlas
- Additional tools in brief: Patch-seq, gene atlases, and anatomy

2:15-2:45 | SEA-AD tools for exploring changes in AD (Irika / Jeremy)

2:45-3:00 | break

3:00-3:30 | external use cases – Rachel

3:30-4:40 | user exploration of allen institute tools

4:40-5:00 | feedback survey (all – please fill out the survey!)

github.com/AllenInstitute/SPOCTT/tree/main/2026_Additional_Presentations

workshop introduction/

- welcome
- logistics
- allen institute overview

allen institute/

about us/

allen institute

Jeremy Miller
senior scientist

Rachel Hostetler
scientist I

Irika Sinha
scientist I

Johns Hopkins University

Sarada Viswanathan
scientific director, Kavli NDI

allen institute/

- founded September 16, **2003**
- founded by Paul G. Allen & Jody Allen
- **~900** employees
- **non-profit** institution
  - no university affiliation (or degrees)
  - no traditional labs with single PI
- **foundational** science

understanding life
advancing health

allen institute/ funding & structure

- **allen institute/** funding types
  - majority private funding, particularly "Fund for Science & Technology" – part of Paul G. Allen's estates
  - some public funding (primarily NIH Grants)
    - no K series (training grants)
    - mainly U & R series

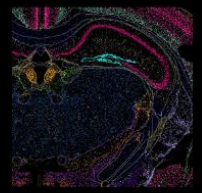
we are privileged in our funding support so we want to ensure all our resources can be used by the community

- **allen institute/** structure
  - structure is very different from standard academic structure
    (e.g., one PI with lab of 3-10, all focus on project related to PI's interest or field)
  - multiple teams who all work on same big project
  - projects are determined by multiple scientific leaders, along with input from our scientific advisory council

this workshop is funded by a **U24 research dissemination grant** from the **NIH** (NINDS 1U24NS133077)

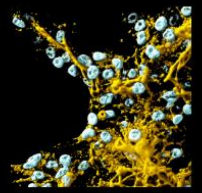
allen institute/ scientific accelerators

explore our accelerators/



brain science/

Mapping every cell, connection, and circuit in the brain—openly shared with the world.

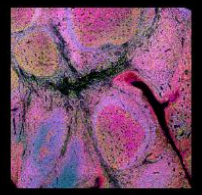


cell science/

Decoding how cells become tissues, then programming that knowledge into powerful new research tools.

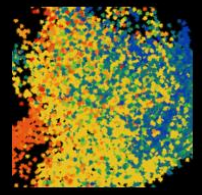
neural dynamics/

Revealing the brain's hidden algorithms that transform neural activity into real-world behavior.



immunology/

Creating the deepest open reference for the healthy human immune system ever built.

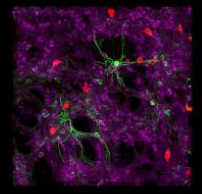


synthetic biology/

Engineering cells to record their own histories, transforming how we understand disease over time.

ai biodesign/

Advancing evolution to explore biology's full design space and advance human health

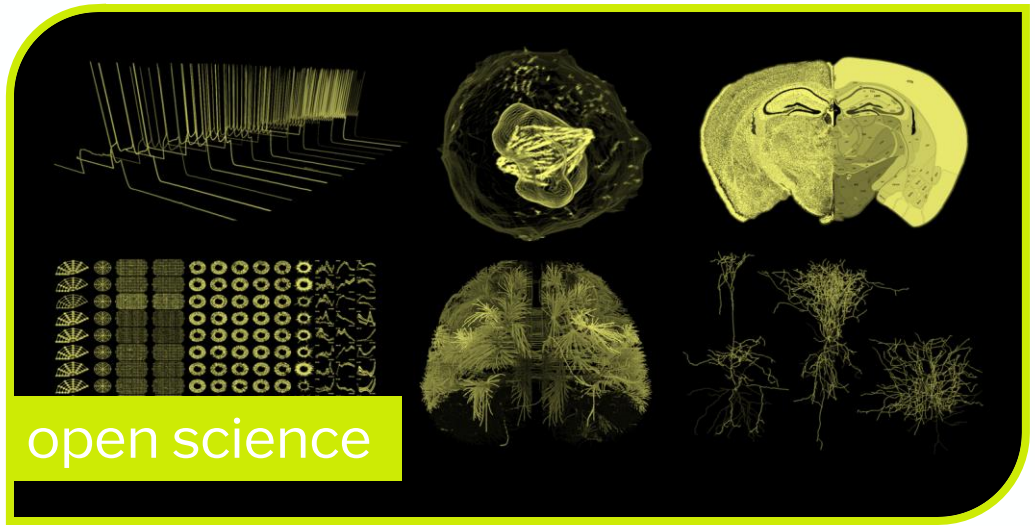


brain health/

Accelerating human brain research and new therapeutic strategies for neurological conditions

alleninstitute.org/our-research

allen institute/ our core principles



big science/

brain science

human brain atlas

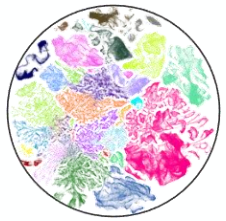
cell science

Paul G. Allen Frontiers Group

immunology

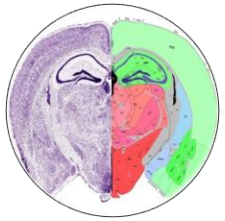
neural dynamics

allen brain cell atlas

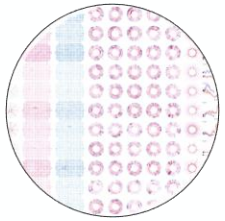


brain health new

2003 2026

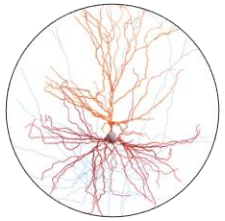


mouse brain atlas

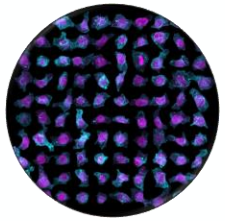


MindScope

connectivity atlas



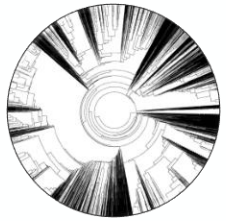
cell types database & brain observatory



allen cell explorer

Alzheimer's disease brain cell atlas

human immune system explorer



Seattle Hub for Synthetic Biology

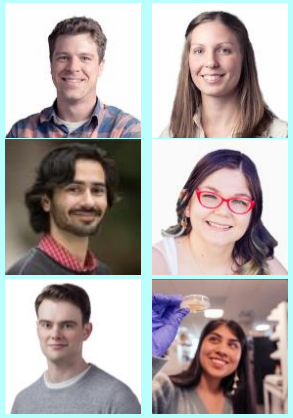
AI biodesign new

brain science is a major focus

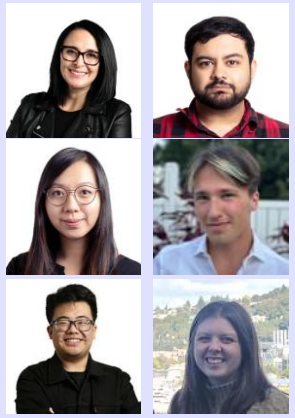
team science/ spatial transcriptomics example

animal care teams

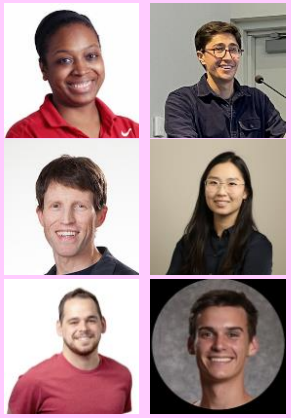
tissue processing



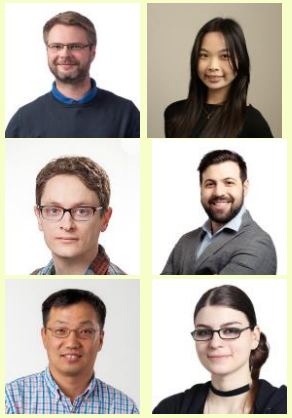
histology



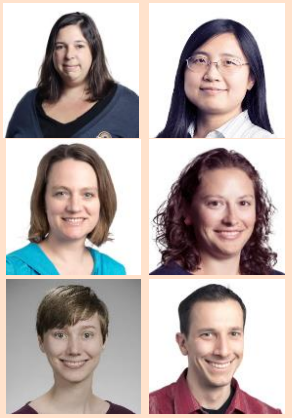
imaging



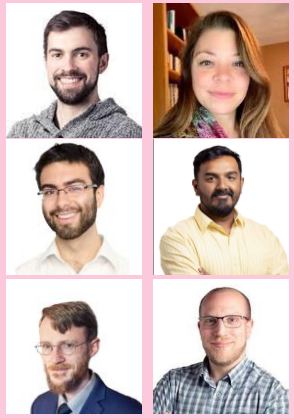
spatial bioinformatics



informatics & analysis



data & Technology



additional teams

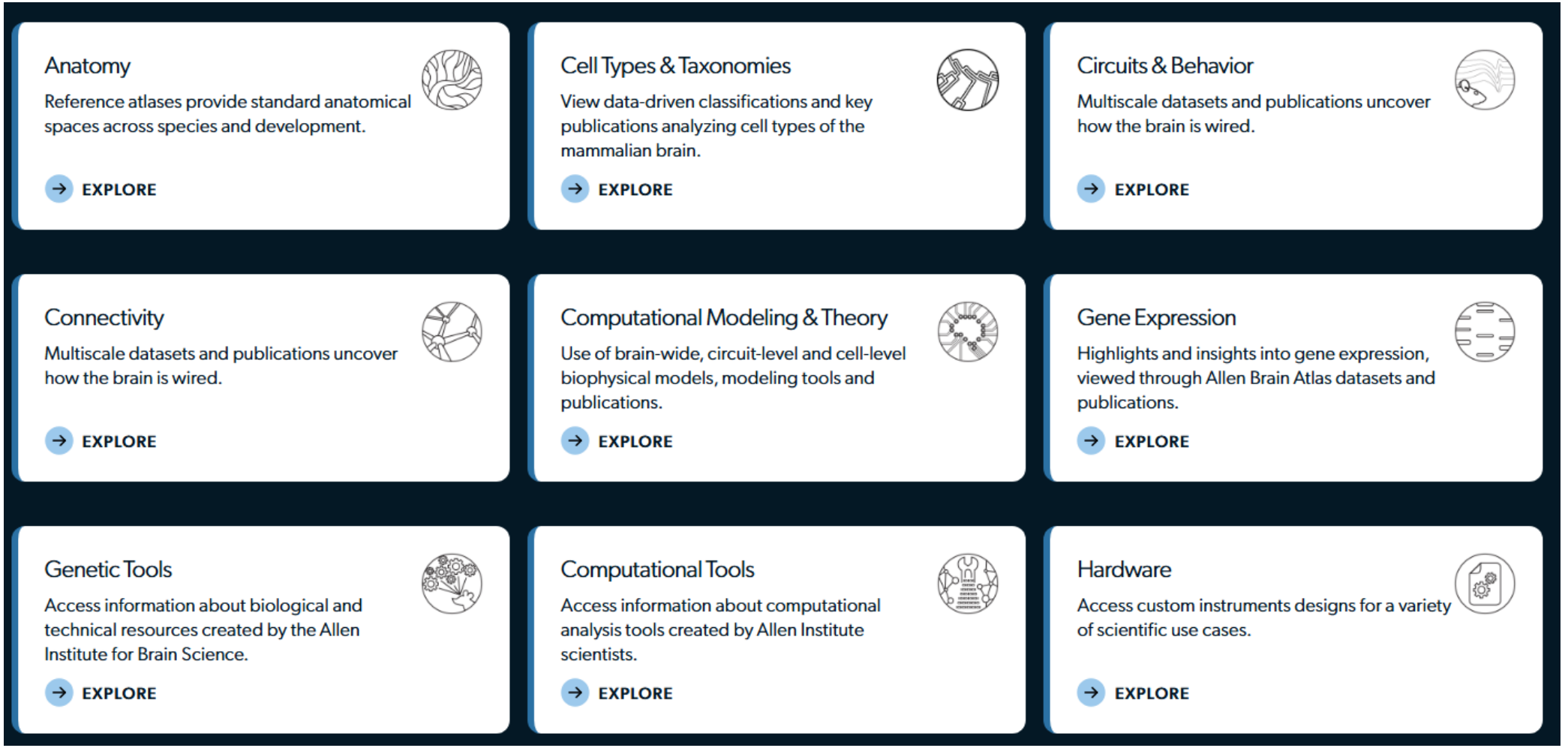
- **project management**
- **scientific support**
- **research compliance**
- **finance**
- **shipping & receiving**

\+ **MANY** more folks & teams not pictured

open science/

- **goal**: make scientific data, tools, and knowledge available and accessible to the public
- this is *more* than just publication!
  - data is available and fully public
  - published protocols (find us on protocols.io)
  - develop open-access tools for data use and access
  - provide resources & trainings to help use the data & tools
  - following FAIR (Findable, Accessible, Interoperable, Reproducible) Principles

all FREE on Allen Brain Map/ brain-map.org



careers/

- we hire straight out of graduate school – no post-doc required!
  - this is typically a Scientist I

    Scientist I - Neurophysiology – Global Brain Observatory | Scientist I, Synthetic Biology | Scientist I – Advanced Light Microscopy | Scientist I - Brain Proteome Organization
  - average “Scientist I” salary is about $80k-$100k USD
  - some **non-research** post-PhD roles (e.g., PM)
  - each position will be clear if it supports Visa or not - it is position dependent
- hiring is year-round, although sometimes more open spots during the start of the new year
- we often look for specific skill sets to fill open positions; the skill set depends on the position.

all careers are posted online
alleninstitute.org/careers

NOTE: We don’t have any undergraduate internships available for international students.

As of now, we also do not have any PhD-level internships.

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introduction/ transcriptomics for cell typing

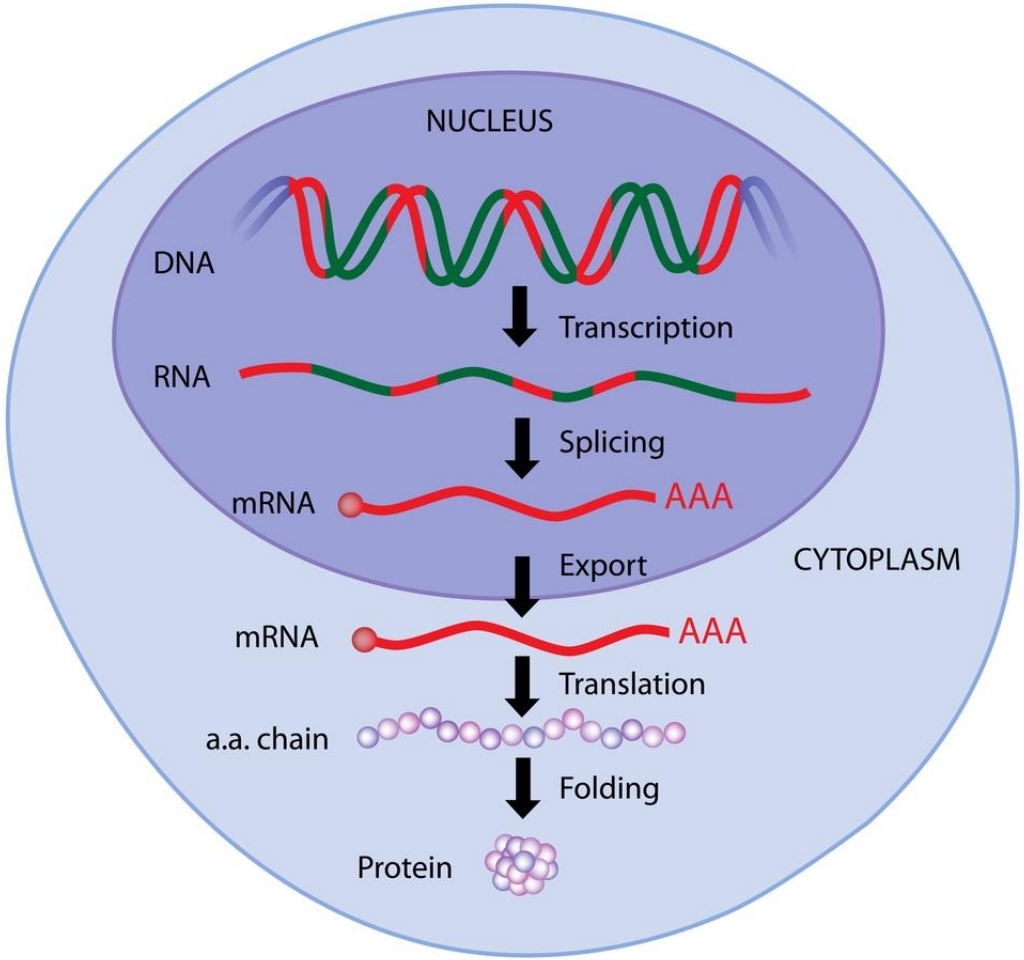
- transcriptomics for gene expression atlases
- single cell/nucleus RNA-sequencing
- spatial transcriptomics
- other –omics modalities

alleninstitute/

why use transcriptomics to define cell types/

- define cell types based on gene expression (transcriptomics)
  - high throughput (millions of cells)
  - many features (~30,000+ genes)
  - direct links to genome (disease/genetic tools)
  - works with postmortem tissue
  - universal across organs and species
  - many analysis tools

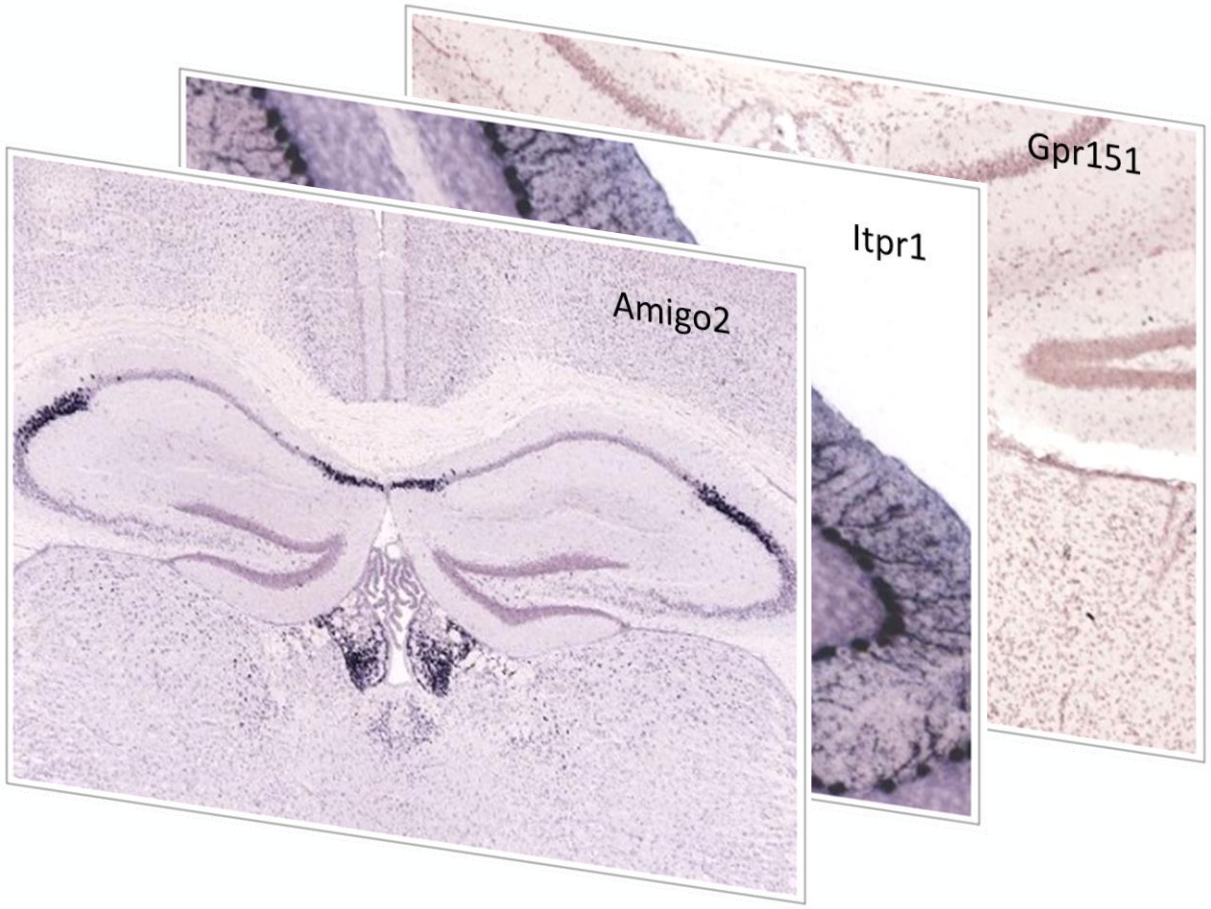
- map knowledge from other modalities onto these transcriptomics-based definitions



context/ genome-wide, brain-wide atlases in adult mouse and human/

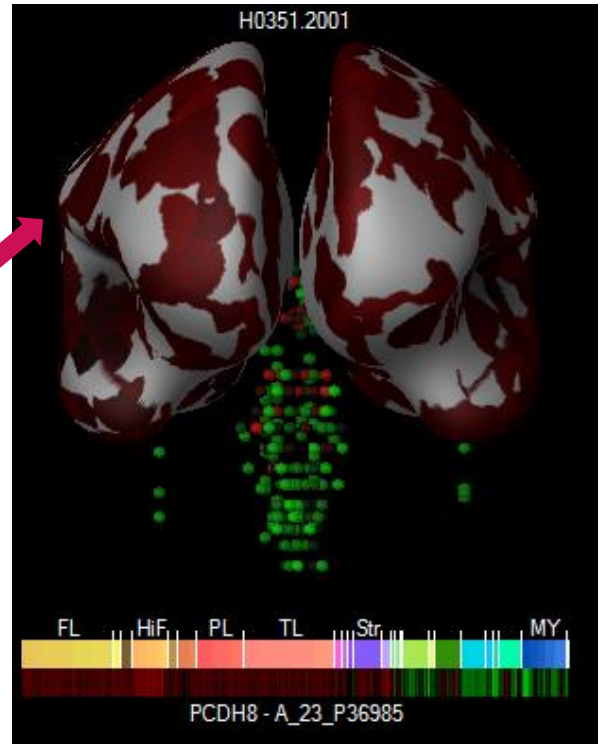
+ atlases of development and non-human primate (not shown)

mouse



human

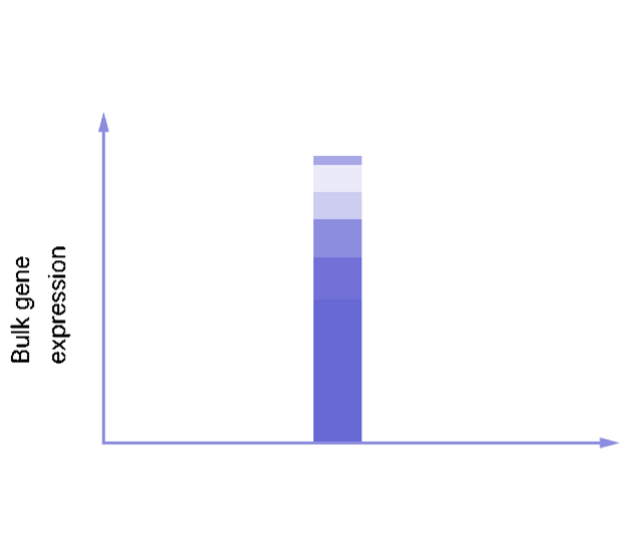
Gene Symbol	Probe Name
ACTA1	A_23_P1102
ACTA1	CUST_15163_PI416261804
ACTA2	A_23_P150053
ACTA2	CUST_15154_PI416261804
ACTB	A_23_P135769
ACTB	A_23_P31323
ACTB	A_24_P226554
ACTB	A_32_P118686
ACTB	A_32_P137939
ACTBL2	A_24_P6903
ACTBL2	CUST_4789_PI416261804
ACTC1	A_23_P205894
ACTC1	A_24_P121535
ACTG1	A_23_P44037



Initial studies focused on localizing gene expression in the brain outside the context of cell types.

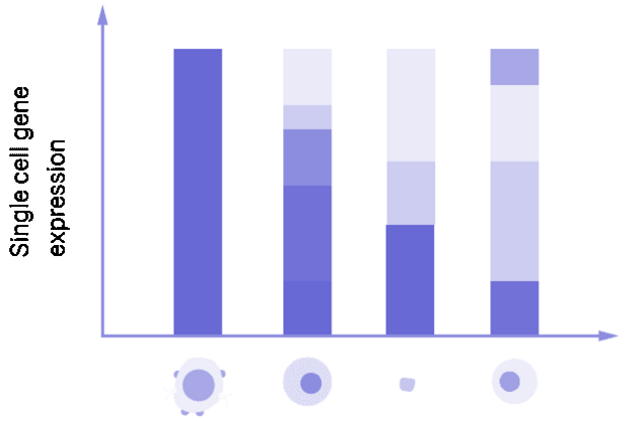
transcriptomics/ key methods/ rna-sequencing

bulk rna-seq/



Average gene expression from all cells

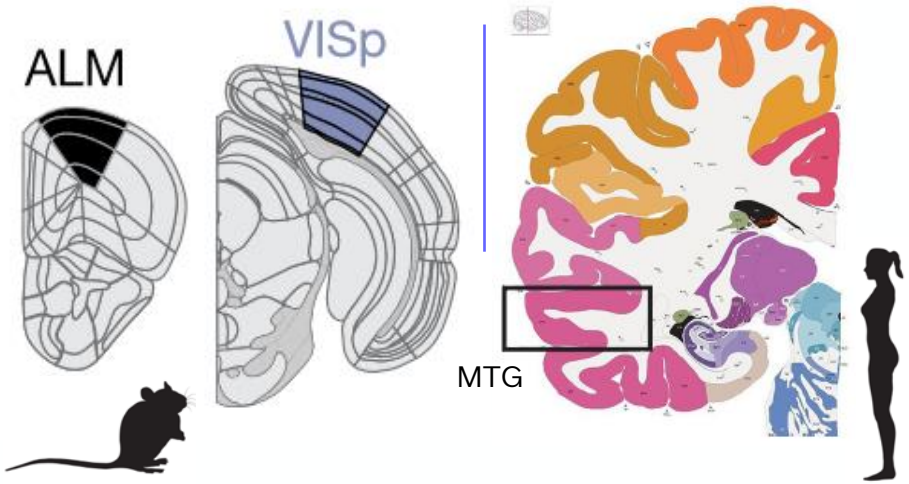
single cell or nucleus rna-seq/



Each cell type has a distinct expression profile

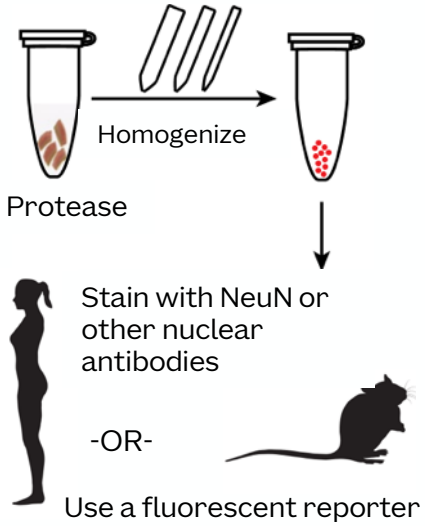
cell type atlasing/ key methods/ sc & snRNA-seq

1. Dissect region & (maybe) layers

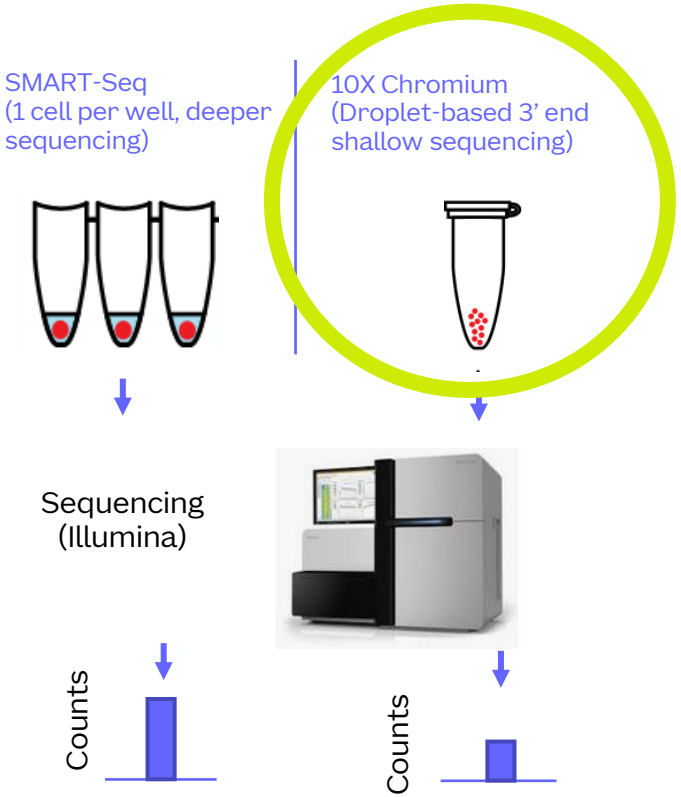


(Note: same strategy used for other brain regions and species)

2. Select cells or nuclei of interest



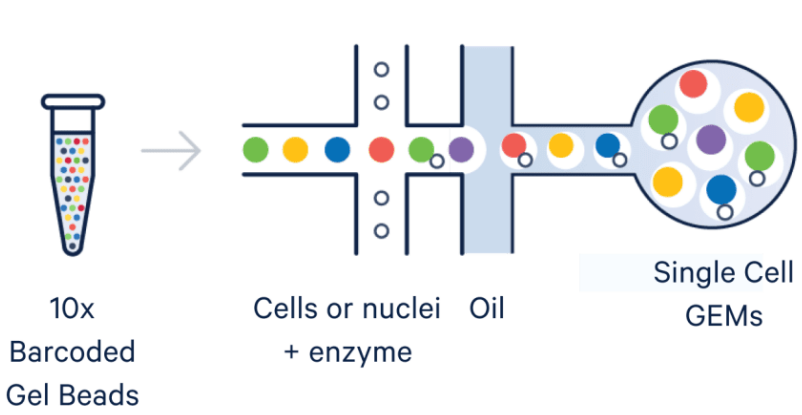
3. Perform sequencing on isolated cells or nuclei

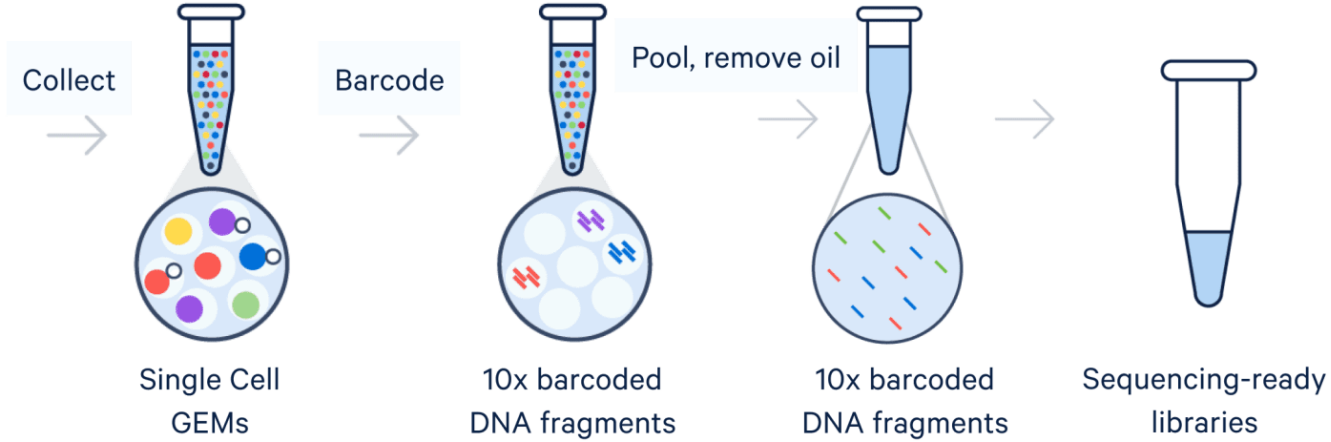


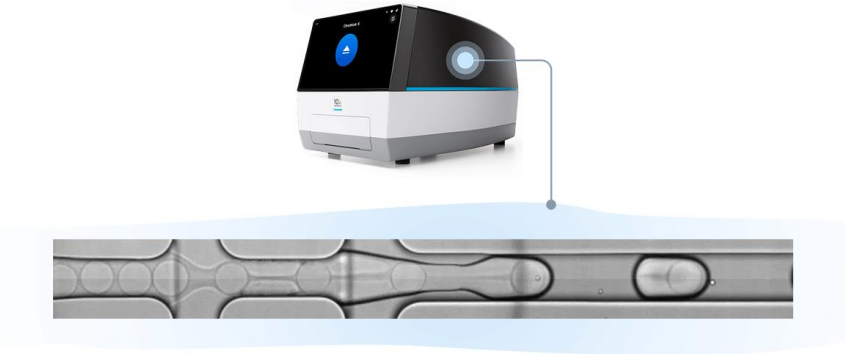
We use this method almost exclusively nowadays... how does it work?

(Tasic et al 2018, Nature; Hodge*, Bakken*, et al 2019, Nature)
(Yao et al 2021, Cell; Bakken, et al 2021, Nature; and MANY more!)

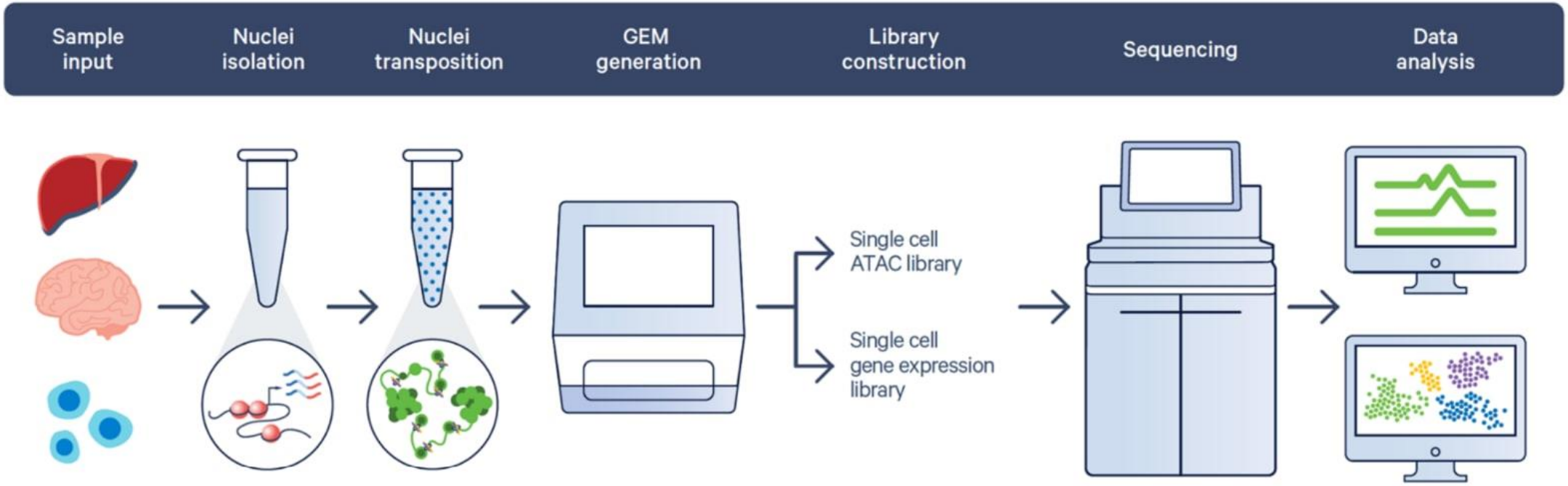
10X Genomics/ droplet-based barcoding







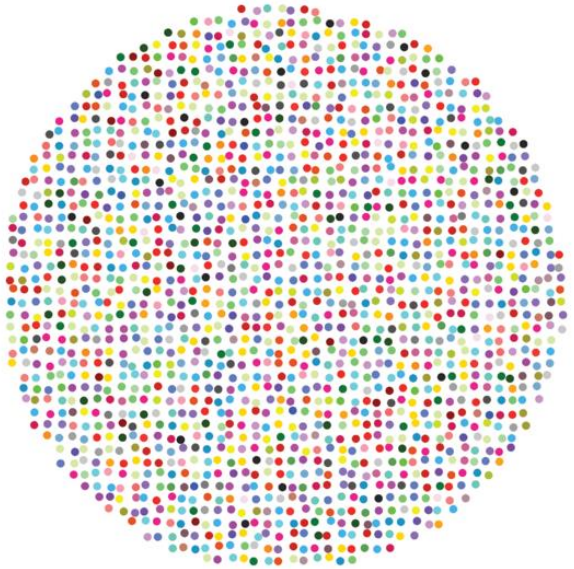
10X Genomics/ sn-multiome RNA-seq



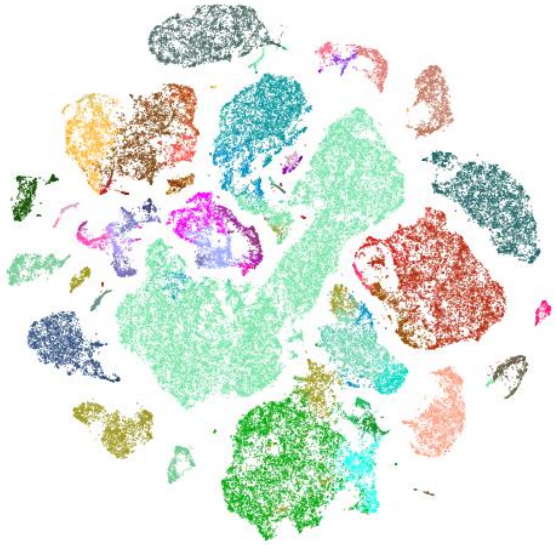
*Multiomics gives us transcriptomics (used for **defining** cell types) and chromatin accessibility (used for **targeting** cell types) from the same cell*

defining cell types/ using gene expression

collect cells/nuclei

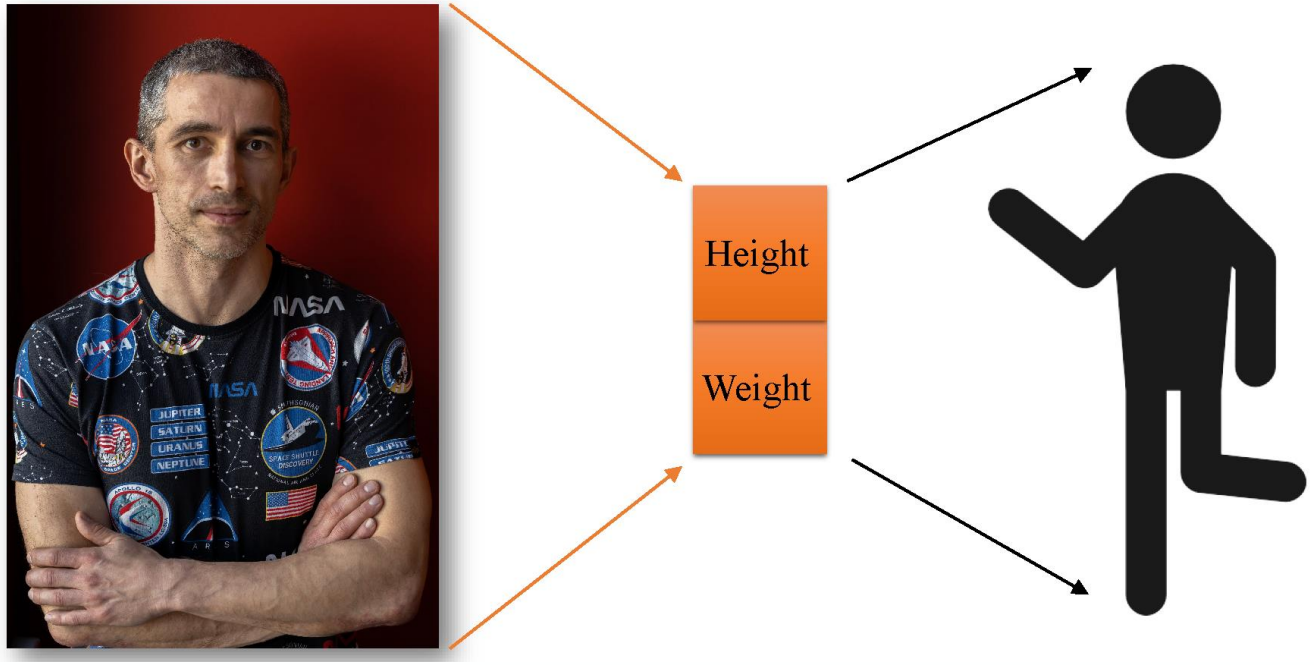


perform unsupervised clustering



What is a UMAP?

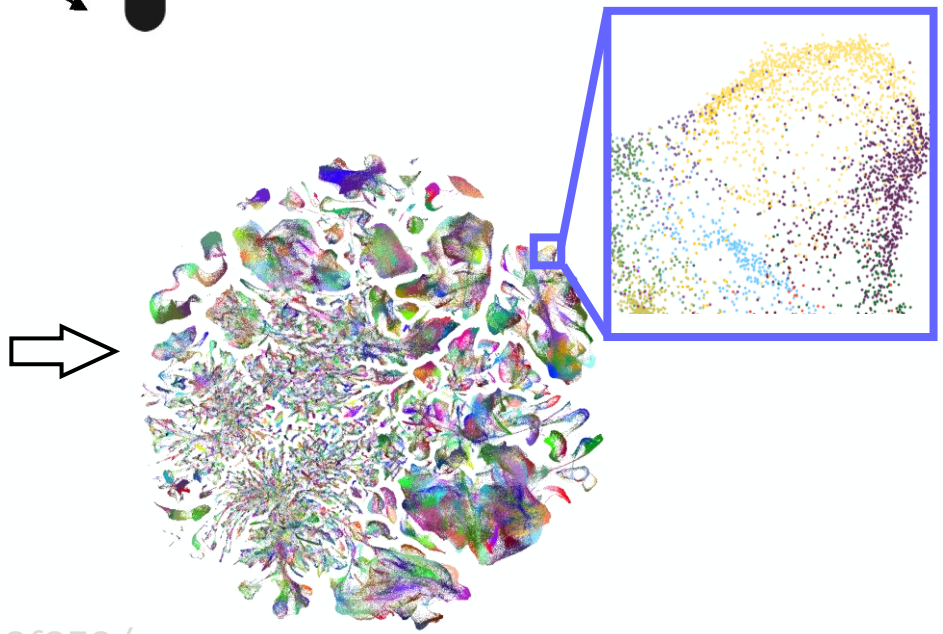
UMAP (uniform manifold approximation and projection) plot



of transcripts in each cell

	gene 1	gene 2	[...]	gene N
cell 1	...	...	...	...
cell 2	...	...	...	...
[...]	...	...	...	...
cell N	...	...	...	...

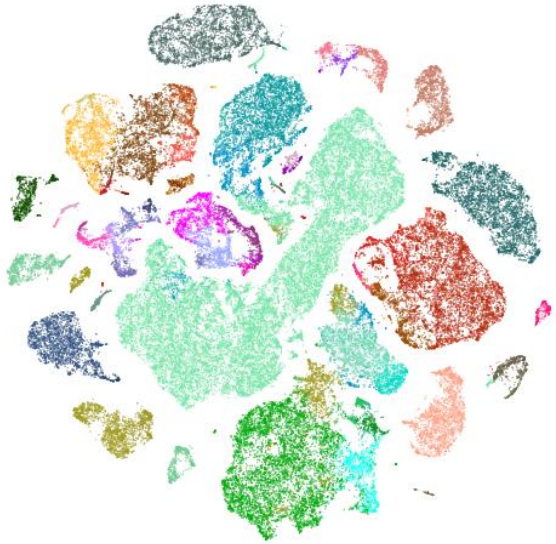
4+ million cells; 30,000+ genes

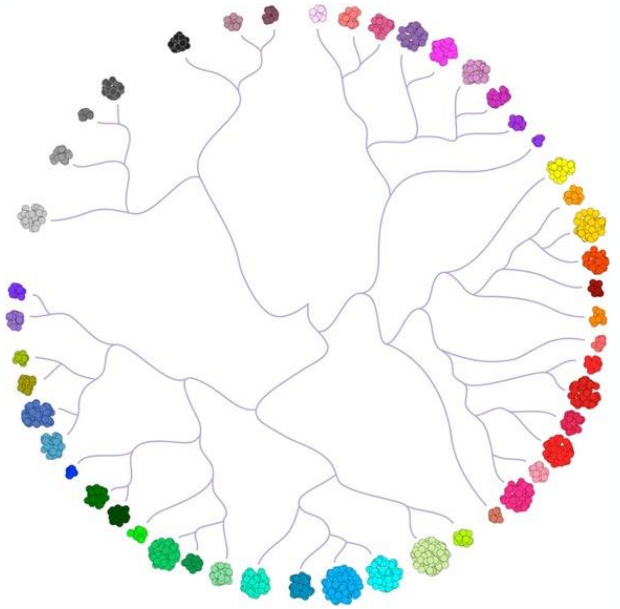


https://towardsdatascience.com/what-is-a-latent-space-065eb8e3f859/

defining cell types/ using gene expression

collect cells/nuclei → perform unsupervised clustering → organize clusters

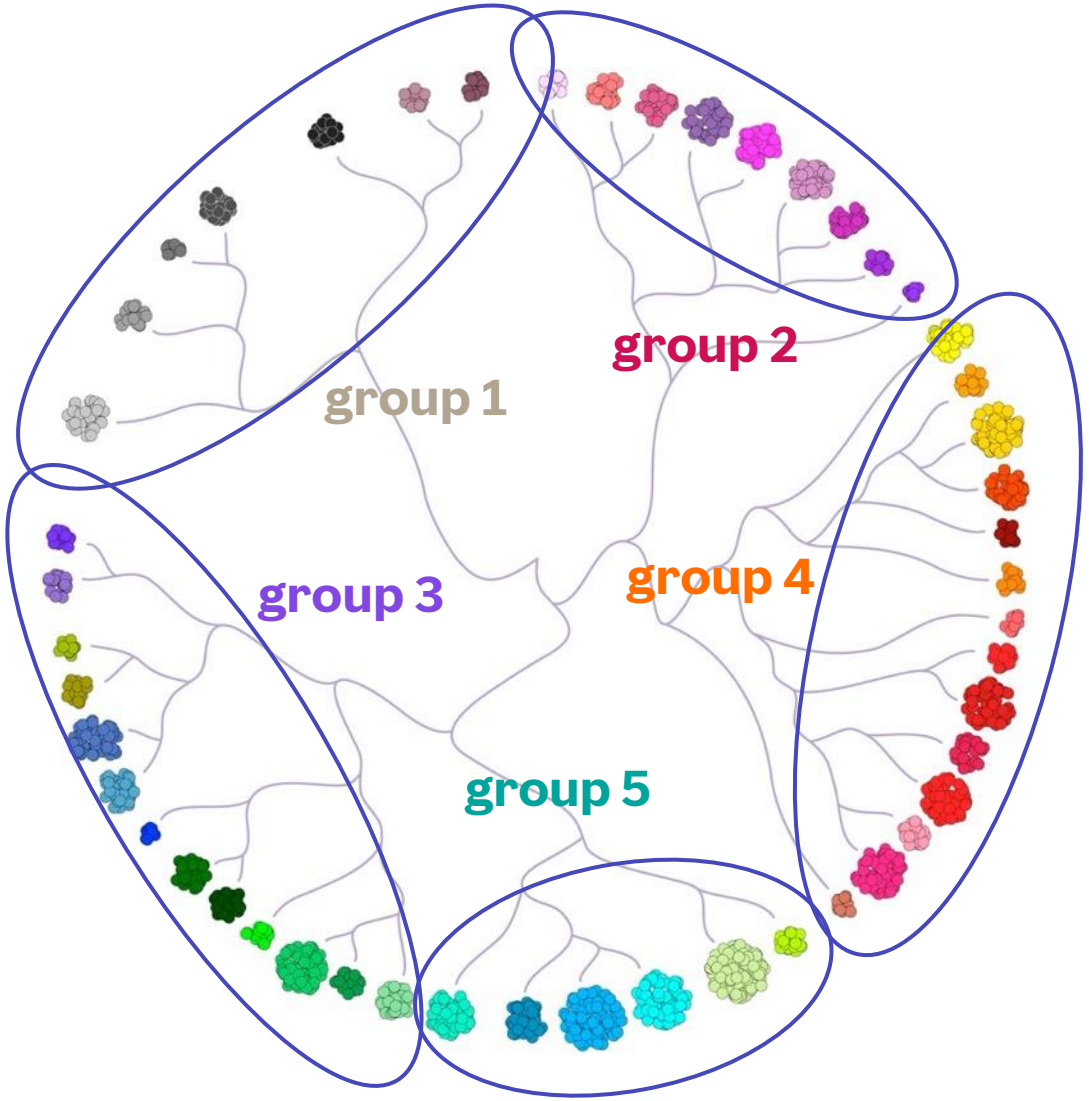




these steps are done quantitatively

defining cell types/ using gene expression

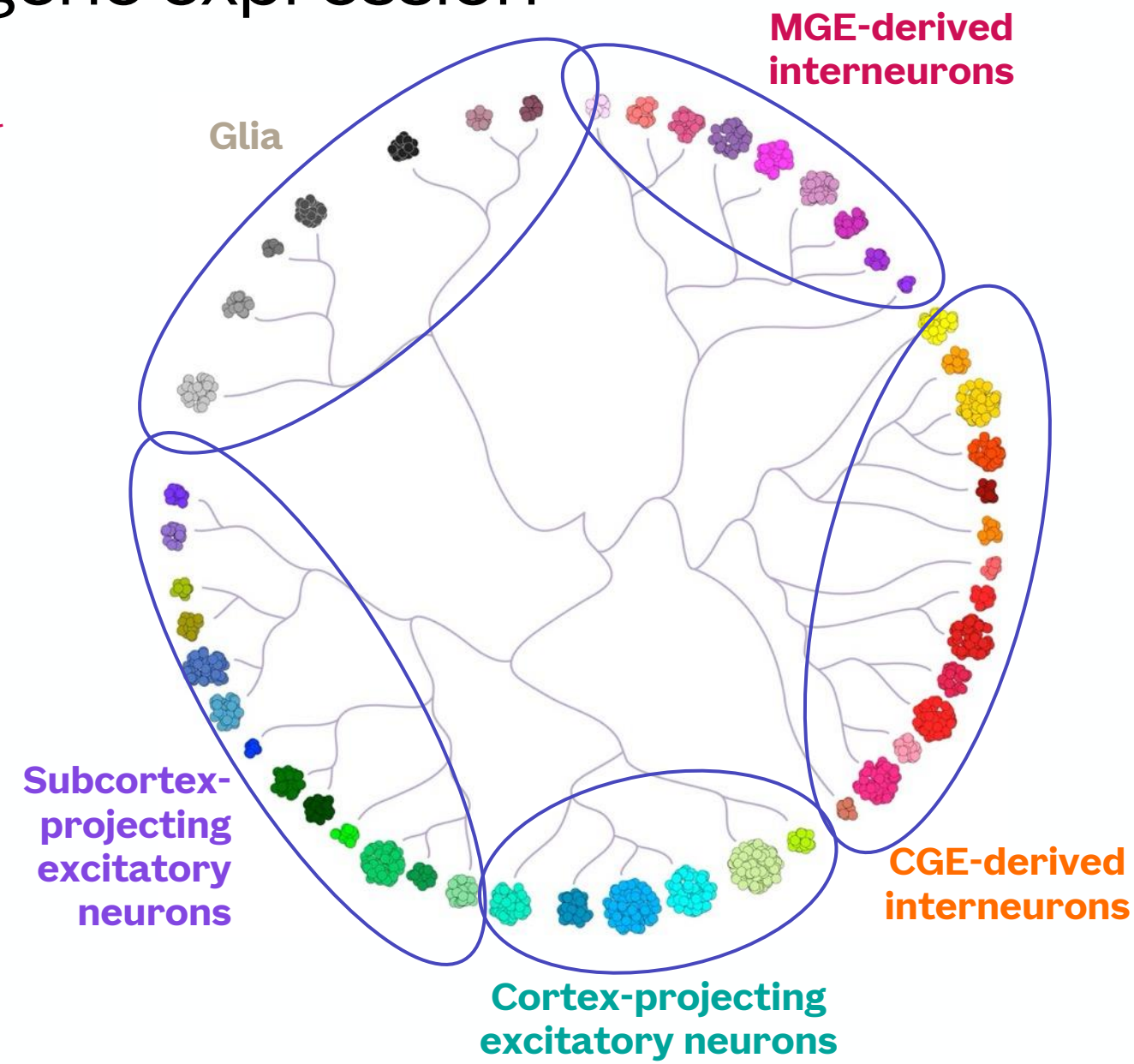
- organized clusters represent groups of cell types related quantitatively



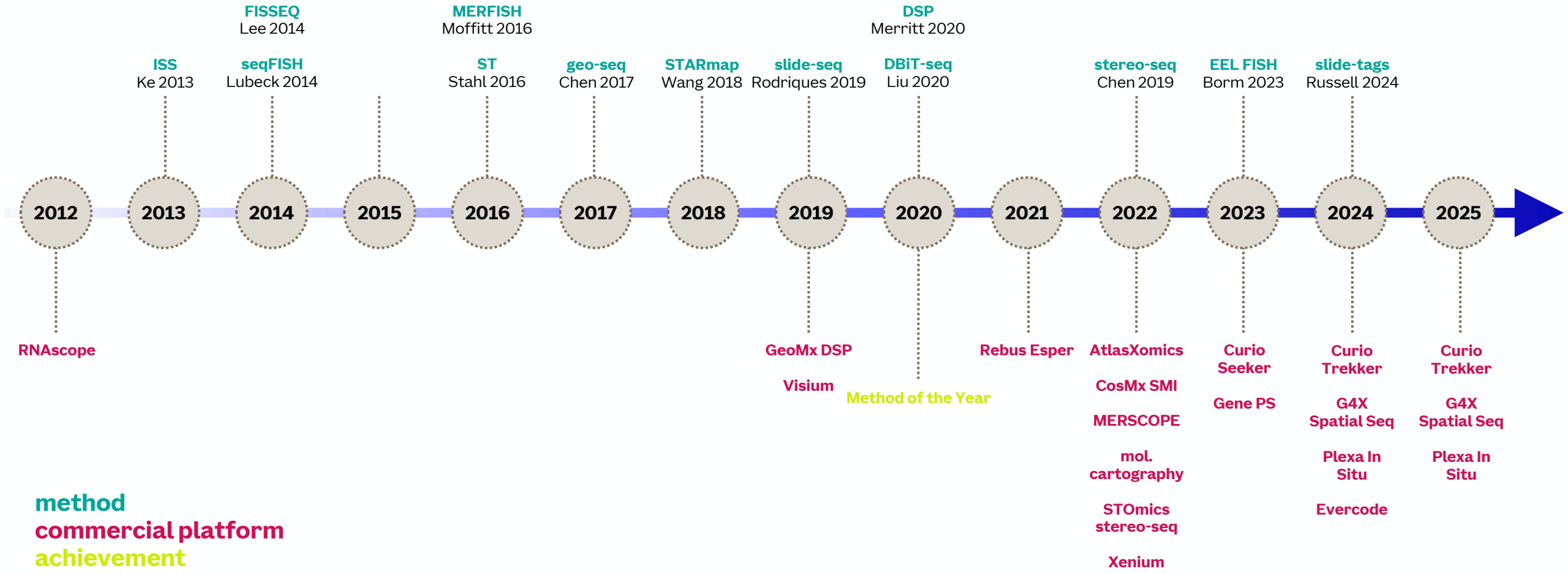
defining cell types/ using gene expression

- organized clusters represent groups of cell types related quantitatively
- prior knowledge and other data modalities help us connect cell types
  - links the math with what is already known about brain cells
  - **spatial transcriptomics** can provide context about the actual location of cells in heterogeneous samples as we annotate the cell types

NOTE: the data, analysis, and associated meta-data stored in a standard format defines a cell type taxonomy



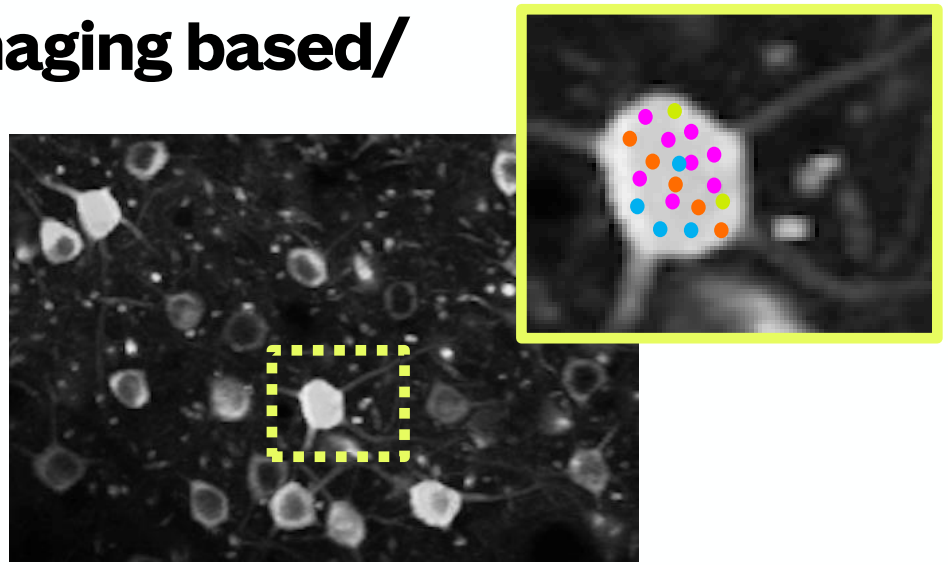
spatial transcriptomics/ timeline



method
commercial platform
achievement

spatial transcriptomics/ methods

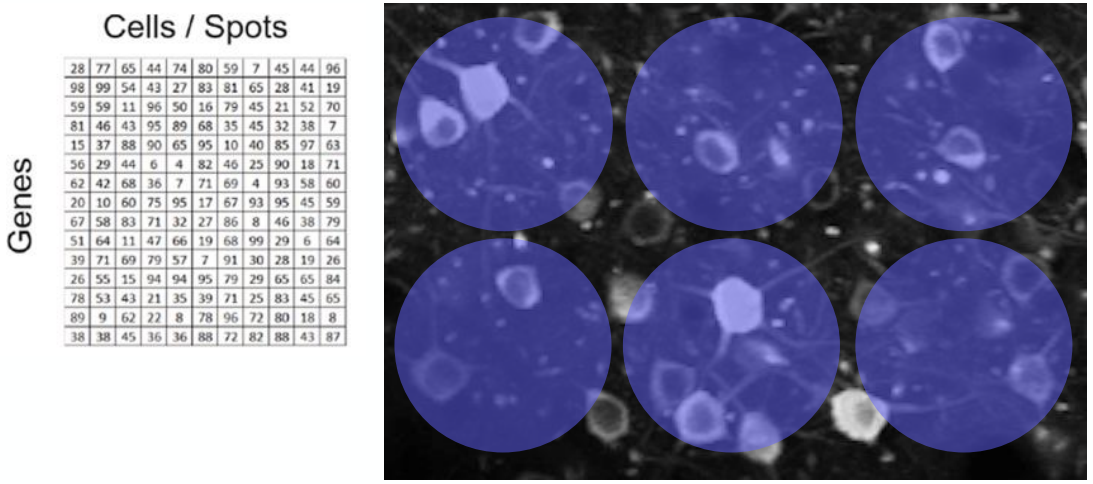
imaging based/



- measure expression of preselected genes in every individual cell
- like multiplexed in situ hybridization
- data challenges: cell segmentation, noisy data, junk proteins (aged human)

- **We do imaging based at the Allen Institute! (MERFISH and Xenium)**

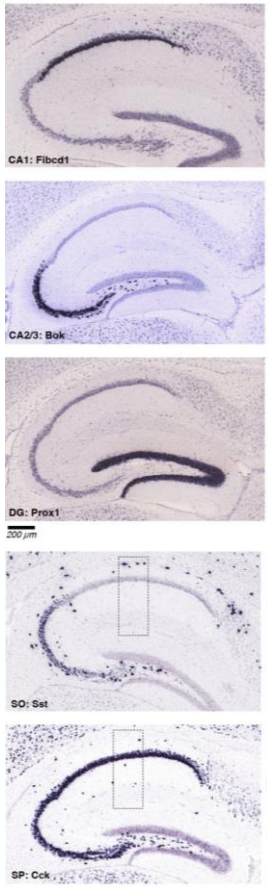
sequencing based/



- measure expression of every** genes in capture spots
- microdissection + droplet sequencing
- data challenges: spots often larger than cells, **low gene recovery

cell type atlasing/ key methods/ spatial profiling

***in situ* hybridization/**



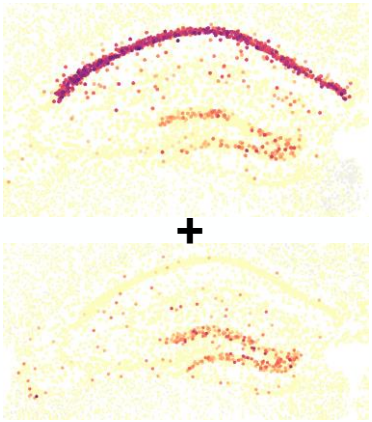
(...)

measure a **single** gene at a time

created by microsoft powerpoint ai image editor (DALL-E 3)

Kaulich et al 2024

spatial transcriptomics/



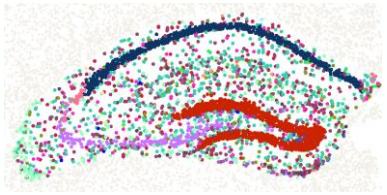
\+

\+

(100-1000 other genes)

=

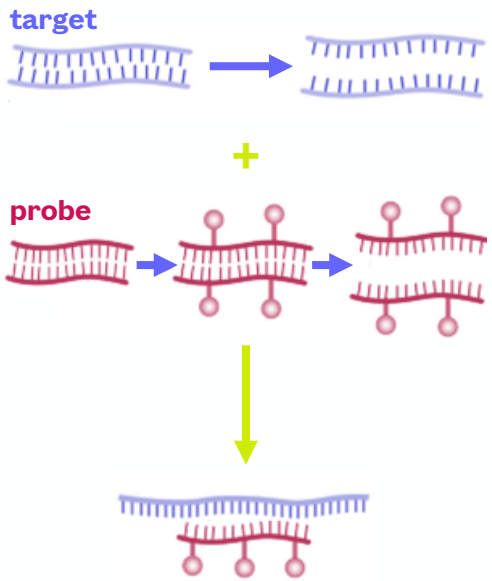
Cell type assignments

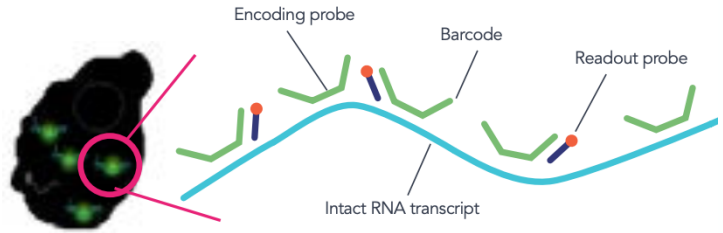


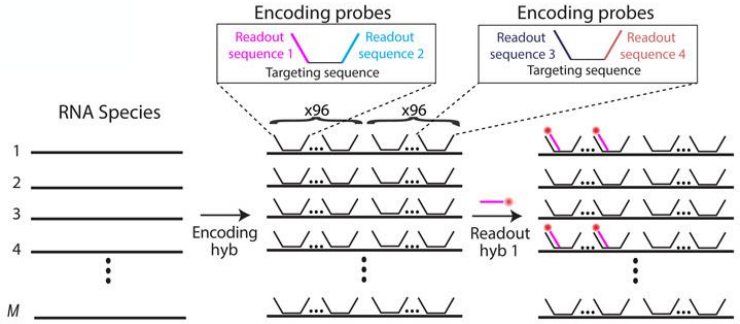
measure **100s-1000s** of genes in **one** experiment

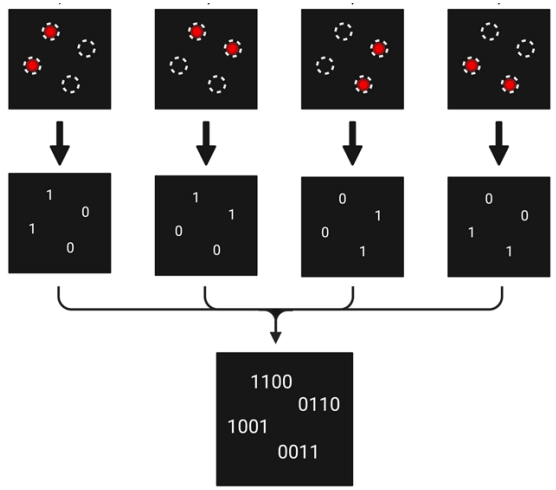
MERFISH (multiplexed error robust in situ hybridization)/

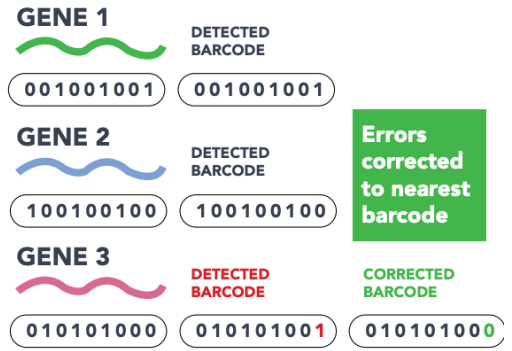
gene panel design → **ISH & imaging** → **sequential imaging** → **barcode identification**





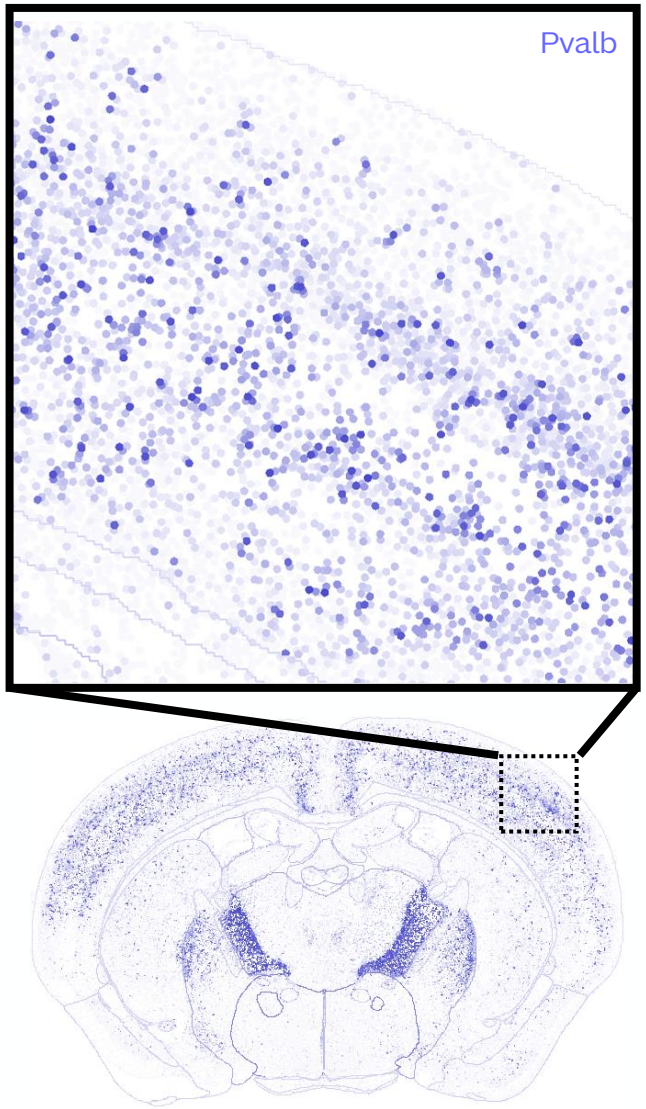




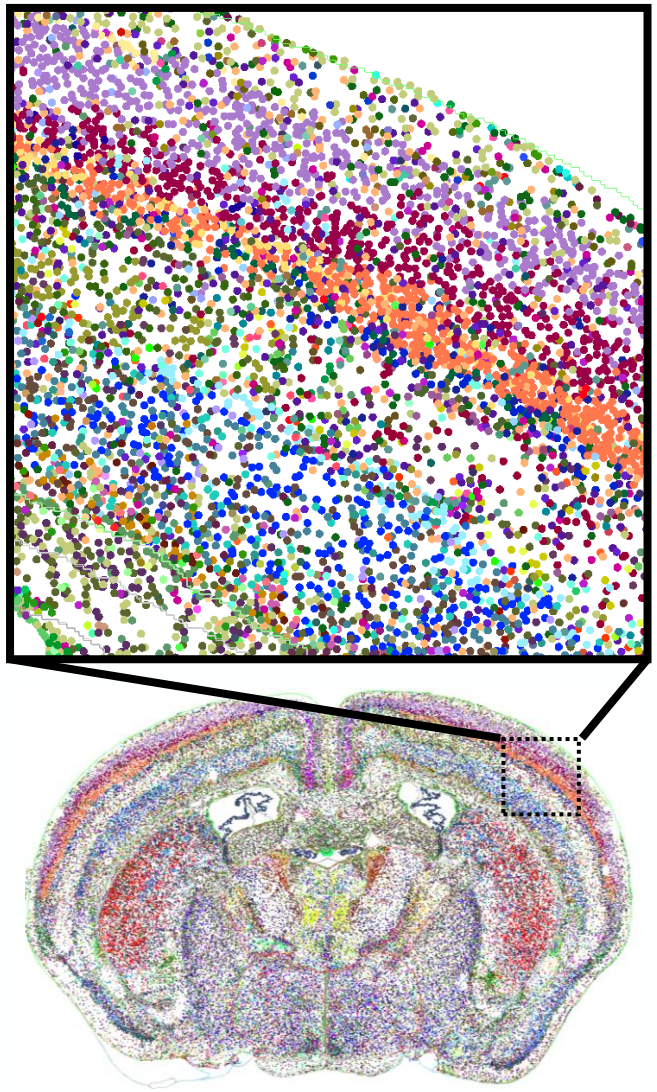


Figures modified frm Chen, K. H. *et al. Science* (2015).; Hu, M., Wu, W. *Clinical Molecular Diagnostics* (2021).; Fortner, A., & Bucur, O. Front. Cell Dev. Biol. (2024); VizGen.

MERFISH/ spatial location of cell types

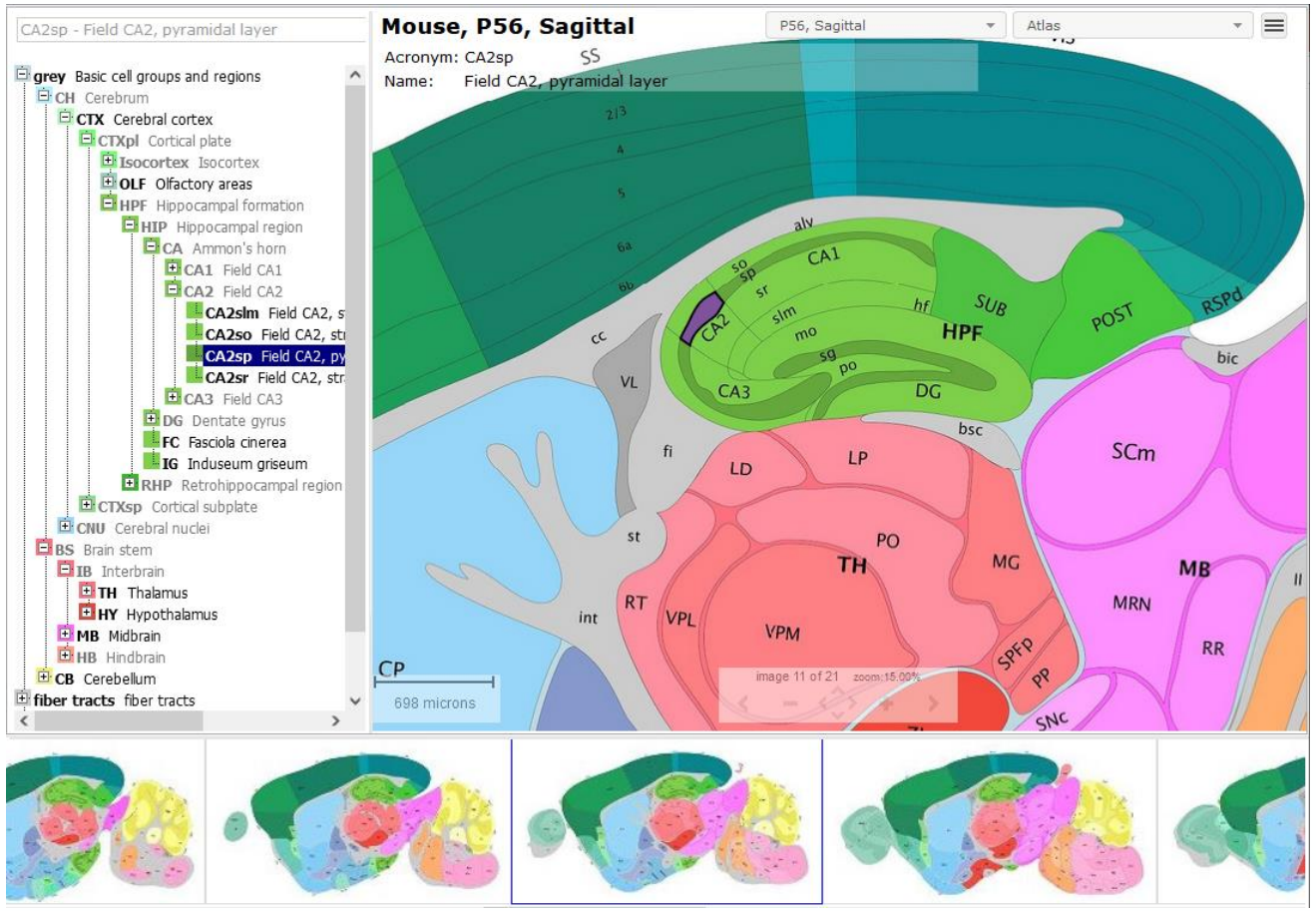


how do we **link** these spatial locations to **brain anatomy**?



Yao et al., 2023

anatomic atlases/ maps of brain regions

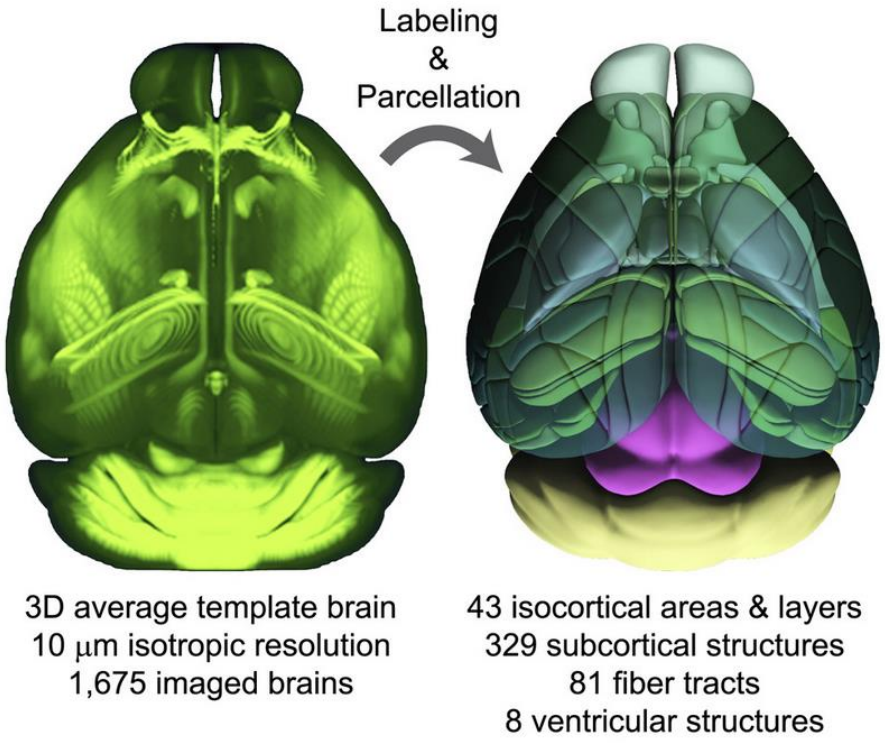


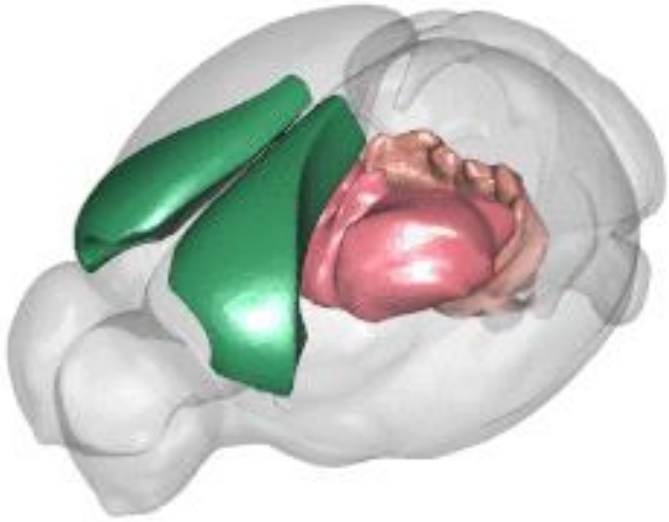
- Hierarchy of brain regions
- Mouse and human
- Adult and development
- 2D and 3D versions

Ng et al 2009, Nature Neuroscience

common coordinate framework (CCF)/

3D space for aligning anything in the brain
(imagine the coordinates of a globe)





compare gene expression, cell type, and anatomy in the 3D space of the CCF

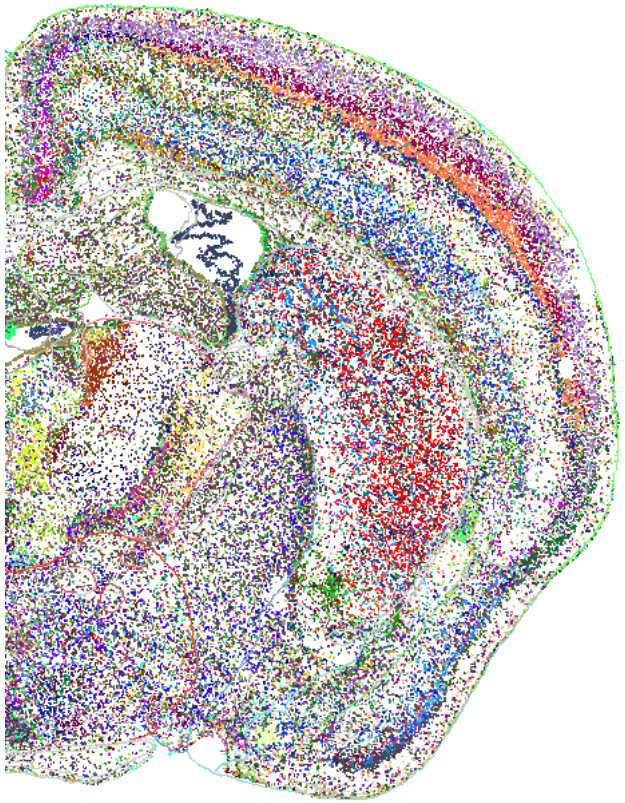
NOTE: For human/NHP, CCF is in MRI space

cell type taxonomy/ annotation

single cell RNA-seq/
what are the cell types

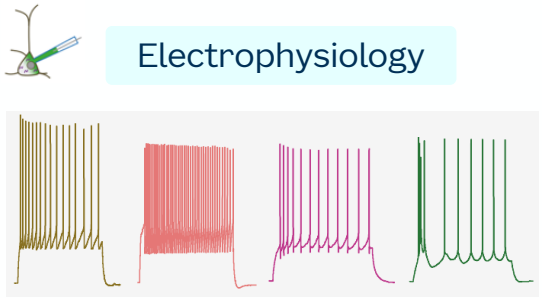
iterate on analysis for quality control & cell type naming

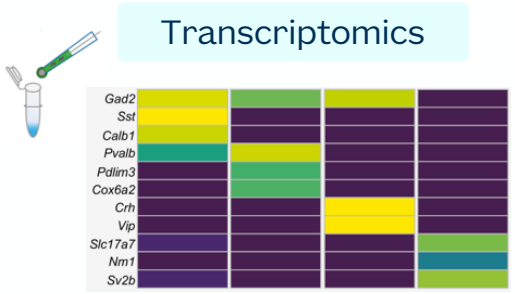
spatial transcriptomics/
where are the cell types

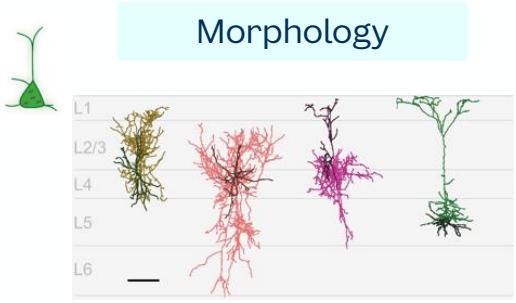


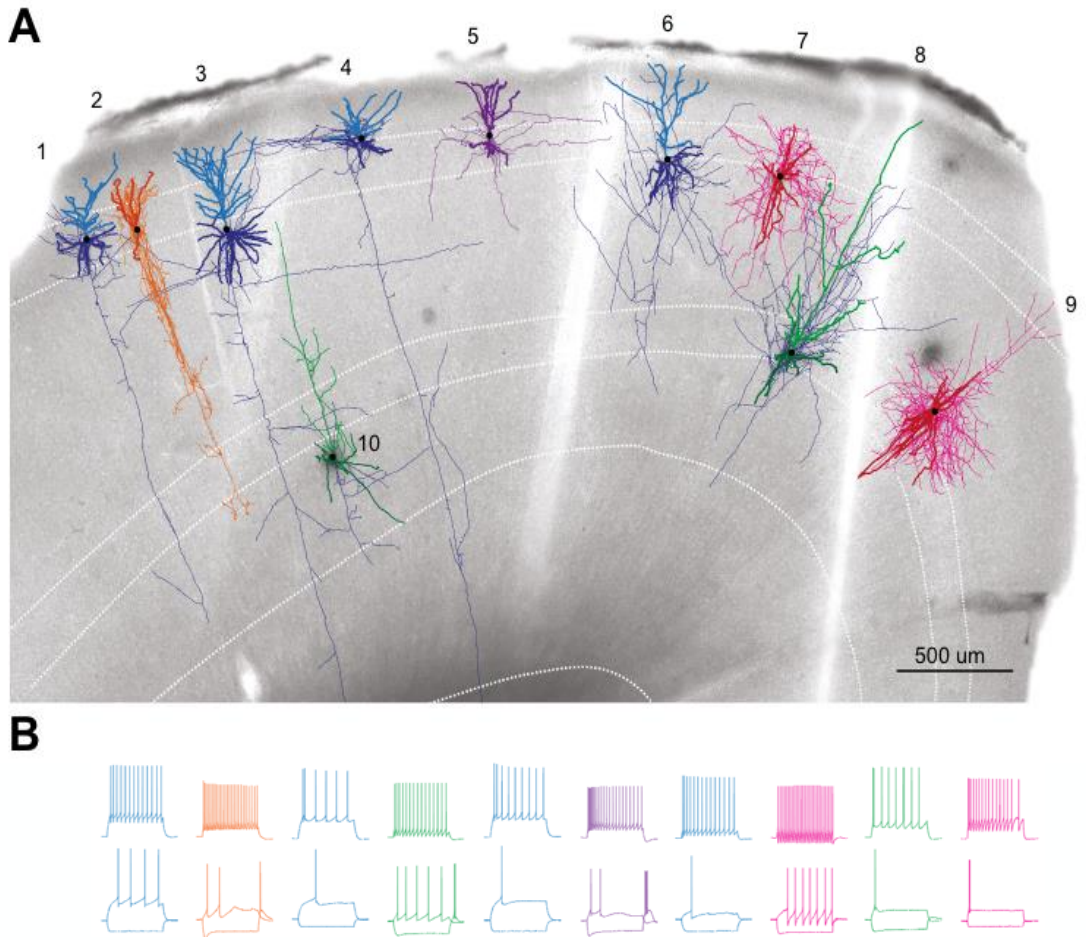
other modalities add context!

patch-seq/ relate morpho-electric diversity to transcriptomic diversity





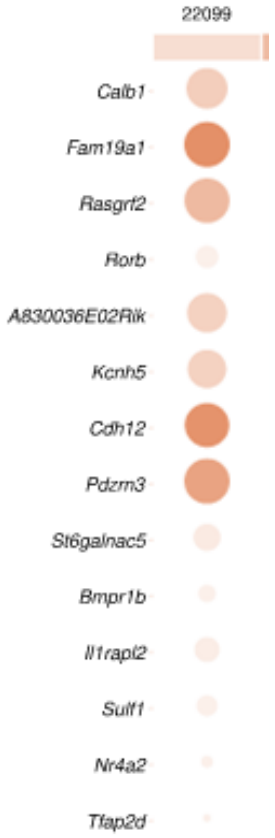




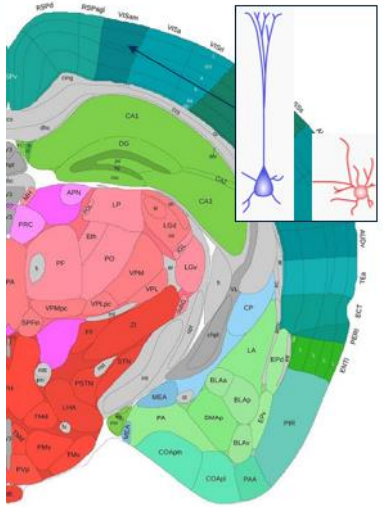
link cell shape and firing properties to cells types defined by gene expression

other modalities/ anchor to gex and location

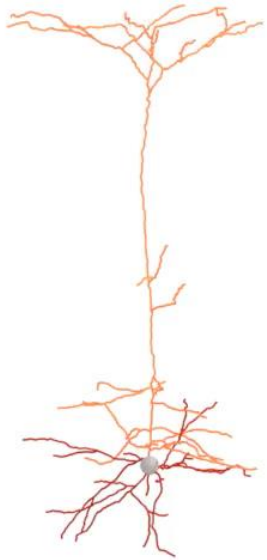
gene expression
(transcriptomics)



cell location
(anatomy)

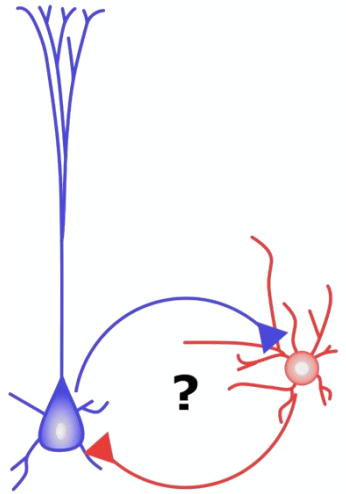


cell shape
(morphology)



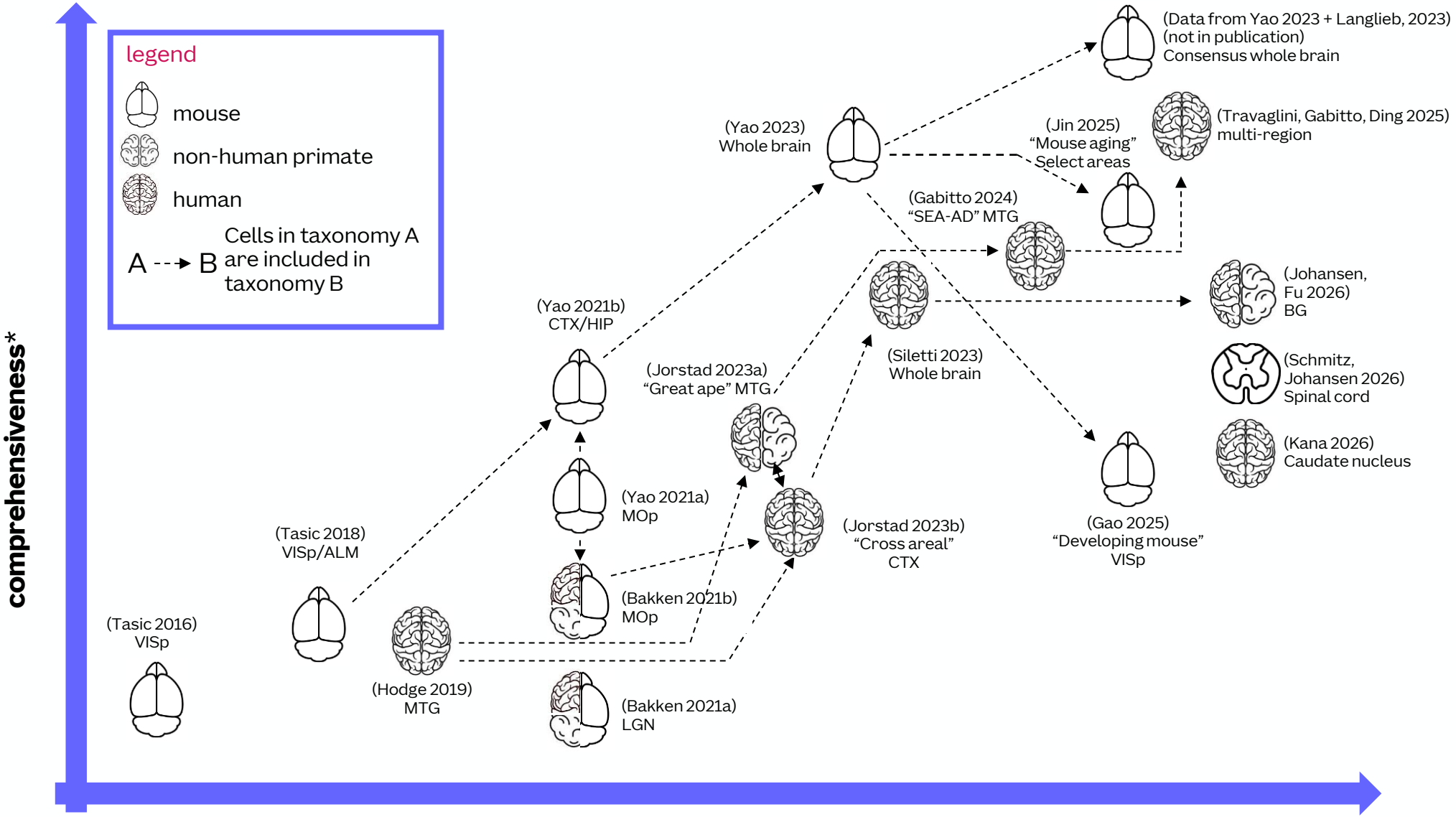
firing properties
(electrophysiology)

cell communication
(connectivity)
+
cell function
(in response to behavior)



chromatic accessibility
(epigenetics / targeting)

allen institute/ cell type taxonomies/ over the last decade



brain-map.org/our-research/cell-types-taxonomies/taxonomies-index

abbreviations
ALM: anterolateral motor cx
BG: basal ganglia
CTX: cortex
HIP: hippocampal formation
LGN: lateral geniculate nucleus
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VISp: primary visual cx

*Comprehensiveness is a qualitative metric combining the number of cells, regions, modalities, and species included in a cell type taxonomy.

Some icon positions are slightly adjusted for readability.

agenda/morning

9:00-9:30 | check-in / breakfast

9:30-9:45 | welcome, logistics, and overview of allen institute (Rachel)

9:45-10:30 | introduction to transcriptomics for cell typing (Jeremy)

10:30-10:45 | break

10:45-12:00 | cell type data sets on allen brain map

- overview of mouse data sets – Rachel
- cross-primate healthy brain atlases – Jeremy
- cell typing in aged humans, SEA-AD and beyond - Irika

12:00-1:00 | lunch

github.com/AllenInstitute/SPOCTT/tree/main/2026_Additional_Presentations

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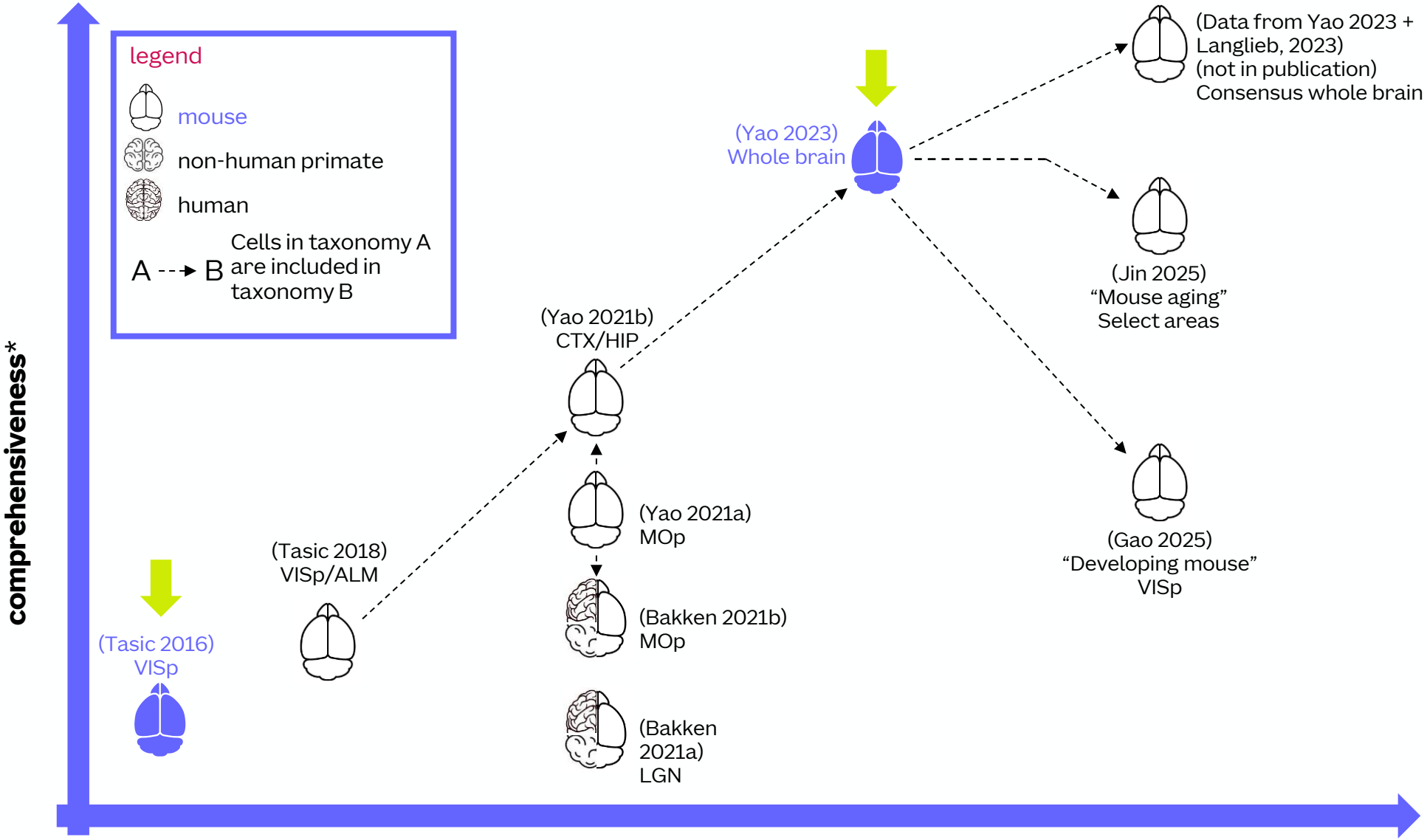
github.com/AllenInstitute/SPOCTT/tree/main/2026_Additional_Presentations

overview/ cell type data/ mouse

- Adult mouse brain
- Aging and development

alleninstitute/

allen institute/ cell type taxonomies/ mouse



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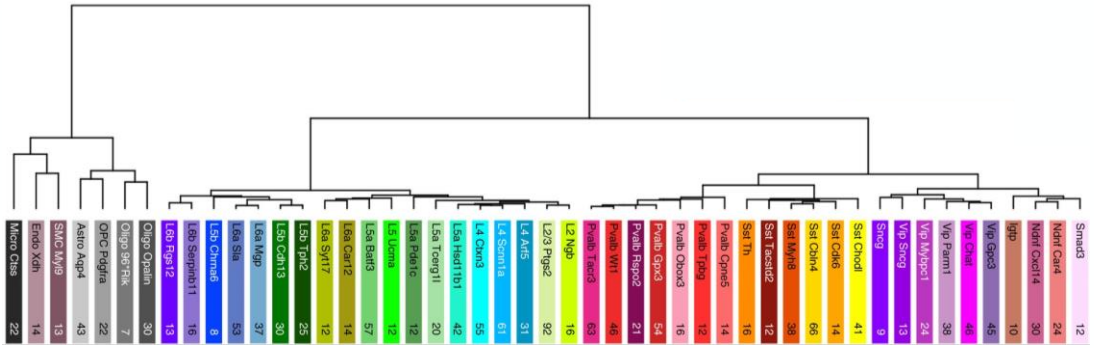
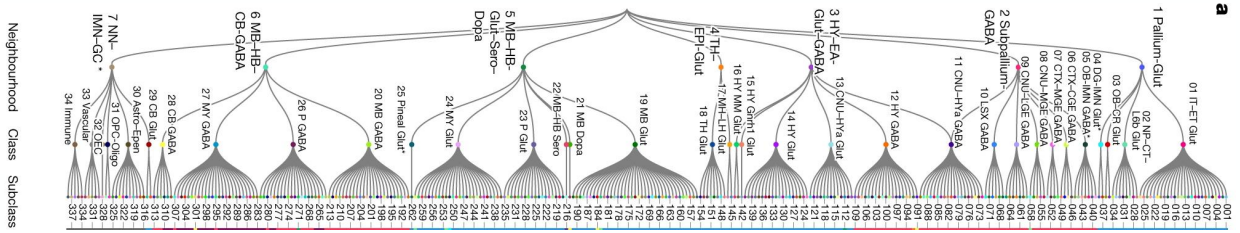
Tasic et al., 2016 → Yao et al., 2023

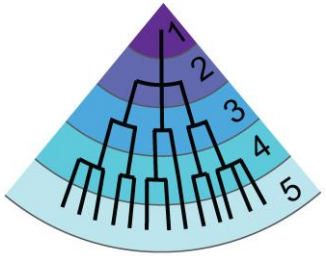
~ 1,600 cells
49 cell types
mouse visual cortex



~4 million cells
5,322 cell types
mouse whole brain

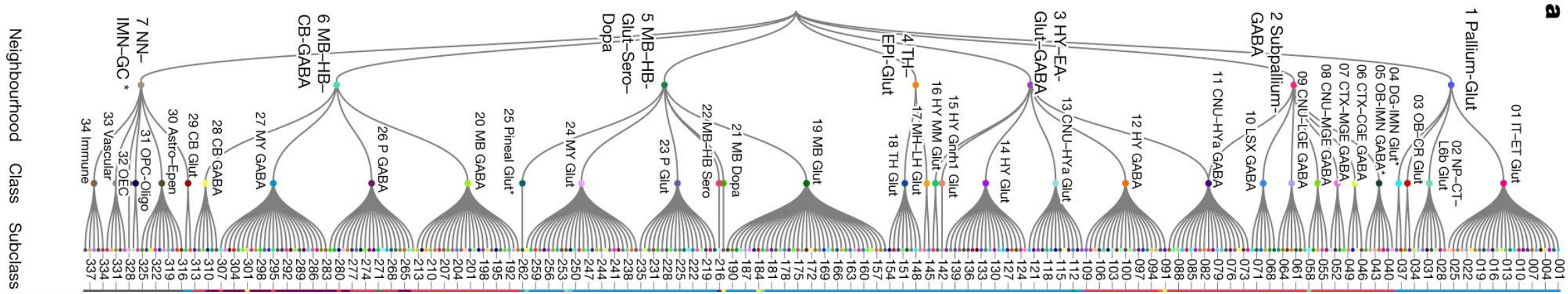


only first 3 levels shown



- Level 1: Neighborhood (n=7)
- Level 2: Class (n=34)
- Level 3: Subclass (n=338)
- Level 4: Supertype (n=1201)
- Level 5: Cluster (n=5322)

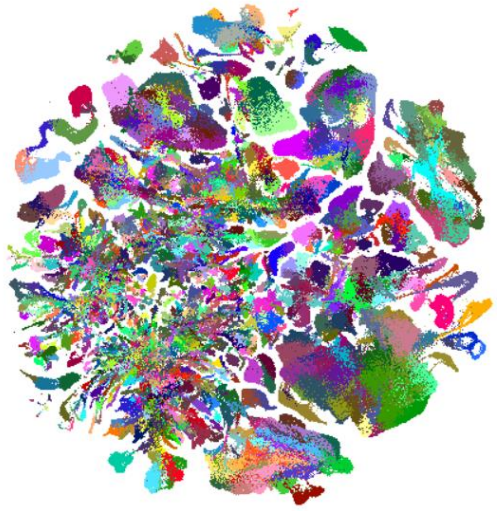
mouse whole brain taxonomy/



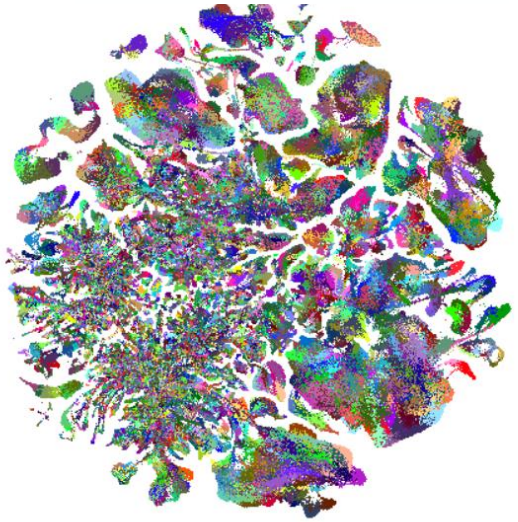
34 classes

338 subclasses

1,201 supertypes



5,322 clusters



plots made with ABC Atlas!

Yao et al., 2023

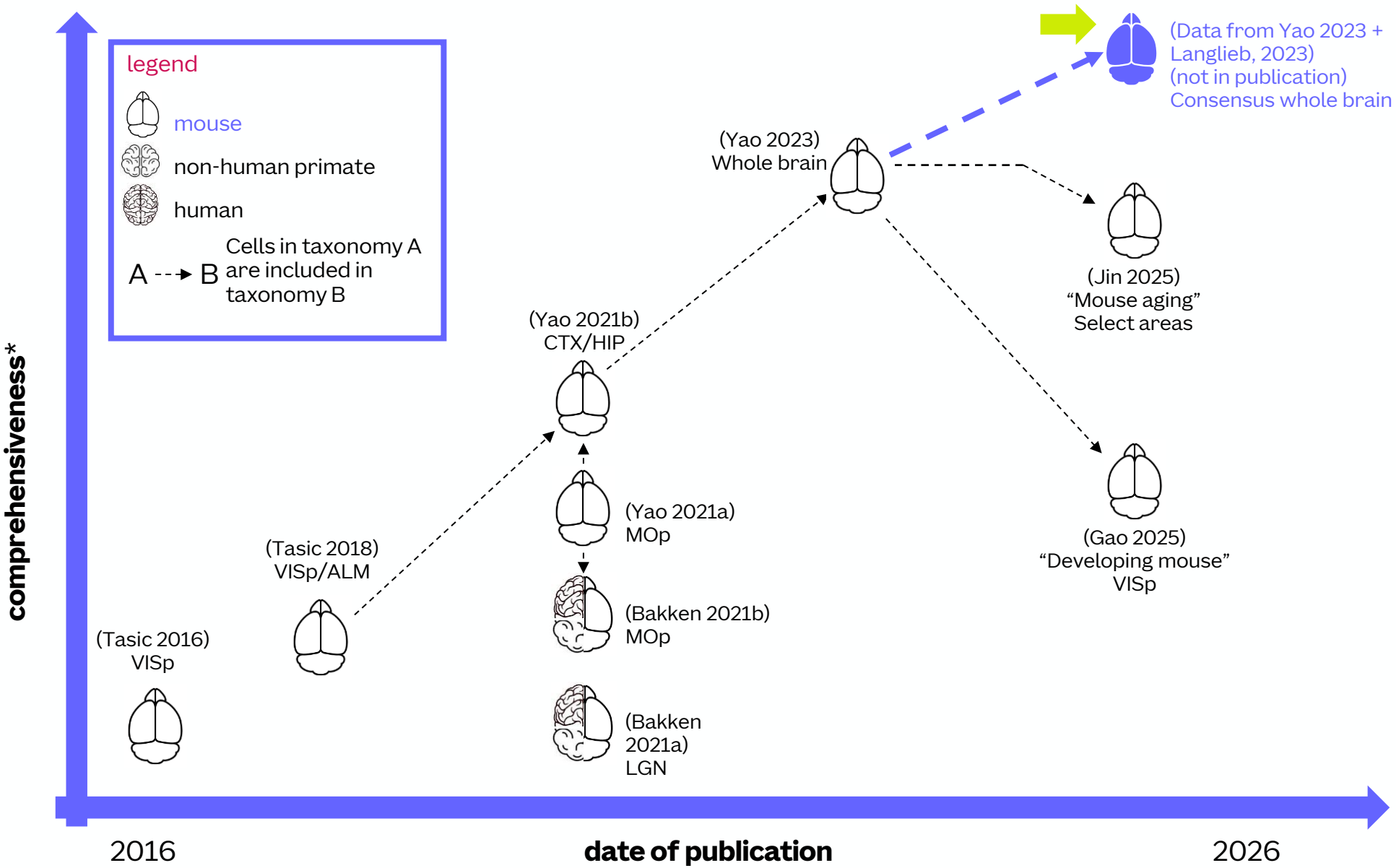
available data/ whole mouse brain

method	institution	Description	paper	ABC Atlas dataset	# genes	# donors
10X Genomics scRNA-seq	allen institute	Mouse whole-brain transcriptomic cell type atlas (Hongkui Zeng)	Yao et al., Nature (2023)	"10x ScRNAseq whole brain"	31,113	317
MERFISH				"MERFISH-C57BL6J-638850"	500	1 (coronal)
				"MERFISH-C57BL6J-638850 with Imputed genes"	8460 (scRNAseq)	
MERFISH	Harvard University	A molecularly defined and spatially resolved cell atlas of the whole mouse brain (Xiaowei Zhuang)	Zhang et al., Nature (2023)	"Zhuang-ABCA-1/2/3/4"	1,122	4 (coronal & sagittal)

NOTE: both MERFISH datasets were integrated with the 10x scRNAseq data and have the same taxonomy in the ABC Atlas (34 classes, 338 subclasses, 1,201 supertypes and 5,322 clusters)

Yao, Langlieb, and Zhang papers are part of a 10-paper Nature package, published December 2023 nature.com/collections/fgihbeccbd

allen institute/ cell type taxonomies/ mouse



brain-map.org/our-research/cell-types-taxonomies/taxonomies-index

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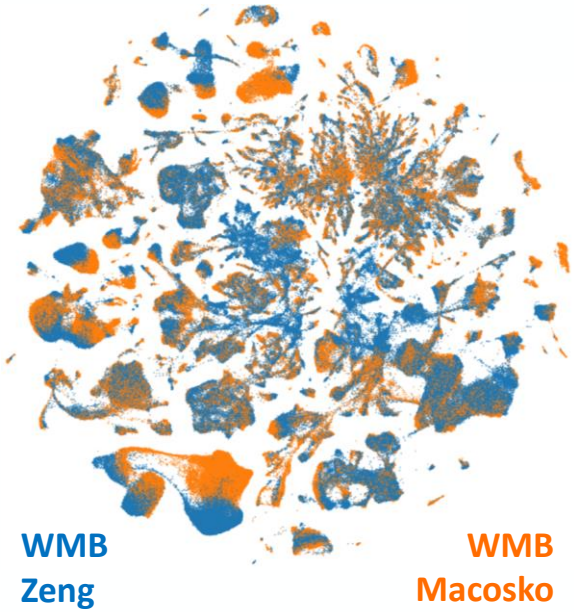
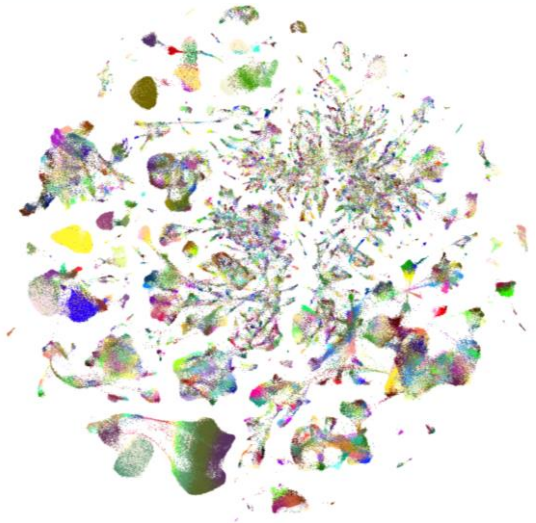
available data/ consensus/ mouse whole brain

method	institution	description	paper	ABC Atlas dataset	# genes	# donors
10X Genomics scRNA-seq	allen institute	Consensus Whole Mouse Brain dataset (Macosko, Zeng)	Yao et al., 2023 – Nature	"Consensus-WMB-integrated-taxonomy"	31,113	317
10X Genomics snRNAseq	Broad Institute		Langlieb et al., 2023		21,205	55



43 classes
414 subclasses
1,386 supertypes
6,721 clusters

~7 million cells



6,721 clusters

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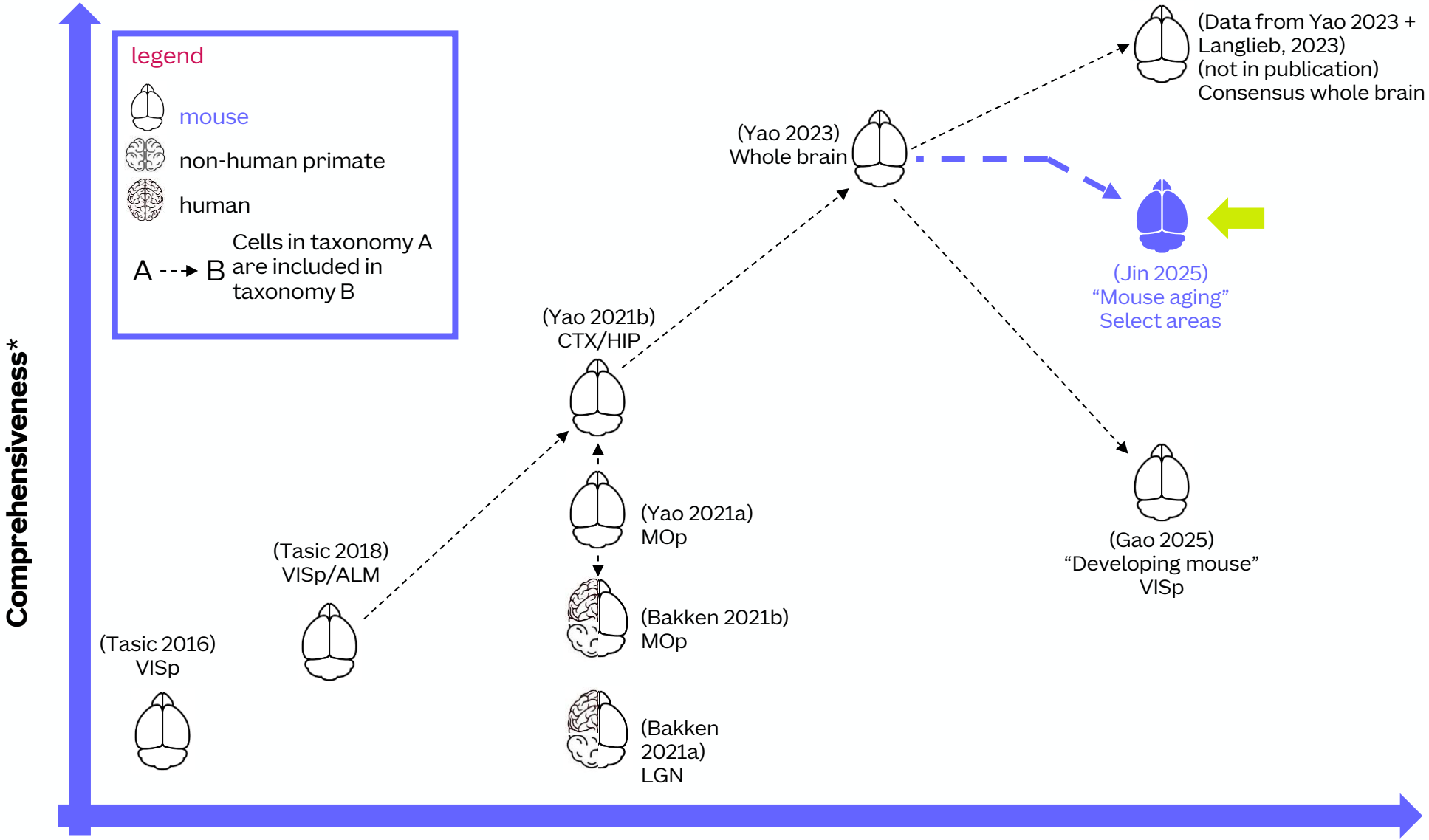
allen institute/ cell type taxonomies/ mouse



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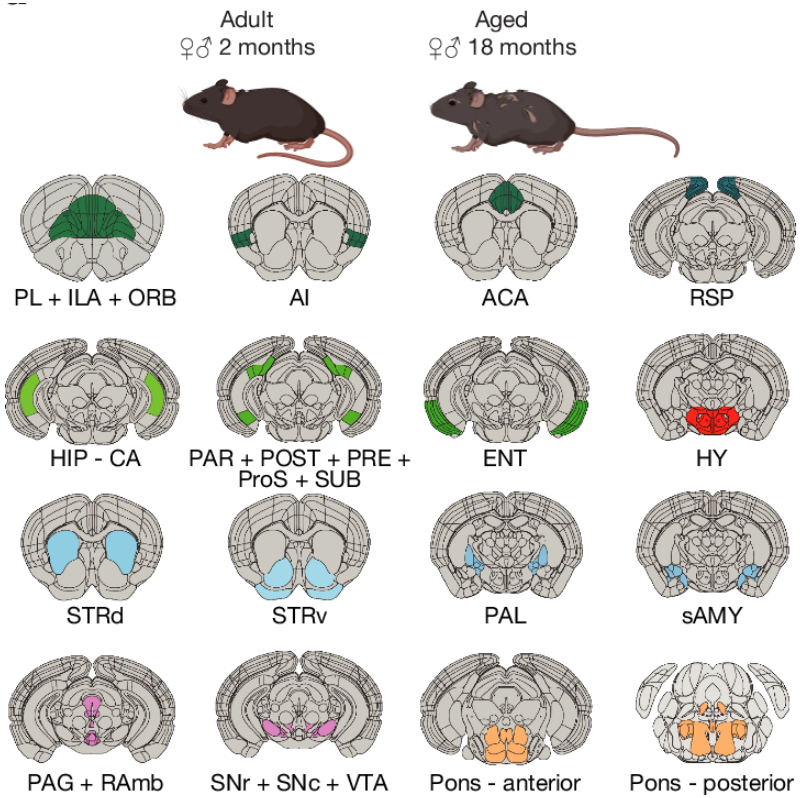
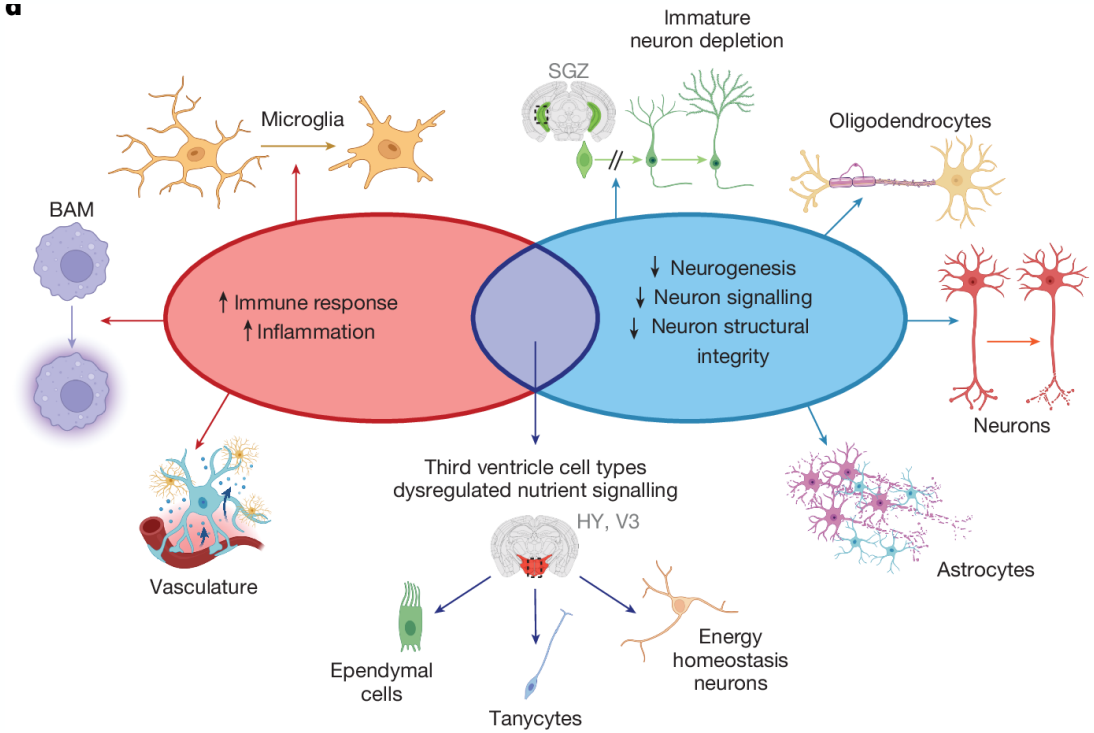
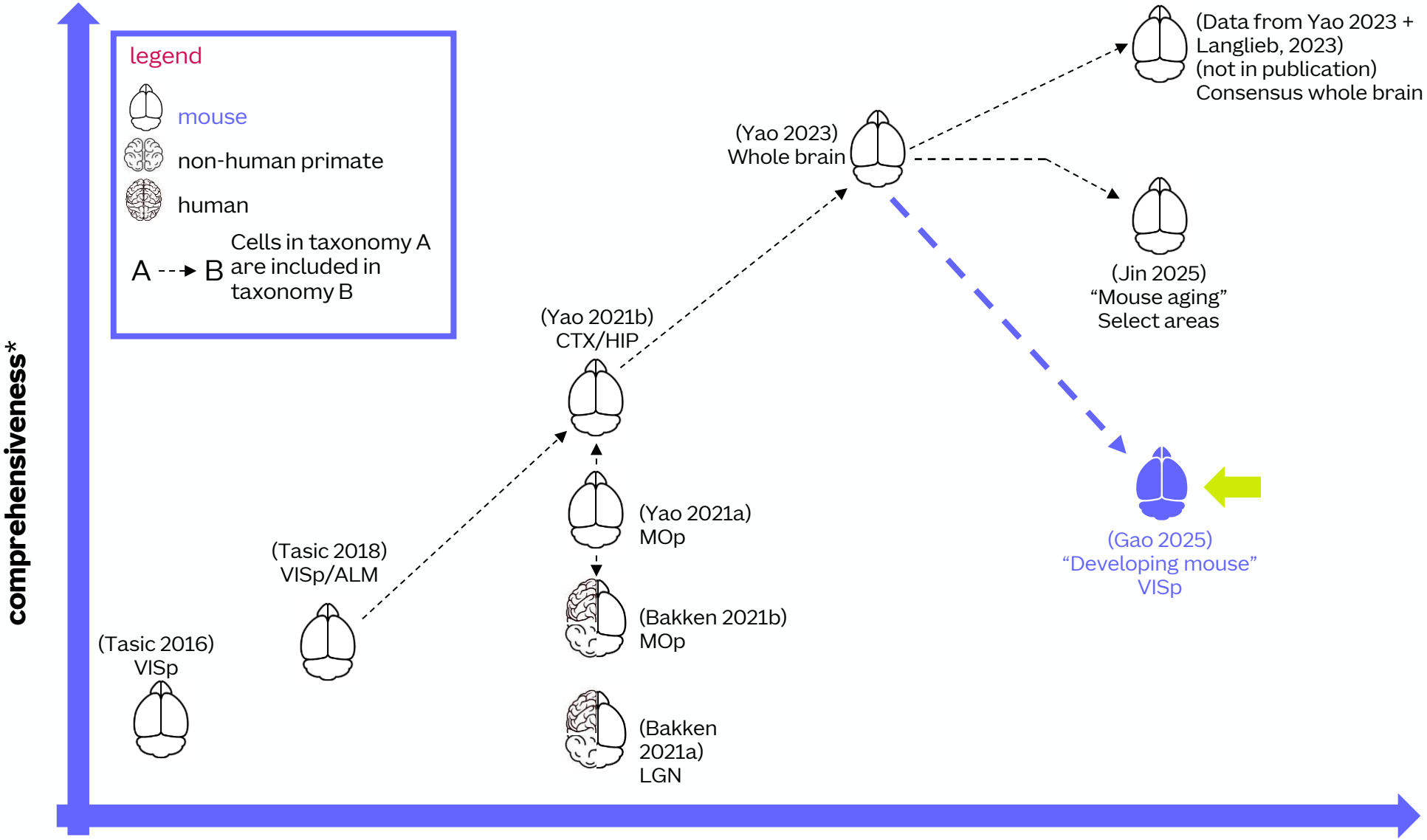
available data/ aging mouse brain

method	institution	description	paper	ABC Atlas dataset	# genes	# donors
10X Genomics scRNA-seq	allen institute	Cellular and Molecular Characterization of the Aged Mouse Brain	Jin et al., Nature (2025)	“10x-scRNAseq-aged-adult”	30,824	44 aged (18 months), 64 young adult mice (from whole mouse brain data Yao et al., 2023)





allen institute/ cell type taxonomies/ mouse



brain-map.org/our-research/cell-types-taxonomies/taxonomies-index

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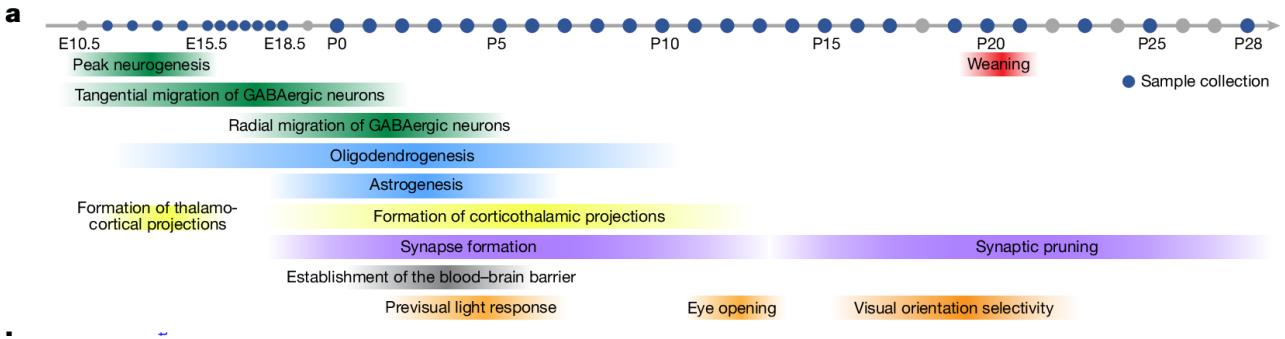
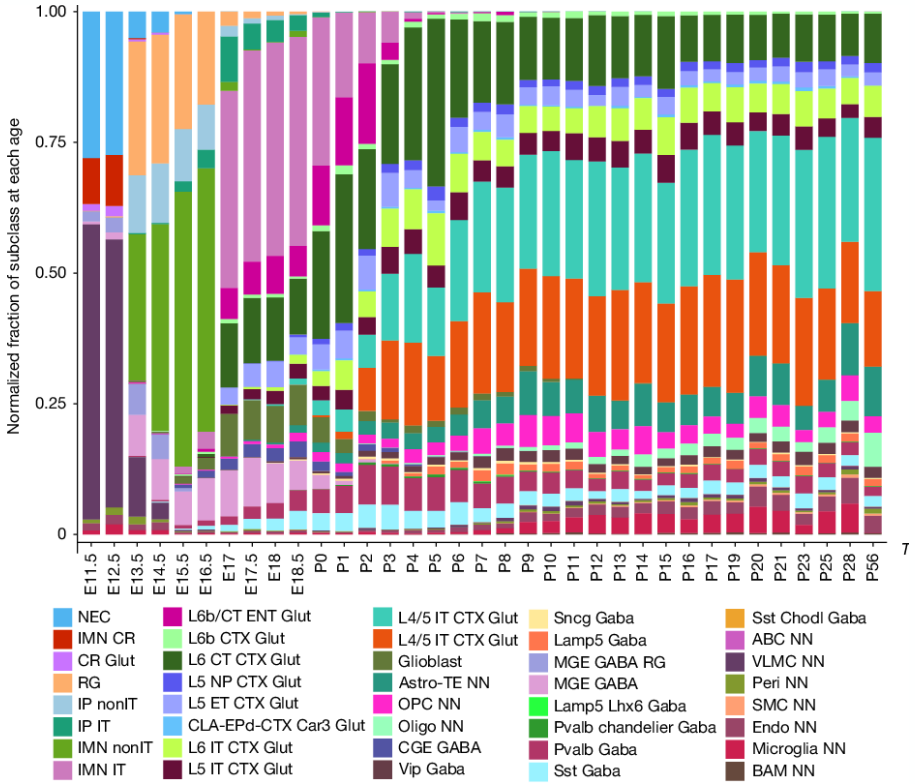
available data/ developing mouse visual cortex

method	institution	description	paper	ABC Atlas dataset	# genes	# donors
10X Genomics scRNA-seq	allen institute	Developing Mouse - Visual Cortex	Gao et al., Nature (2025)	“Dev-Mouse-Vis-Cortex-taxonomy”	32,285	53 (from E11.5 to young adult P56 (from WMB Zeng data))



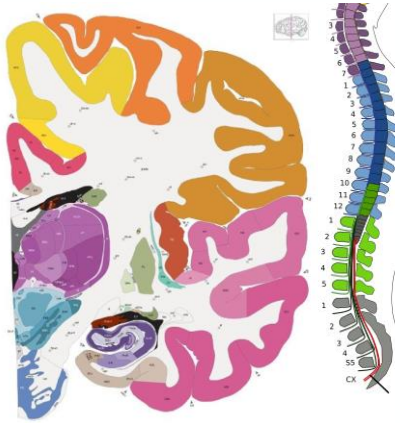


overview/ cell type data/ cross primate (healthy)

- studies in healthy human
- cross-species basal ganglia
- extending to whole brain and spinal cord

alleninstitute/

cross-species brain cell atlases as foundational references/ building on NIH Brain Initiative



>3000 cell types survey of cell type diversity in ~100 regions of the human brain and spinal cord in 3 individuals

Siletti, Hodge...Lein, Linnarsson (Science, 2023)

HMBA

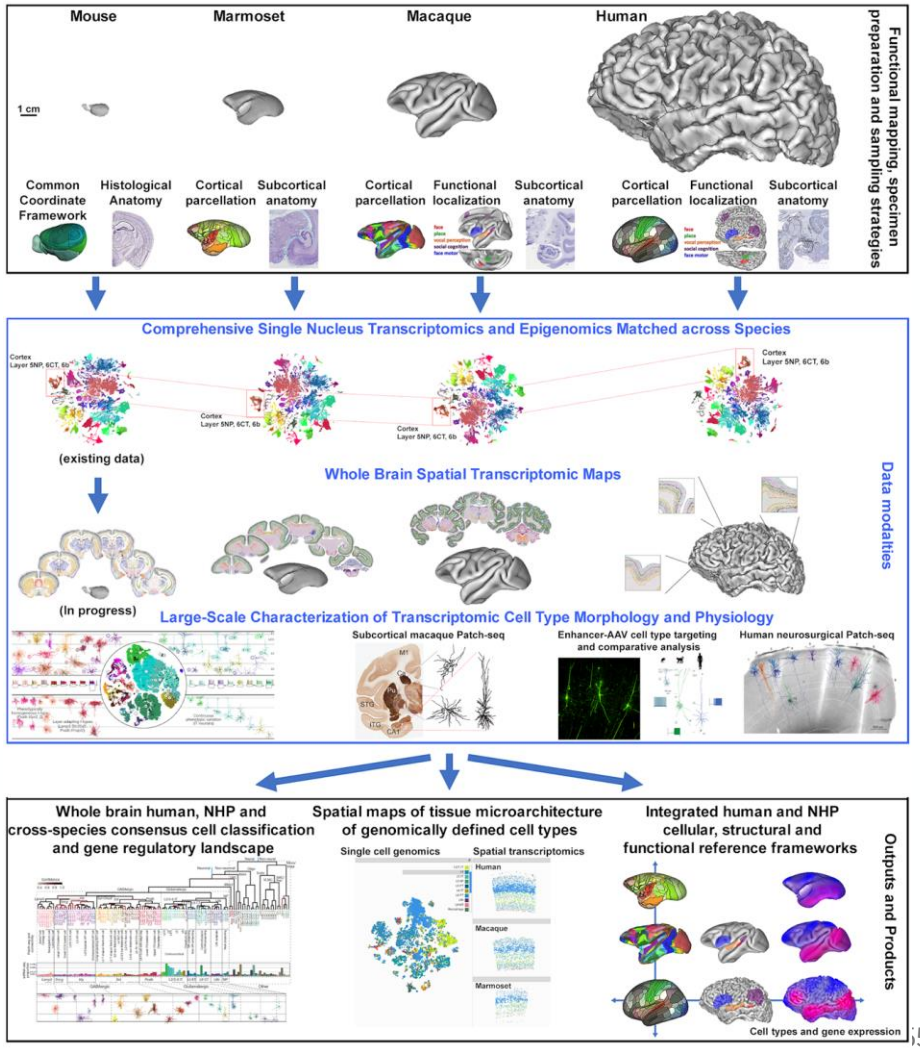
fMRI

single cell/nucleus genomics

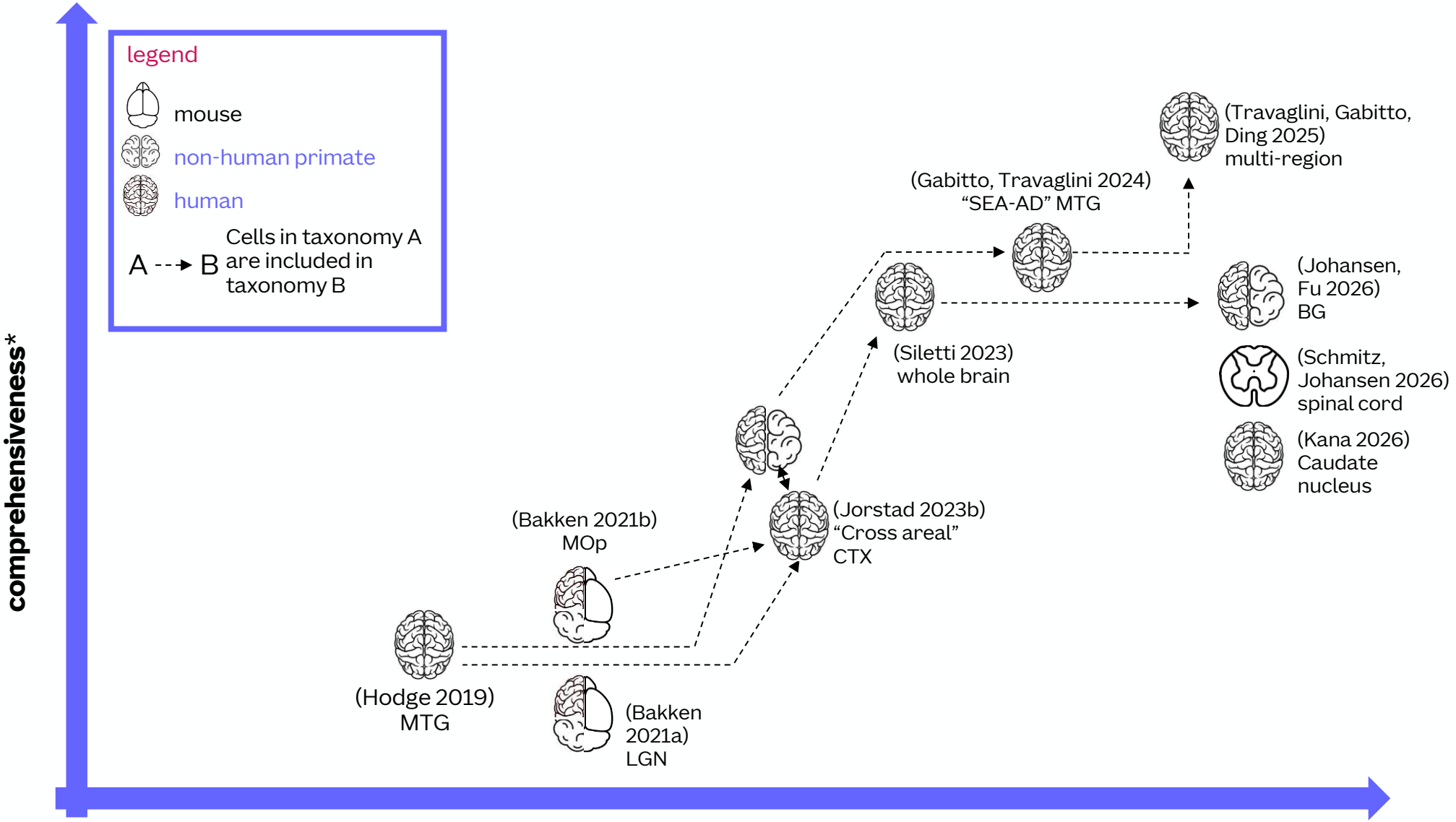
spatial transcriptomics

patch-seq

products (tools/data for scientist)



allen institute/ cell type taxonomies/ NHP & human



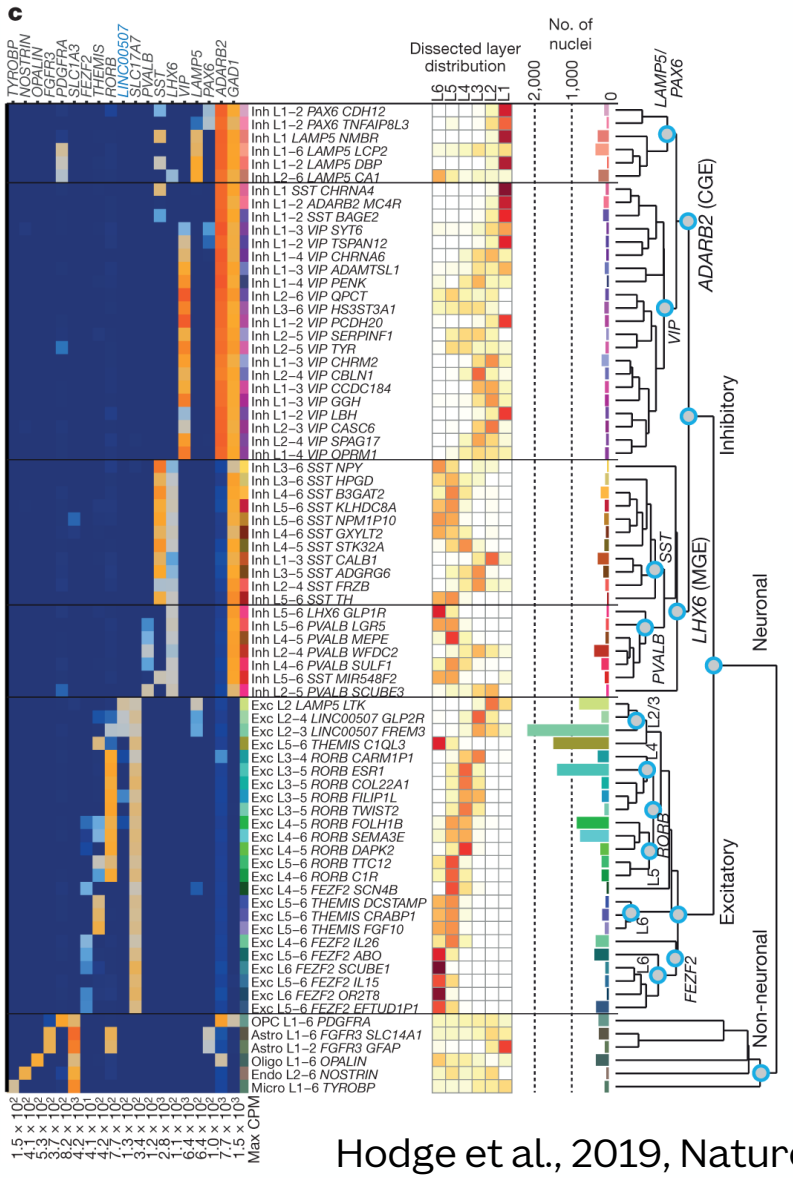
brain-map.org/our-research/cell-types-taxonomies/taxonomies-index

abbreviations
ALM: anterolateral motor ctx
BG: basal ganglia
CTX: cortex
HIP: hippocampal formation
LGN: lateral geniculate nucleus
MOp: primary motor ctx
VISp: primary visual ctx

*Comprehensiveness is a qualitative metric combining the number of cells, regions, modalities, and species included in a cell type taxonomy.

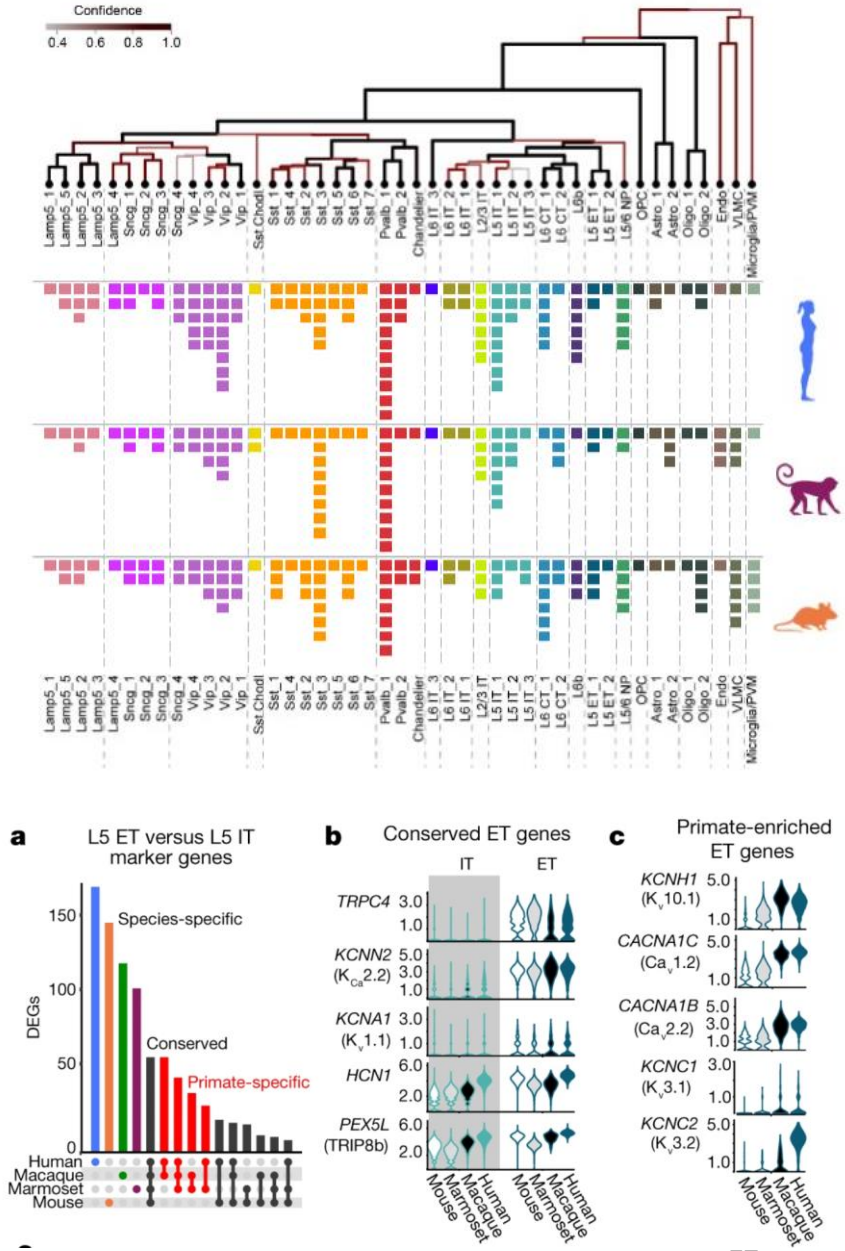
Some icon positions are slightly adjusted for readability.

historic studies/



Hodge et al., 2019, Nature

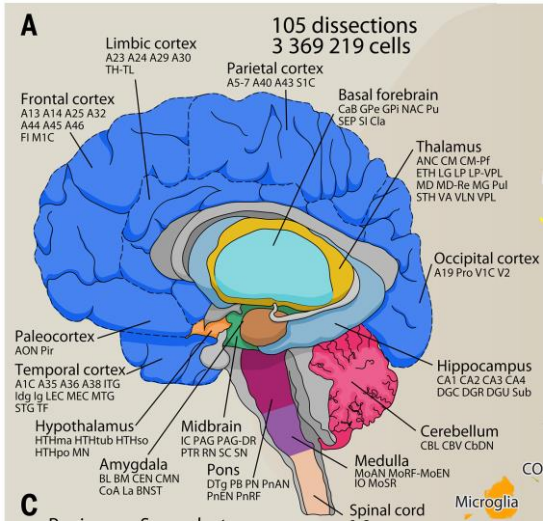
- First snRNA-seq study from the Allen Institute
- ~15,000 cells spanning 75 cell types
- Middle temporal gyrus selected for biological (important for diseases) and practical reasons (neurosurgical tissue available)
  - Part of major cross-species, multi-modal look at a cortical area (primary motor cortex)
  - ~45 conserved cross-species types, with some notable species differences
  - Cell Type Knowledge Explorer includes these data.



Bakken et al., 2021, Nature

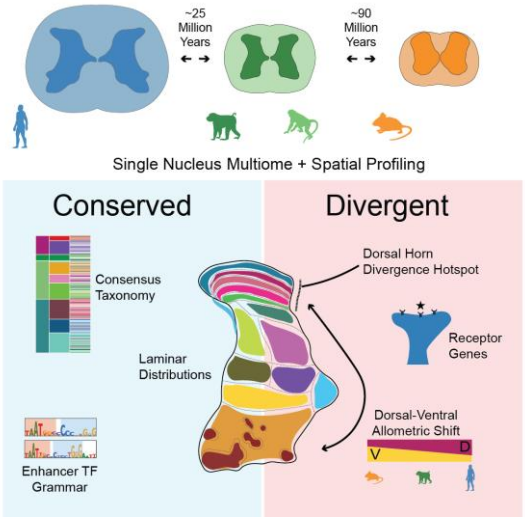
available data/ whole brain & spinal cord

method	institution	description	paper	ABC Atlas dataset	# genes	# donors
10X Genomics snRNA-seq	allen institute	human neurons	Siletti et al., Science (2023)	“Human Brain Cell Type Diversity – Neurons”	51,820	4 human adults (3 whole brain, 1 M1 only)
		non-neuronal cells		“Human Brain Cell Type Diversity – Non-Neuronal Cells”		
10X Genomics snRNA-seq	allen institute	human & NHP spinal cord	Schmitz, Johansen et al. Preprint (2026)	TBA	TBA	3 human; 7 NHP



Siletti paper is part of a 21-paper Science package, published October 2023

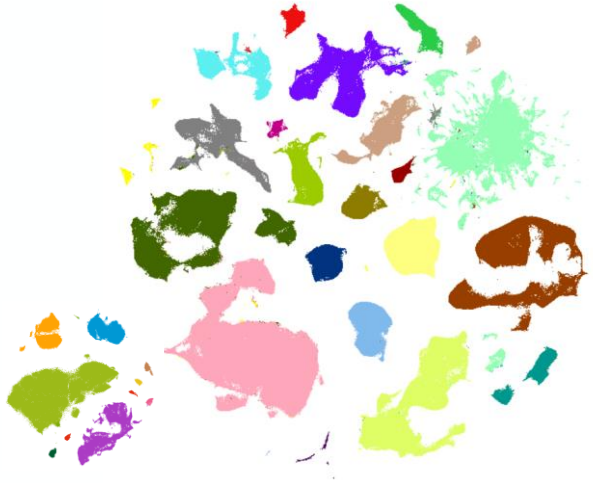
science.org/collections/brain-cell-census



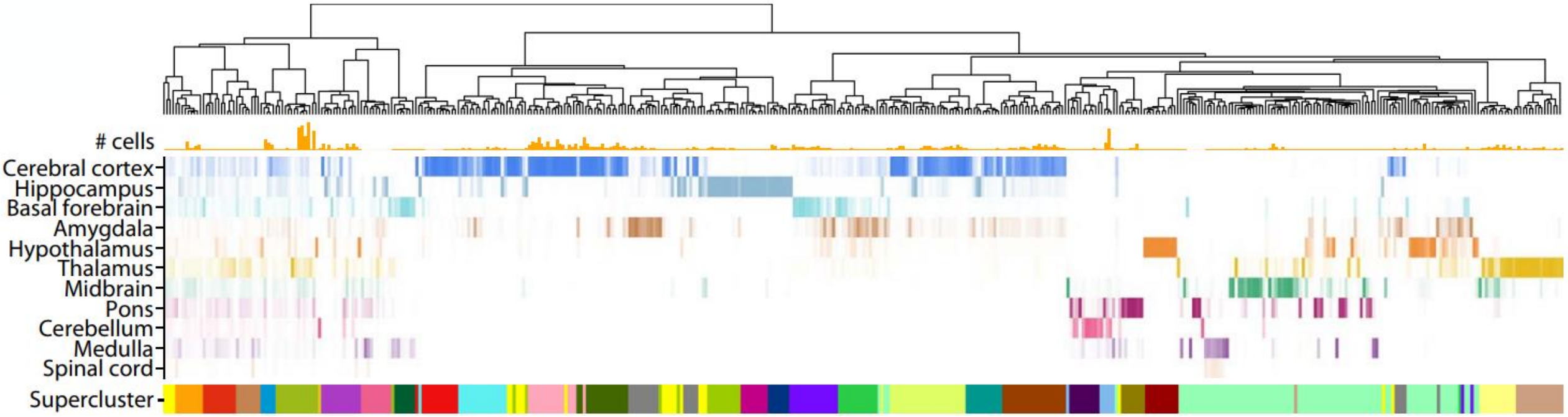
Schmitz, Johansen paper is part of a submitted publication package, 2026

keep an eye out! to be released soon

whole human brain taxonomy/



~3.3 million nuclei from 105 anatomical dissections



cross-species HMBA/ basal ganglia

keep an eye out! publication package will be released soon

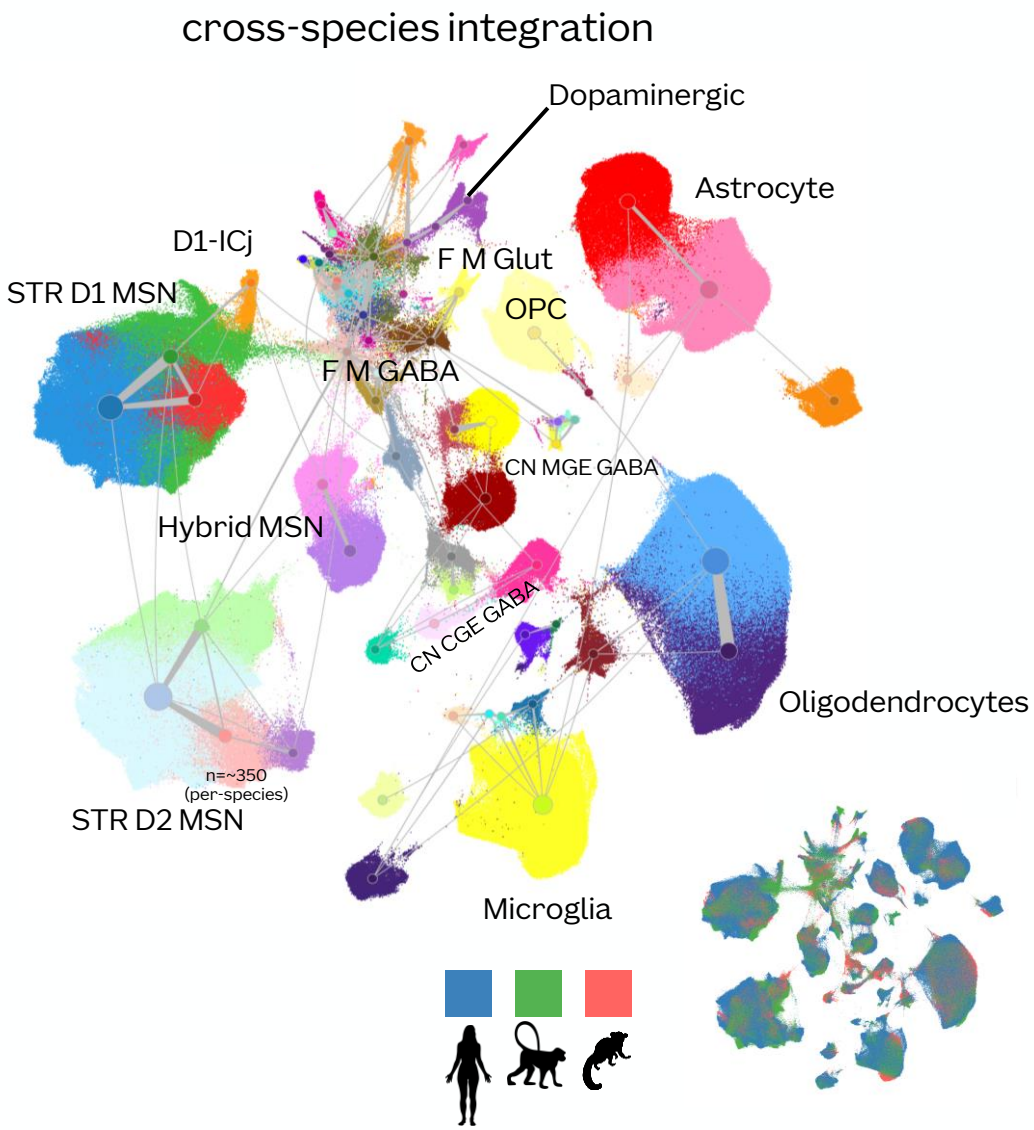
species	method	paper	ABC Atlas dataset	# genes	# donors
human, macaque, marmoset	10X Genomics snMultiome-seq	Johansen & Fu et al. bioRxiv (2026)	"Pre-print HMBA-10XMultiome-BG-Aligned"	16,630 orthologs*	10 Hu, 8 Mac, 4 Mar
human	MERFISH	Hewitt & Turner et al. bioRxiv (2026)	"Preprint Human BG spatial Atlas"	299	1
macaque	MERFISH		"Preprint Macaque BG spatial Atlas"	300	1
marmoset	MERFISH		"Preprint Macaque BG spatial Atlas"	300	1

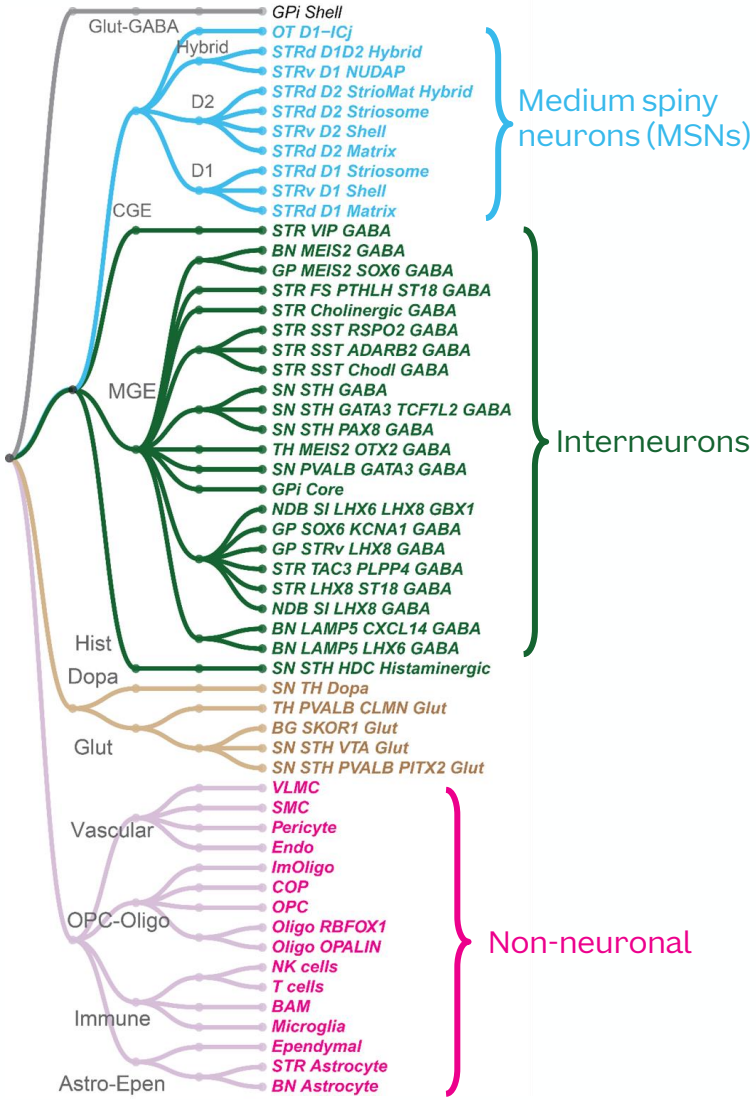
* Files with complete 35K+ gene sets also available

All MERFISH datasets were integrated with the multiome data and have the same taxonomy in the ABC Atlas (4 neighborhoods, 11 classes, 33 subclasses, 55 groups, hundreds of species-specific clusters)

preprints are part of a 20+ paper package under review in *Cell* journals and available on *bioRxiv*

human & non-human primate consensus atlas/ BG taxonomy





distinct localization & functional properties of MSN types/

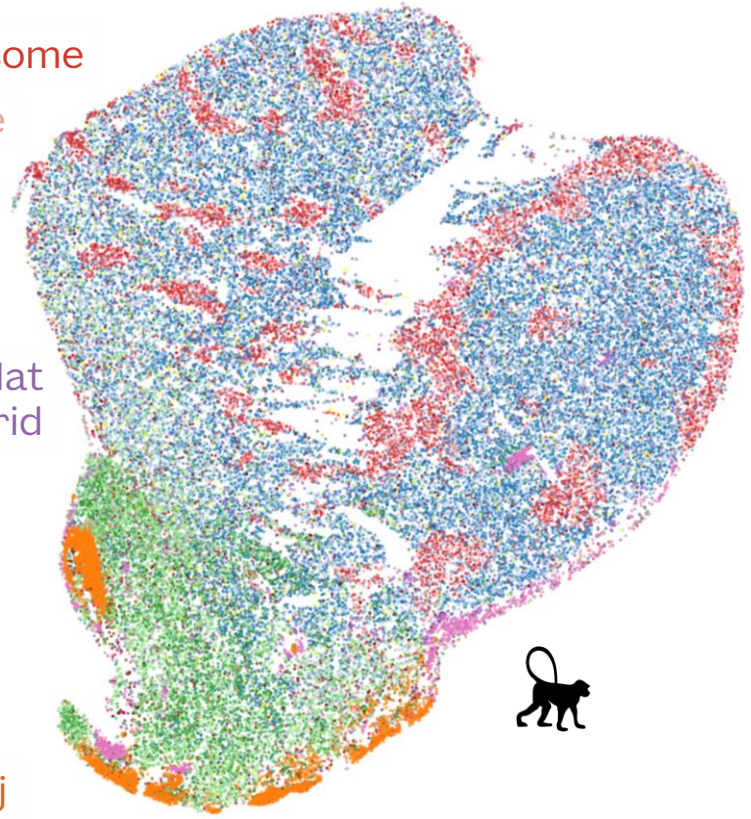
Dorsal Striatum

- STRd D1 Striosome
- STRd D2 Striosome
- STRd D1 Matrix
- STRd D2 Matrix
- STRd D1D2-Hybrid
- STRd D2-StrioMat Hybrid

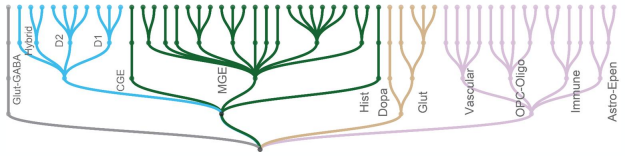
Ventral Striatum

- STRv D1 MSN
- STRv D2 MSN
- D1-NUDAP

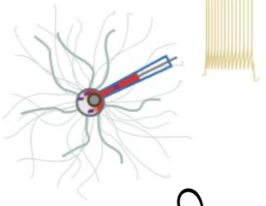
Islands of Calleja (Granular Cells) OT D1-ICj

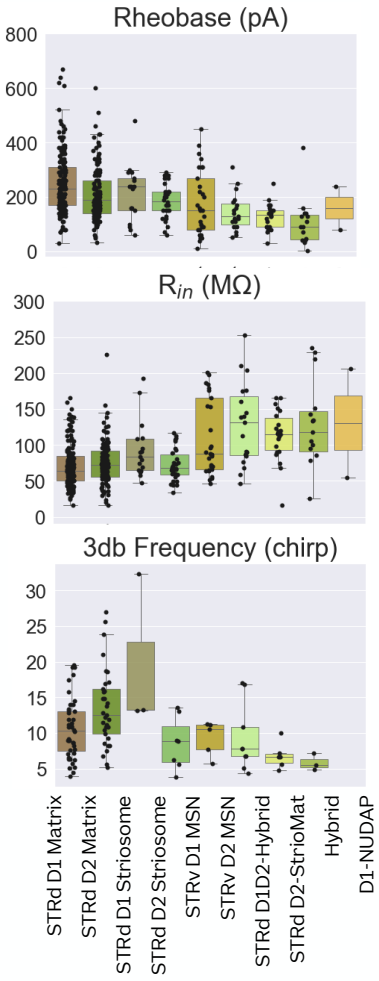


Spatial mapping



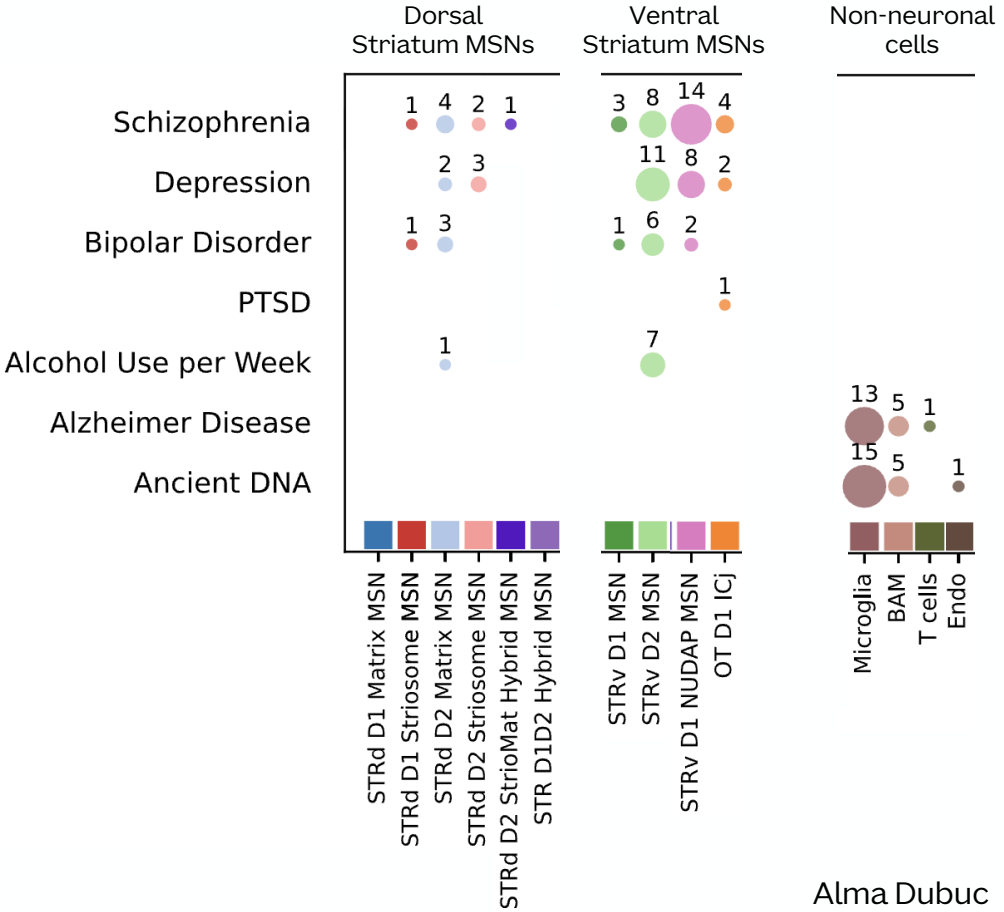
Patch-seq





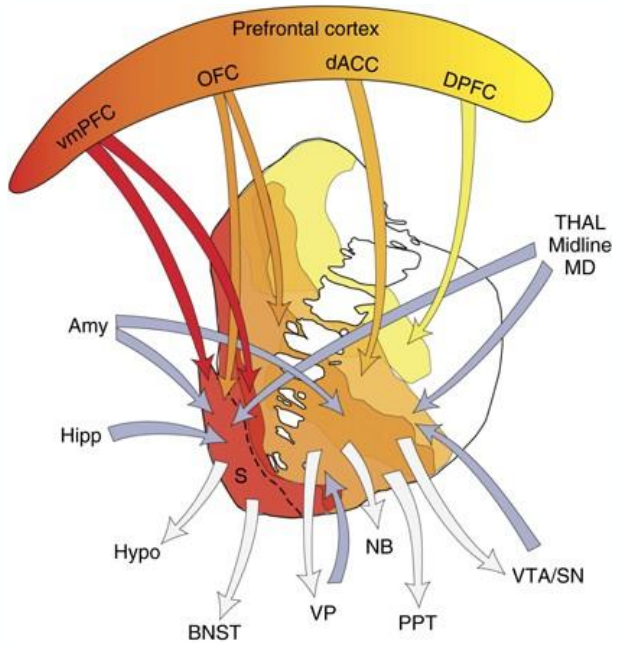
psychiatric disease risk linked to ventral MSN types/

disease-associated SNPs enriched near cell-type marker genes



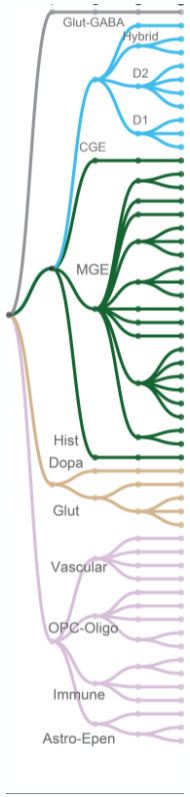
Alma Dubuc

ventral striatum integrates reward, motivation, and emotion, guiding adaptive behavior



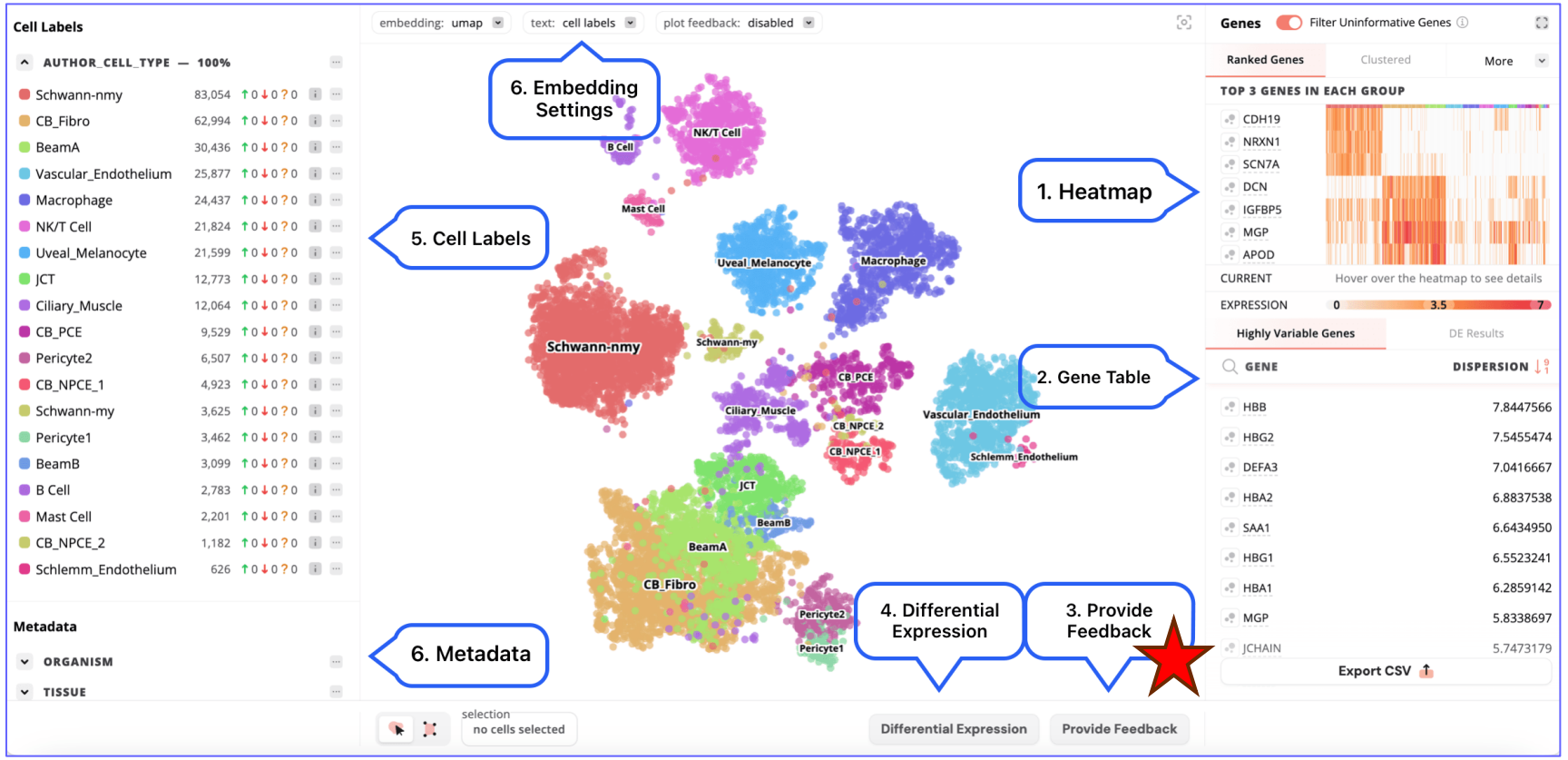
Haber & Knutson 2010

consensus brain atlas/ community refinement



Cell Annotation Platform

HUMAN CELL ATLAS



An effort by the Villani, Kharchenko, Gehlenborg, Theis, Marioni, and Regev labs.

so what's next in healthy brain atlasing?

celltype.info/project/637

HMBA overview/ building on previous datasets to create comprehensive, multiomic atlases of primate brains

Analysis-driven interative sampling
Refinement of sampling depths
Increase coverage of low frequency types
Targeted sampling of rare types

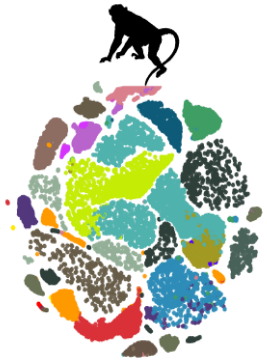
Map to reference datasets
Mouse whole brain taxonomy
Human 100 brain regions

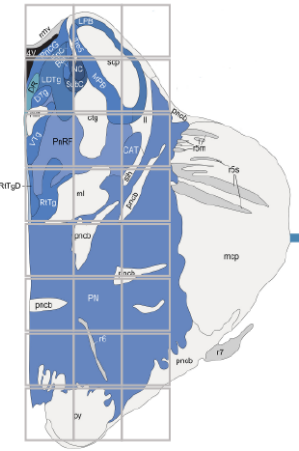
Cortex

Parcellation guided dissections

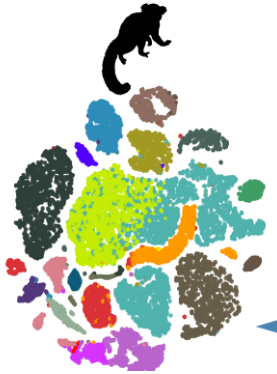
Subcortex

Gross anatomy and parcellation guided dissections





Tiling over anatomically complex structures



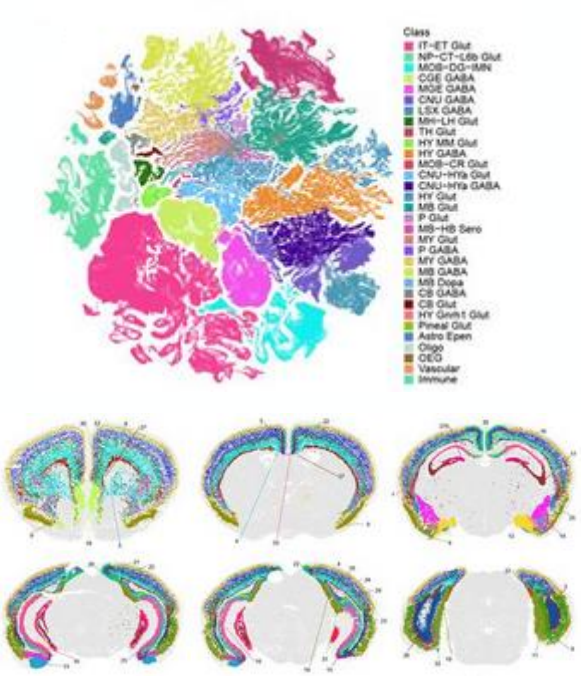
Gene expression
Open chromatin

snMultiome libraries

HMBA

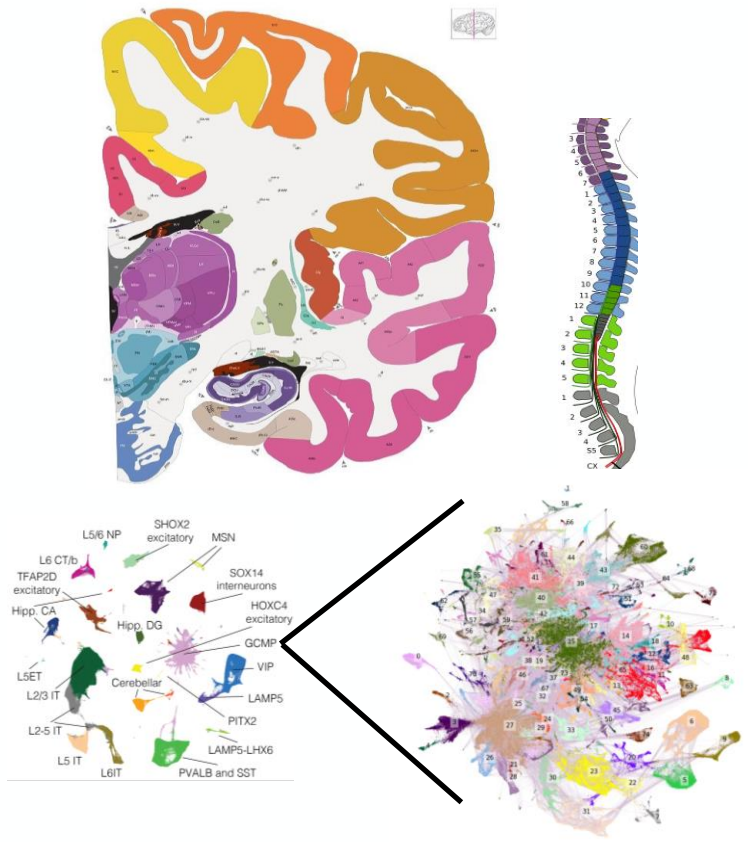
HMBA overview/ building on previous datasets to create comprehensive, multiomic atlases of primate brains

Mouse whole brain atlas



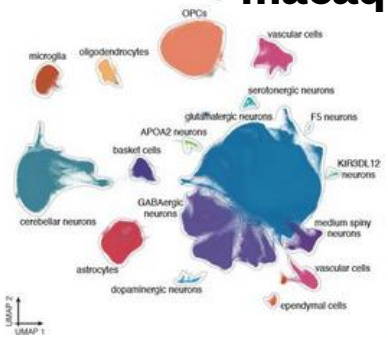
~7M cells profiled with scRNA-seq, spatially resolved dataset of ~4.3M cells with MERFISH, 5200 clusters

Human brain survey



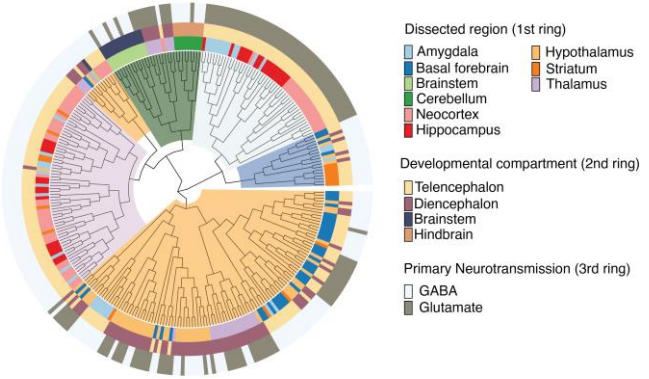
100 dissections across forebrain, midbrain, hindbrain, ~3M nuclei, >3000 clusters

Single cell multiomic atlas of rhesus macaque brain



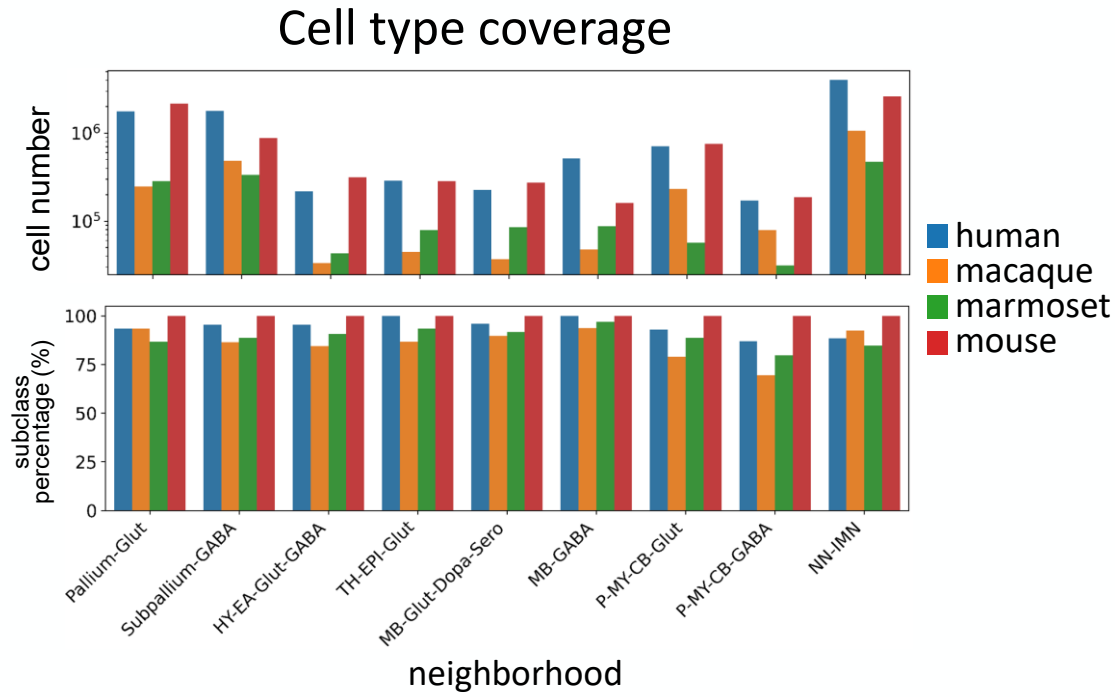
30 brain regions, transcriptomic and epigenomic profiling, 4.2M cells

Marmoset brain cell census

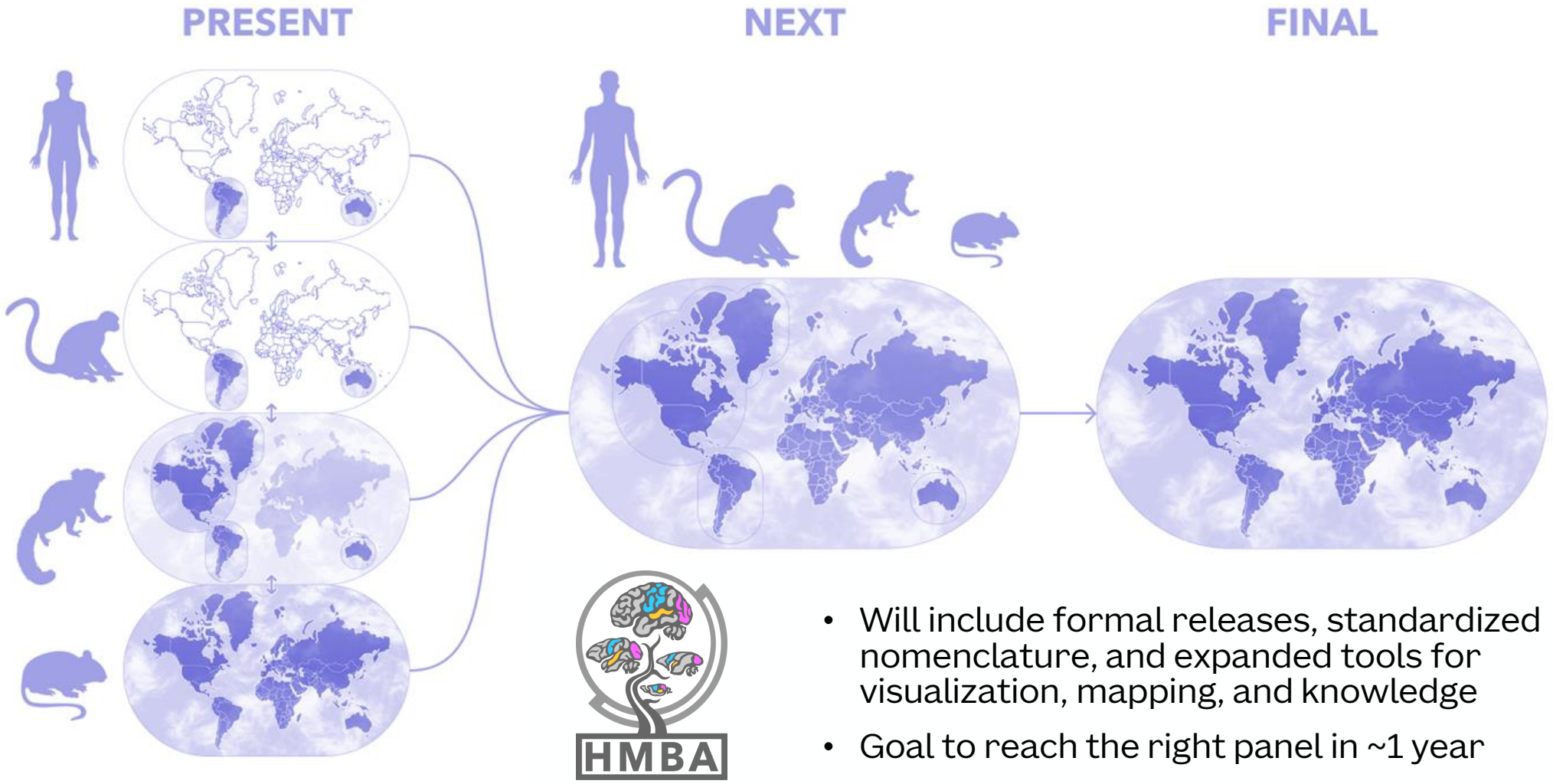


2.4M nuclei, 8 cortical and 8 subcortical areas, smFISH quantification of striatal & cortical interneurons

HMBA atlas v1/ integrated cross-species subcortical data with sparse cortical coverage



brain-wide cell atlases as foundational references/



- Will include formal releases, standardized nomenclature, and expanded tools for visualization, mapping, and knowledge
- Goal to reach the right panel in ~1 year

overview/ cell type data/ aged humans

- Seattle Alzheimer's Disease Brain Cell Atlas (SEA-AD)
- Extension Parkinson's disease and other neuro-degenerative disorders

alleninstitute/

brain health accelerator/ characterization and treatment of neurodegenerative disorders

- broader neurodegenerative disease focus
  - Alzheimer's disease
  - Parkinson's disease
  - Lewy body disease
  - Huntington's disease
  - ALS
- broader **multiomic** profiling
  - well characterized donor cohorts
  - available biofluids & neuroimaging
- major focus on **data modeling** and **AI**
  - identify actionable early **biomarkers**
  - identify actionable **therapeutic targets**
- **GOAL**: create a cell and circuit-targeted therapeutic for a clinical trial

available data/ aged human/ summary

method	institution	description	paper	ABC Atlas dataset	# genes	# donors
10X Genomics snMultiome-seq	allen institute	SEA-AD multiregion	Travaglini KT, et al. bioRxiv (2026)	”SEA-AD Multiregion”	36,378	84
10X Genomics snMultiome-seq	allen institute	SEA-AD caudate	Kana O, et al. bioRxiv (2026)	“SEA-AD CaH”	36,006	84
10X Genomics snRNA-seq	allen institute	SEA-AD MTG & DLPFC	Gabitto MI, et al. Nat Neuro (2024)	“SEA-AD snRNAseq – MTG and DLPFC”	36,266	84
MERFISH	allen institute	SEA-AD MTG	Gabitto MI, et al. Nat Neuro (2024)	“SEA-AD MTG"	140	24
10X Genomics snRNA-seq	Aligning Science Across Parkinson’s (ASAP)	human postmortem-derived brain sequencing (PMDBS)	N/A	“10x-scRNAseq-Parkinsons-cohort”	36,139	212 from 5 integrated studies

bha/ precision health/ diseases of interest

Alzheimer’s disease

Parkinson’s disease

Amyotrophic lateral sclerosis

Huntington’s disease

Lewy body dementia

healthy adult brain cells act as anchor for defining cell types in development, aging, and disease/

cell type atlases in healthy, developing, & adult mouse, human, and NHP brains

BICCN

BICAN

The BRAIN Initiative®

HMBA

extend atlases to the aged human brain to understand early molecular and cellular changes in Alzheimer's disease

SEA-AD

team science/ SEA-AD consortium

The Seattle Alzheimer's Disease Brain Cell Atlas (SEA-AD) consortium is supported by the National Institutes on Aging (NIA) grant 2U19AG060909.

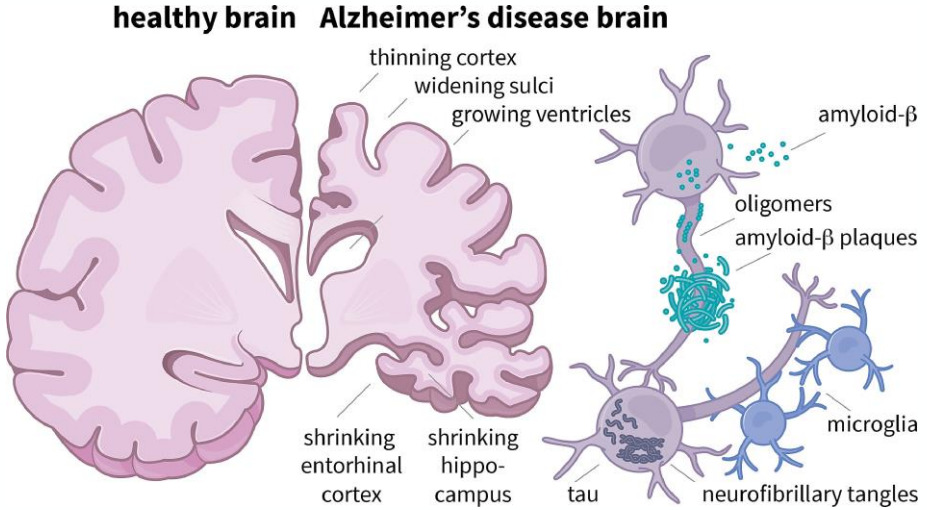
NIH National Institute on Aging

Shanahan Family Foundation

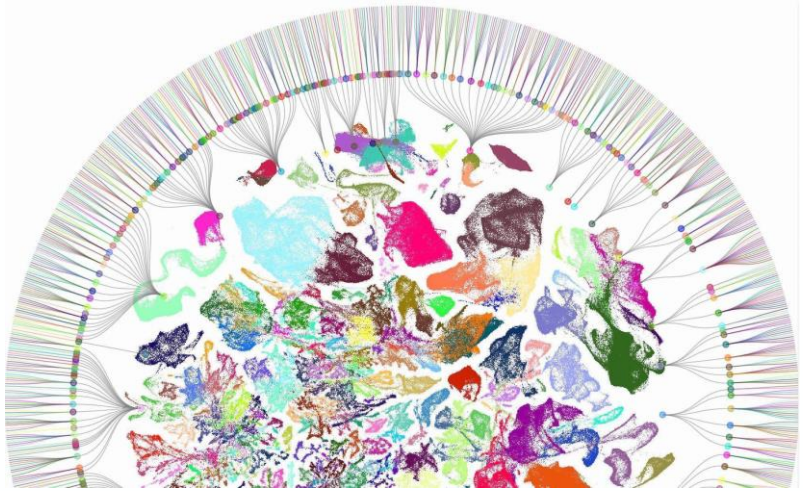
sea-ad.org

what is SEA-AD?/

- **OVERARCHING GOALS**
  - create a **high-resolution brain cell atlas of Alzheimer's disease** using cutting edge single-cell & spatial omics, and neuropathology
  - map the cellular and molecular events over the course of disease to **identify vulnerable cell types**
  - **share data and knowledge** by creating a data resource for the scientific community
- **multi-consortium** project funded until **2030**
- hope to **accelerate** development of **diagnostic** and **therapeutic** strategies

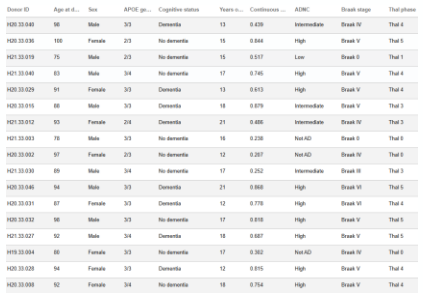


Brain Multiphysics 2: 10039 (2021) (4), (CC BY 4.0)

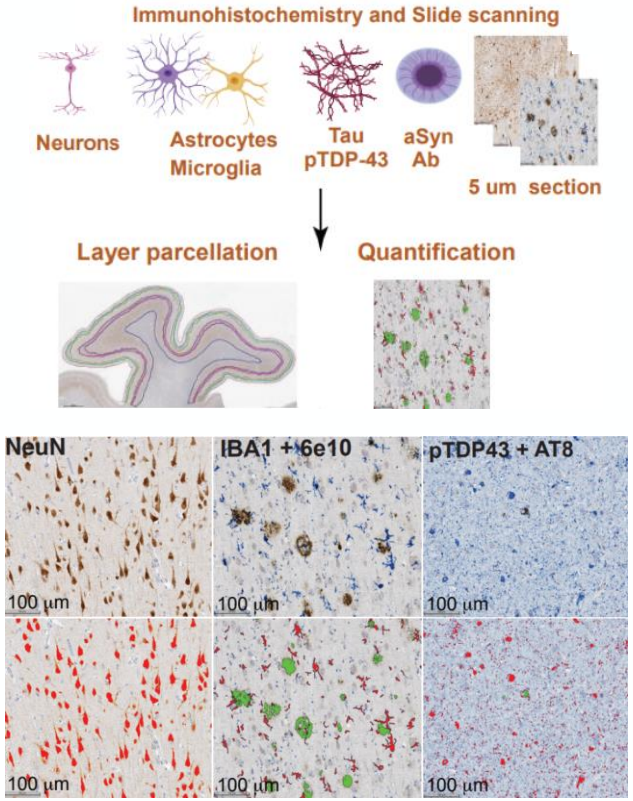


SEA-AD/ data generation, analysis, and the search for vulnerable cell types

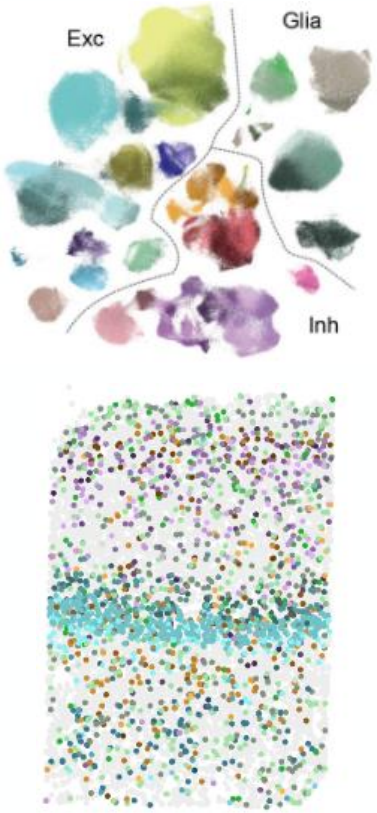
donor metadata/



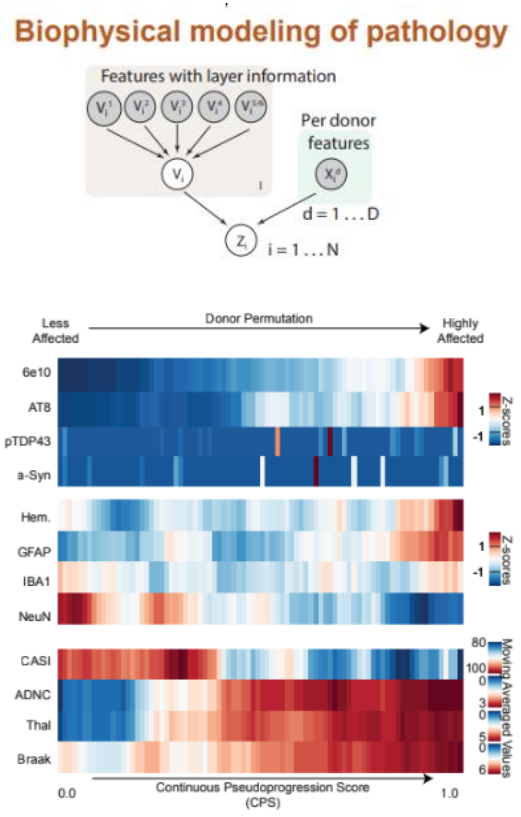
quantitative neuropath/



single-cell & spatial omics/



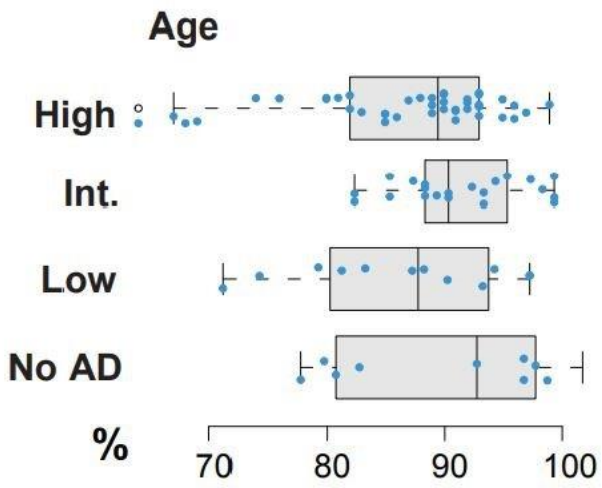
AI modeling/

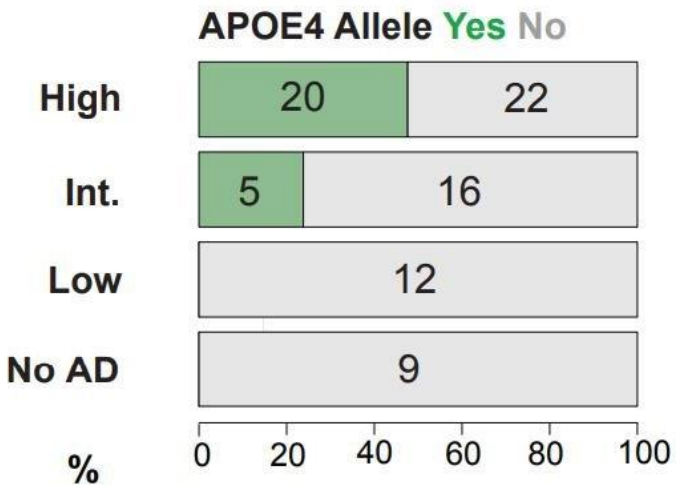


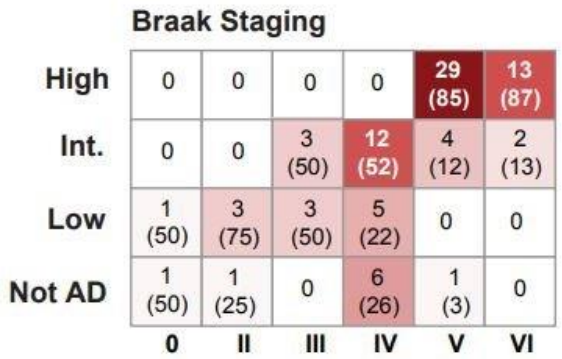
Gabitto et. al., Nature Neuroscience 2024

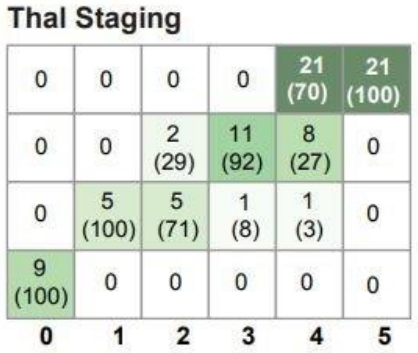
sea-ad.org

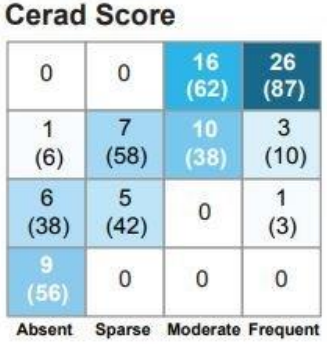
SEA-AD/ cohort/ moving beyond case-control to capture AD pathology spectrum

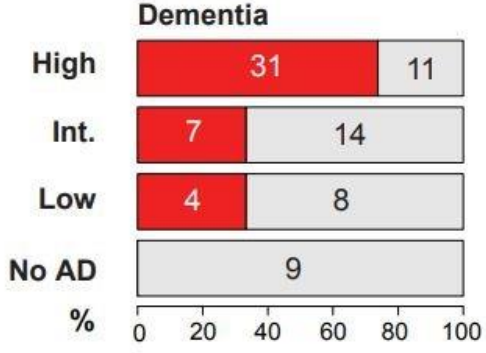


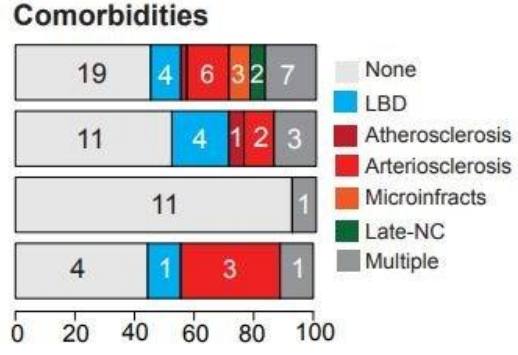






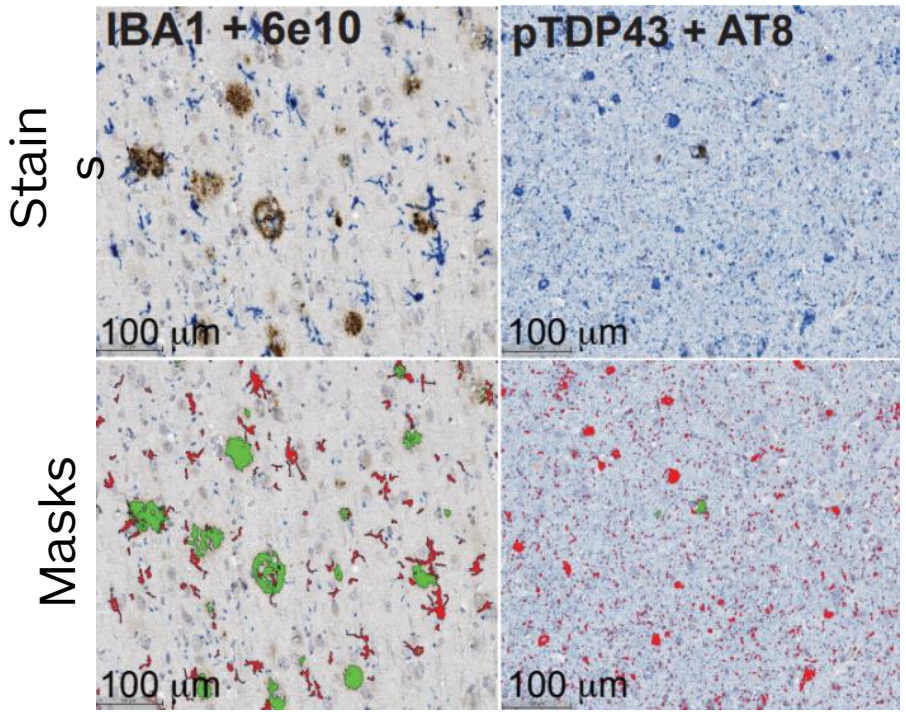


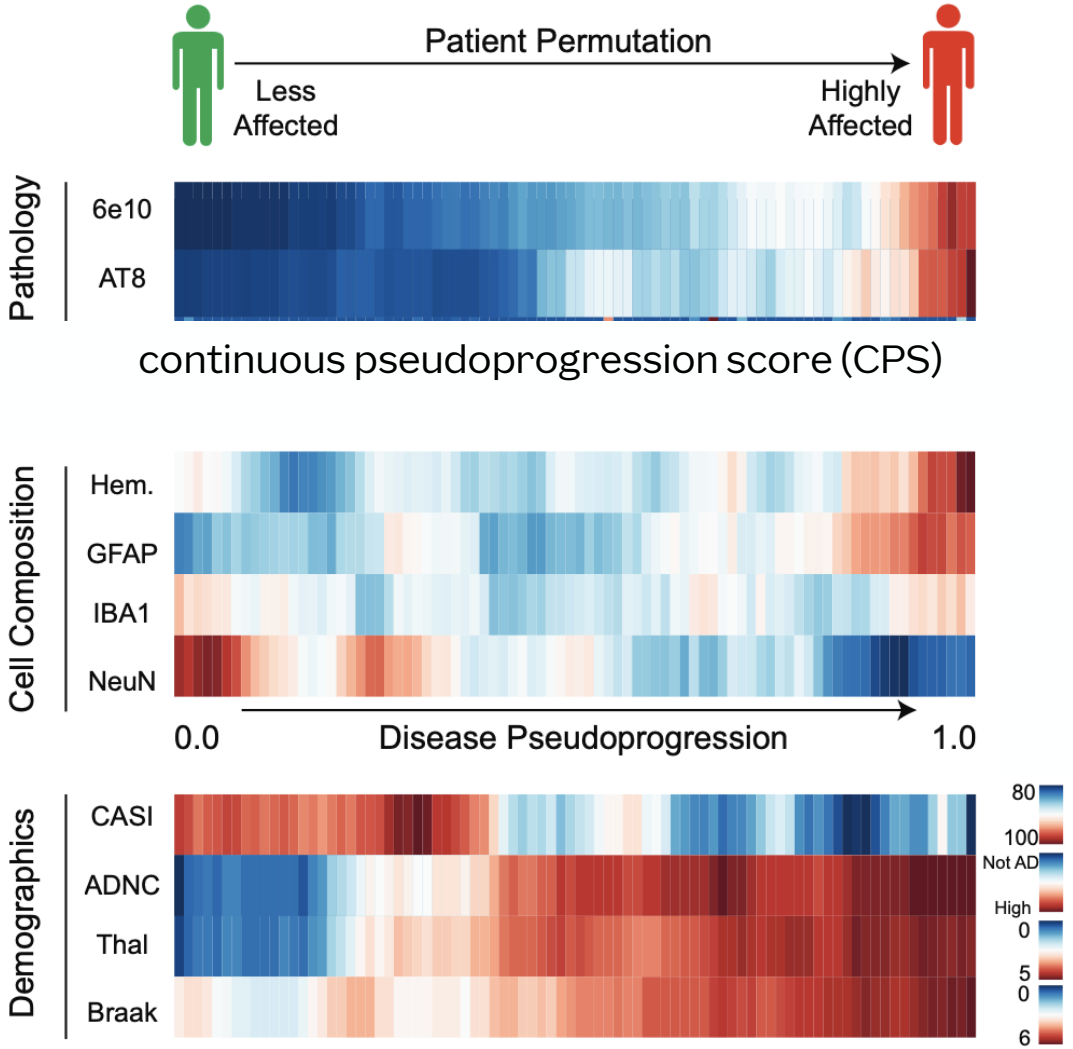




SEA-AD/ cohort/ organize by neuropathology

1. collect neuropathology images
2. quantify the amount of each stain using machine learning
3. organize donors from least to most overall pathology

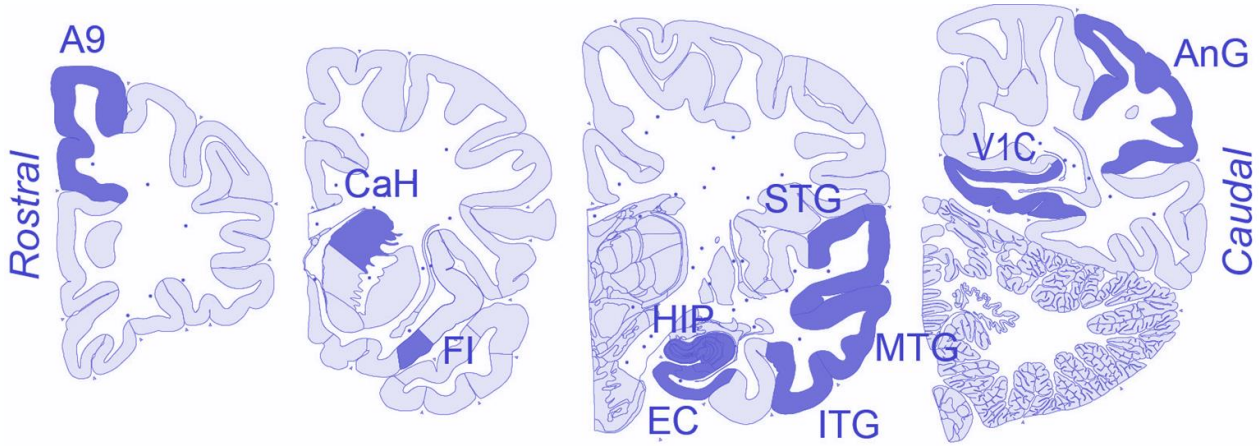


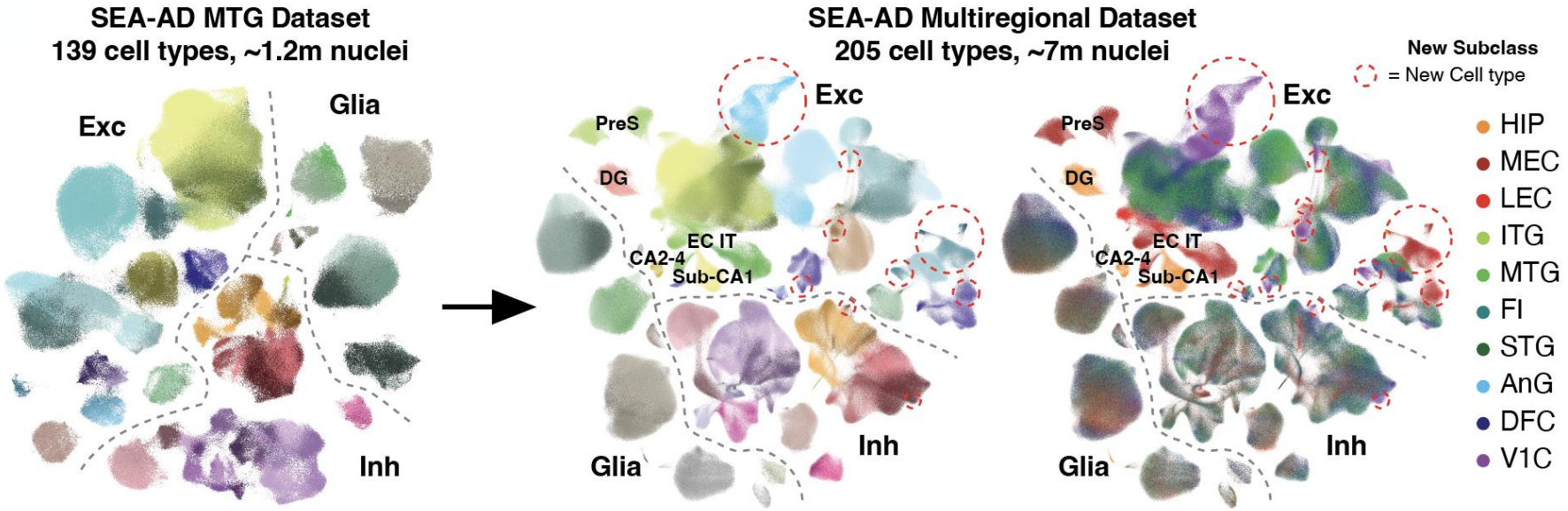


continuous pseudoprogression score (CPS)

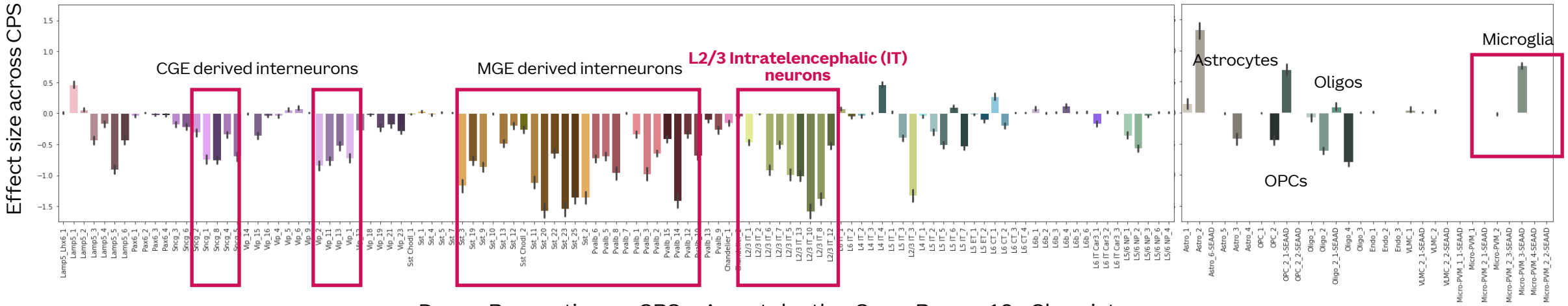
sanity check: CPS relates to common metrics

SEA-AD/ multi-regional atlas

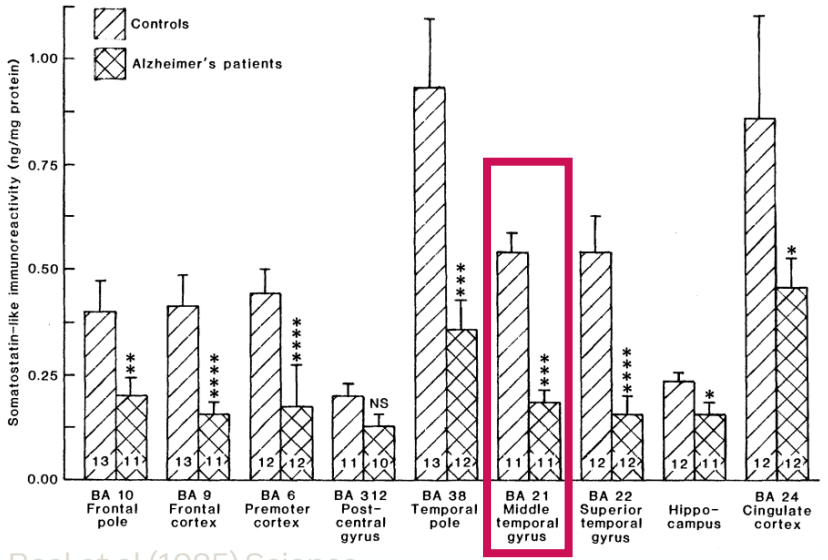




MTG/ cell type abundance/ changes span all populations

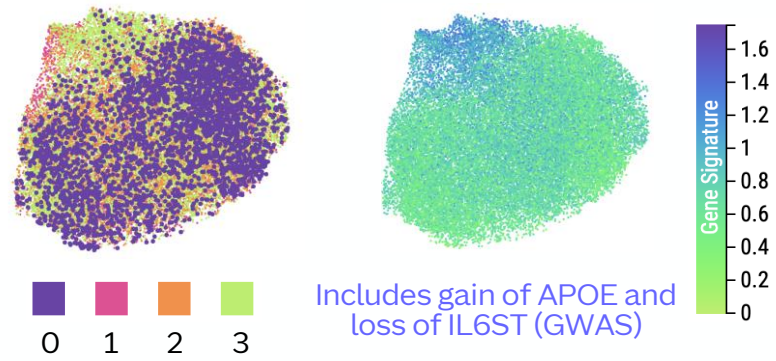


Donor Proportions ~ CPS + Age at death + Sex + Race + 10x Chemistry



Beal et al (1985) Science
Roberts et al (1985) Nature

Microglia-PVM_1 AD Pathology

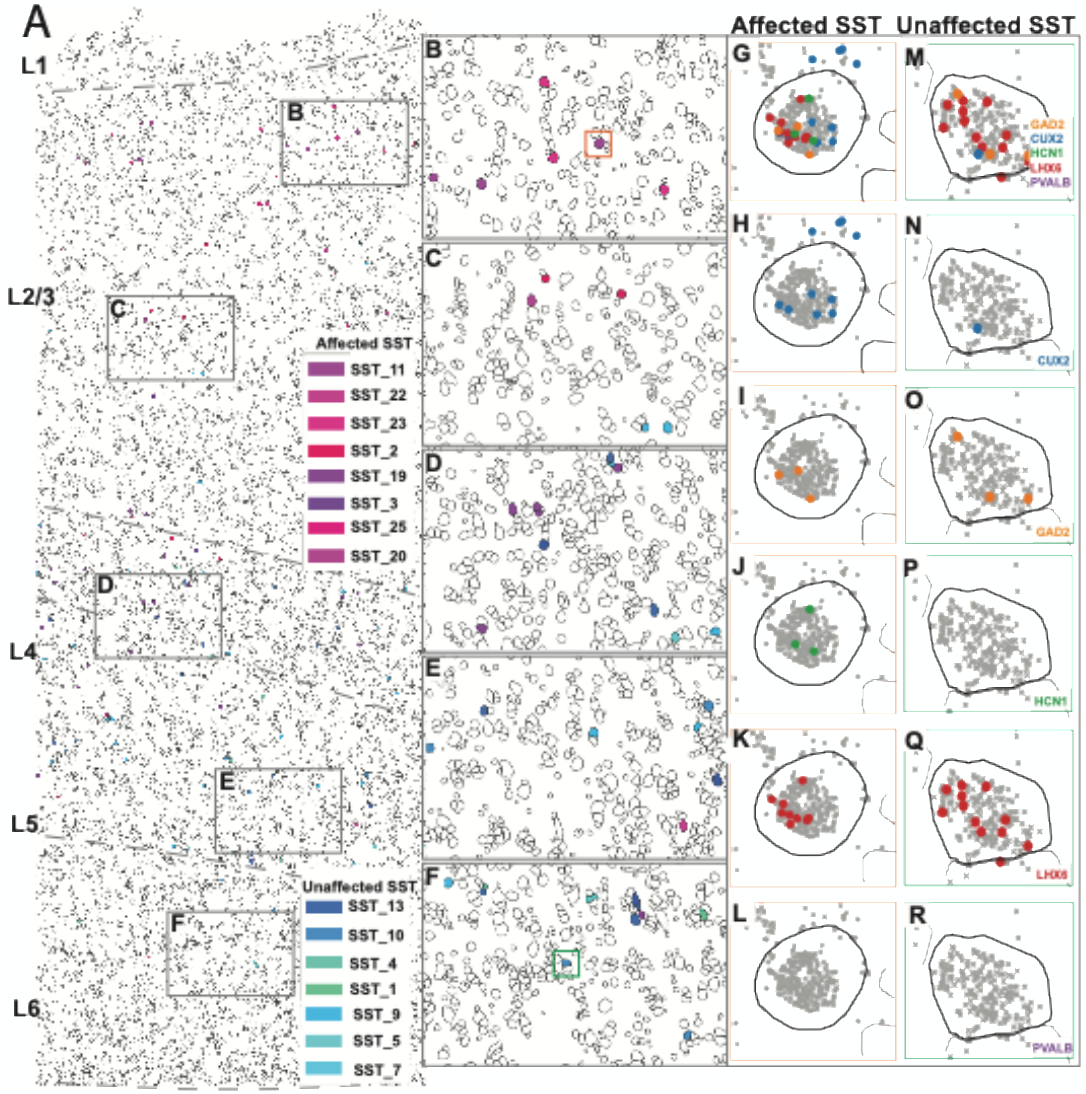


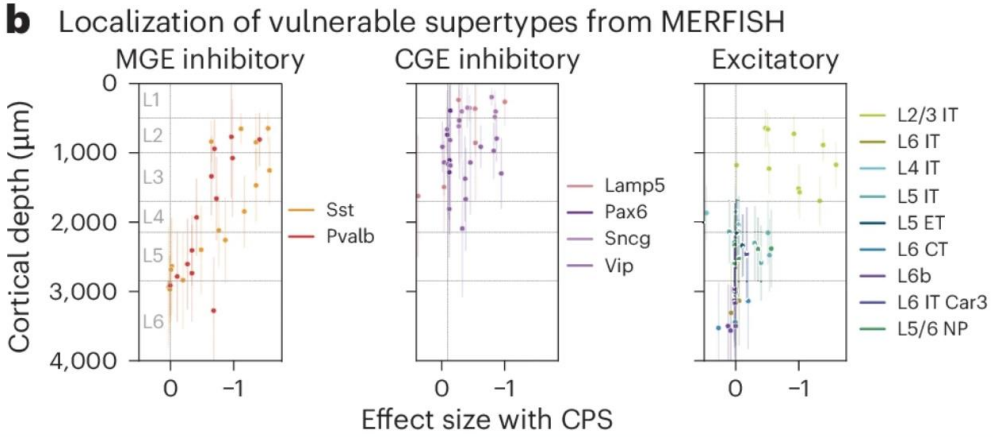
Includes gain of APOE and loss of IL6ST (GWAS)

Mathys et al (2019) Nature

sea-ad.org

vulnerable cell types localize to superficial cortical layers/



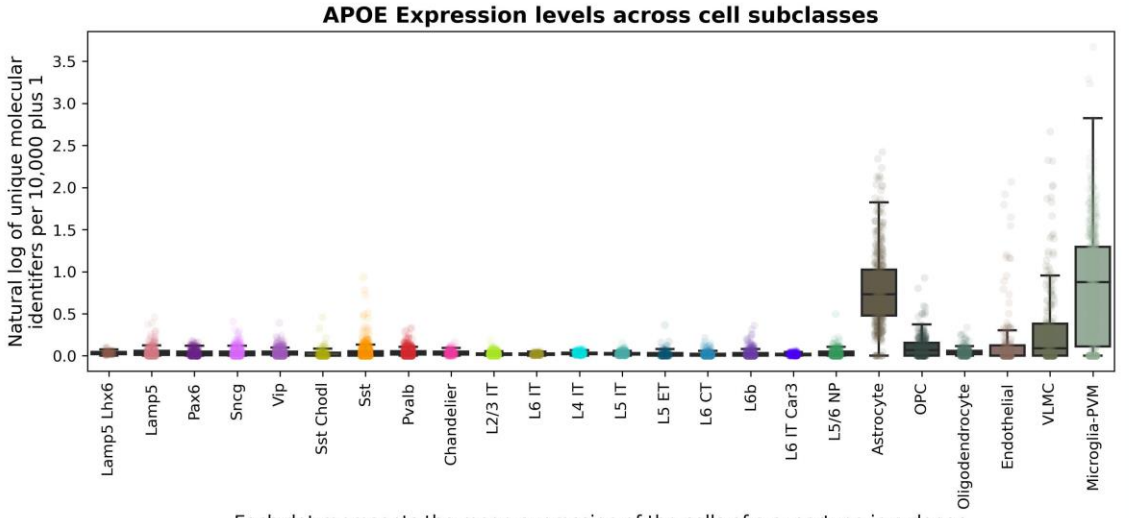


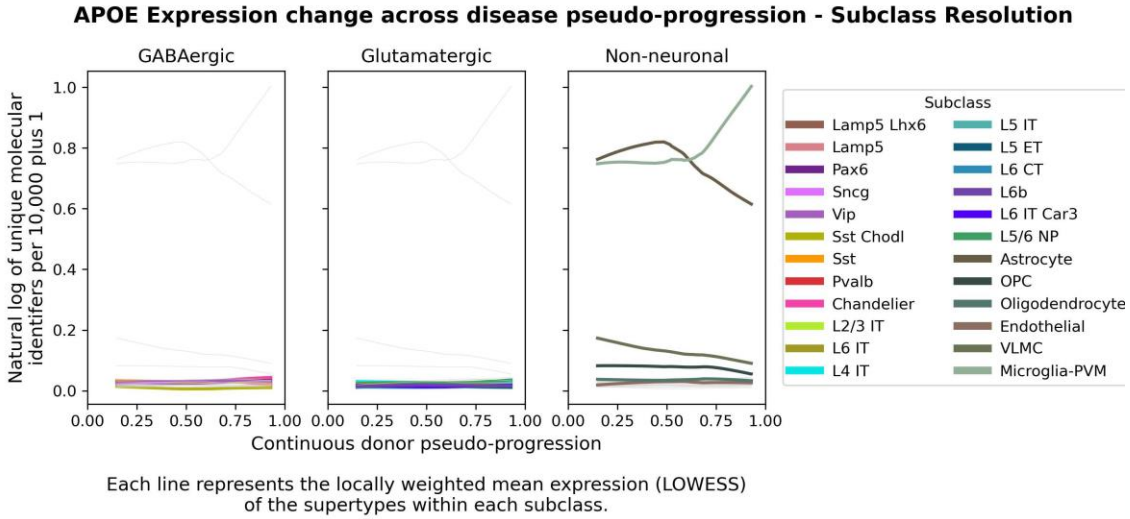
- spatial transcriptomics (MERFISH) confirmed that vulnerable neurons localize to upper layers
- cell type vulnerabilities in snRNA-seq largely confirmed via MERFISH as well

gene expression changes/ cell type-specific/ AD impact

Example: APOE (known AD risk factor)

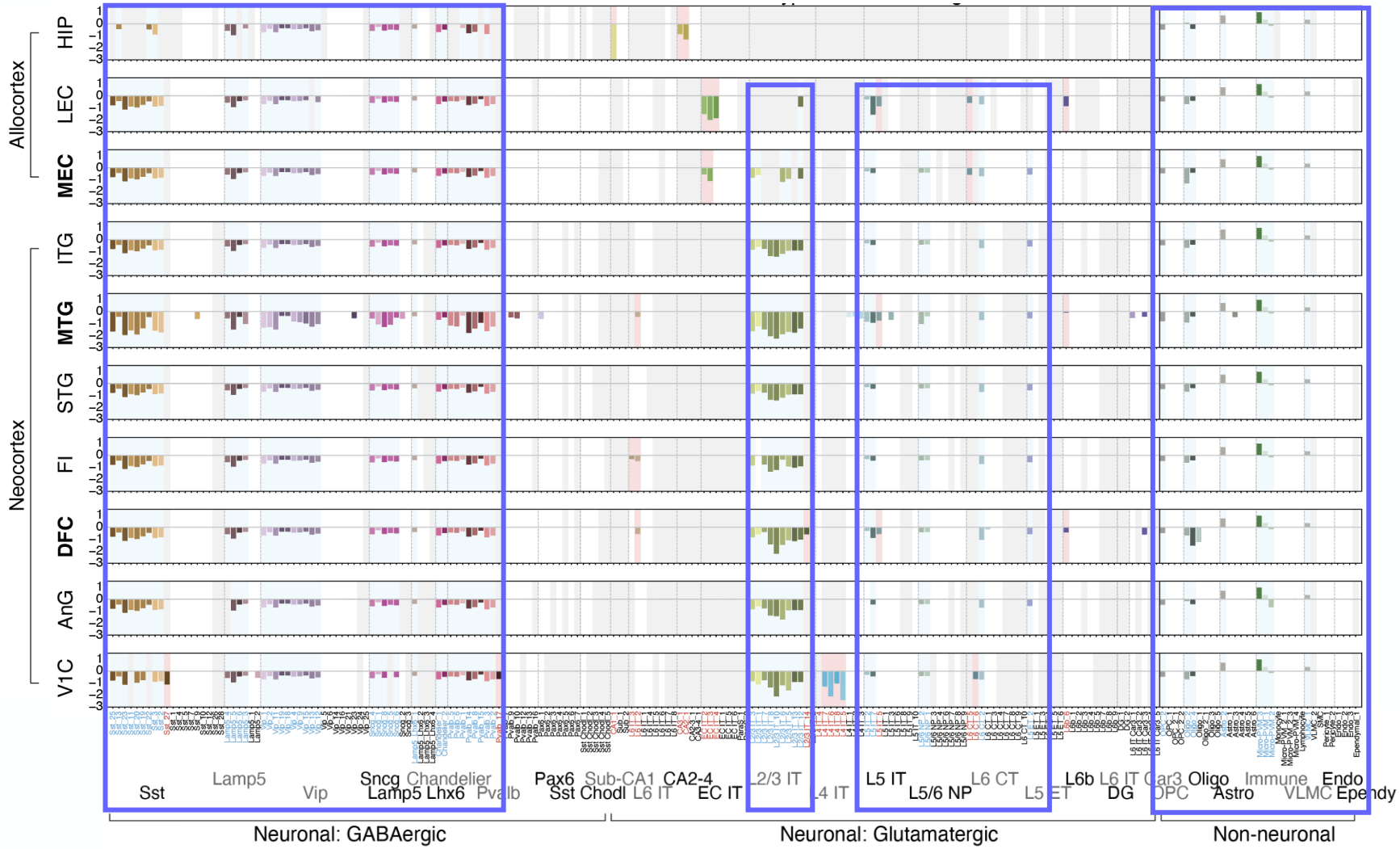
- found in astrocytes and microglia
- decreased expression in AD in some astrocyte types
- increased expression in AD in some microglial types





gene expression changes/ shared

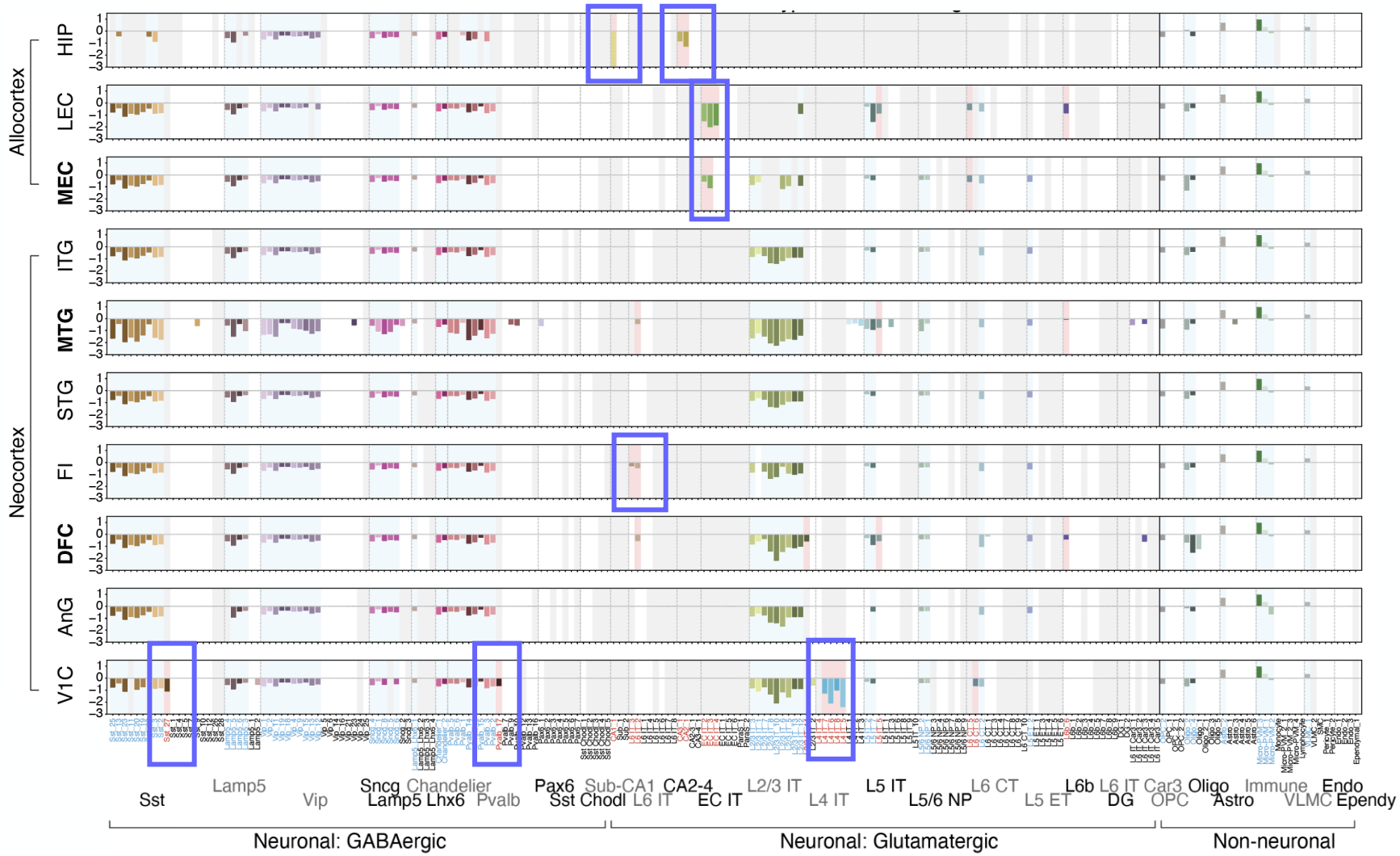
Cellular Abundances ~ CPS_Local + Sex + Age at Death + APOE4 Status + PMI



AD Spectrum+
AD Spectrum only
type not in region
regional change
shared change

gene expression changes/ region-specific

Cellular Abundances ~ CPS_Local + Sex + Age at Death + APOE4 Status + PMI



AD Spectrum+
AD Spectrum only
type not in region
regional change
shared change

SEA-AD/ read more at sea-ad.org

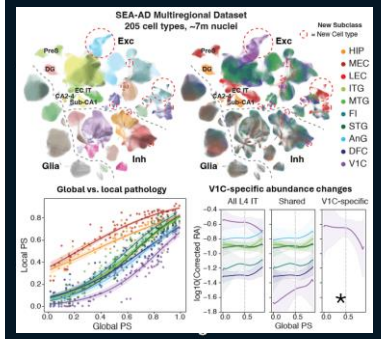
Our Science (SEA-AD)

The Seattle Alzheimer's Disease Brain Cell Atlas (SEA-AD) consortium aims to provide scientific insight on the pathogenesis of Alzheimer's disease using data generated by SEA-AD and others. Access our latest published work describing cellular and molecular changes in AD progression.

Note: Code for reproducing analyses from some of these papers is now available via the "Our Code" landing page.

Platform papers

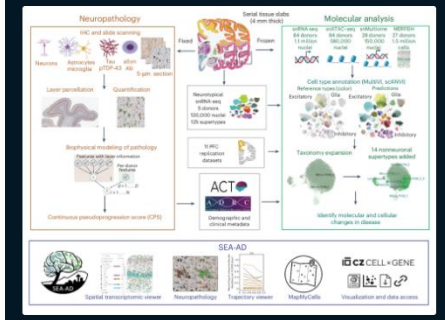
Large studies from the whole SEA-AD consortium describing major data generation efforts and associated knowledge from these data sets.



Multiregional single-cell profiling reveals shared andspecialized cellular vulnerability in Alzheimer's disease (Travaglini,Gabitto, Ding et al., bioRxiv, 2026)

This paper extends the SEA-AD atlas across ten neo- and allocortical brain regions to determine whether the cellular changes associated with Alzheimer's disease are shared across the brain. By analyzing the molecular profiles of roughly seven million cells and relating them to regional pathology, the researchers found that vulnerability is remarkably consistent across regions, with the same subsets of inhibitory neurons, oligodendrocytes, and glia changing in a stereotyped sequence that begins before symptoms emerge. The study also uncovered unexpected vulnerability in specialized cell populations, including primary visual cortex layer 4 excitatory neurons previously thought to be resilient, and identified neuronal hyperexcitability as a potential common mechanism linking both early- and late-vulnerable cell types.

READ THE PAPER | Download the data



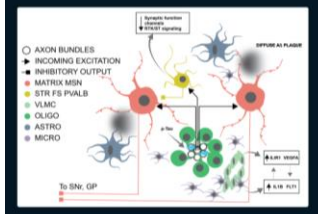
Integrated multimodal cell atlas of Alzheimer's disease (Gabitto, Travaglini, et al, Nature Neuroscience, 2024)

This paper describes the creation of a massive molecular and cellular atlas of middle temporal gyrus to better frame and understand how Alzheimer's disease progresses. By analyzing the genetic activity of millions of individual cells, the researchers discovered that the disease actually unfolds in two distinct phases: an early, slow-moving stage where specific Sst+ interneurons are lost and inflammation begins, followed by a later stage of accelerated neuropathology and loss of excitatory neurons and Pvalb+ and Vip+ inhibitory neuron subtypes. These findings were replicated in prefrontal cortex in SEA-AD and other major AD studies.

READ THE PAPER | Download the data | Explore interactive resources

Focused studies

Other papers with SEA-AD researchers as primary authors focused on SEA-AD data sets or novel computational algorithms.



The Caudate Nucleus Exhibits Distinct Pathology and Cell Type-Specific Responses Across Alzheimer's Disease (Kana et al, bioRxiv, 2026)

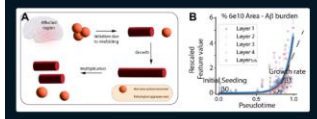
This paper represents the characterization of the Alzheimer's pathological spectrum within the caudate nucleus. Utilizing digital pathology, researchers were able to confirm a less pronounced phosphorylated tau and amyloid β burden when compared to other cortical areas. Phosphorylated tau and amyloid β were also shown to localize separately into white and gray matter, respectively. Both, in part, explain the lack of neuronal cell death observed in the caudate even in high burden parts of the spectrum. However, researchers were able to identify glimmers of a global phosphorylated tau response in microglia.

READ THE PAPER | Download the data

SEA-AD is a multimodal cellular atlas and resource for Alzheimer's disease (Hawrylycz et al, Nature Aging, 2024)

This commentary on Gabitto, Travaglini, et al 2024 presents SEA-AD as a multifaceted open community resource designed to identify cellular and molecular pathologies that underlie Alzheimer's disease. Integrating neuropathology, single cell and spatial genomics, and longitudinal clinical metadata, SEA-AD is a unique resource for studying the pathogenesis of Alzheimer's and related dementias.

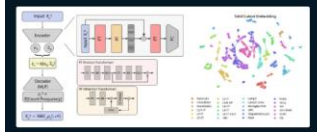
READ THE PAPER | Download the data | Explore interactive resources



B-BIND: Biophysical Bayesian Inference For Neurodegenerative Dynamics (Agrawal et al, bioRxiv, 2024, accepted at the Annals of Applied Statistics)

This paper introduces B-BIND, a new mathematical framework designed to track how Alzheimer's disease progresses by analyzing the buildup of various pathological proteins. Because researchers often only have "snapshots" of the brain from different individuals, this model uses a "pseudotime" scale to rank patients along a continuous timeline of disease severity. Ultimately, this framework lays the groundwork for identifying the hidden biological stages of neurodegeneration and the specific cellular changes that drive the disease over time.

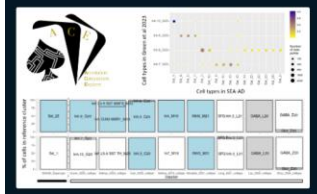
READ THE PAPER



TabVI: Leveraging Lightweight Transformer Architectures to Learn Biologically Meaningful Cellular Representations (Chandrashekar et al, bioRxiv, 2026)

This paper introduces TabVI, a lightweight AI model designed to analyze single-cell genomic data more effectively than standard language-based AI. By adapting "transformer" technology—the same tech behind ChatGPT—to better fit the non-sequential, hierarchical nature of genes, TabVI can more accurately identify cell types and biological patterns. This tool provides a more efficient and interpretable way for researchers to study how diseases like Alzheimer's affect individual cells across large datasets.

READ THE PAPER



Annotation Comparison Explorer (ACE): connecting brain cell types across studies of health and Alzheimer's Disease (Miller et al, bioRxiv, 2025)

This paper introduces Annotation Comparison Explorer (ACE), a web-based tool developed by the Allen Institute to address the challenge of comparing cell type classifications across different brain studies. ACE allows researchers to map their own single-cell data to established taxonomies, such as the Seattle Alzheimer's Disease Brain Cell Atlas (SEA-AD), and compare annotations like donor demographics and disease-related changes. By applying ACE to multiple published Alzheimer's datasets, the authors identified consistent signatures of the disease, such as a decrease in specific somatostatin interneurons, demonstrating the tool's utility in unifying diverse datasets to advance the understanding of brain health and neurodegeneration.

READ THE PAPER | Explore ACE

Collaboratory studies

Projects led by other folks in the AD community, but with significant SEA-AD input or using SEA-AD data resulting in co-authorship by SEA-AD researchers.

Somatic cancer driver mutations are enriched and associated with inflammatory states in Alzheimer's disease microglia (Huang et al, bioRxiv, 2024, accepted at Cell)

Using transcriptomic datasets (including SEA-AD), this study reveals that somatic mutations—genetic changes occurring after birth—are twice as common in Alzheimer's brains than in healthy, age-matched controls. These mutations are concentrated in the brain's immune cells (microglia) and often affect "cancer driver" genes, causing the cells to multiply and trigger inflammation. This suggests that Alzheimer's may be driven by a tumor-like process where mutated immune cells expand throughout the brain and accelerate neurodegeneration.

Read the paper | Use RNA-MosiacHunter

Single Cell Landscape of Sex-specific Drivers of Alzheimer's Disease (Wu et al, Alzheimer's & Dementia, 2025)

This study analyzed single nucleus transcriptomics data (including from SEA-AD) to understand how Alzheimer's disease affects men and women differently at a genetic level. Researchers found female-specific gene associations providing "protective" effects in neurons of dorsolateral prefrontal cortex, while specific inflammatory signals in microglia may increase risk for protein buildup in women. These findings highlight that the molecular path to cognitive decline is sex-specific, pointing toward new, targeted genes for future treatment research.

Read the paper | Access the SEA-AD data | Access the ROS/Map data

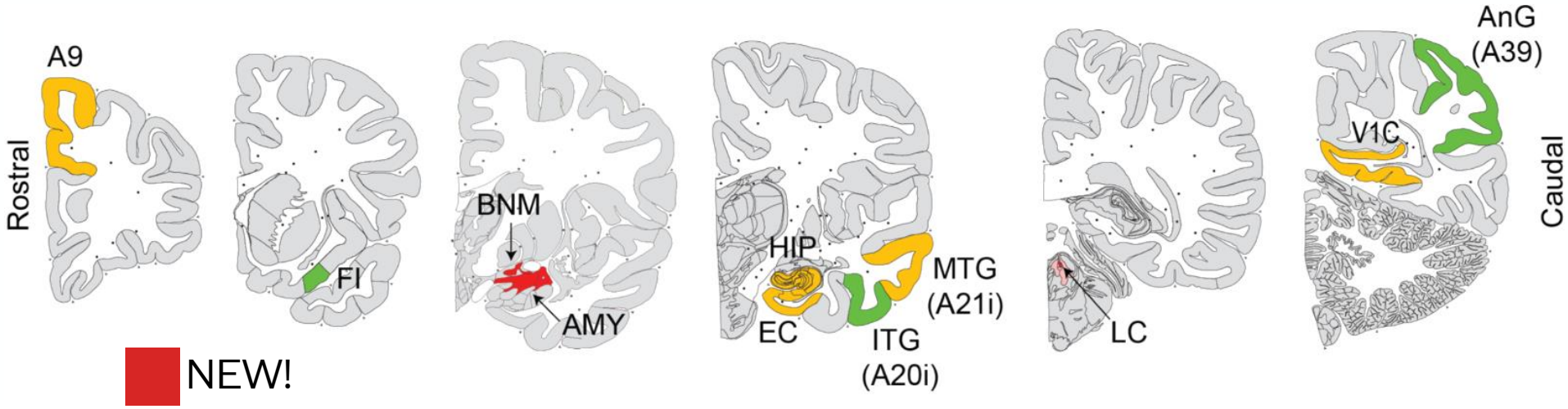
CRISPR screens in iPSC-derived neurons reveal principles of tau proteostasis (Samelson et al, bioRxiv, 2025)

This study utilizes genome-wide CRISPRi screens in human iPSC-derived neurons to identify cellular factors regulating the accumulation of tau aggregates, a primary driver of Alzheimer's disease. The researchers identified the ubiquitin ligase CUL5-SOCS4, alongside pathways such as UFMylation and mitochondrial function, as critical modifiers of tau stability and potential therapeutic targets. Furthermore, the team leveraged SEA-AD data to validate these findings, demonstrating that these tau-regulating genes exhibit altered expression within the specific cell types most vulnerable to Alzheimer's in human donors.

Read the paper | Explore CRISPR resources

SEA-AD/ future plans

- more **brain regions** (e.g., key subcortical and neuromodulatory)
- more donors via an **expanded neuropathology core network**
  - 300 heterogeneous donors spanning AD and AD/ADRD polypathologies
  - varied genetic backgrounds, topological and cognitive subtypes, cognitive resilience, and resistance to pathology
- more data types, including **imaging**, **large format assays**, and **proteomics**
- improved analysis and modeling of our data and others
- expanded web portal for scientific community to mine and explore SEA-AD tools and data



bha/ precision health/ diseases of interest

Alzheimer’s disease

Parkinson’s disease

Amyotrophic lateral sclerosis

Huntington’s disease

Lewy body dementia

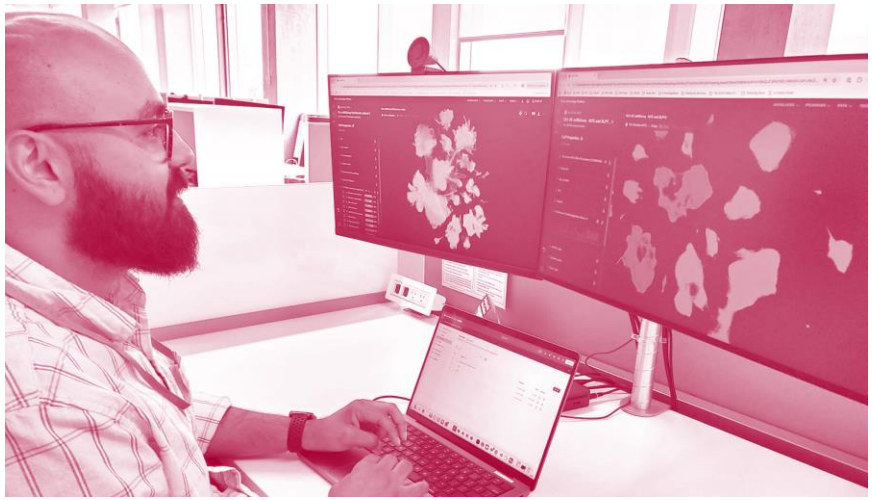
available data/ human/ Parkinson's disease

method	description	paper	ABC Atlas dataset	# genes	# donors
10X Genomics snRNA-seq	ASAP Human Postmortem BrainSeq	N/A	"10x-scRNAseq-Parkinsons-cohort"	36,139	212 from 5 integrated studies

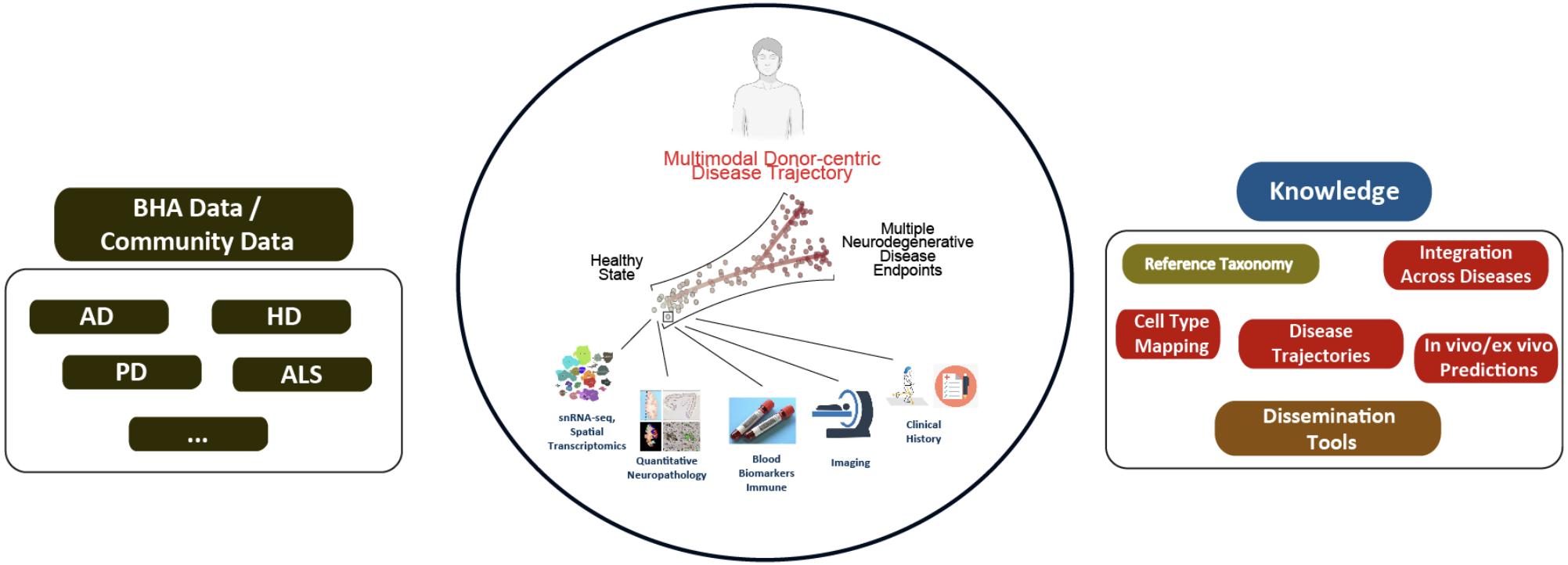
study	# donors
Lee	25*
Hafler	12
Hardy	64
Jakobsson	25
Scherzer	94*

* some donor overlap

- data were integrated and clustered but not fully annotated (e.g., cluster_001)
  - clusters 0 and 15 are likely low-quality cells (and therefore omitted from the visualization)
- data **aligned** to whole human brain & SEA-AD taxonomies
- alignments between clustering and mapping are available via Annotation Comparison Explore (discussed later)



brain health accelerator/ AI modeling



GOAL: To bridge postmortem ground truth and in vivo disease biomarkers to identify disease phenotypes and their underlying molecular and cellular consequences

agenda/morning

9:00-9:30 | check-in / breakfast

9:30-9:45 | welcome, logistics, and overview of allen institute (Rachel)

9:45-10:30 | introduction to transcriptomics for cell typing (Jeremy)

10:30-10:45 | break

10:45-12:00 | cell type data sets on allen brain map

- overview of mouse data sets – Rachel
- cross-primate healthy brain atlases – Jeremy
- cell typing in aged humans, SEA-AD and beyond - Irika

12:00-1:00 | lunch

github.com/AllenInstitute/SPOCTT/tree/main/2026_Additional_Presentations

agenda/afternoon

12:00-1:00 | lunch

1:00-2:15 | walkthrough of allen brain map tools (Rachel / Jeremy)

walkthrough of allen brain map
Brain Knowledge Platform: ABC Atlas, MapMyCells, Genetics Tools Atlas
Additional tools in brief: Patch-seq, gene atlases, and anatomy

2:15-2:45 | SEA-AD tools for exploring changes in AD (Irika / Jeremy)

2:45-3:00 | break

3:00-3:30 | external use cases – Rachel

3:30-4:40 | user exploration of allen institute tools

4:40-5:00 | feedback survey (all – please fill out the survey!)

github.com/AllenInstitute/SPOCTT/tree/main/2026_Additional_Presentations

Allen Brain Map tools/

- Walkthrough of Allen Brain Map
- ABC Atlas
- MapMyCells
- Genetics Tools Atlas

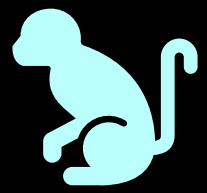
alleninstitute/

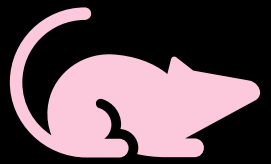
brain knowledge platform/

- open + AI-powered neuroscience tool developed by the allen institute
- harmonizes data from over 34 million brain cells across 22 species into a single, searchable map, providing scientists with a universal language to study both healthy and diseased mammalian brains
- combines extensive data catalogs with advanced analytical software and genetic tools to help researchers accelerate discoveries in neurological conditions like Alzheimer's and Parkinson's diseases

integrated “uber atlas” for cells/

species

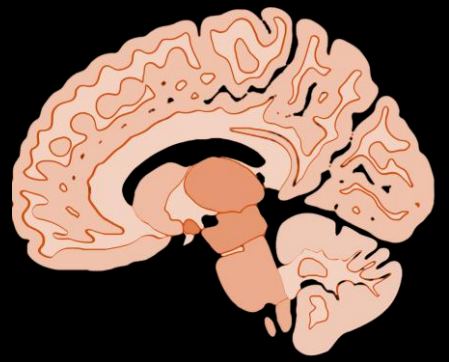


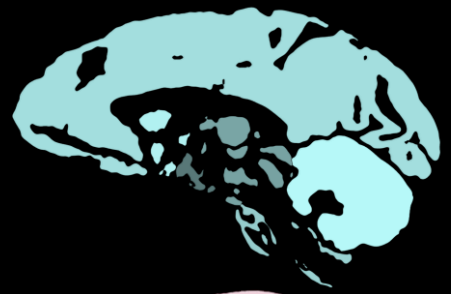


modalities

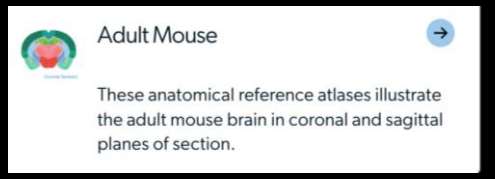
- transcriptomics
- epigenomics
- spatial omics
- genomics
- proteomics
- morphology
- connectivity
- physiology

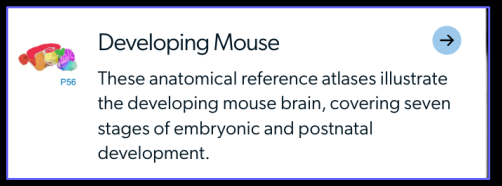
regions

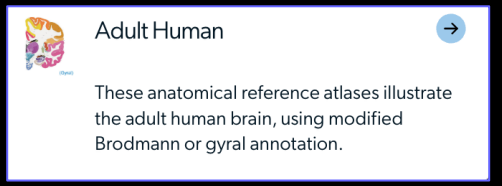


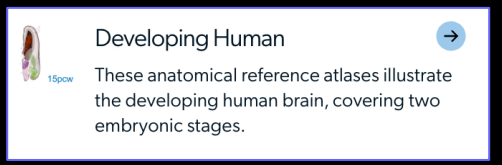


time









brain knowledge platform (BKP)/

brain-map.org/bkp

browse

Find data in the Data Catalog

explore

Visualize and compare data in ABC Atlas

analyze

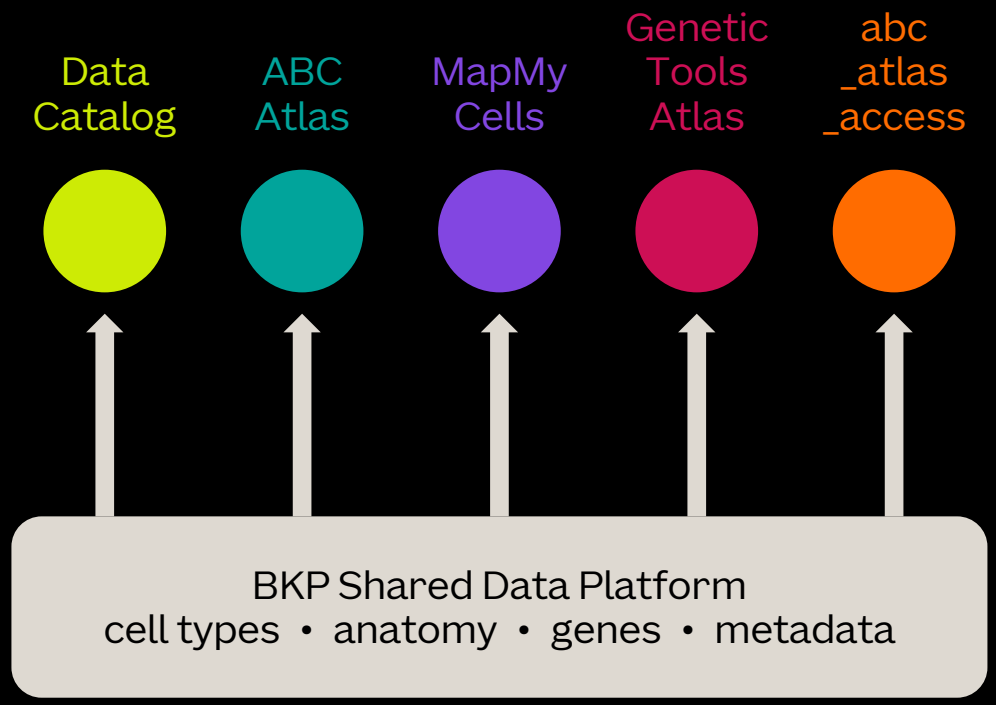
Label and align your data with MapMyCells

experiment

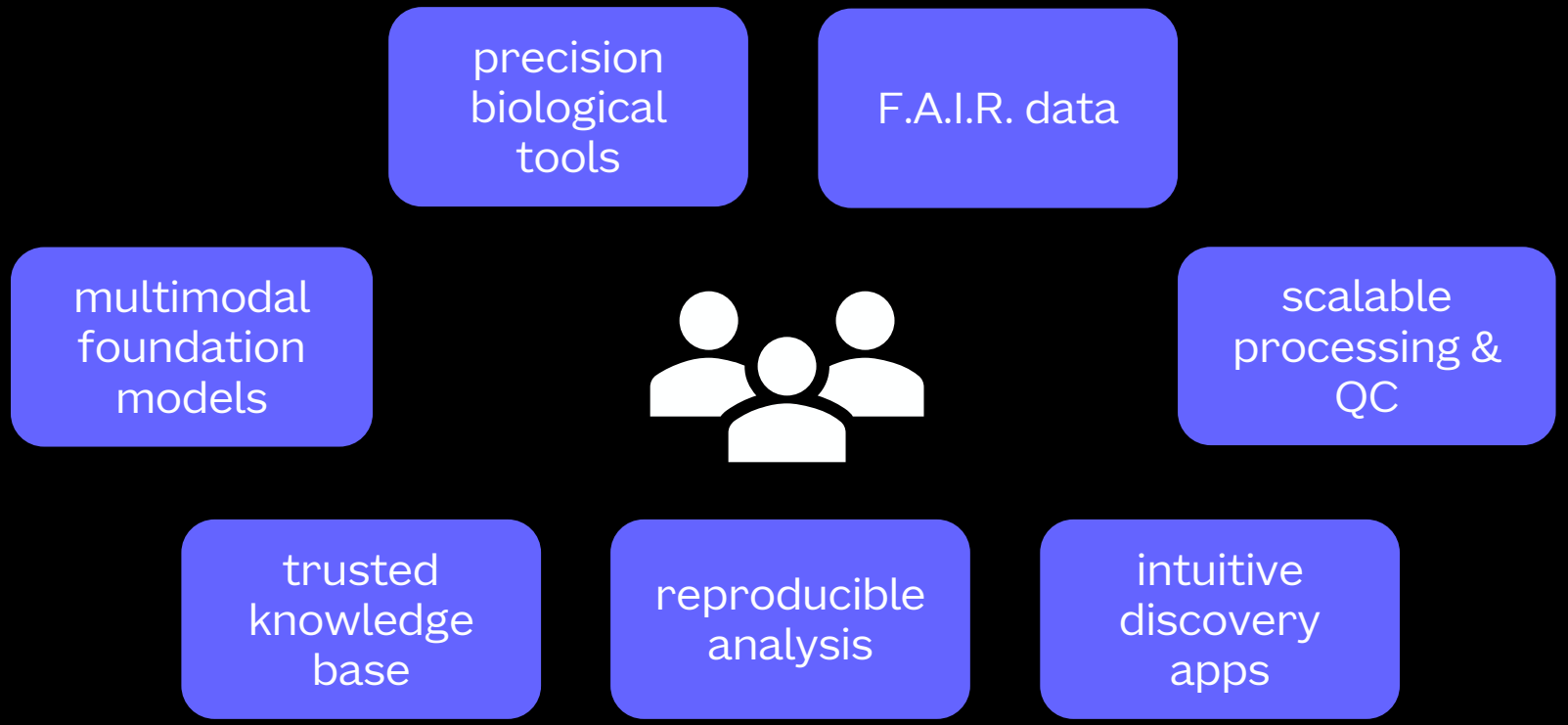
Design experiments with the Genetic Tools Atlas

develop

Download and work with data programmatically with abc_atlas_access



bkp/ technology components



AI-integrated dynamic infrastructure

- optimized data flow and computation
- fast feedback loops
- shared semantics
- provenance driven
- resource governance
- programmatic access
- AI-assisted coordination
- support data sharing and reuse

bkp/ browse

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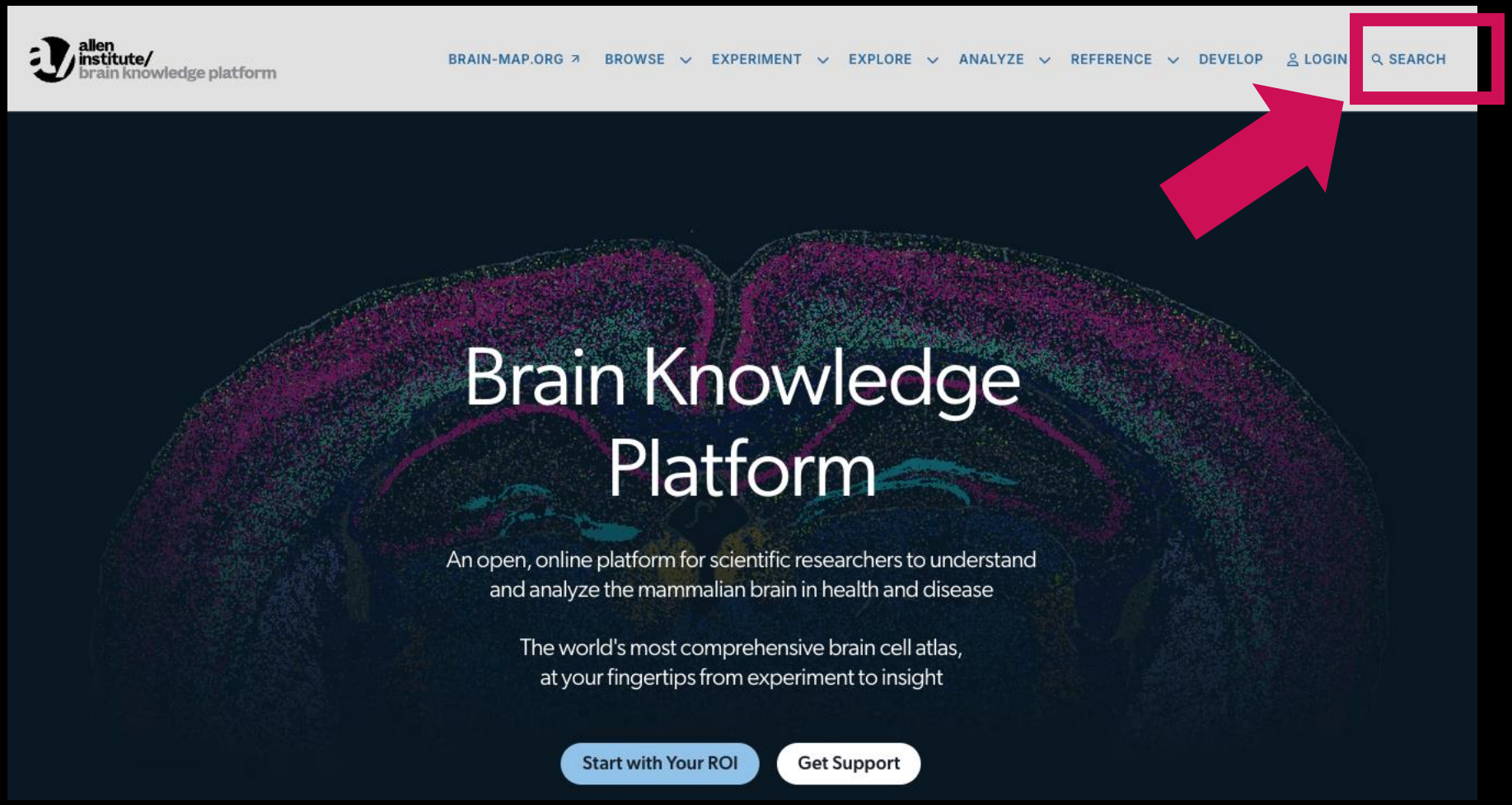
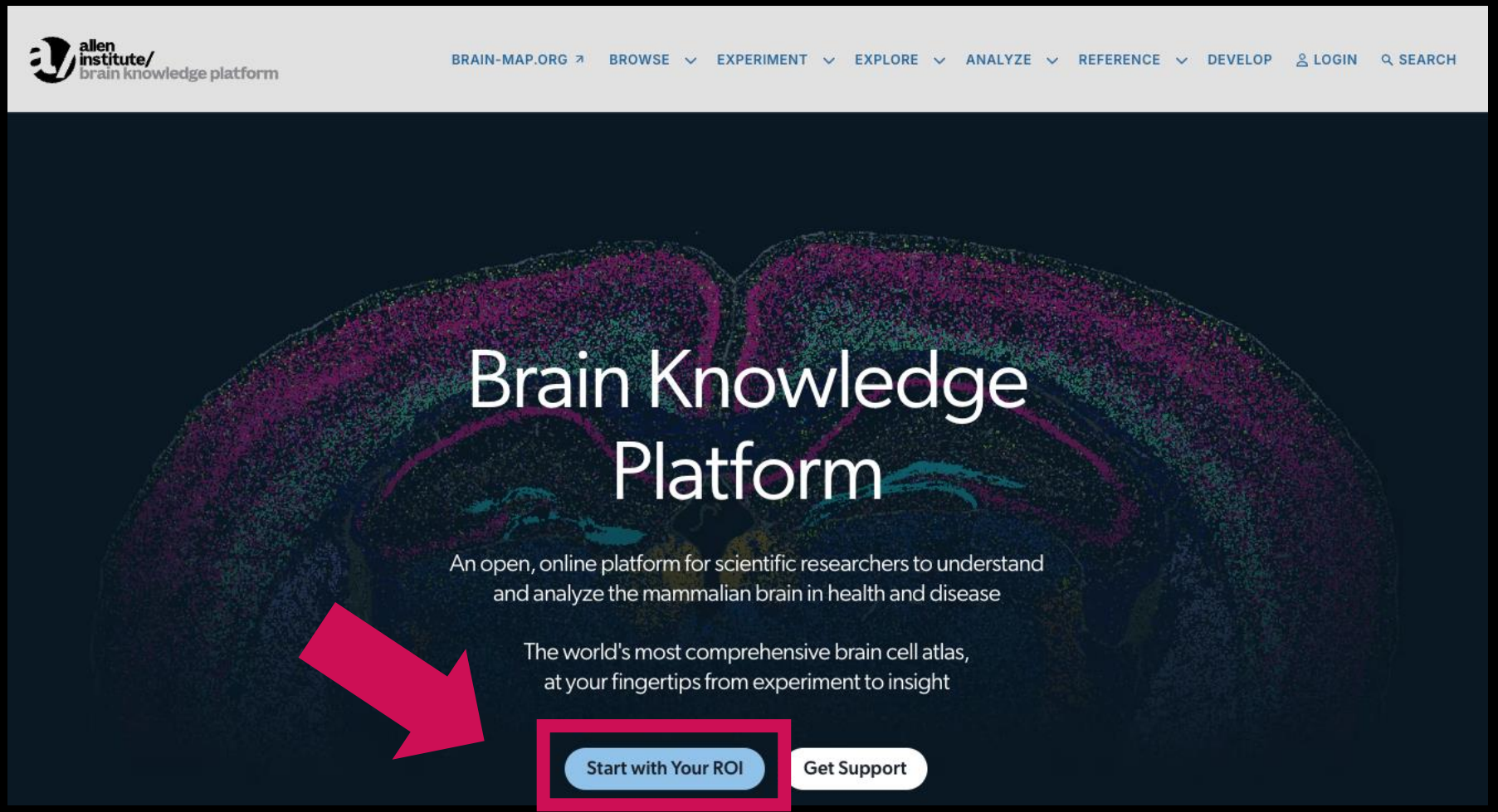


brain-map.org/bkp

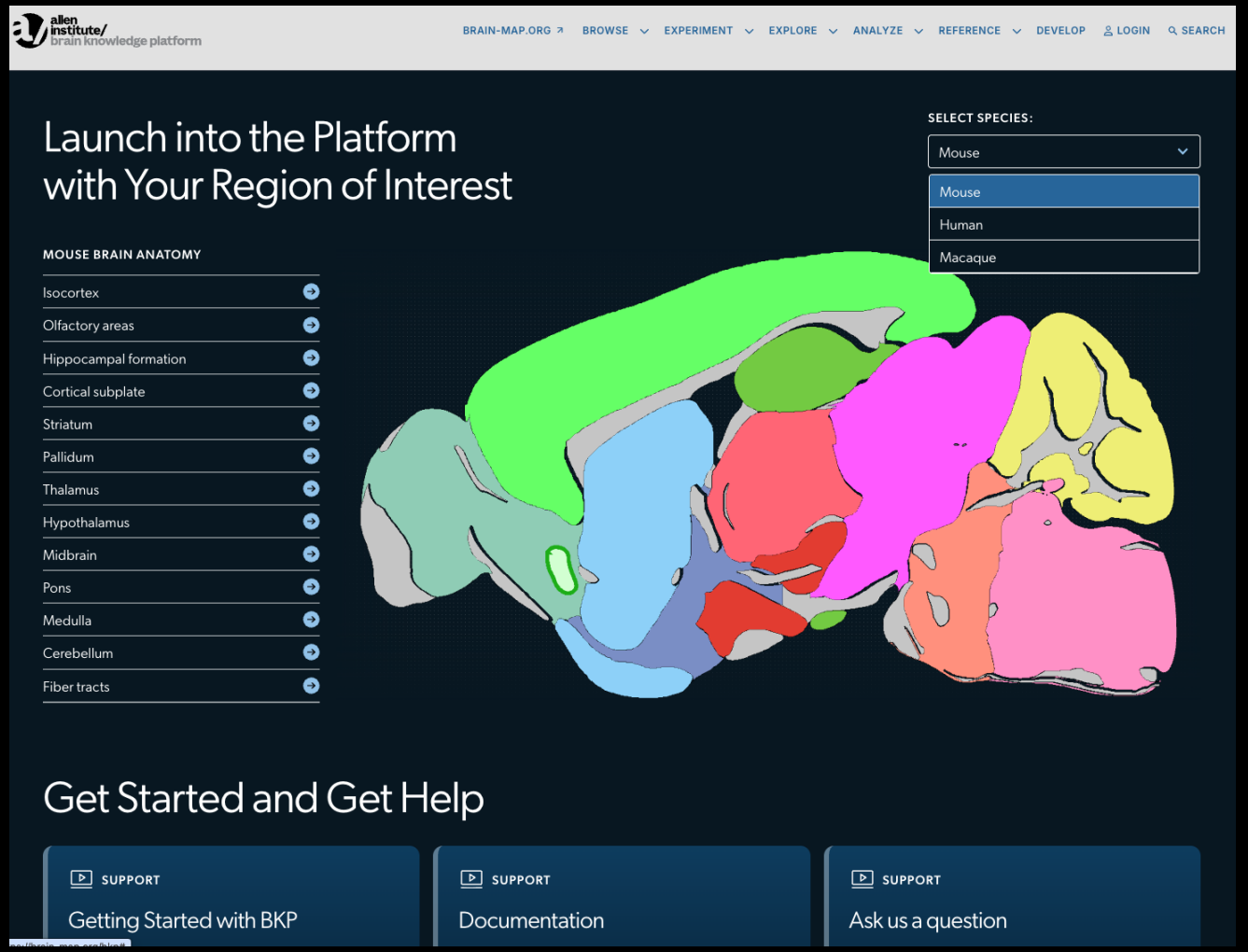
bkp/ browse



brain-map.org/bkp

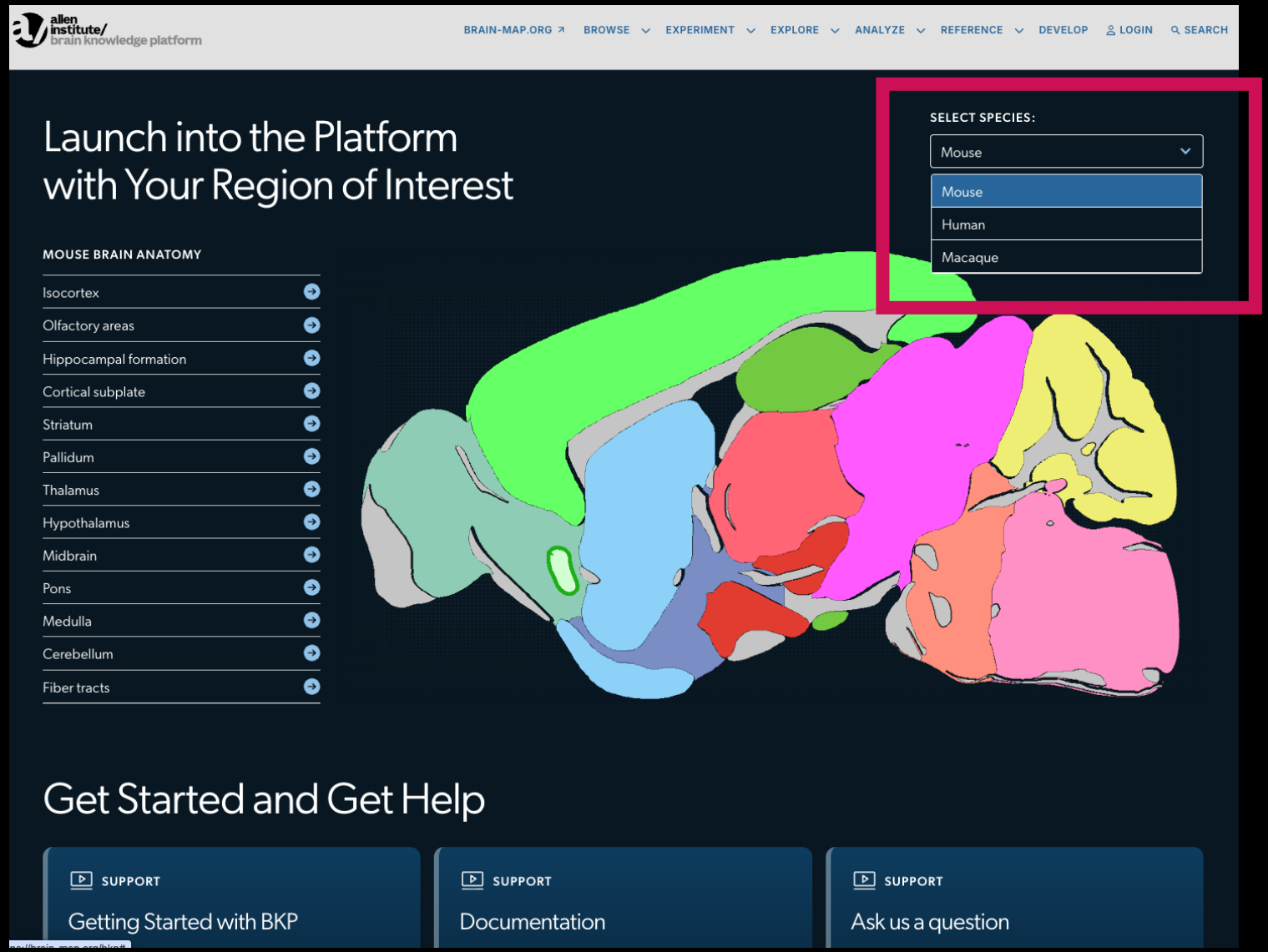
bkp/ browse

- The Brain Knowledge Platform's Data Catalog offers access to a wide range of projects & data collections
- Find relevant-to-you datasets to accelerate your research
- Filter and search by species, modality, technique, author, specimen type, and more



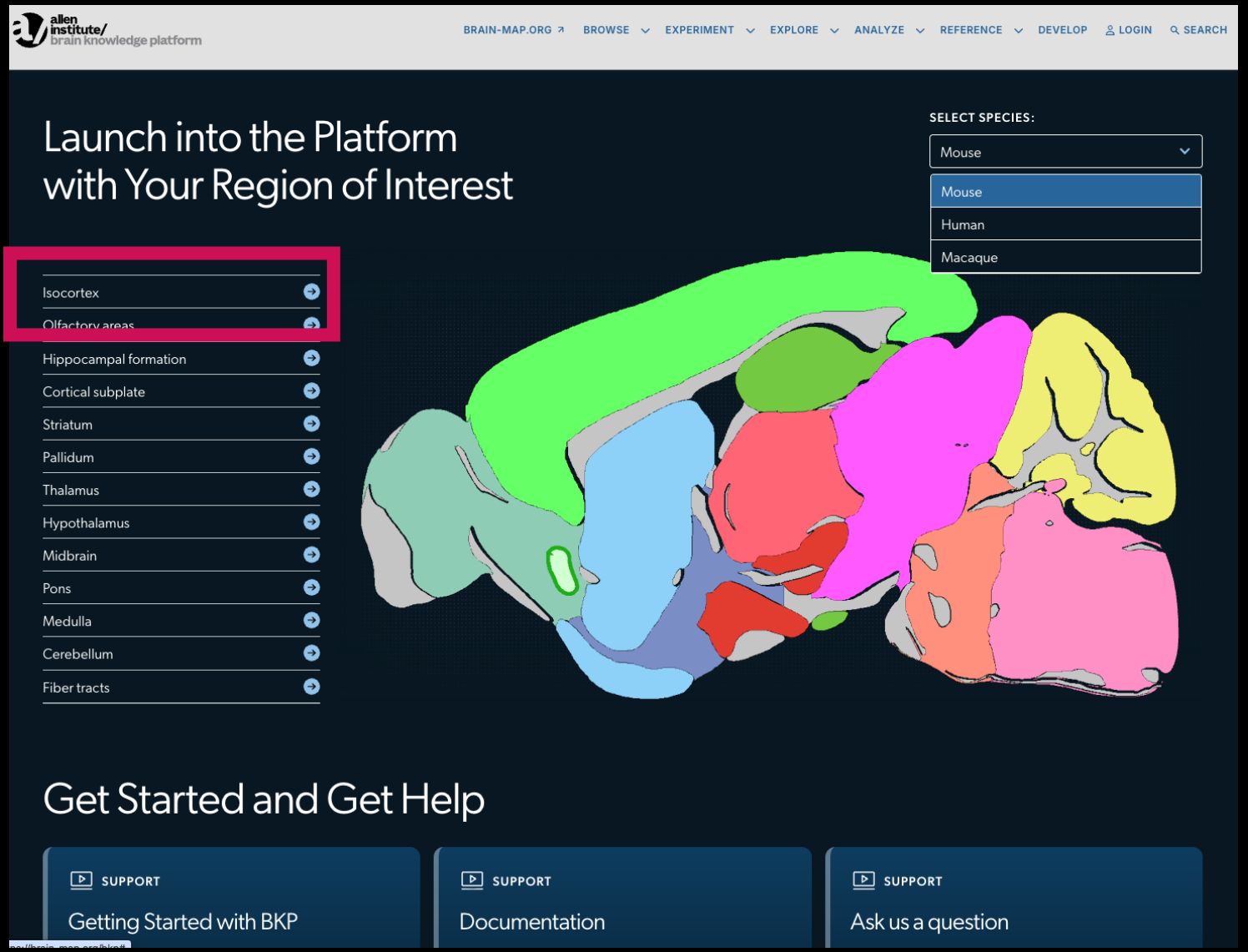
brain-map.org/bkp

bkp/ browse



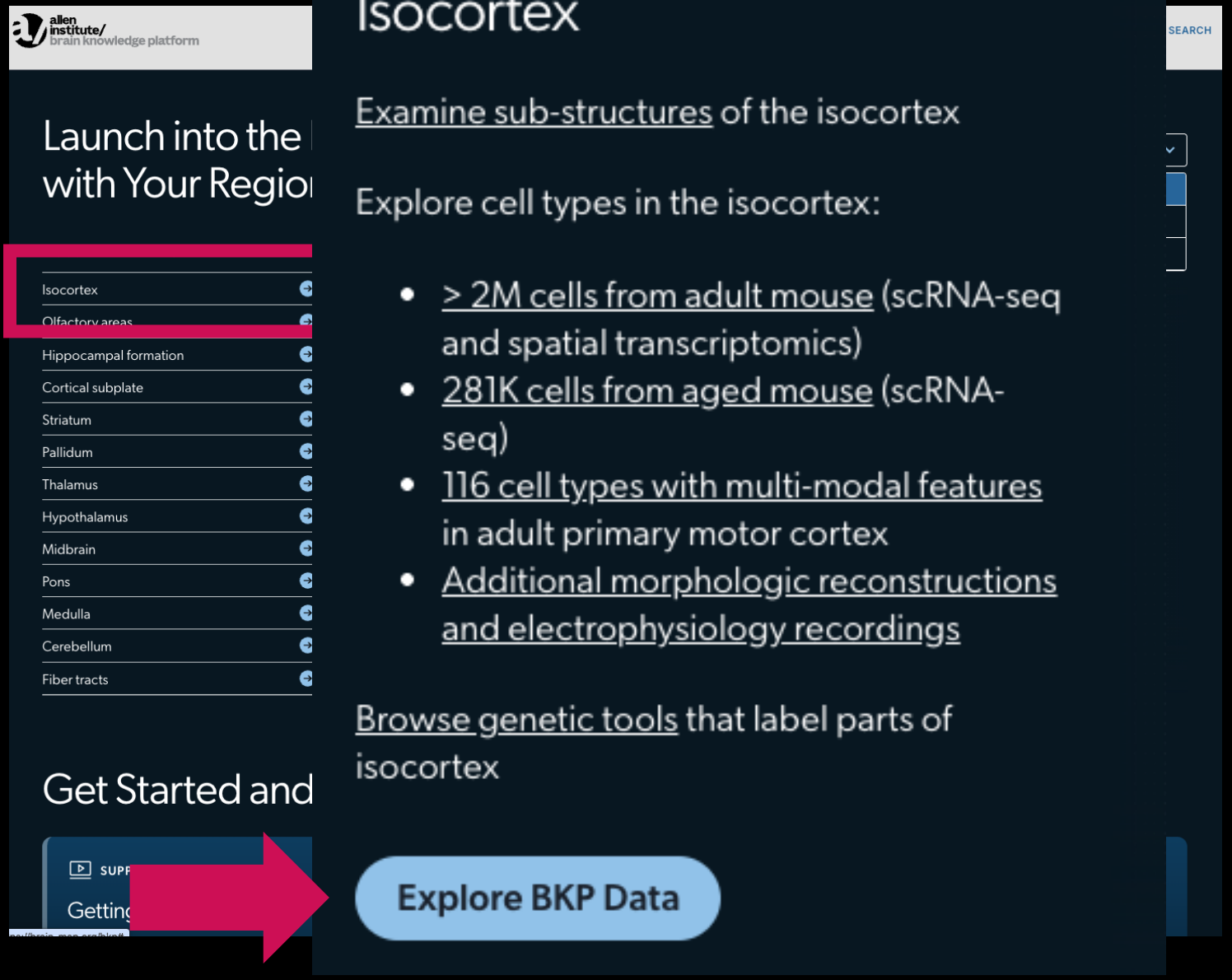
brain-map.org/bkp

bkp/ browse



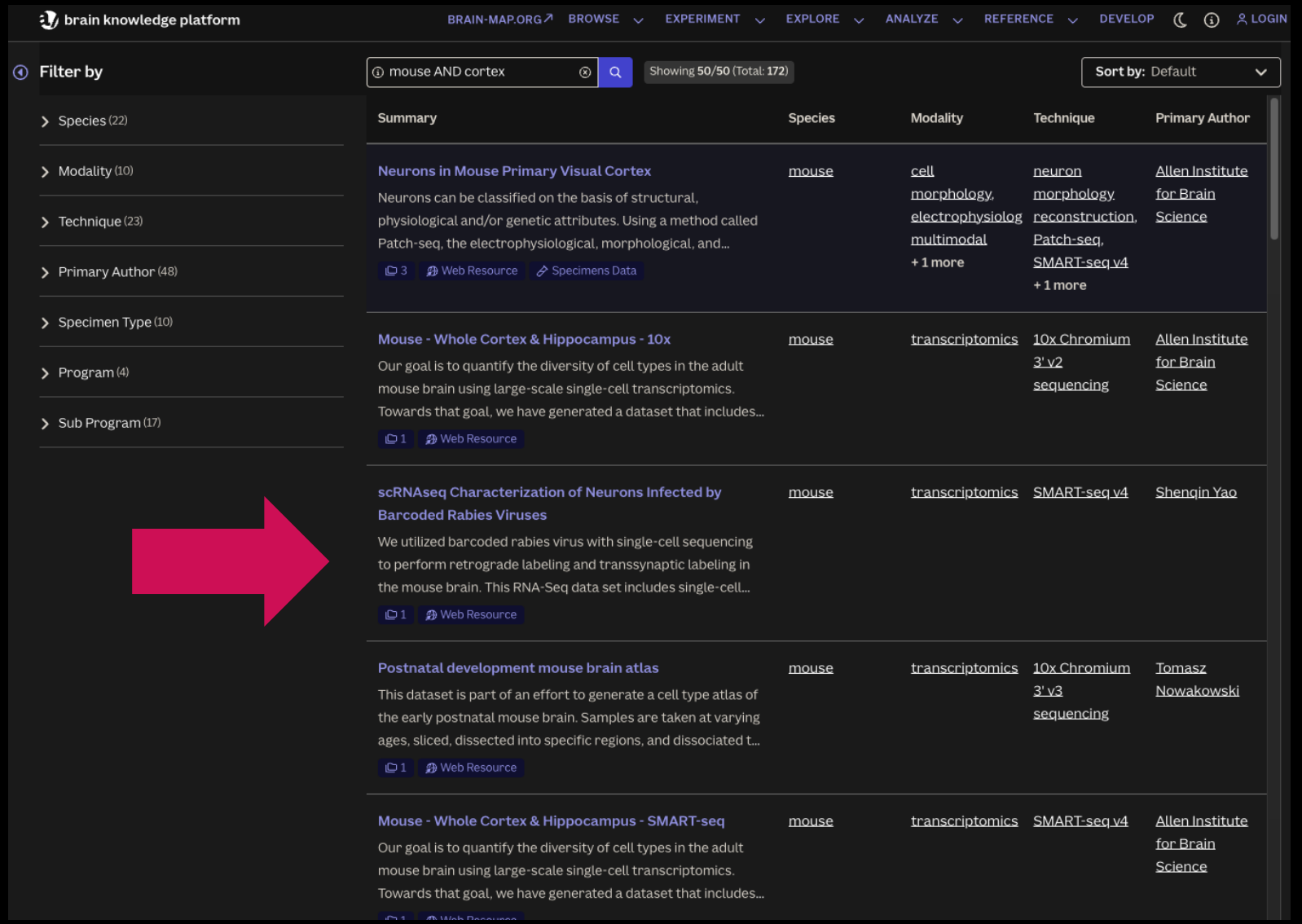
brain-map.org/bkp

bkp/ browse



brain-map.org/bkp

bkp/ browse



brain-map.org/bkp

bkp/ explore

browse

Find data in the Data Catalog

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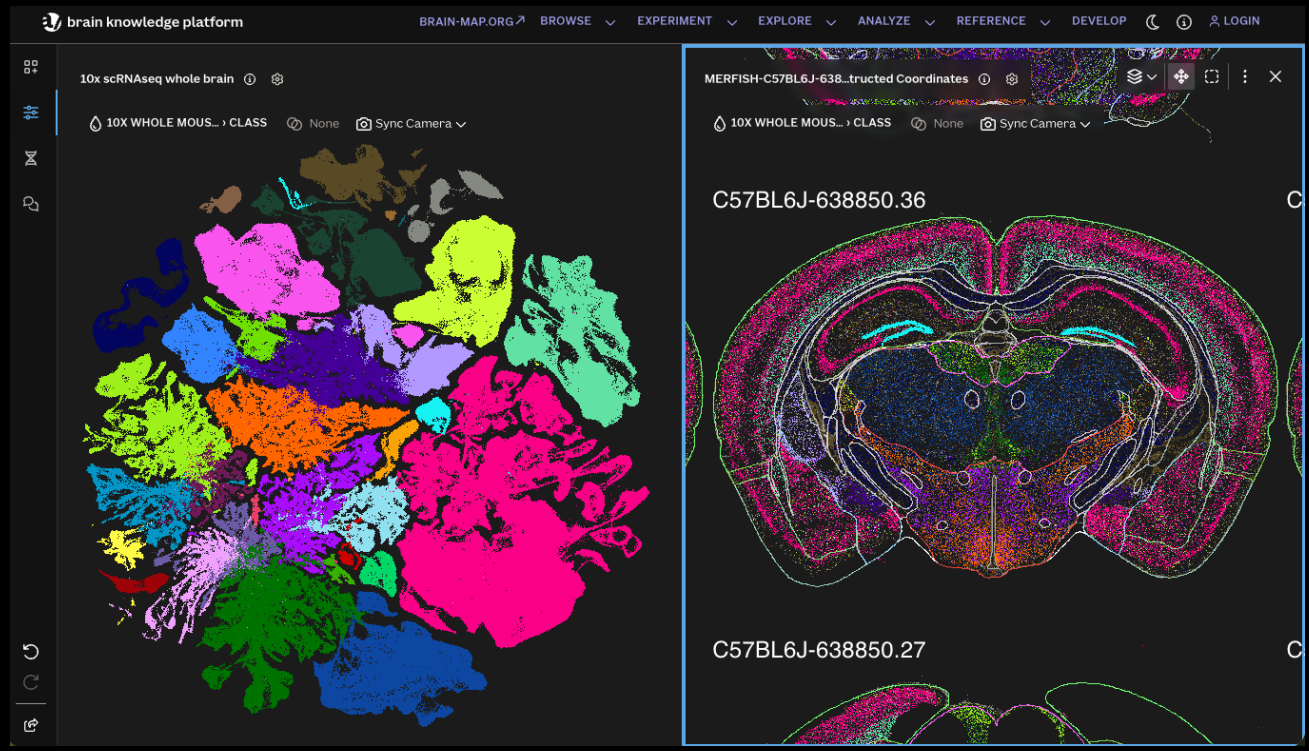
ABC atlas/ what is it?

a comprehensive data visualization platform/

28 high-quality data sets

20+ million cells



performance at scale

supports viewing hundreds of millions of points simultaneously and many thousands of cell type clusters

integrated

visualize connected data across species, modality, development, and condition, linked by shared metadata

accessible

free, web-based access serving ~3,500 monthly users

brain-map.org/explore

ABC atlas/ what's in it?

mouse/

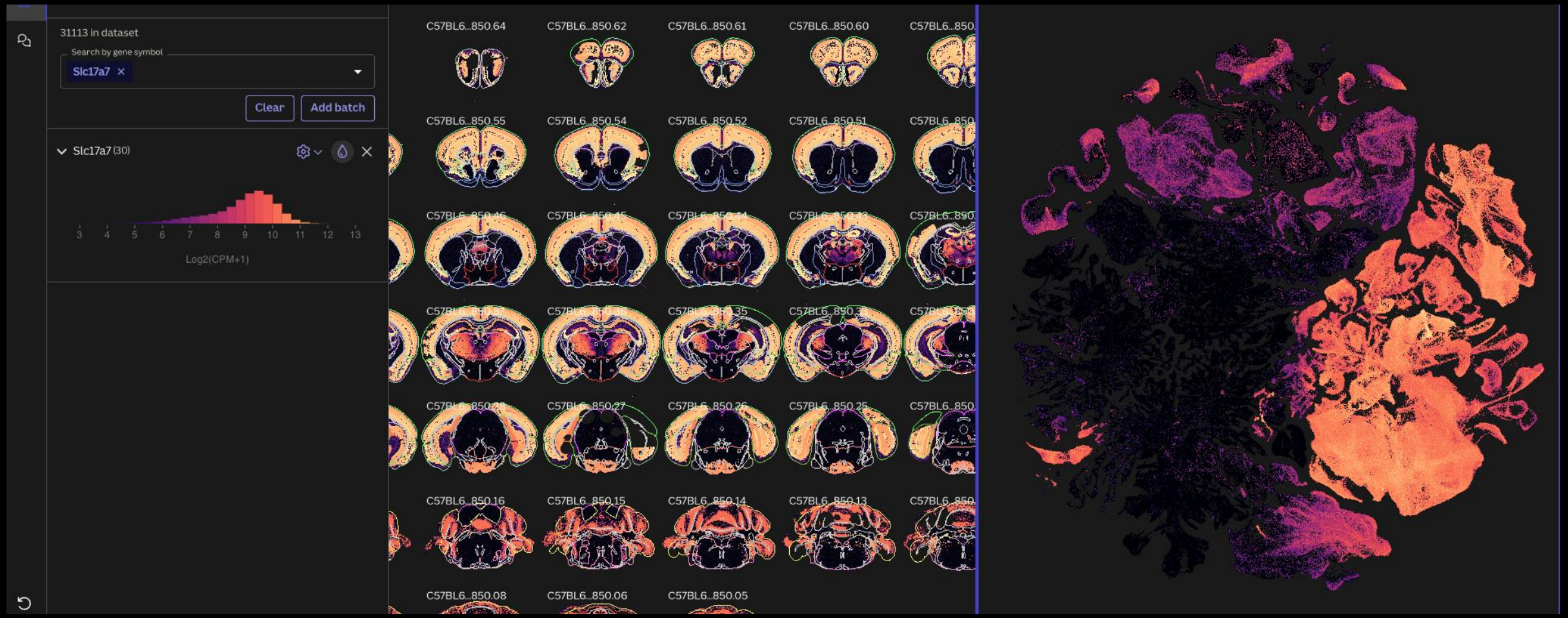
- Whole Mouse Brain (scRNAseq and MERFISH data)
  - Yao et al., 2023, Zhang et al., 2023 – Nature
- Consensus Whole Mouse Brain (scRNAseq)
  - Yao et al., 2023, Langlieb et al., 2023 – Nature
- Aging Mouse (scRNAseq)
  - Jin et al., 2025 – Nature
- Developing Mouse Visual Cortex (scRNAseq)
  - Gao et al., 2025 – Nature

human & NHP/

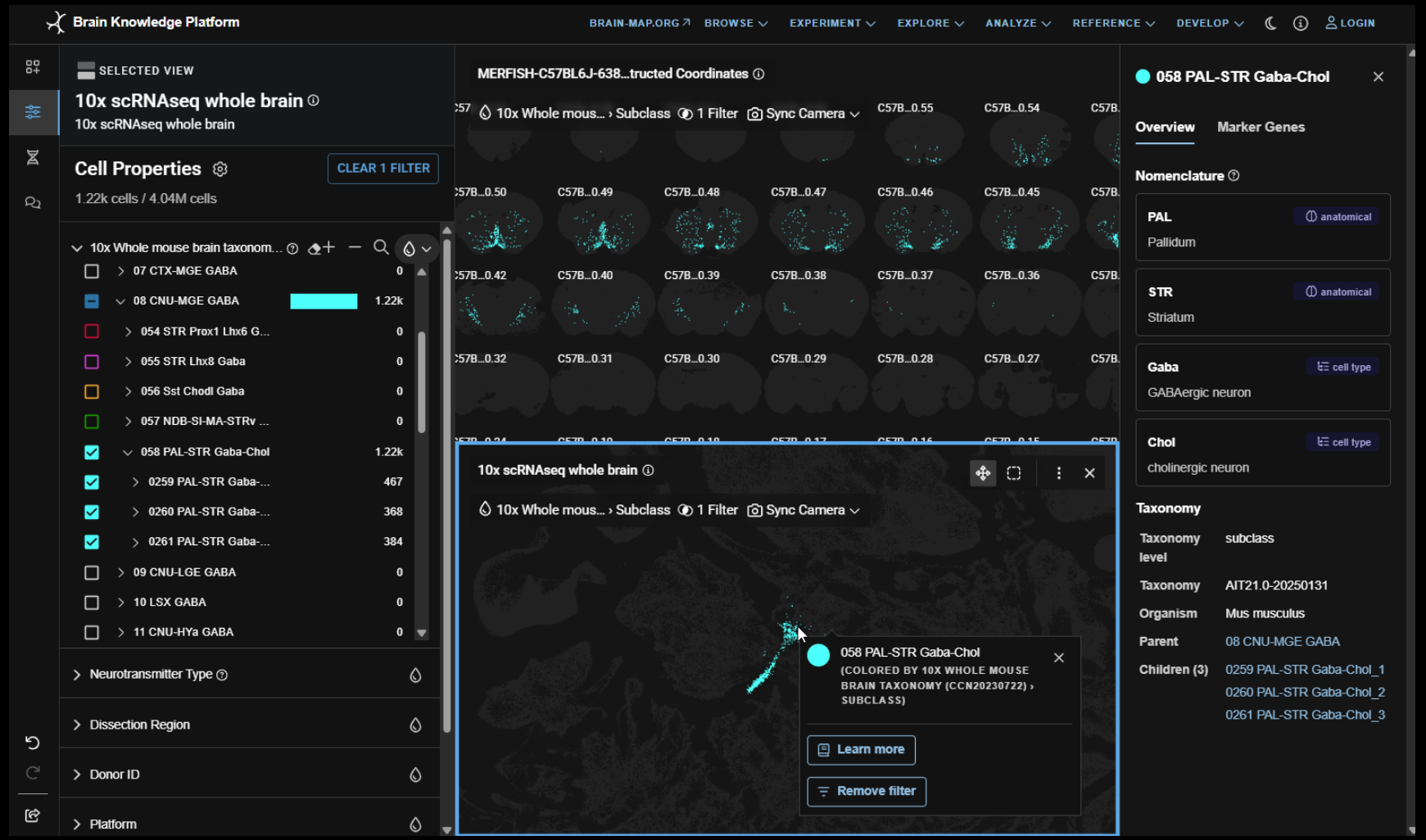
- Whole Human Brain (snRNAseq)
  - Siletti et al., 2023 – Science
- Human Alzheimer's Disease – MTG and DLPFC (SEA-AD) (snRNAseq and MERFISH data)
  - Gabitto & Travaglini et al., 2024 – Nature Neuroscience
  - Travaglini, Gabitto, Ding, et al 2026 – bioRxiv
  - Kana et al 2026 - bioRxiv
- Human Parkinson's Disease (ASAP Human Postmortem BrainSeq) (snRNAseq)
  - (No publication—collaboration with MJFF/ASAP)
- Cross-species HMBA Basal Ganglia (snRNAseq and MERFISH data)
  - Johansen & Fu et al, 2026
  - Hewitt & Turner et al, 2026 - bioRxiv

brain-map.org/explore

ABC atlas/ what can you do/ view gene expression



ABC atlas/ what can you do/ explore cell types



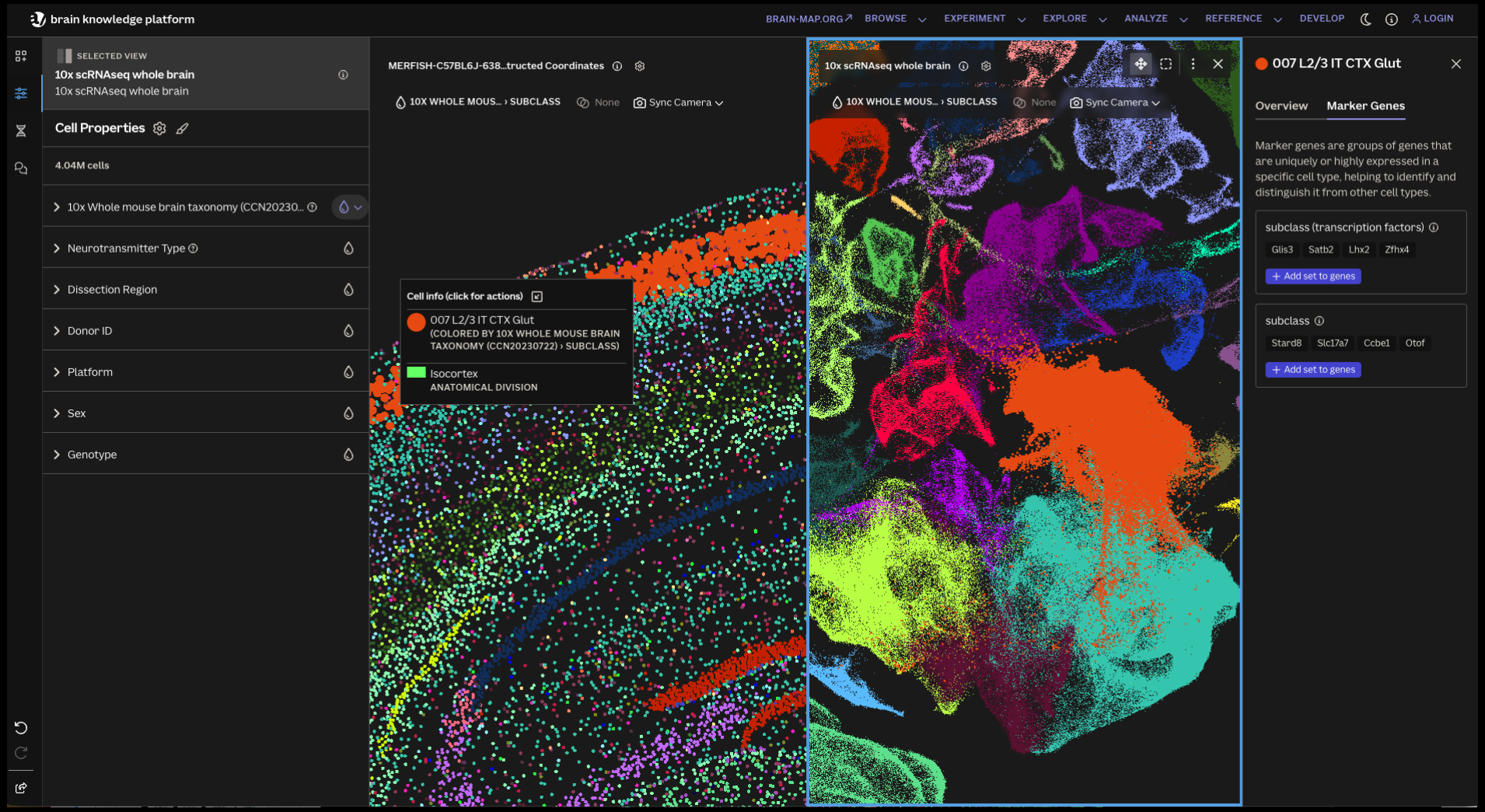
What do the acronyms mean?

How is this cell type defined?

What else is known about this cell type?

brain-map.org/explore

ABC atlas/ what can you do/ explore cell types



brain-map.org/explore

ABC atlas/ what can you do/ compare data

- unlock new comparative visualization and analysis
- Parkinson's data (MJFF/ASAP) was mapped using MapMyCells
  - whole human brain cell types (Siletti et al)
  - well-defined Alzheimer's cell types (SEA-AD)

View in ABC Atlas

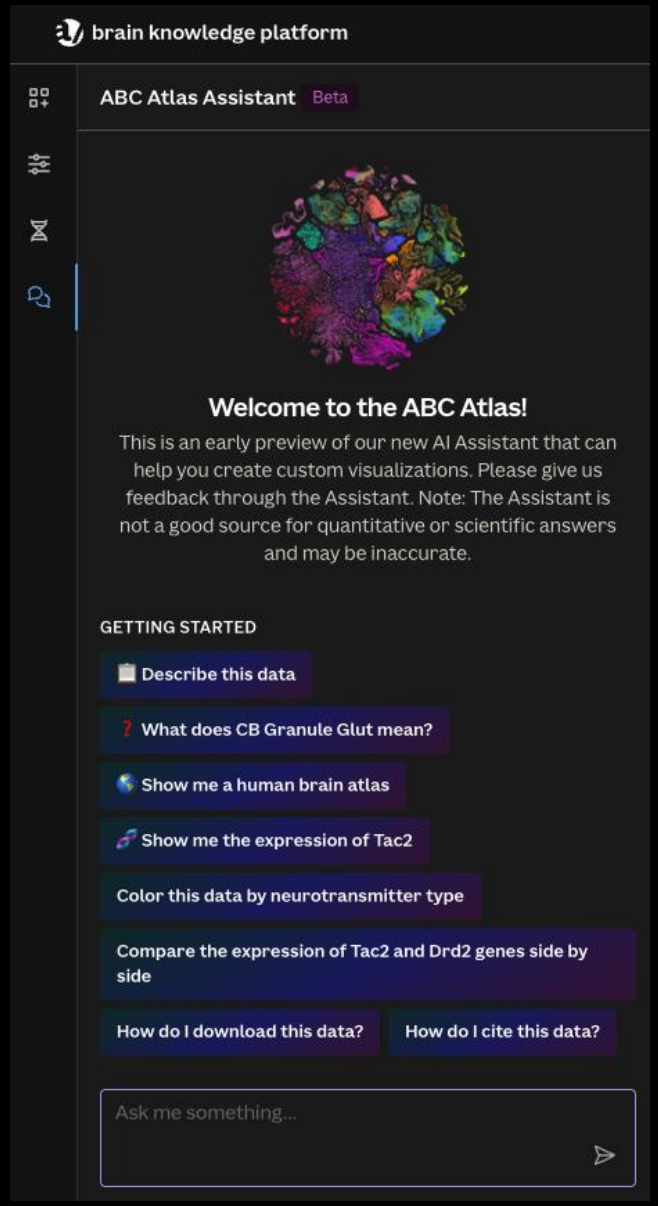
ABC atlas/ ask the ABC atlas assistant!

the AI Assistant uses natural language input to help users create custom visualizations faster

- simplifies complex tasks for new users
- accelerates advanced controls for experienced users

the Assistant can:

- describe, suggest, and open data sets
- adjust colors and filters across multiple data sets based on user instruction
- offer “how to” instructions



brain-map.org/explore

bkp/ explore

brain-map.org/bkp

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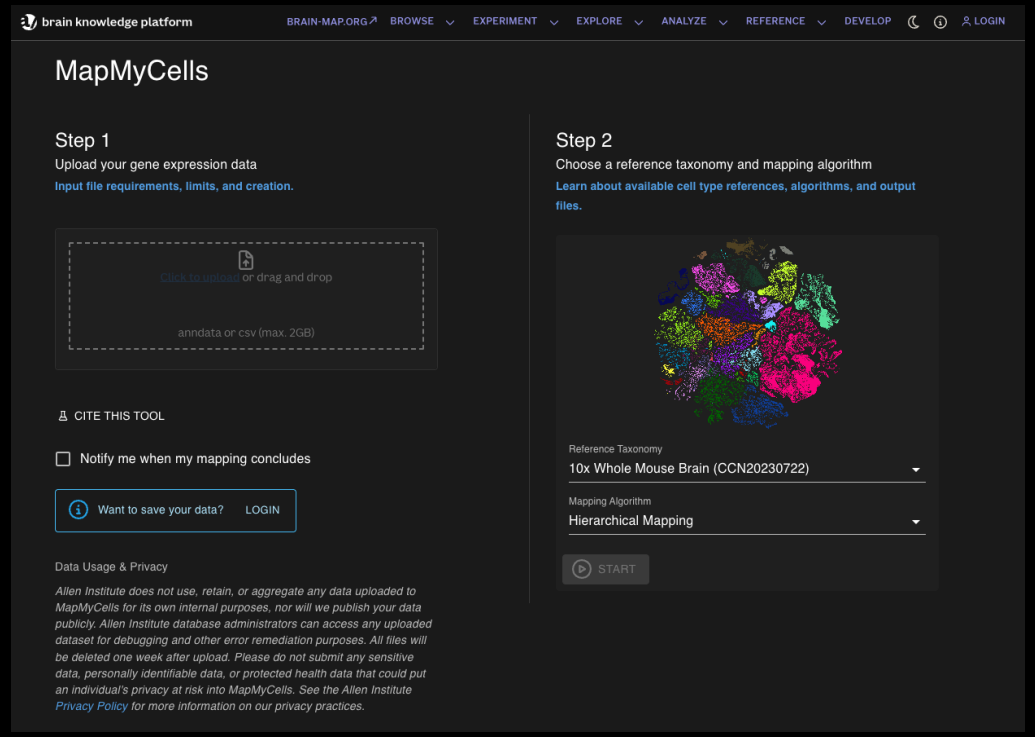
MapMyCells/ map your own RNAseq data onto an Allen reference taxonomy

6 cell type taxonomies

3 mapping algorithms



input

.h5ad file, .csv file, or cellxgene matrix

output

- .csv file
- cell IDs with allen institute taxonomy labels & probabilities
- .JSON file (metadata)

also available for offline programmatic access:
github.com/AllenInstitute/cell_type_mapper

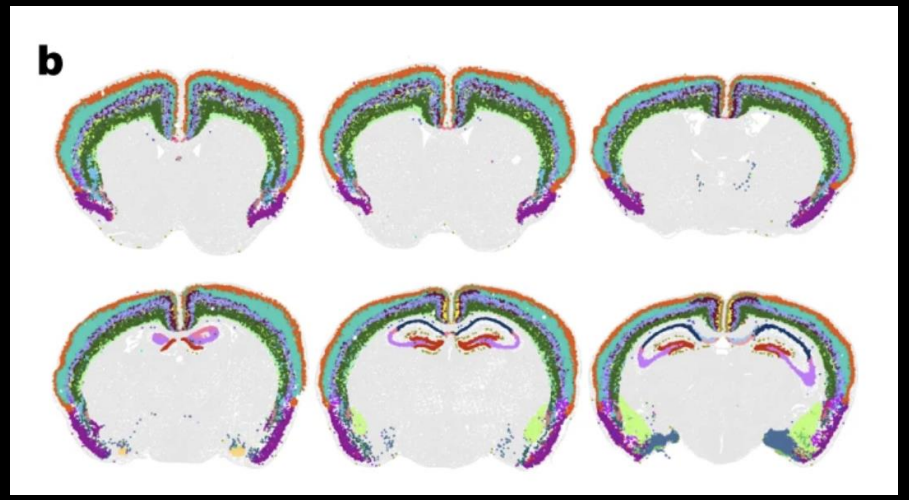
brain-map.org/bkp/analyze

MapMyCells/ map your own RNAseq data onto an Allen reference taxonomy

Purpose: "I have unlabeled cell-by-gene data. What cell types are in my data, according to the Allen Institute?"

Example use cases:

I have spatial transcriptomic data. What cell types are represented there? Do cell types segregate spatially? (actual use case in Yao et al.)



I have data annotated with cell types by [some other method]. What cell types does the Allen Institute think I have? How do the two annotations align?

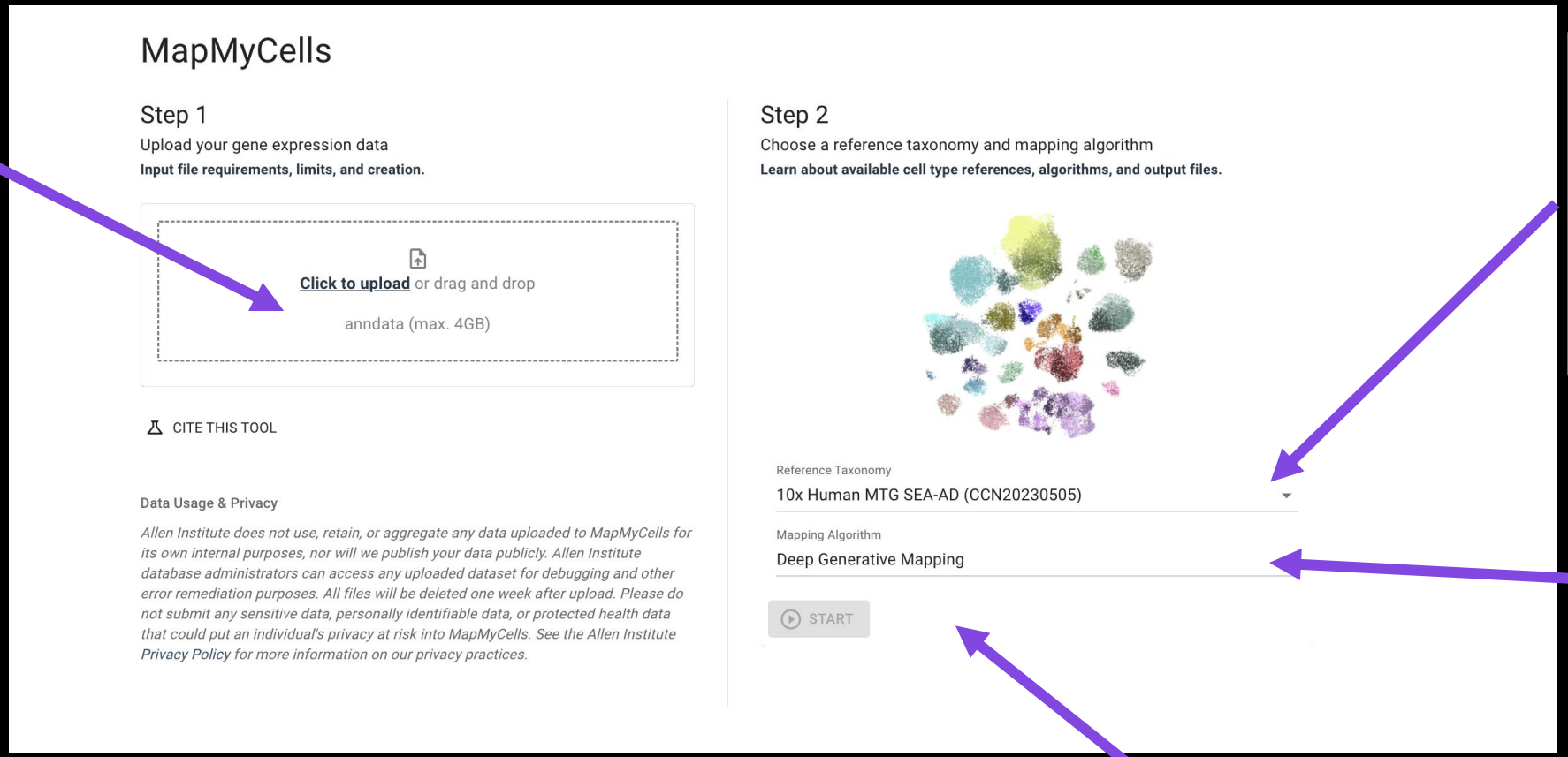
I have developmental or disease model data. How does the distribution of cell types (according to MapMyCells) change with time?

brain-map.org/bkp/analyze

MapMyCells/ how does it work?

https://knowledge.brain-map.org/mapmycells/process/

1) Drag-and-drop input file here



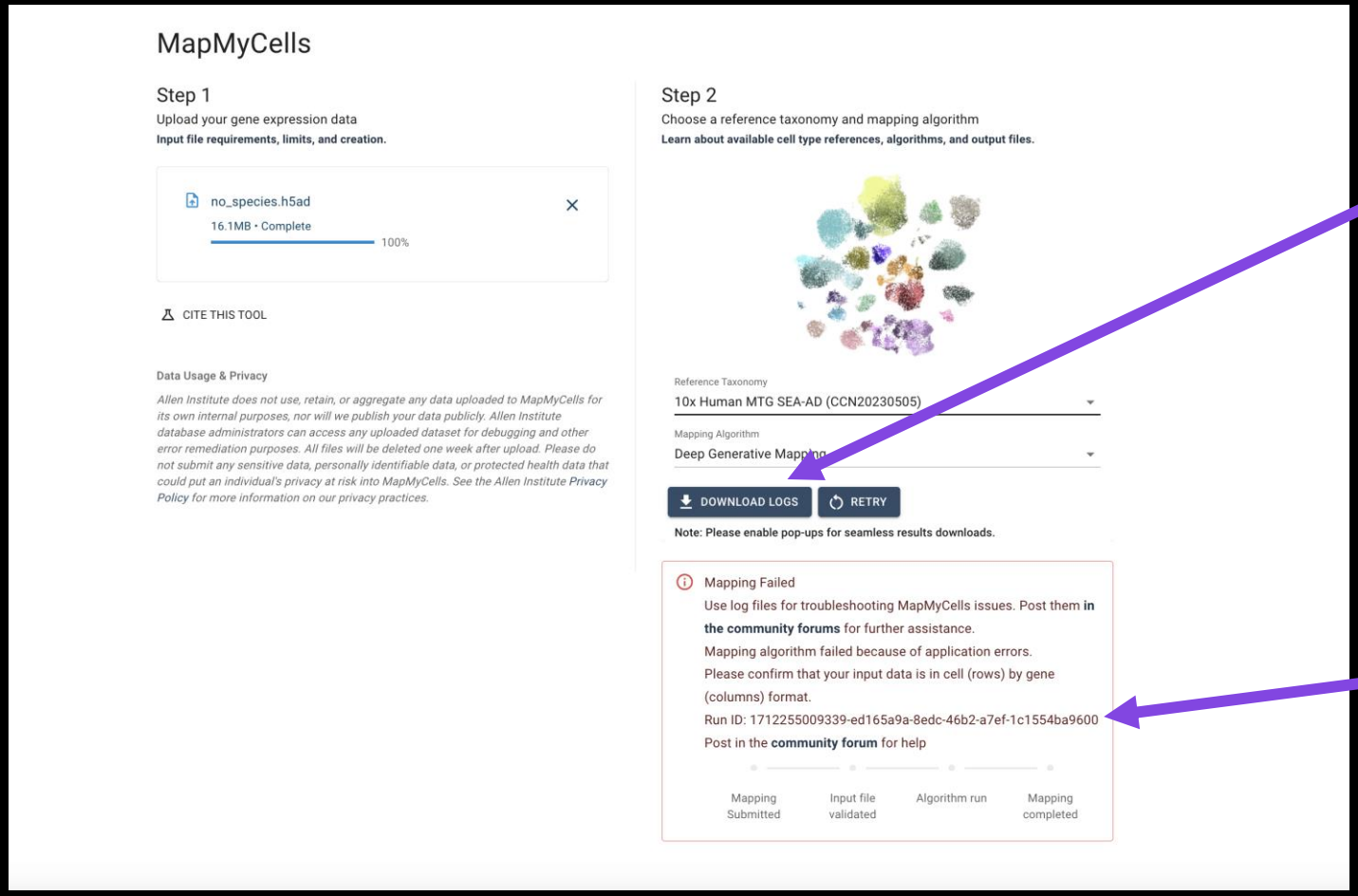
2) Select taxonomy to which to map
- Whole Mouse Brain
- SEA-AD (human MTG)

3) Select mapping algorithm

4) Click "Start"

MapMyCells/ failed mapping

https://knowledge.brain-map.org/mapmycells/process/



If run fails, can still download log file with potentially helpful error message.

Note Run-ID. Can post to Allen community forums and we can look up your run and help you debug it.

MapMyCells/ what is the output?

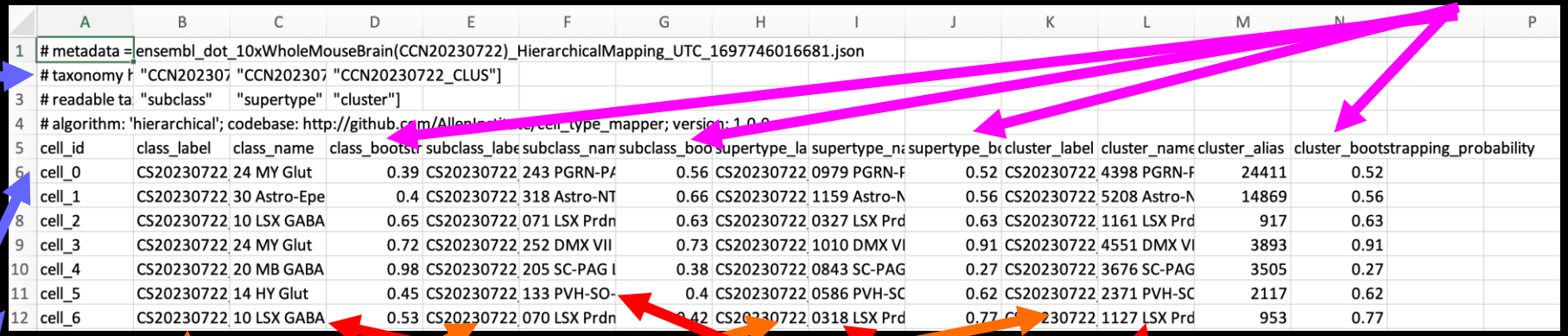
https://knowledge.brain-map.org/mapmycells/process/

CSV output

Confidence in mapping

Rows denoted by '#' are metadata

Each row is a cell from the input file.



	A	B	C	D	E	F	G	H	I	J	K	L	M	N	P
1	# metadata =	ensembl_dot_10xWholeMouseBrain(CCN20230722)_HierarchicalMapping_UTC_1697746016681.json													
2	# taxonomy h	"CCN202307	"CCN202307	"CCN20230722_CLUS"]											
3	# readable ta	"subclass"	"supertype"	"cluster"]											
4	# algorithm: 'hierarchical'; codebase: http://github.com/AllenInstitute/cell_type_mapper; version: 1.0.0														
5	cell_id	class_label	class_name	class_bootstr	subclass_labe	subclass_nam	subclass_boo	supertype_la	supertype_na	supertype_bo	cluster_label	cluster_name	cluster_alias	cluster_bootstrapping_probability	
6	cell_0	CS20230722	24 MY Glut	0.39	CS20230722	243 PGRN-PA	0.56	CS20230722	0979 PGRN-F	0.52	CS20230722	4398 PGRN-F	24411	0.52	
7	cell_1	CS20230722	30 Astro-Epe	0.4	CS20230722	318 Astro-NT	0.66	CS20230722	1159 Astro-N	0.56	CS20230722	5208 Astro-N	14869	0.56	
8	cell_2	CS20230722	10 LSX GABA	0.65	CS20230722	071 LSX Prdn	0.63	CS20230722	0327 LSX Prd	0.63	CS20230722	1161 LSX Prd	917	0.63	
9	cell_3	CS20230722	24 MY Glut	0.72	CS20230722	252 DMX VII	0.73	CS20230722	1010 DMX VI	0.91	CS20230722	4551 DMX VI	3893	0.91	
10	cell_4	CS20230722	20 MB GABA	0.98	CS20230722	205 SC-PAG I	0.38	CS20230722	0843 SC-PAG	0.27	CS20230722	3676 SC-PAG	3505	0.27	
11	cell_5	CS20230722	14 HY Glut	0.45	CS20230722	133 PVH-SO-	0.4	CS20230722	0586 PVH-SC	0.62	CS20230722	2371 PVH-SC	2117	0.62	
12	cell_6	CS20230722	10 LSX GABA	0.53	CS20230722	070 LSX Prdn	0.42	CS20230722	0318 LSX Prd	0.77	230722	1127 LSX Prd	953	0.77	

Machine-readable assignments

Human-readable assignments

MapMyCells/ what are the mapping algorithms?

https://knowledge.brain-map.org/mapmycells/process/

Available algorithms:

- **Deep Generative mapping**
  - Use variational autoencoders to map labeled and unlabeled data to the same latent space. Assign cell types in that space.
  - Supported taxonomies:
    - SEA-AD (human MTG)
- **Hierarchical mapping**
  - Use Pearson's correlation coefficient between unlabeled cells and cell types to assign cell types hierarchically (e.g. class, then subclass, then cluster)
  - Supported taxonomies:
    - Whole Mouse Brain
    - SEA-AD (human MTG)
- **Flat mapping**
  - Hierarchical mapping, but maps directly to leaf node of taxonomy
  - Supported taxonomies:
    - Whole Mouse Brain
    - SEA-AD (human MTG)

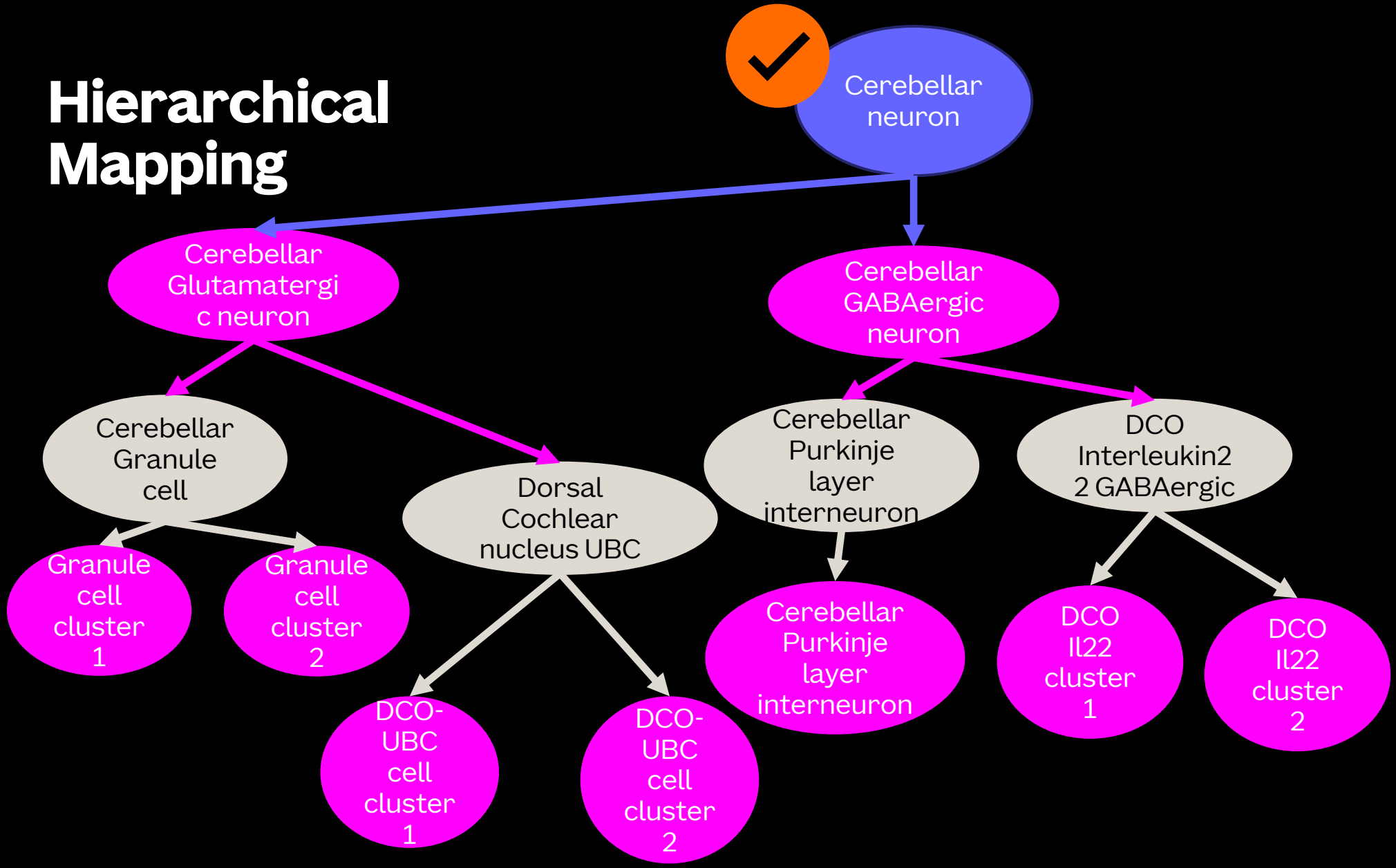
MapMyCells/ hierarchal mapping example

Depends on two summaries of taxonomy data

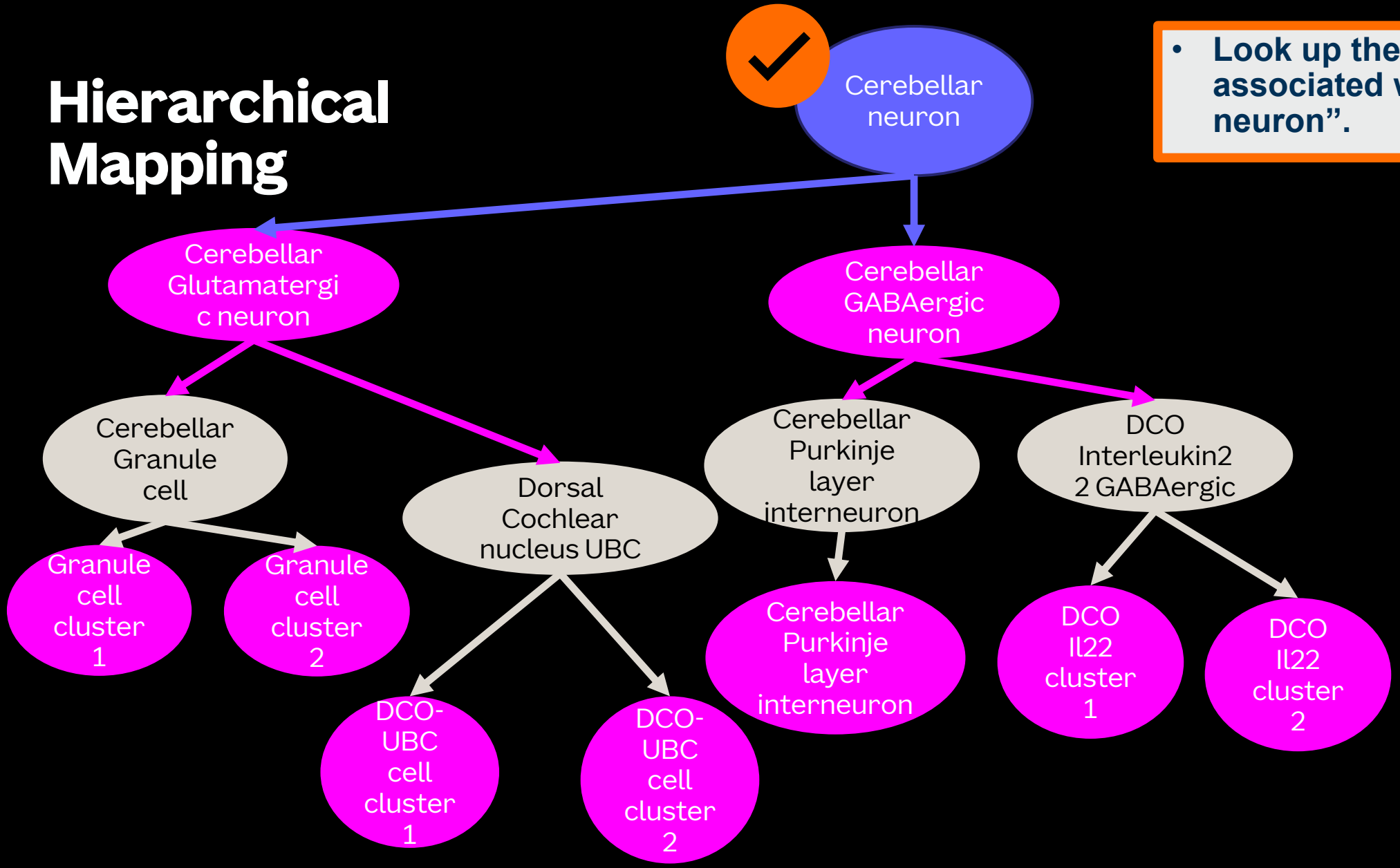
- Average gene expression profile of each cell type in the taxonomy
- Marker gene lookup table
    - For each cell type in the taxonomy, give me a list of marker genes that are good at discriminating between that cell type's children

In broad strokes: the algorithm takes each cell and correlates its gene expression profile to the average expression profile of cell types using only the designated marker genes.

Hierarchical Mapping

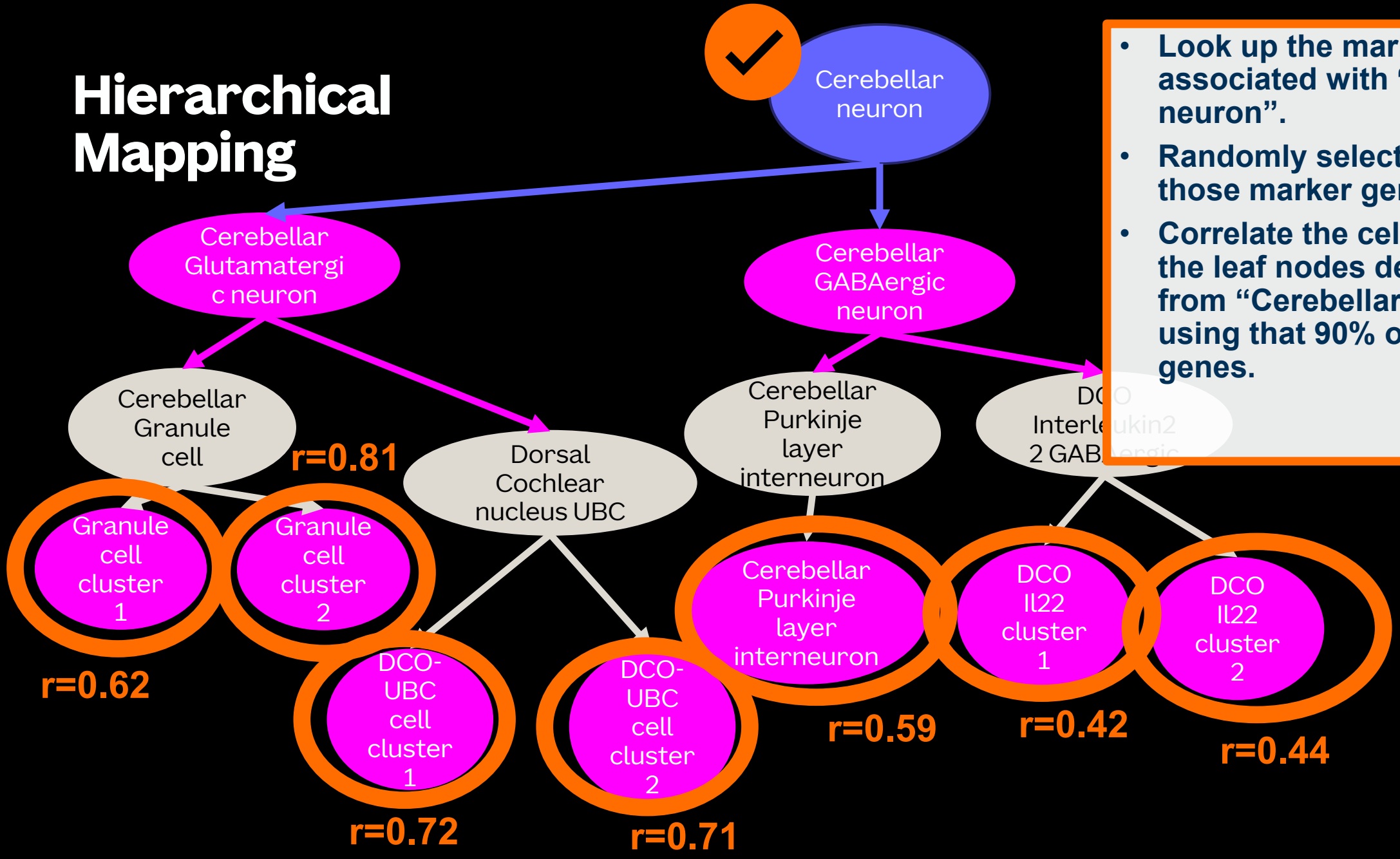


Hierarchical Mapping



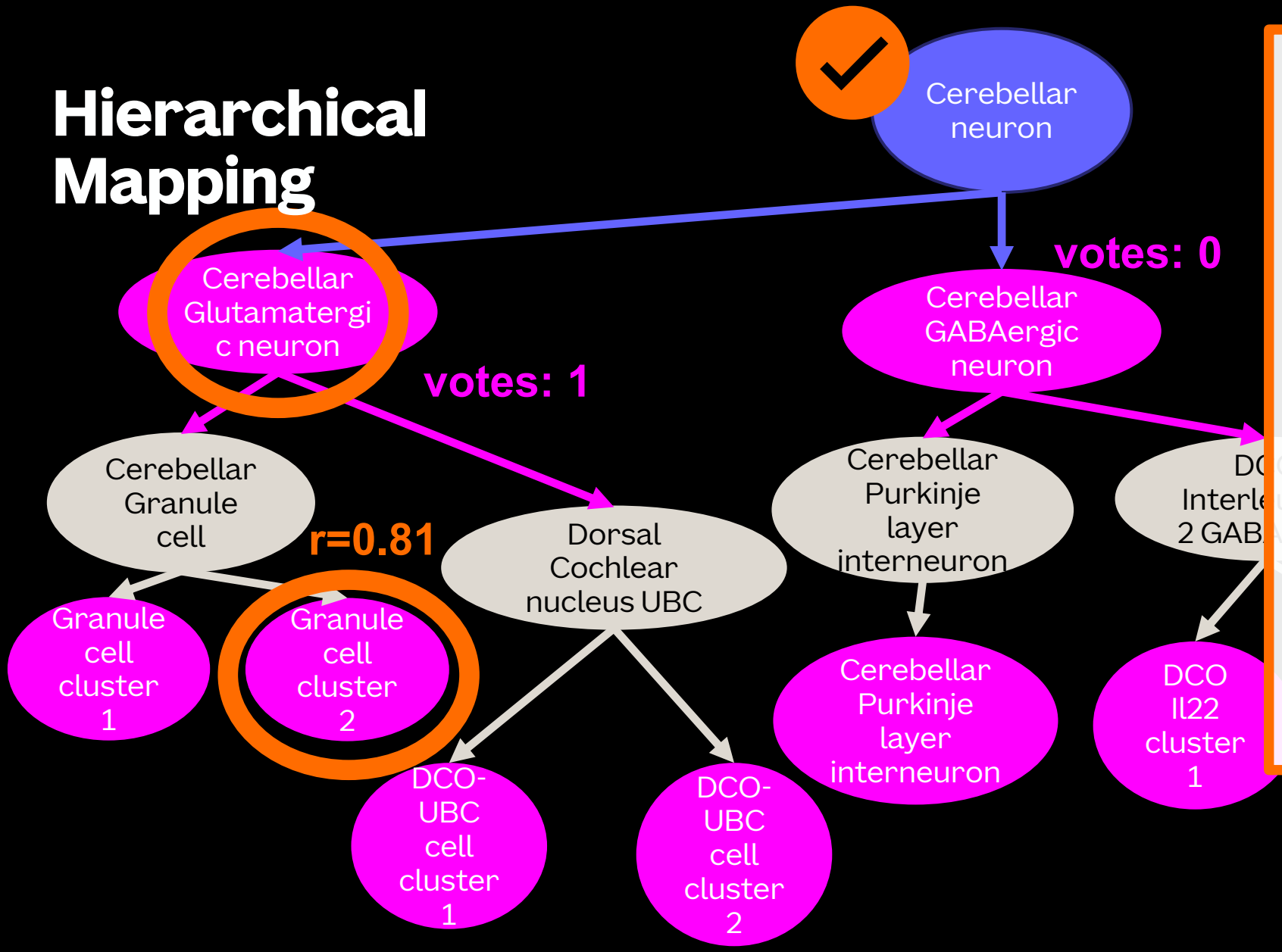
- **Look up the marker genes associated with “Cerebellar neuron”.**

Hierarchical Mapping



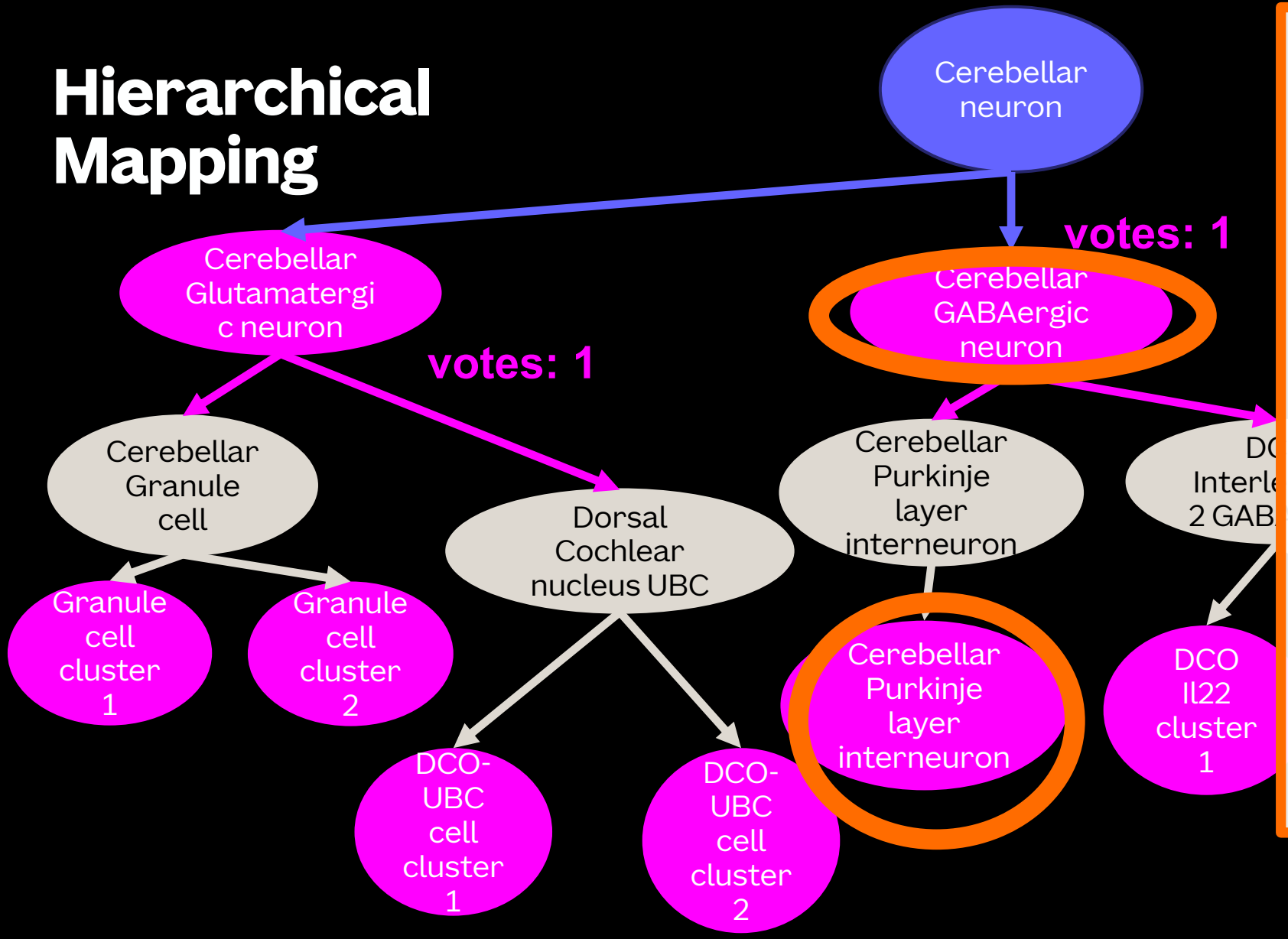
- Look up the marker genes associated with “Cerebellar neuron”.
- Randomly select 90% of those marker genes.
- Correlate the cell with each of the leaf nodes descended from “Cerebellar neuron” using that 90% of marker genes.

Hierarchical Mapping



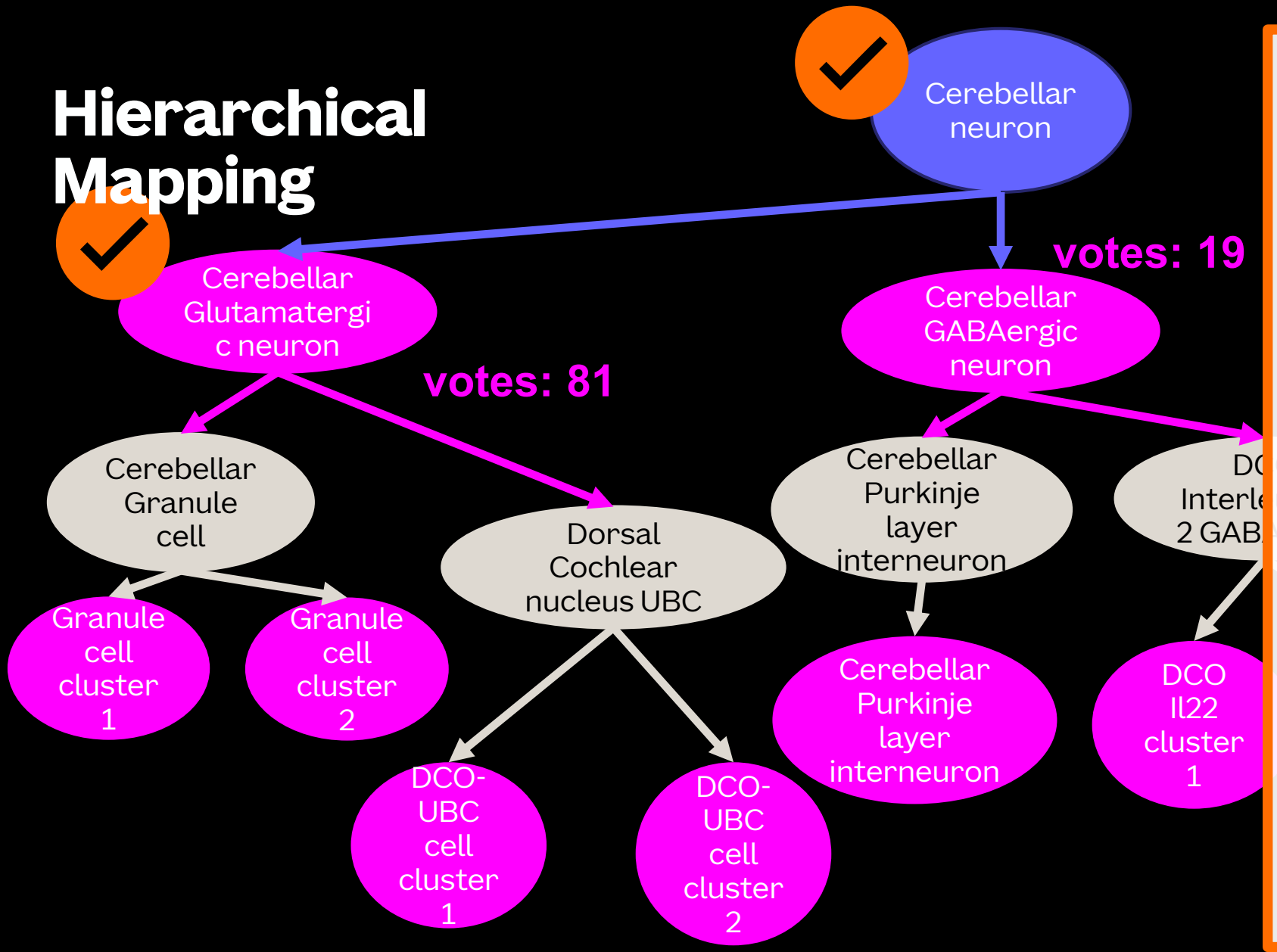
- Look up the marker genes associated with “Cerebellar neuron”.
- Randomly select 90% of those marker genes.
- Correlate the cell with each of the leaf nodes descended from “Cerebellar neuron” using that 90% of marker genes.
- Select the most highly correlated leaf node.
- Treat a vote for that leaf node as a vote for its parent at the level being chosen.

Hierarchical Mapping



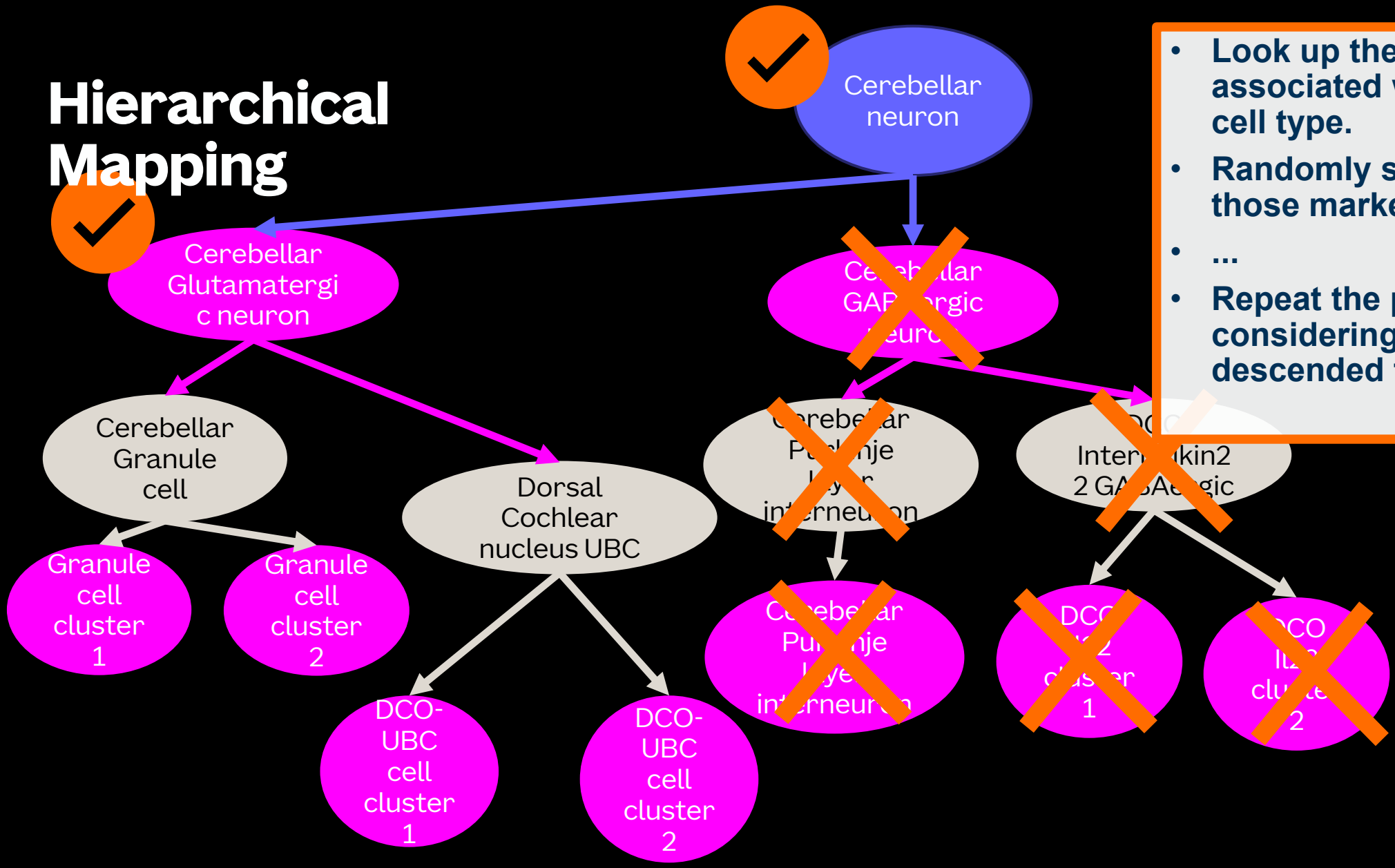
- Look up the marker genes associated with “Cerebellar neuron”.
- Randomly select 90% of those marker genes.
- Correlate the cell with each of the leaf nodes descended from “Cerebellar neuron” using that 90% of marker genes.
- Select the most highly correlated leaf node.
- Treat a vote for that leaf node as a vote for its parent at the level being chosen.
- Repeat the process 100 times.

Hierarchical Mapping



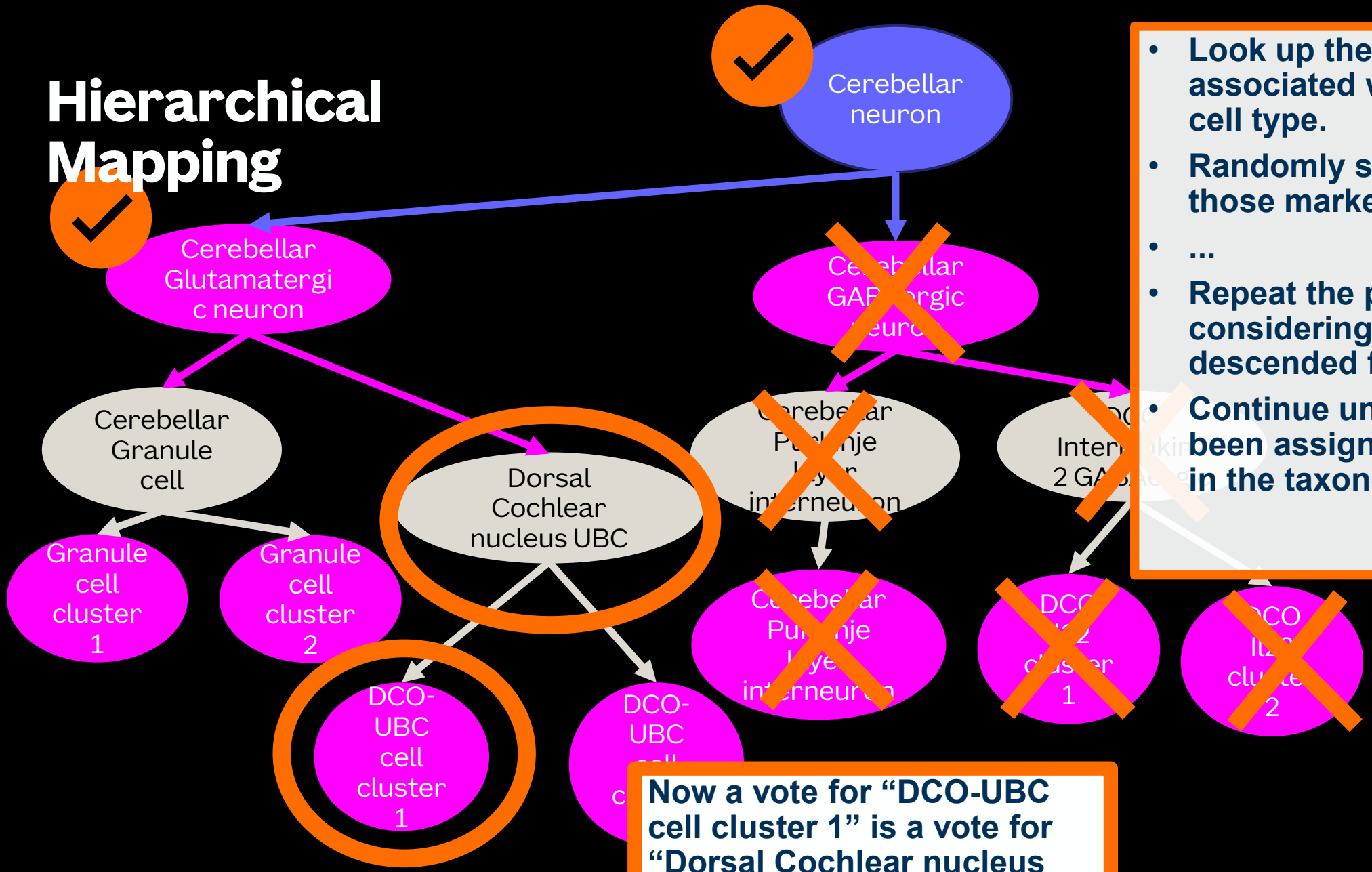
- Look up the marker genes associated with “Cerebellar neuron”.
- Randomly select 90% of those marker genes.
- Correlate the cell with each of the leaf nodes descended from “Cerebellar neuron” using that 90% of marker genes.
- Select the most highly correlated leaf node.
- Treat a vote for that leaf node as a vote for its parent at the level being chosen.
- Repeat the process 100 times.
- Assign the cell to the parent that receives the most votes.

Hierarchical Mapping



- Look up the marker genes associated with the assigned cell type.
- Randomly select 90% of those marker genes.
- ...
- Repeat the process considering only cell types descended from that type.

Hierarchical Mapping



- Look up the marker genes associated with the assigned cell type.
- Randomly select 90% of those marker genes.
- ...
- Repeat the process considering only cell types descended from that type.
- Continue until the cell has been assigned to every layer in the taxonomy.

Now a vote for “DCO-UBC cell cluster 1” is a vote for “Dorsal Cochlear nucleus UBC”, etc.

bkp/ explore

brain-map.org/bkp

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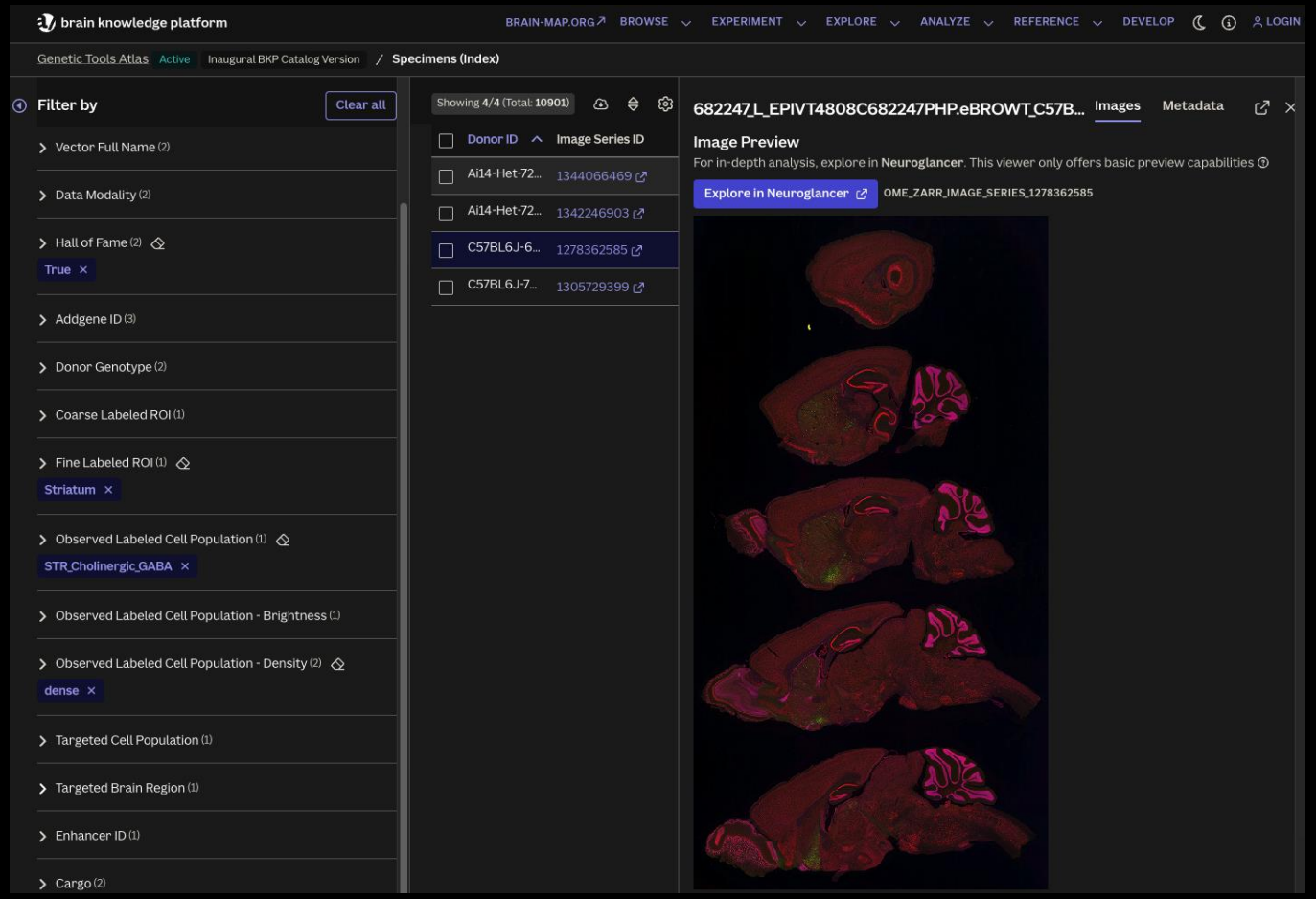
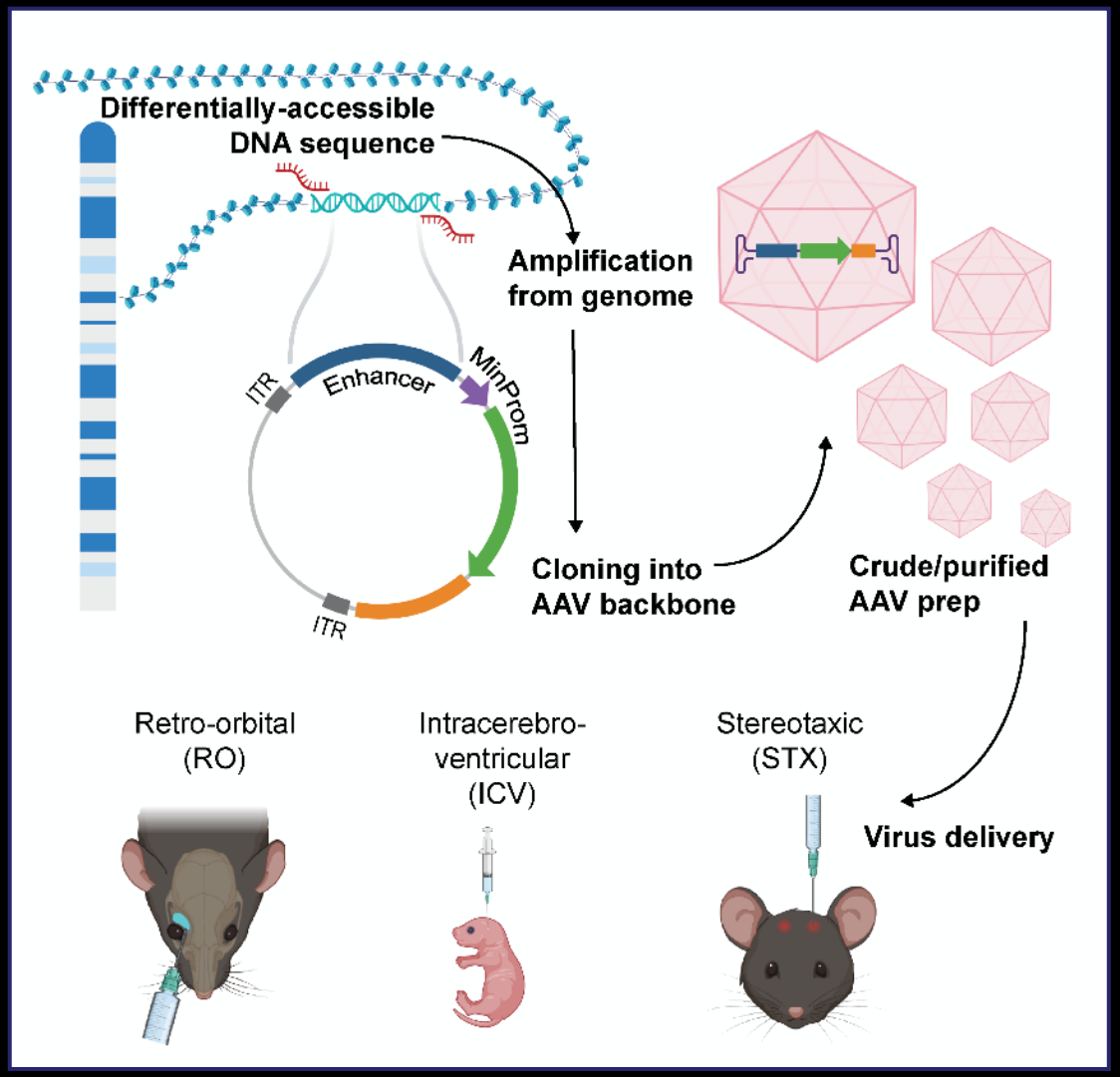
genetic tools atlas (GTA)/

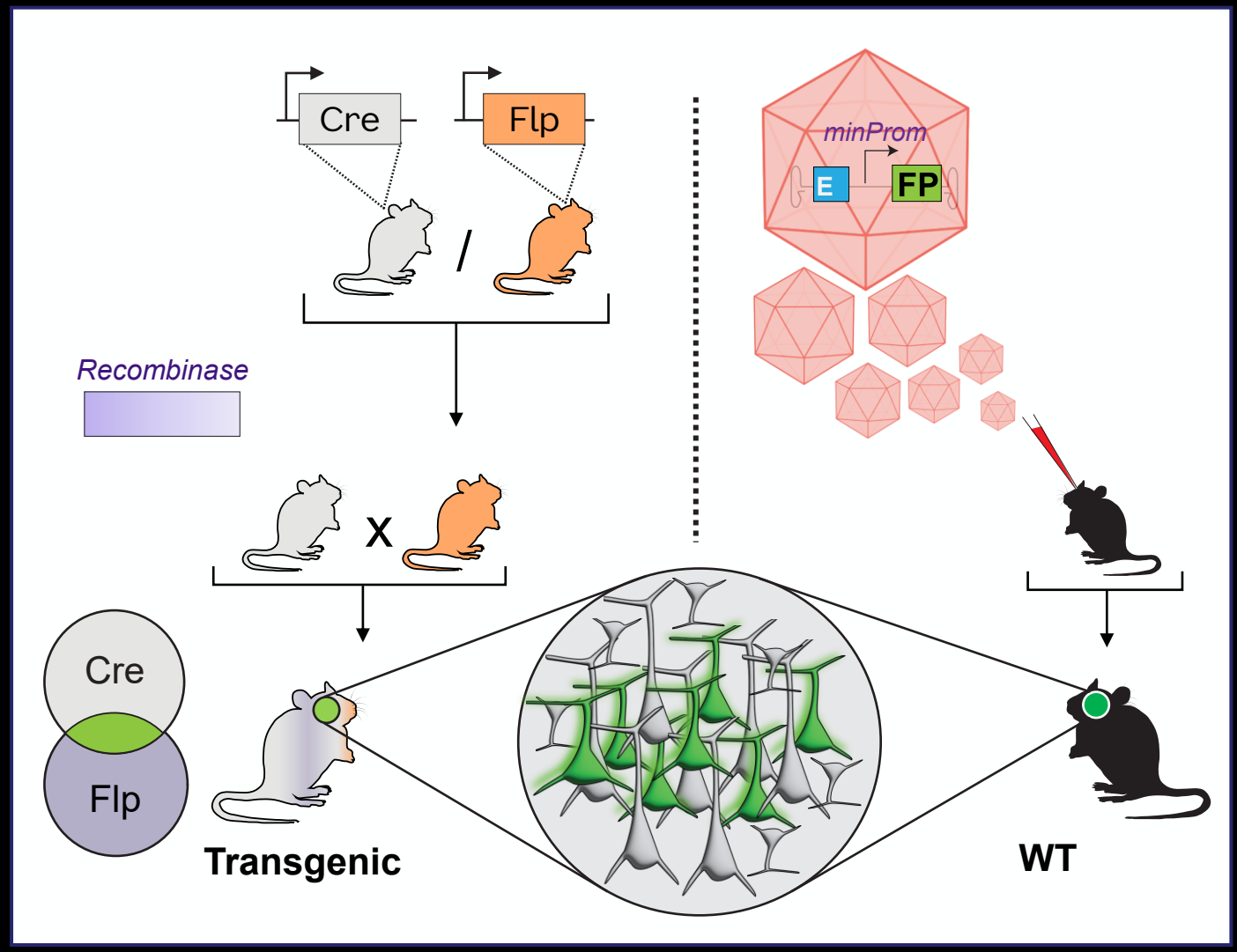
- large genetic toolkit for selective gene expression in brain cell types of interest
- Searchable web tool
  - information and data on enhancer-adeno-associated viruses (enhancer AAVs)
  - information and data on mouse transgenes
- all tools developed and tested at the allen institute
  - Some are available to purchase from Addgene!



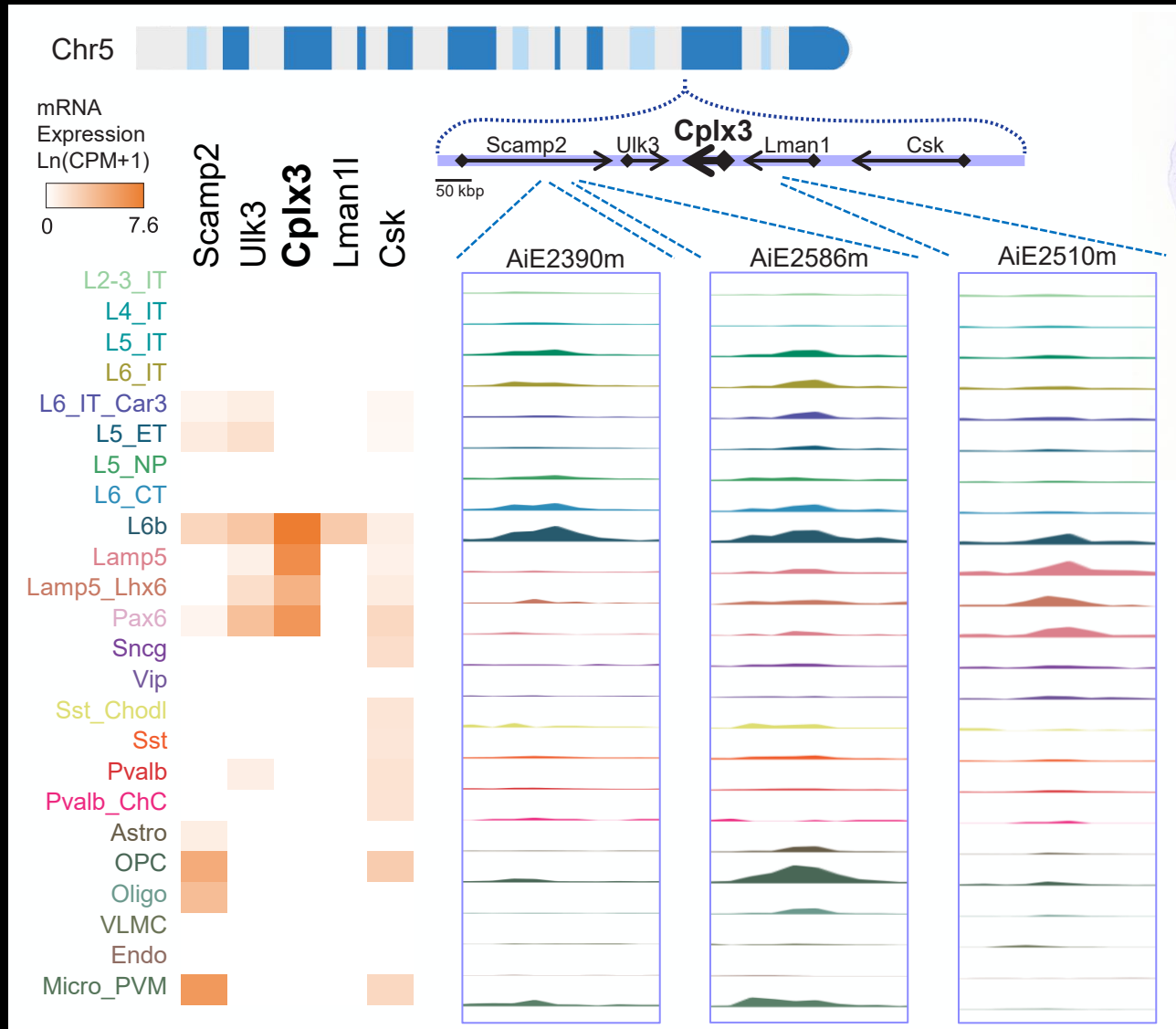
brain-map.org/bkp/experiment

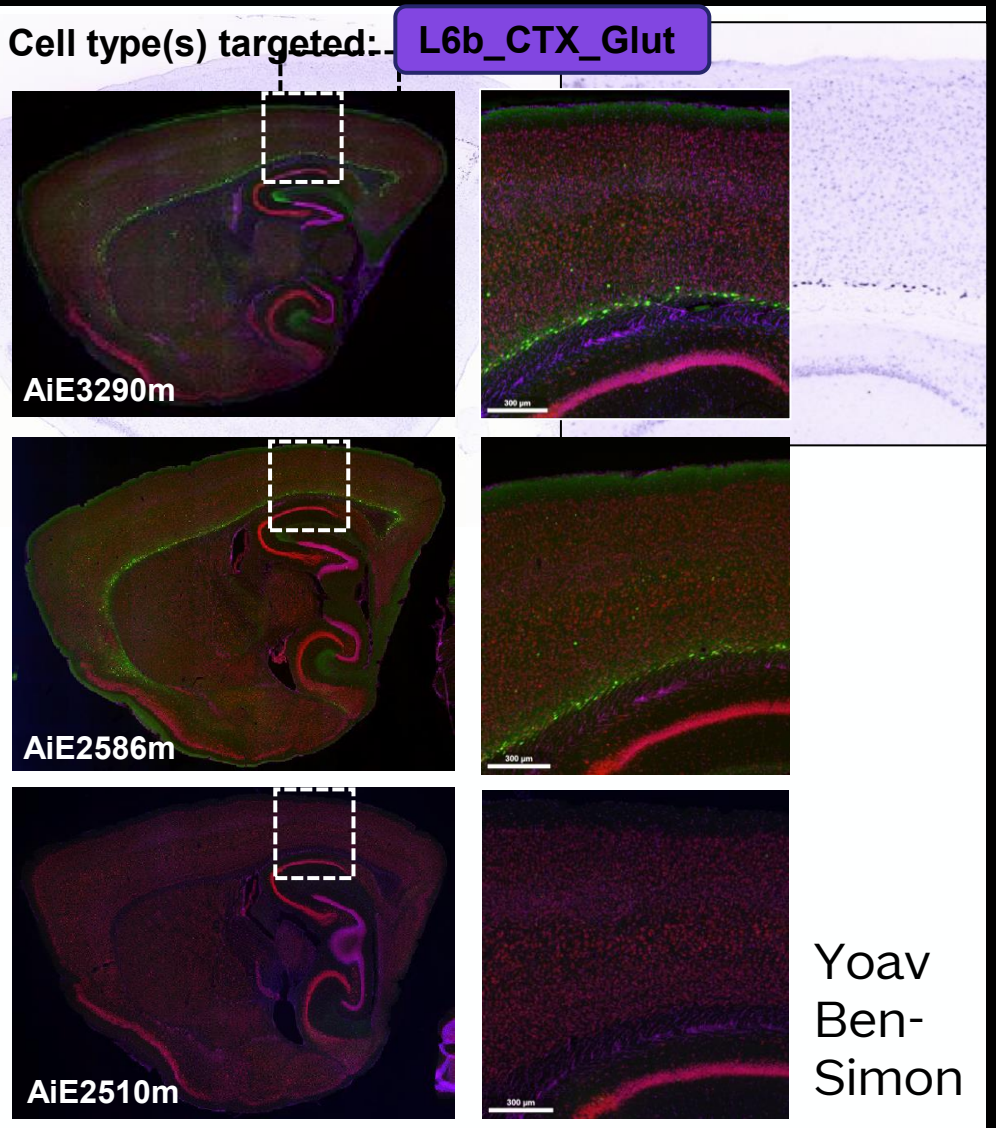
genetic tools atlas (GTA)/ modalities used





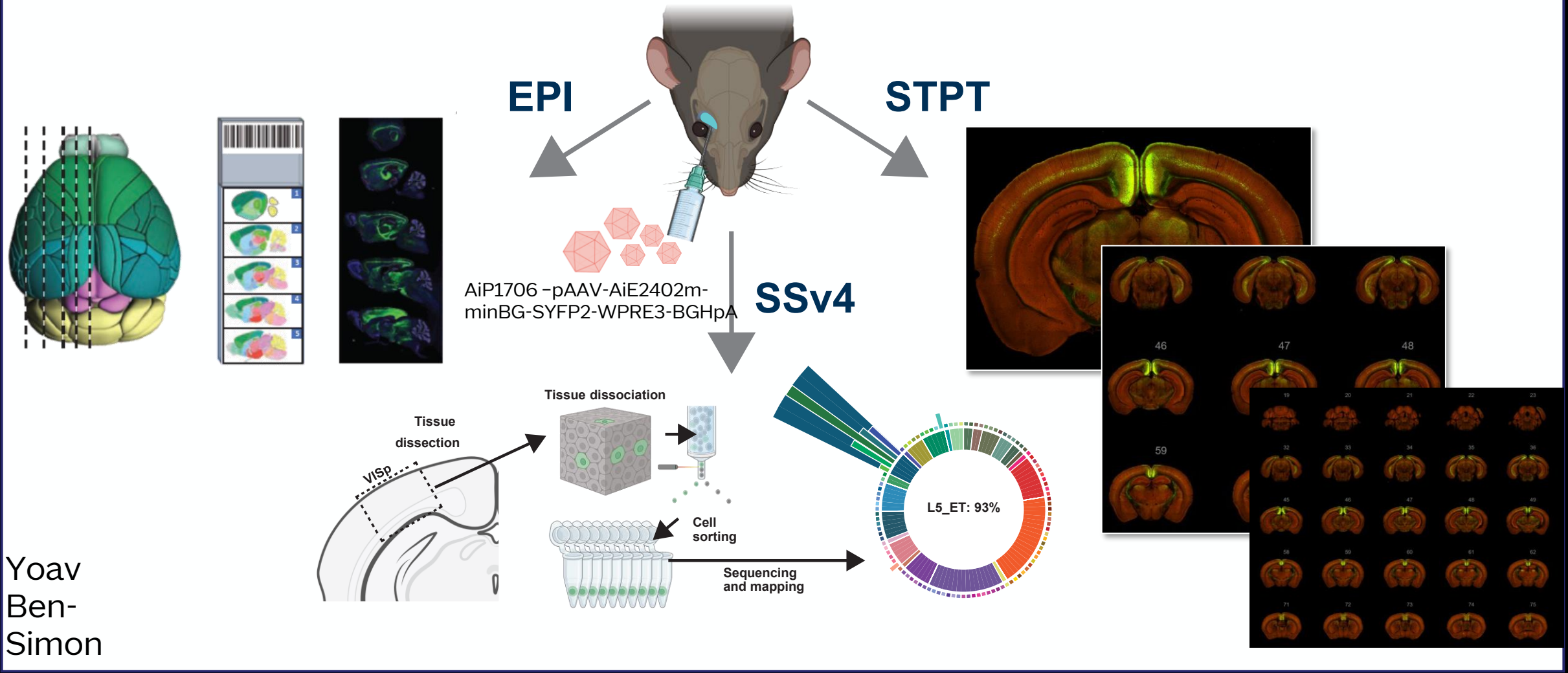
genetic tools atlas (GTA)/ manual discovery of putative enhancers based on proximity to marker genes





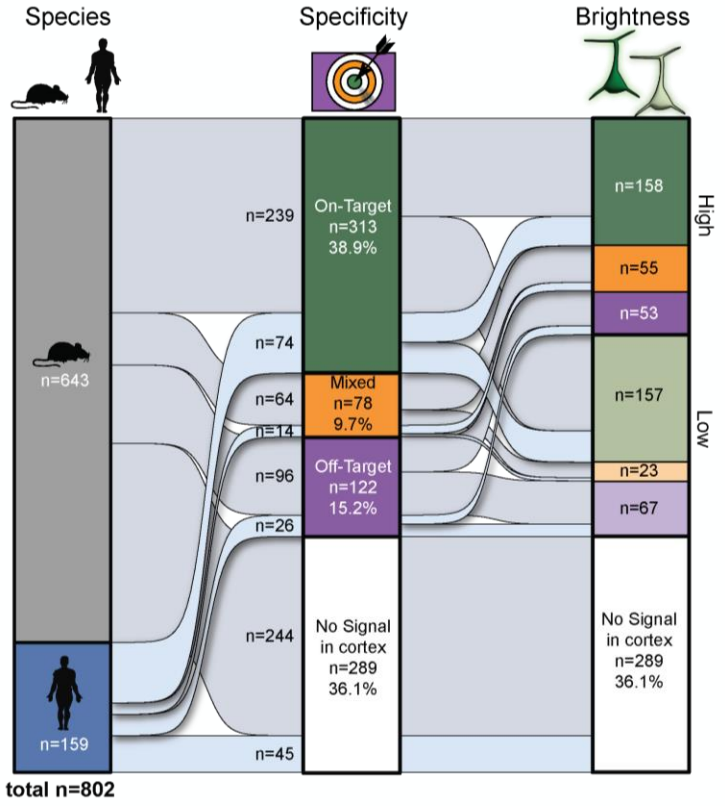
Yoav Ben-Simon

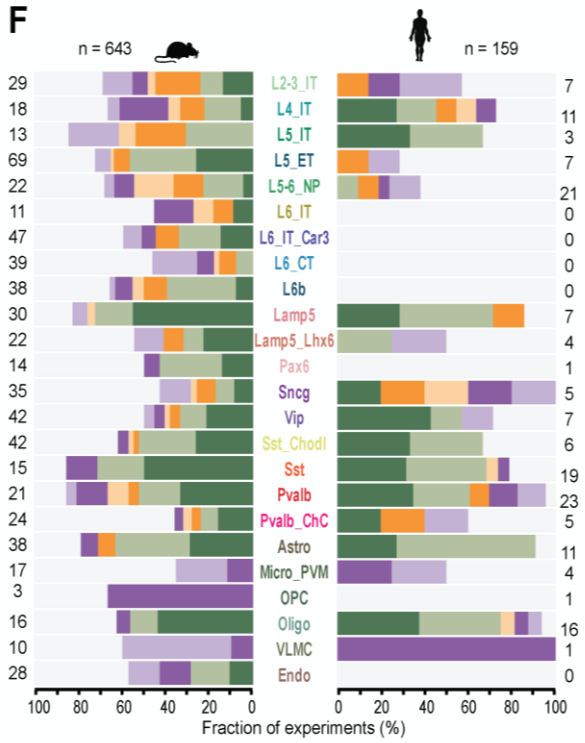
genetic tools atlas (GTA)/ enhancer screening pipeline

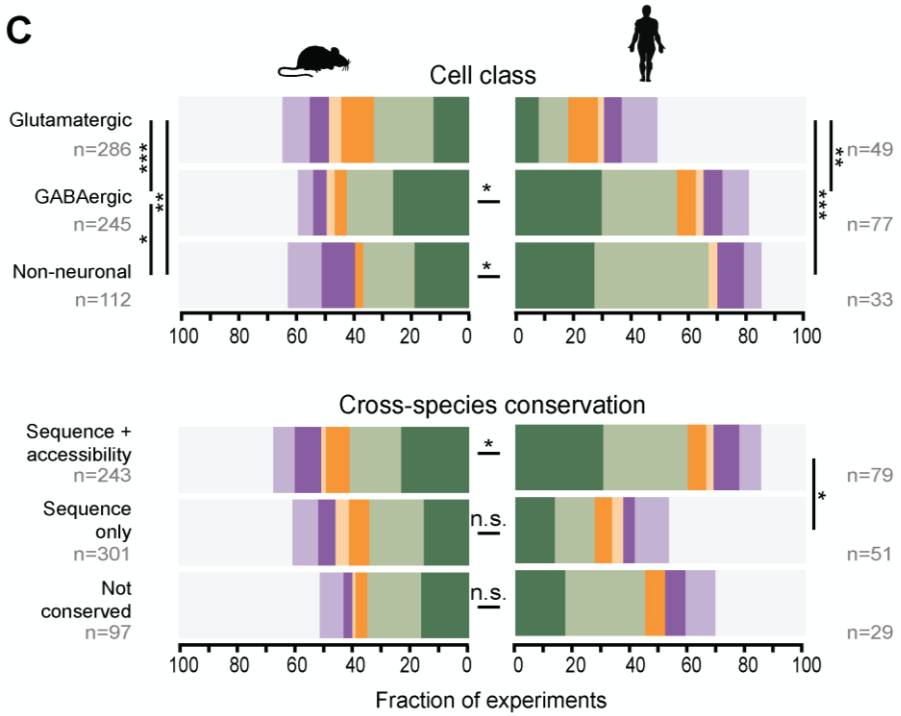


Yoav Ben-Simon

genetic tools atlas (GTA)/ experiment scoring



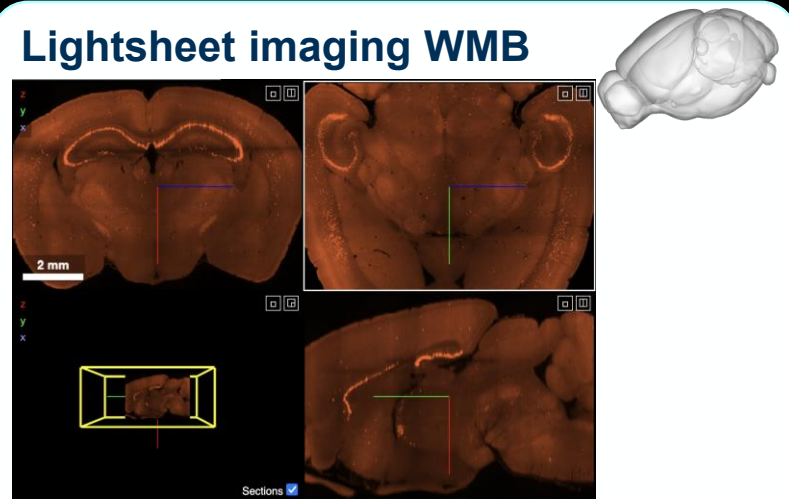


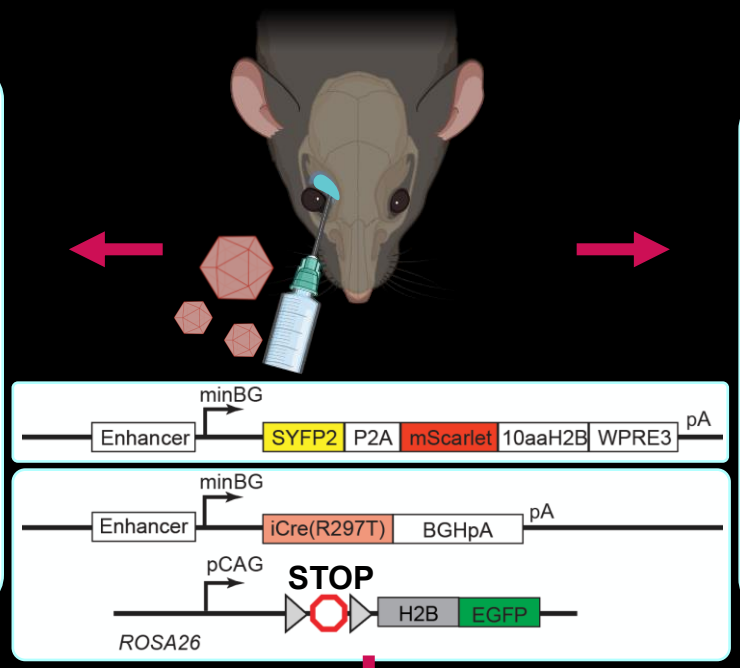


Ben-Simon et al., 2025

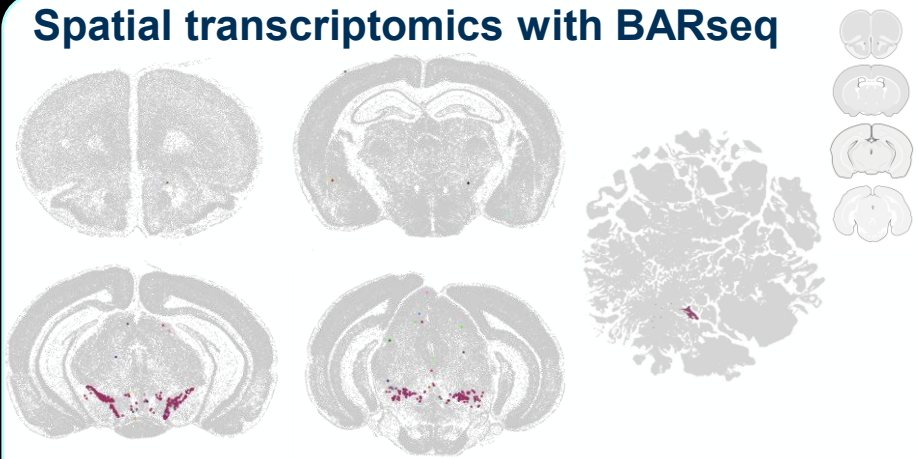
genetic tools atlas (GTA)/ new modalities in v2.0

Lightsheet imaging WMB



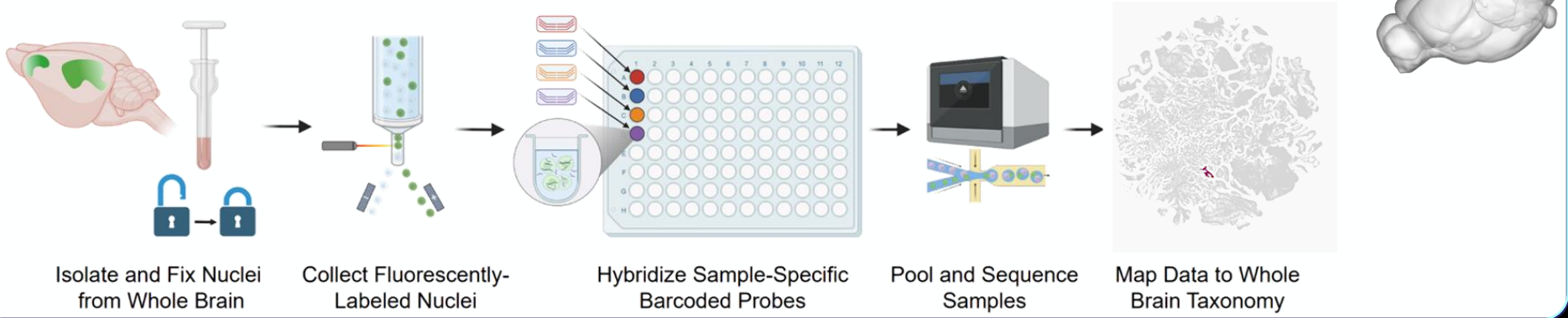


Spatial transcriptomics with BARseq



Map transcriptome to the WMB AiT21

10x Flex on WMB



brain-map.org/bkp/experiment

Yoav Ben-Simon

genetic tools atlas (GTA)/ acknowledgments

Leadership
- Bosiljka Tasic
- Jonathan Ting
- Boaz Levi
- Yoav Ben-Simon
- Tanya Daigle
- Trygve Bakken

Project management
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- Bargavi Thyagarajan

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- Ximena Optiz-Araya
- Jada Roth
- Noah Greisman

Analysis
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- Sujatha Narayan
- Avery Hunker
- John Mich
- Deepanjali Dwivedi
- Morgan Wirthlin
- Nelson Johansen

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- Vector core (Shenqin Yao)
- LAS (Melissa Reding)
- Histology (Jeanelle Ariza)
- Imaging (Jack Waters)
- Tissue processing (Nick Dee)
- MolBio (Kim Smith)

Online resources
- Tyler Mollenkopf
- Sven Otto
- Sean Meharry
- Nicholas Lusk

Data management
- Aaron Oster
- Raymond Sanchez

bkp/ explore

browse

Find data in the Data Catalog

explore

Visualize and compare data in ABC Atlas

analyze

Label and align your data with MapMyCells

experiment

Design experiments with the Genetic Tools Atlas

develop

Download and work with data programmatically with abc_atlas_access

allen institute github repo/ abc_atlas_access

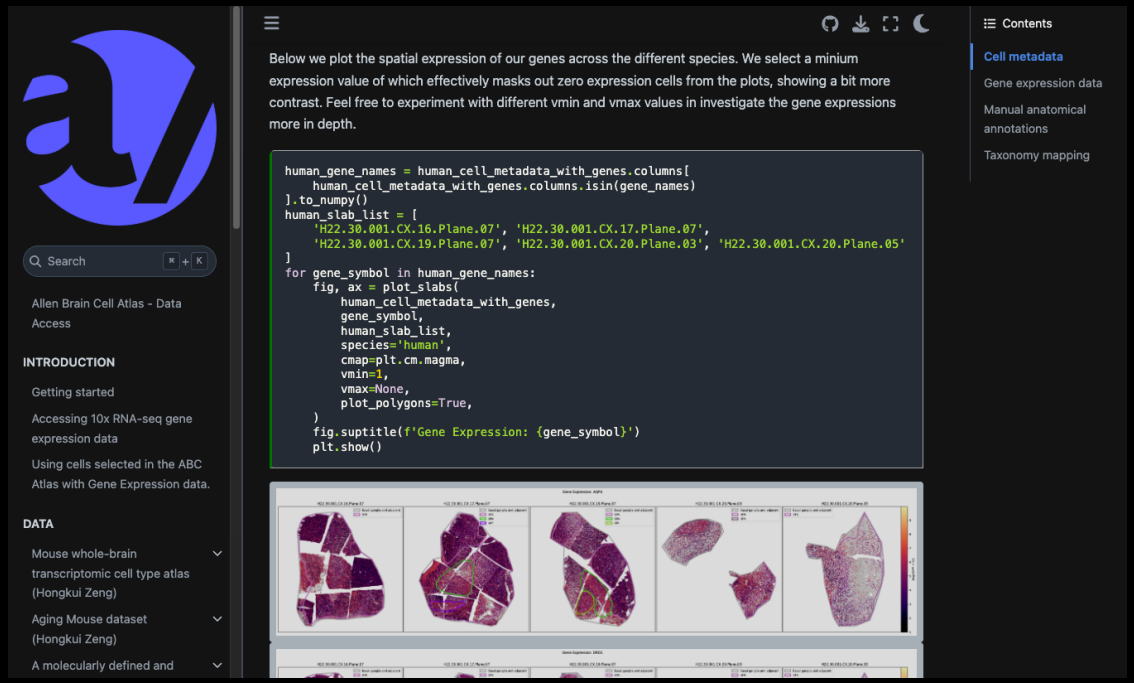
allen institute github repo/ abc_atlas_access

How do I create custom differential gene expression analysis?

How do I create custom visualizations using Python?

allen institute github repo provides an overview of the available data & how to download and use it

- Download full and partitioned data sets
- Access data definitions and schemas
- Tutorial Python notebooks with example queries

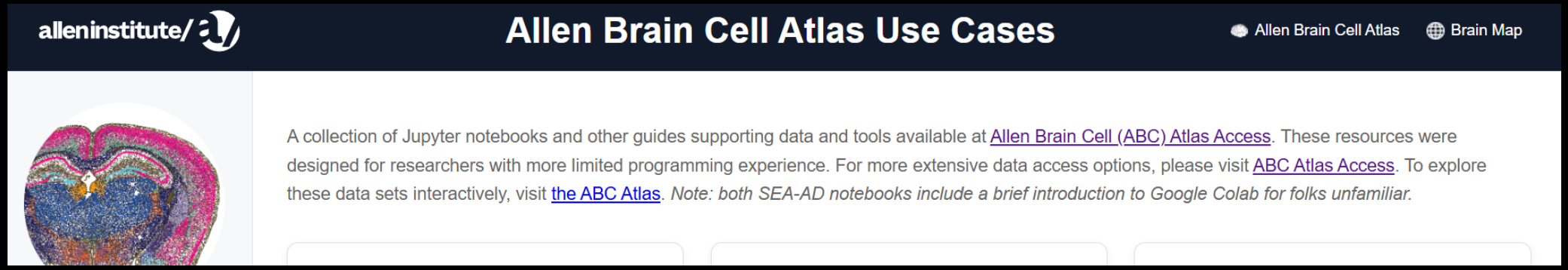


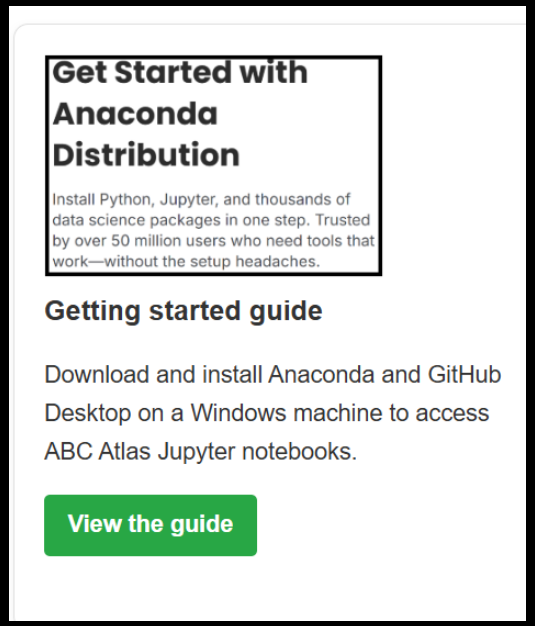
data is hosted as an AWS Public Dataset: arn:aws:s3:::allen-brain-cell-atlas

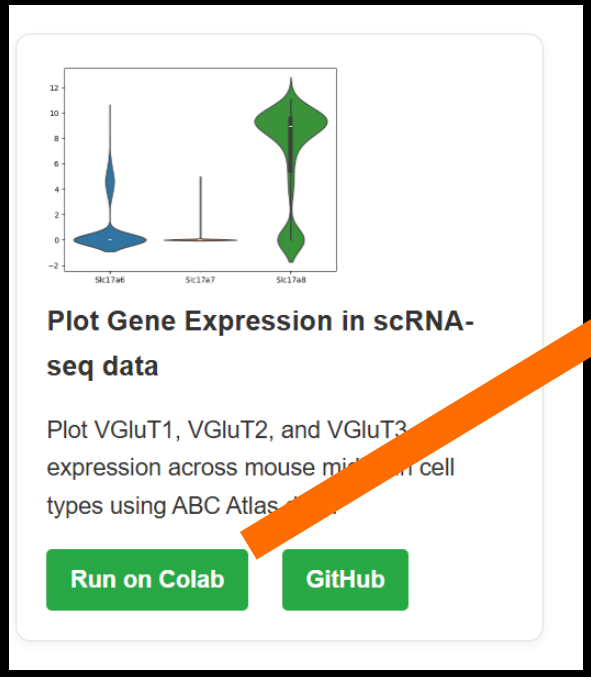
no account or login is required for access

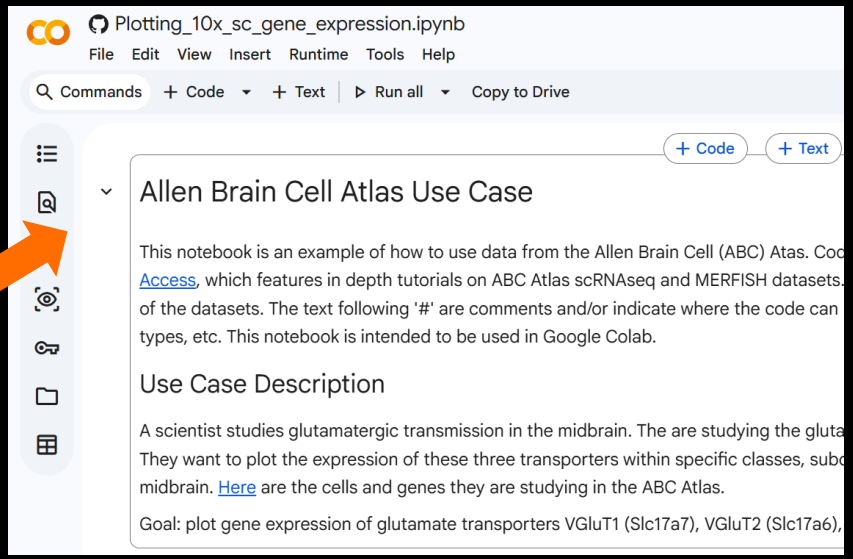
alleninstitute.github.io/abc_atlas_access/intro.html

allen institute github repo/ SPOCTT









https://alleninstitute.github.io/SPOCTT/ABCAtlas_UseCases

now time to find everything on the website!

additional Allen Brain Map tools/

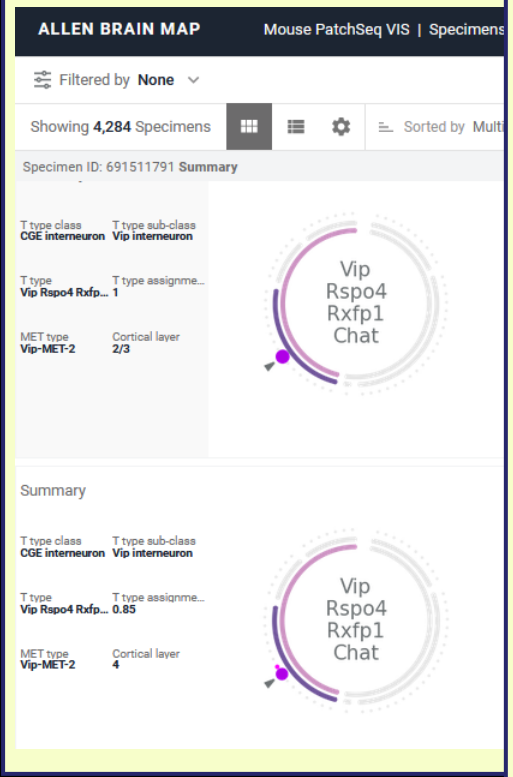
- patch-seq
- gene atlases
- anatomy

alleninstitute/

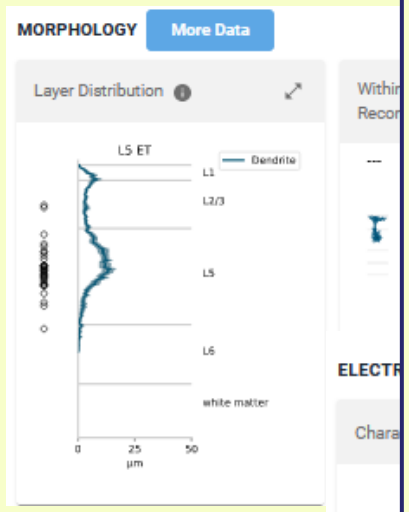
patch-seq/ visualization tools

- patch-seq is a method for collecting electrophysiology, morphology, and transcriptomics (and therefore cell type information) from the same cell

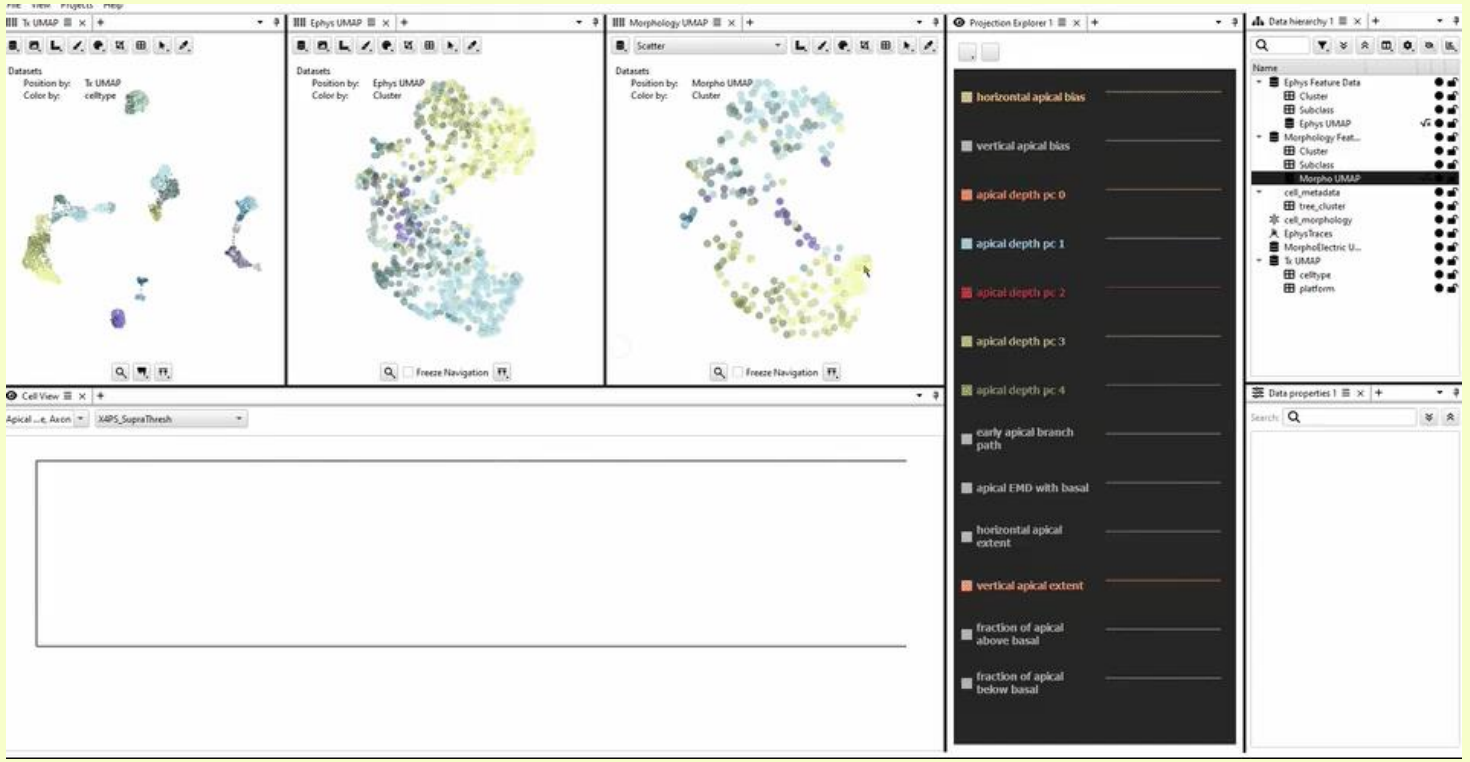
Patch-seq Viewer (mouse VIS GABA)



Cell Type Knowledge Explorer (mouse M1)

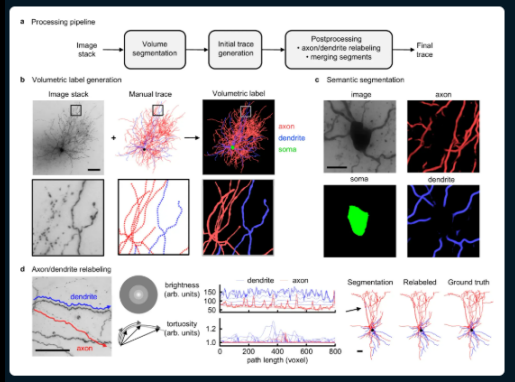


Cytosplore (human neocortex / macaque BG)



brain-map.org/bkp

patch-seq/ viewing and accessing data



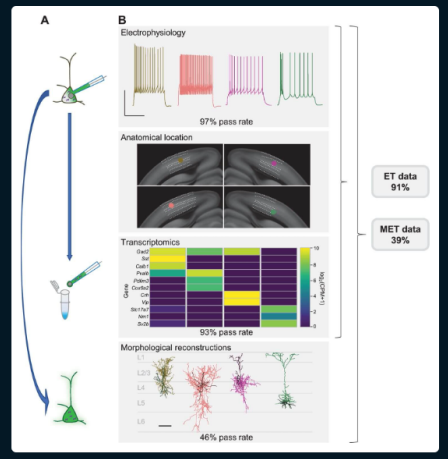
High-throughput analysis of dendrite and axonal arbors reveals transcriptomic correlates of neuroanatomy

Introduces an automated pipeline for high-throughput reconstruction and analysis of dendritic and axonal arbors from brightfield-imaged neurons, enabling large-scale morphological characterization across transcriptomically defined cell types.

Analysis Code (GitHub)

Gliko, Mallory et al. Nature Communications 2024

Protocol



Scaled, high fidelity electrophysiological, morphological, and transcriptomic cell characterization

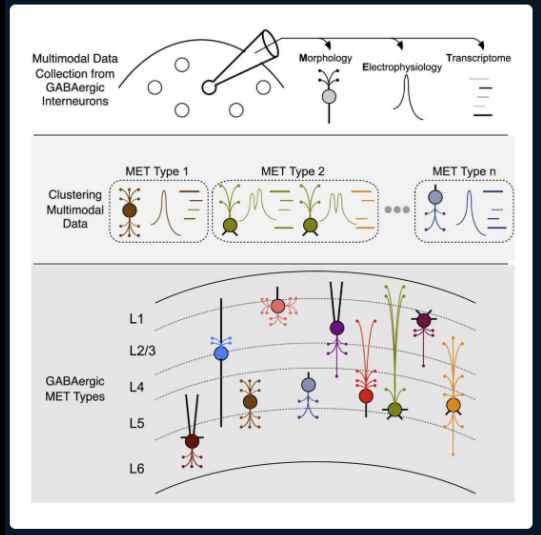
We have optimized the Patch-seq technique to efficiently collect high-quality electrophysiological, morphological and transcriptomic data. Here we describe this optimized technique as well as the publicly available tools that can be used to generate comparable data.

Patch-seq Analysis Tools (GitHub)

Community Forums

Lee, Budzillo et al. eLife 2021

patch-seq/ viewing and accessing data



Integrated Morphoelectric and Transcriptomic Classification of Cortical GABAergic Cells

This dataset includes 4,435 mouse inhibitory neurons from the visual cortex with transcriptomic, intrinsic physiological and, for a subset, morphological data. Neurons with all three modalities were assigned to one of 28 interneuron MET-types that have congruent morphological, electrophysiological, and transcriptomic properties.

Researchers can use the metadata file to find cells of a particular cell type of interest and then use the manifest file to download raw data files for those cells. Instructions and links are below. This video tutorial provides an overview of the data and walks through an example Jupyter notebook.

Explore the Data in the Mouse Patch-seq VIS Viewer

Gouwens, Sorensen et al. Cell 2020

→ DOWNLOAD INSTRUCTIONS

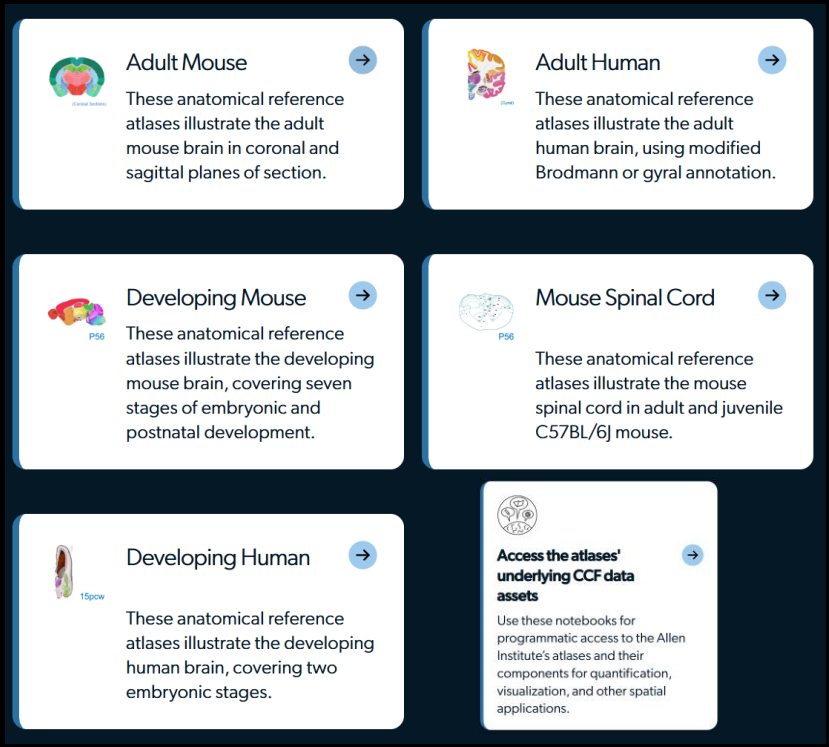
→ DOWNLOAD CELL METADATA

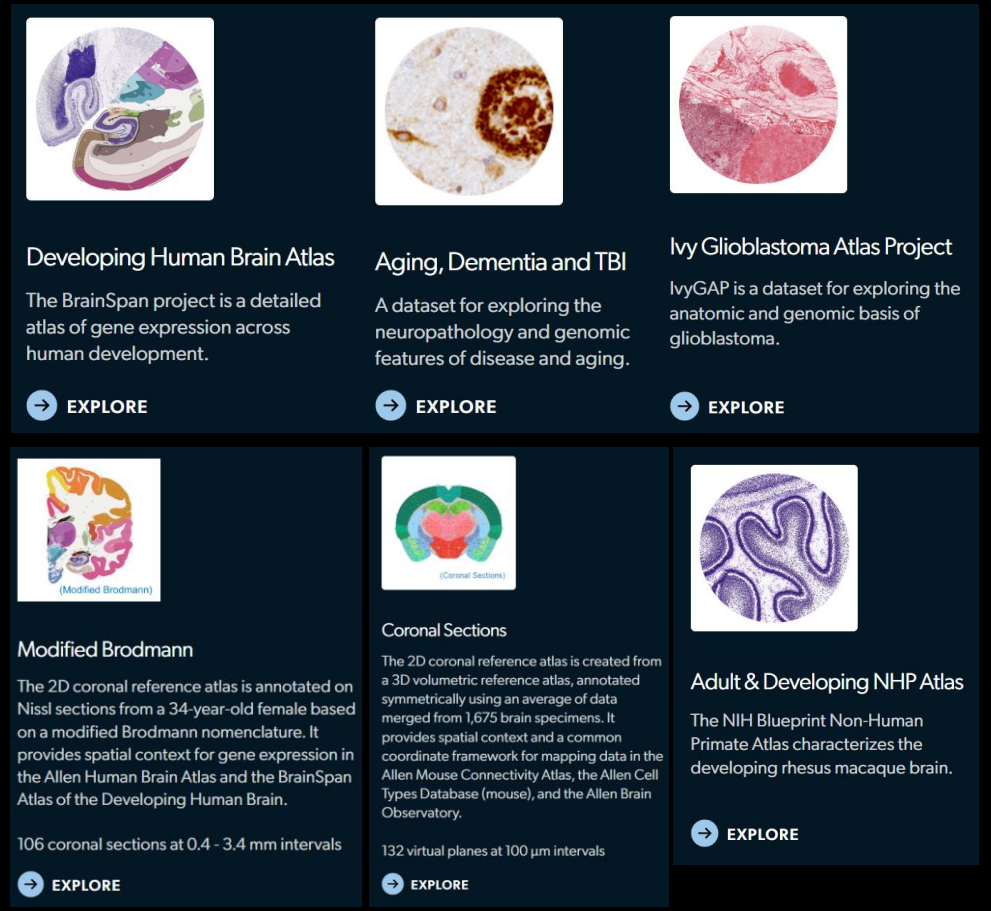
→ DOWNLOAD RAW DATA MANIFEST FOR INDIVIDUAL CELLS

→ MORE DOWNLOADS

atlases/ gene expression and anatomic

- gene expression/





And much more! Please reach out if you are looking for something specific.

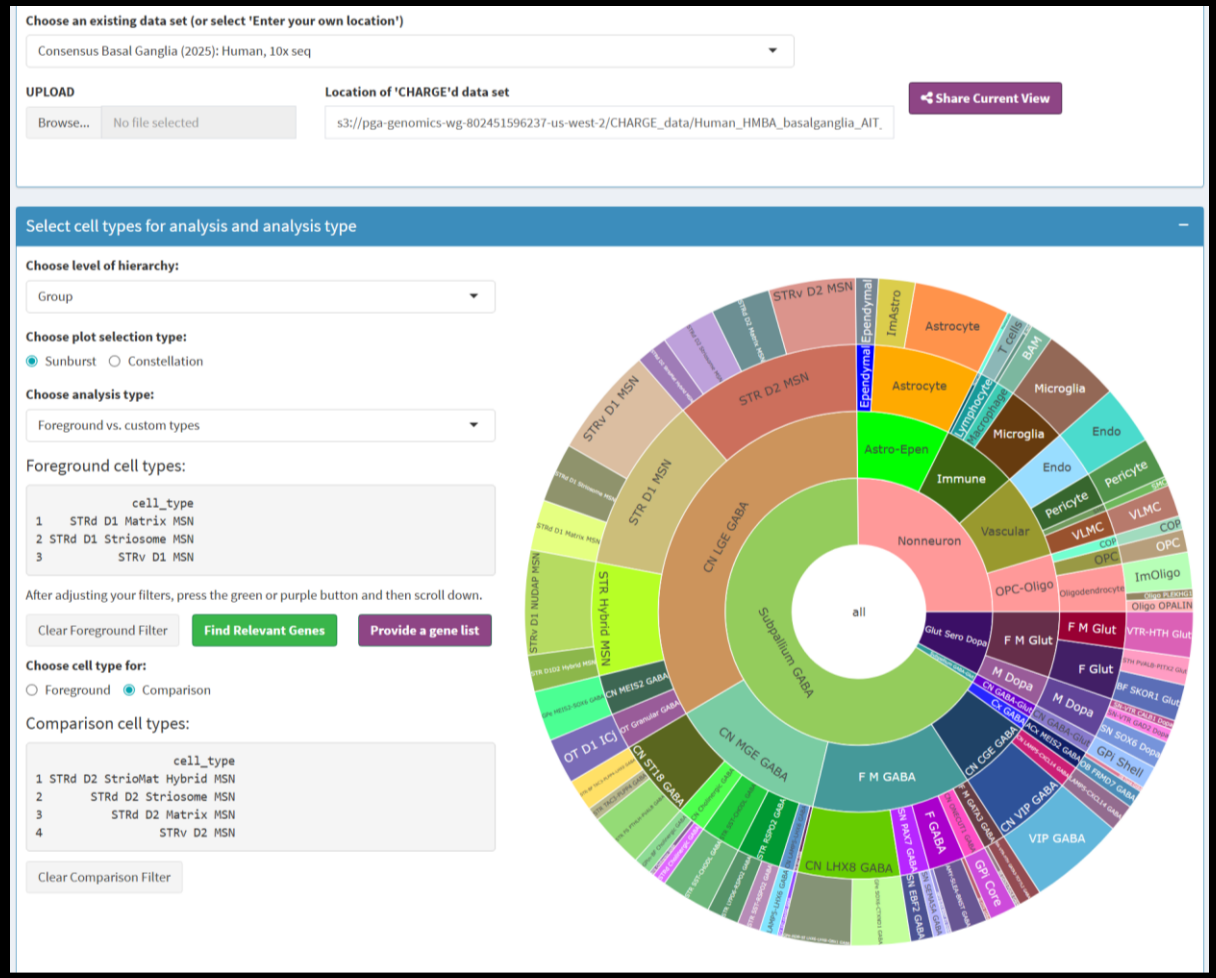
Cell HierARchy Gene Explorer (CHARGE)/

interactive tool for quickly identifying genes of potential biological interest in a cell type taxonomy

input: taxonomy

can run on any AIT file via [CHARGE_taxonomy]

output: visualizations & statistics for interesting genes and gene sets



select Human BG taxonomy

select D1 MSN vs. D2 MSN

molgen-shiny.org/CHARGE/

Cell HierARchy Gene Explorer (CHARGE)/

interactive tool for quickly identifying genes of potential biological interest in a cell type taxonomy

input: taxonomy

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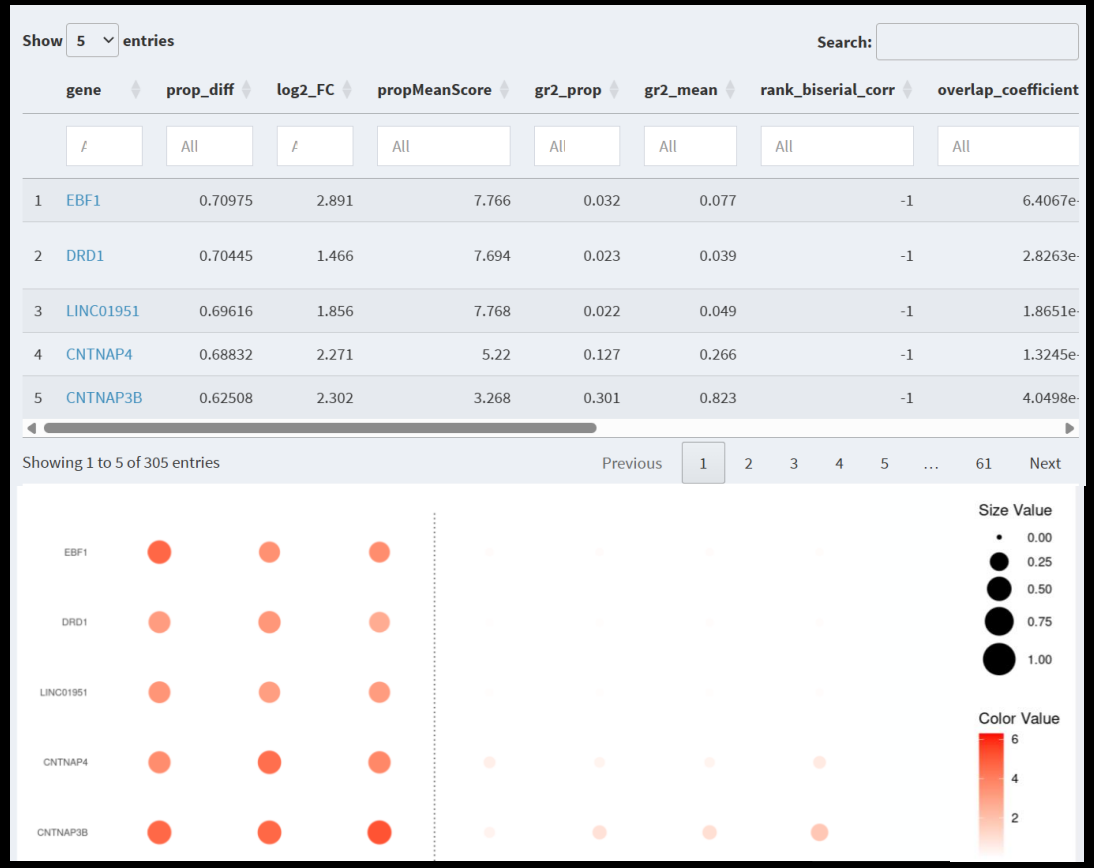

filter and sort using basic metrics

(and link-out to ABC atlas)

see gene plots for top genes

molgen-shiny.org/CHARGE/

Cell HierARchy Gene Explorer (CHARGE)/

interactive tool for quickly identifying genes of potential biological interest in a cell type taxonomy

input: taxonomy

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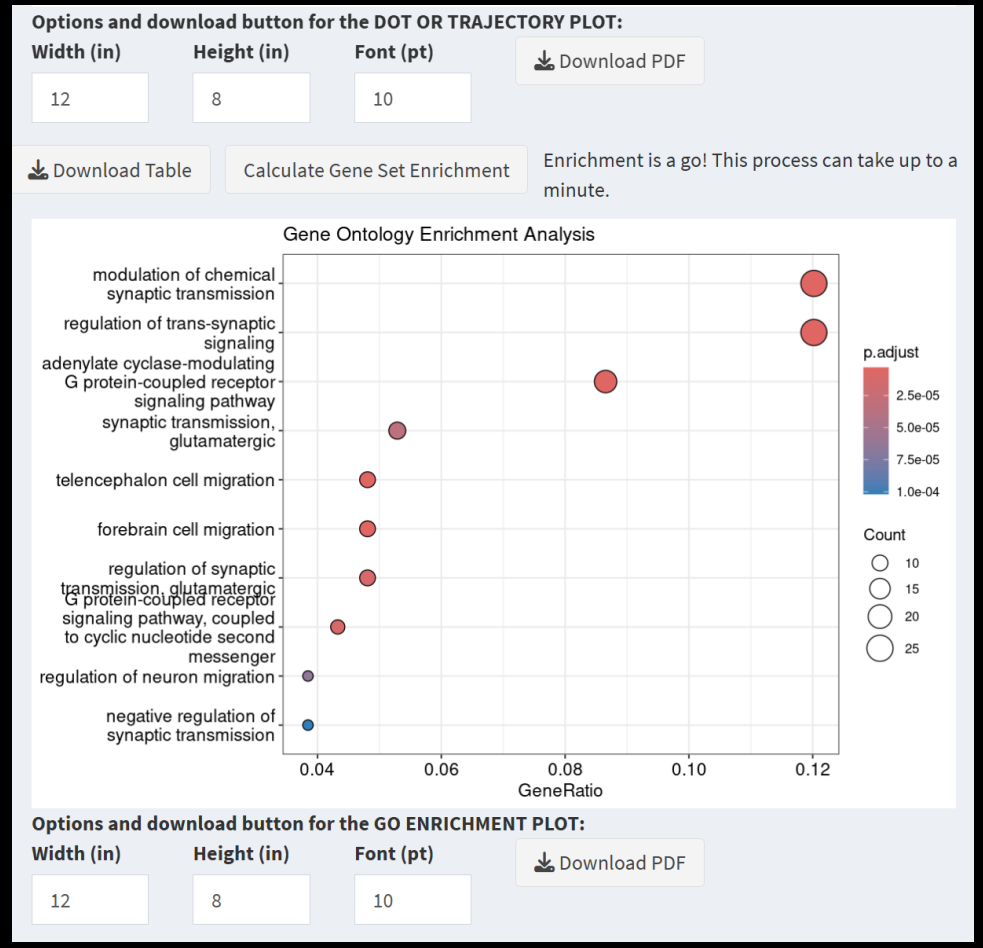


molgen-shiny.org/CHARGE/

download plots/data

calculate (basic) gene set enrichment

NOTE: Not available for whole mouse or human brain taxonomies!

agenda/afternoon

12:00-1:00 | lunch

1:00-2:15 | walkthrough of allen brain map tools (Rachel / Jeremy)

- walkthrough of allen brain map
- Brain Knowledge Platform: ABC Atlas, MapMyCells, Genetics Tools Atlas
- Additional tools in brief: Patch-seq, gene atlases, and anatomy

2:15-2:45 | SEA-AD tools for exploring changes in AD (Irika / Jeremy)

2:45-3:00 | break

3:00-3:30 | external use cases – Rachel

3:30-4:40 | user exploration of allen institute tools

4:40-5:00 | feedback survey (all – please fill out the survey!)

github.com/AllenInstitute/SPOCTT/tree/main/2026_Additional_Presentations

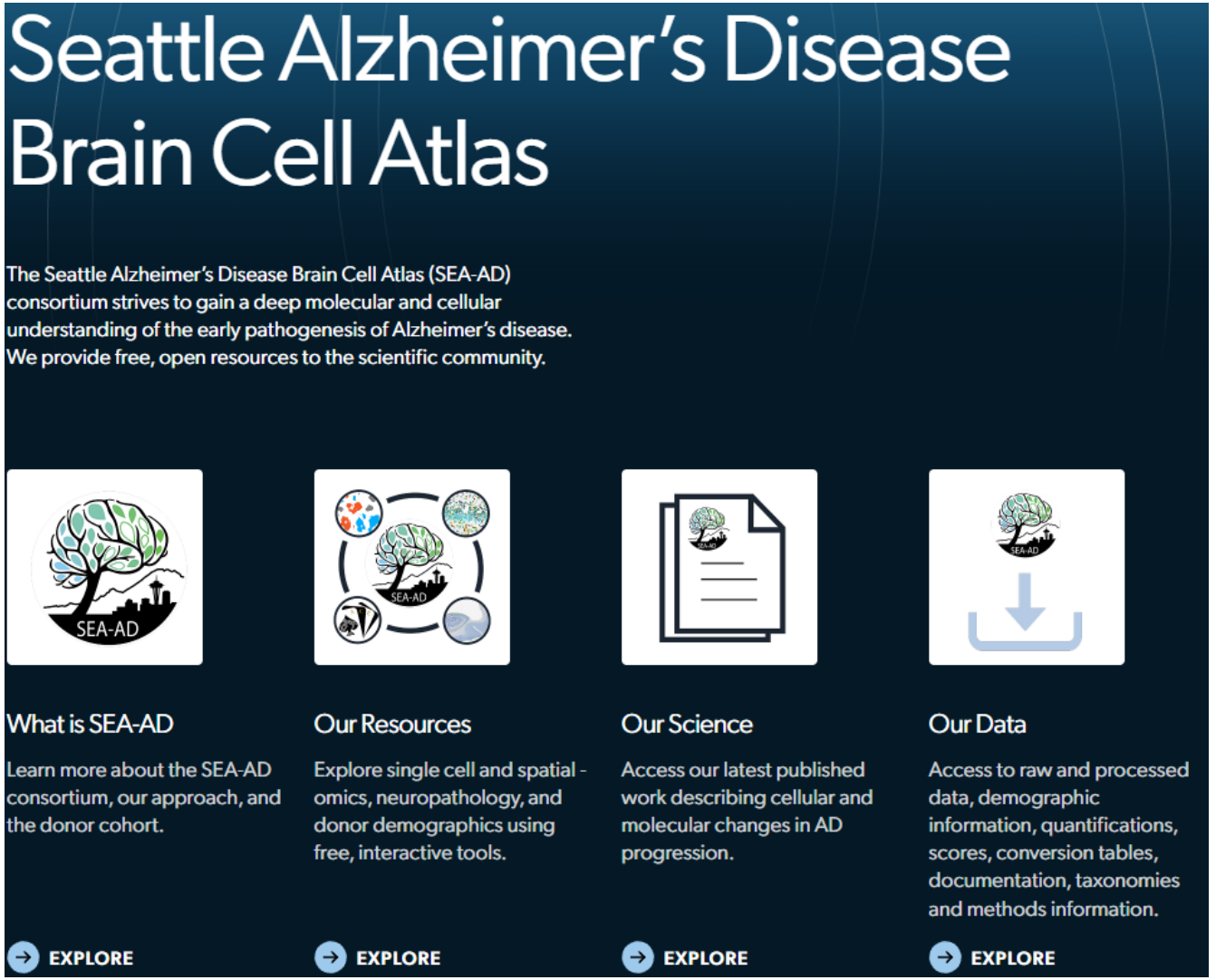
SEA-AD/ tools

- Neuropathology and donor viewers
- Exploring gene expression changes with disease
- Other ways to access SEA-AD data and knowledge

alleninstitute/

sea-ad portal/

- Public repository for SEA-AD data, tools, and science
- Recent major updates to brain-map.org (and sea-ad.org) for improved user experience



our resources/ tools to explore SEA-AD data

Alzheimer's Disease Resources (SEA-AD)

Overview

SEA-AD provides several resources allowing users to actively explore donor metadata, neuropathology, and cell type-specific gene expression, chromatin accessibility, and spatial localization in a cohort of donors with varying disease-associated pathologies. This page summarizes the available resources by data type, with each section summarizing why each data type is important, the kinds of questions that can be answered using the resources, and links to/brief summaries of each resource.

Examples of questions addressed by these resources include:

- In which cell types is gene X expressed?
- How does gene X change with disease in different cell types?
- What kind of cells do I have in my own omics data set?
- What is the spatial relationship between cell type X and cell type Y in AD?
- How does AD pathology look in the brains of donors with or without dementia?
- How do cell types from SEA-AD compare with cell types defined in younger donors and other cohorts?

Know your data type but not your resource—jump to the relevant section below:

→ SINGLE NUCLEUS TRANSCRIPTOMICS → SPATIAL TRANSCRIPTOMICS

→ DONOR & NEUROPATHALOGY → KNOWLEDGE INTEGRATION → EPIGENOMICS

→ WHOLE GENOME SEQUENCING

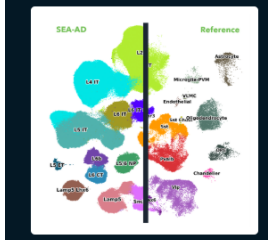
Single Nucleus Transcriptomics

Why single-nucleus transcriptomics data?

- Create a foundational high resolution cell type reference atlas, akin to the human genome, to understand the molecular underpinnings of Alzheimer's disease at the cell type level.
- More than one million nuclei align to over 100 cell types in each cortical area assayed.
- Consistent cell types identified across multiple studies of AD.

Questions you can asking with these data & tools

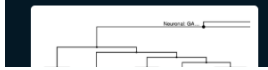
- *What cell types are found in the adult and aged brains?*
- *In which cells types is gene X expressed?*
- *Are genes X and Y co-expressed in the same cell type?*
- *How does gene X change with disease in different cell types?*
- *Which genes are most selectively expressed in a particular cell type or disease state?*
- *What kind of cells do I have in my own omics data set?*



Transcriptomics Comparative Viewer

- Compare gene expression in cells from neurotypical and aged brains side by side for any gene
- Overlay and quantification of expression differences in the context of donor demographics and disease metrics
- Tissue assayed: middle temporal gyrus from all donors

Transcriptomics Comparative Viewer



Transcriptomics Explorer

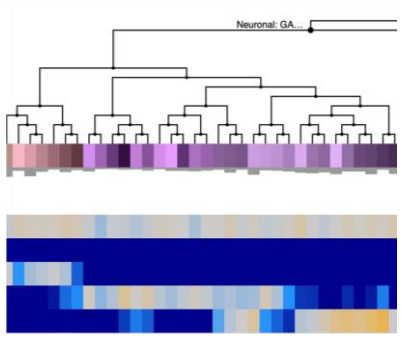
← category

← use cases in category

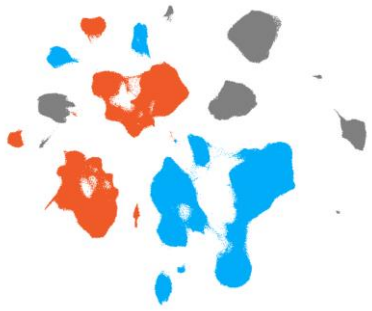
← resources

our resources/ tools to explore SEA-AD data

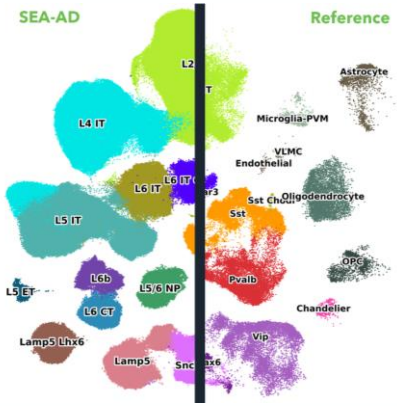
multiple user-friendly resources for entry into our data sets!



Transcriptomics Explorer

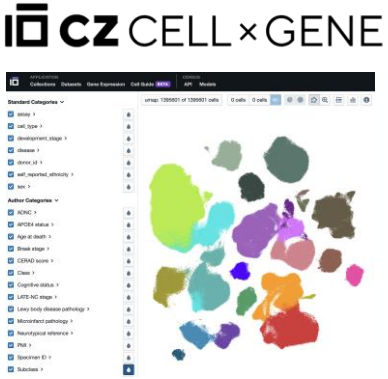


Allen Brain Cell Atlas
Single nucleus RNA-seq

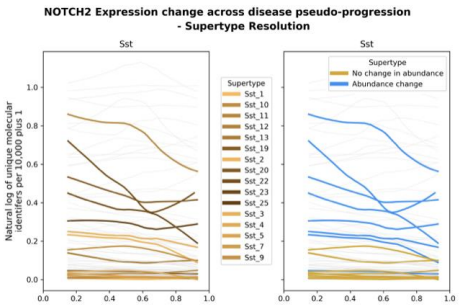


Comparative Viewer

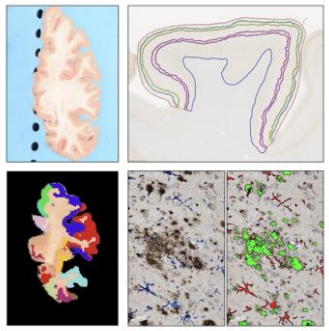
Whole Genome
Sequencing



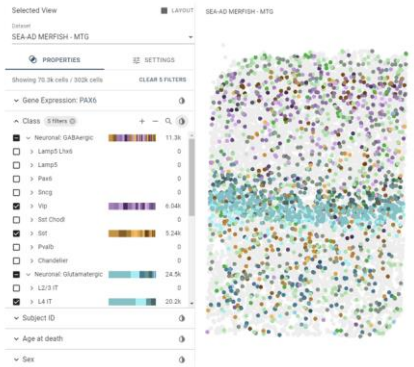
CZI Cell x Gene



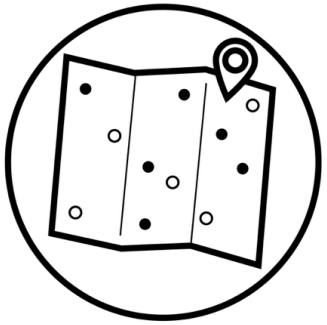
gene trajectory viewer



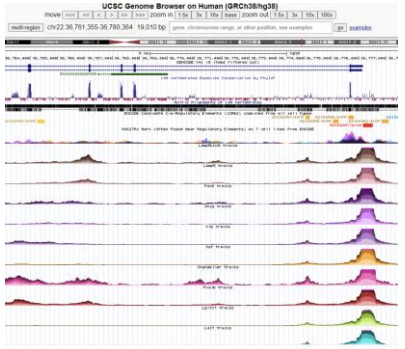
Donor index and
Neuropathology Viewer



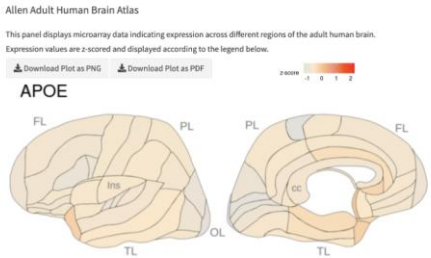
Allen Brain Cell Atlas
Spatial Transcriptomics



MapMyCells



Epigenomics Browser



Gene Knowledge
Cards

Annotation
Comparison Explorer

explore donors & neuropathology/

Brain Knowledge Platform

SEA-AD - Metadata and Neuropath > Donors (Index)

Filter by

- Donor ID
- Age at death
- Sex
- APOE genotype
- Cognitive status
- Years of education
- Continuous Pseudo-progression Score
- ADNC
- Braak stage
- Thal phase
- CERAD score
- Lewy body disease pathology

Showing 50/84 (Total: 84)

Donor ID	Age at death (years)	Sex	APOE genotype	Cognitive status	Years of education
H19.33.004	80	Female	3/3	No dementia	17
H20.33.001	82	Male	3/3	No dementia	16
H20.33.002	97	Female	2/3	No dementia	12
H20.33.004	86	Male	3/4	Dementia	15
H20.33.005	99	Female	2/3	No dementia	12
H20.33.008	92	Female	3/4	No dementia	18
H20.33.011	93	Female	3/4	Dementia	16
H20.33.012	91	Female	3/3	No dementia	18
H20.33.013	94	Male	3/3	No dementia	14
H20.33.014	82	Female	3/3	No dementia	16
H20.33.015	88	Male	3/3	Dementia	18
H20.33.016	93	Female	2/3	Dementia	14
H20.33.017	69	Male	3/3	Dementia	14
H20.33.018	81	Female	3/4	Dementia	21
H20.33.019	87	Female	3/4	No dementia	15
H20.33.020	81	Male	4/4	Dementia	12
H20.33.024	90	Male	3/3	No dementia	19

H20.33.008 Metadata **Images**

Regi... Medial Entorhinal Cortex and Hippocampus (...

Slide: Abeta(6E10) and IBA1

Medial Entorhinal Cortex and Hippocampus (MEC-HIP)

Middle Temporal Gyrus (MTG) and Superior Temporal Gyrus (STG)

Prefrontal Cortex (PFC)

Abeta(6E10) and IBA1

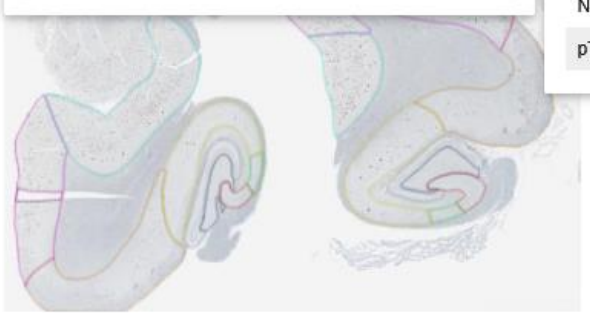
a-Synuclein

GFAP

H&E-LFB

NeuN

pTau(AT8) and pTDP43

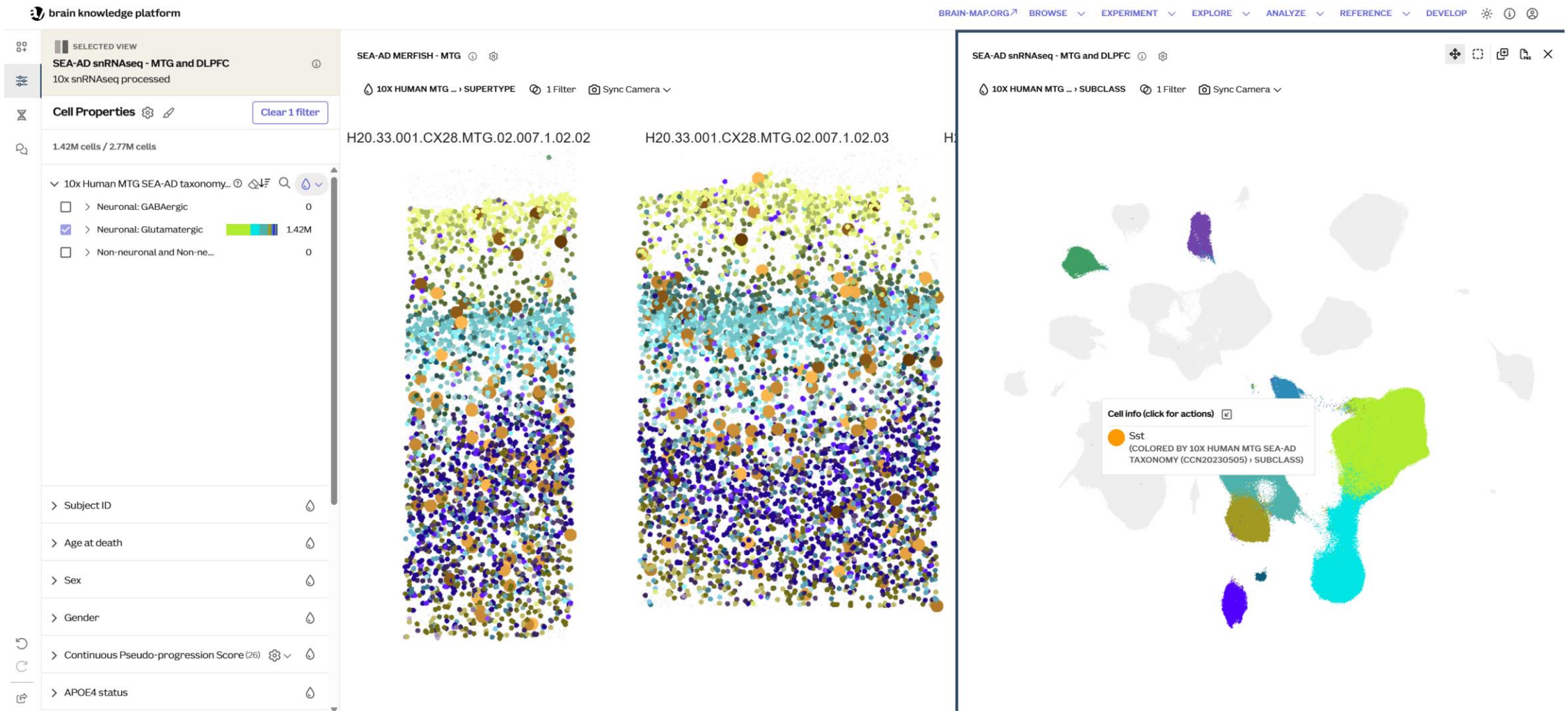


segment), Subiculum (SB, orange segment), Entorhinal cortex (EC, purple segment) and Trans-entorhinal cortex (TEC, cyan segment). Dotted lines reflect analysis area subtractions due to background staining / blood vessels.

Positive markup image

HALO WSI digital image from Object Colocalization module application. Object segmentation is applied only for the positive stain in Abeta-plaques (green) above background, cell segmentation for IBA1 positive microglia cells (red) above background, and area of colocalization of both markers (yellow).

ABC atlas/ explore spatial & sn-omics



ABC atlas currently includes single nucleus and/or spatial data from all three published studies

access and visualize data/ Jupyter notebooks

- ABC atlas access files and Jupyter notebooks provides a standard entry point into multiple single cell and spatial transcriptomics data sets
- SEA-AD multi-region and caudate nucleus data sets are now available via this tool!

alleninstitute.github.io/abc_atlas_access

SEA-AD Multiregion taxonomy and snRNASeq analysis: clustering and annotations

Alzheimer's disease (AD) is characterized by the progressive accumulation of amyloid-beta (Aβ) plaques and hyperphosphorylated tau (pTau) tangles across brain regions. While these regions differ in architecture, function, and susceptibility to pathology, many share common cellular populations.

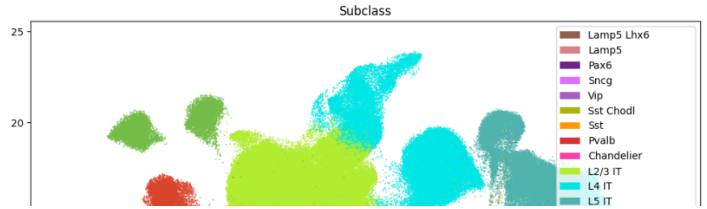
In [10]:
```
clinical = abc_cache.get_metadata_dataframe(
    directory='SEA-AD-Multiregion-10X',
    file_name='clinical'
).set_index('donor_label')
clinical
```

```
clinical.csv: 100%|██████████| 13.3k/13.3k [00:00<00:00, 136kMB/s]
```

Out[10]:

donor_label	Overall AD neuropathological Change	Thal	Braak	CERAD score	Overall CAA Score	Highest Lewy Body Disease	Total Microinfarcts (not observed grossly)	Total microinfarcts in screening sections	Atherosclerosis	Arteriol
H19.33.004	Not AD	Thal 0	Braak IV	Absent	Not identified	Not Identified (olfactory bulb not assessed)	1	1	Mild	
H20.33.001	Low	Thal 2	Braak IV	Sparse	Not identified	Not Identified (olfactory bulb not assessed)	0	0	Mild	

In [39]:
```
fig, ax = plot_umap(
    plot_cell_extended['x'],
    plot_cell_extended['y'],
    cc=plot_cell_extended['Subclass_color'],
    labels=plot_cell_extended['Subclass'],
    term_orders=plot_cell_extended['Subclass_order'],
    fig_width=12,
    fig_height=12
)
res = ax.set_title("Subclass")
plt.show()
```



SEA-AD CaH snRNASeq analysis: clustering and annotations

The SEA-AD CaH taxonomy represents is formed from a combination of taxons from the (HMBA-BG taxonomy)[HMBA-BG-taxonomy-CCN20250428.md] and the SEA-AD MTG Taxonomy. These 66 taxons are built into a new hierarchy with 16 subclasses, and 8 neighborhoods. Of 886,037 total nuclei, 686,439 pass the quality checks to be included in this analysis.

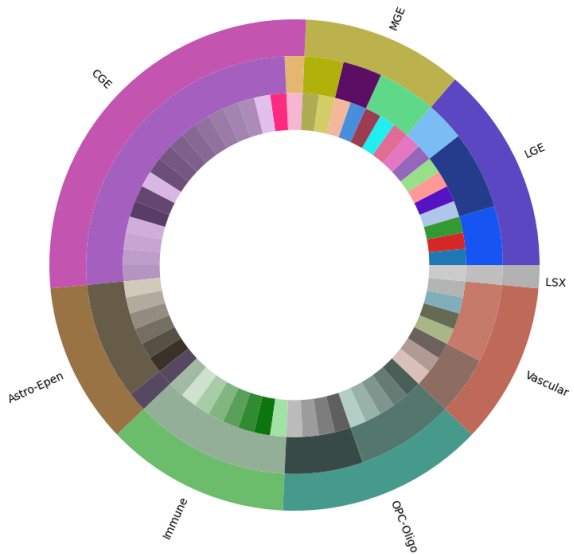
The notebook presented here demonstrates how to access and visualize taxonomy annotations, cell type metadata, donor information, brain region annotations, and AD-associated cellular changes from the SEA-AD CaH dataset. These examples focus on precomputed metadata accessible through the Allen Brain Cell Atlas Access package. Additional data, including links to raw sequencing, spatial transcriptomic, and neuropathology data, are available at SEA-AD.org.

Visualizing the SEA-AD CaH Taxonomy

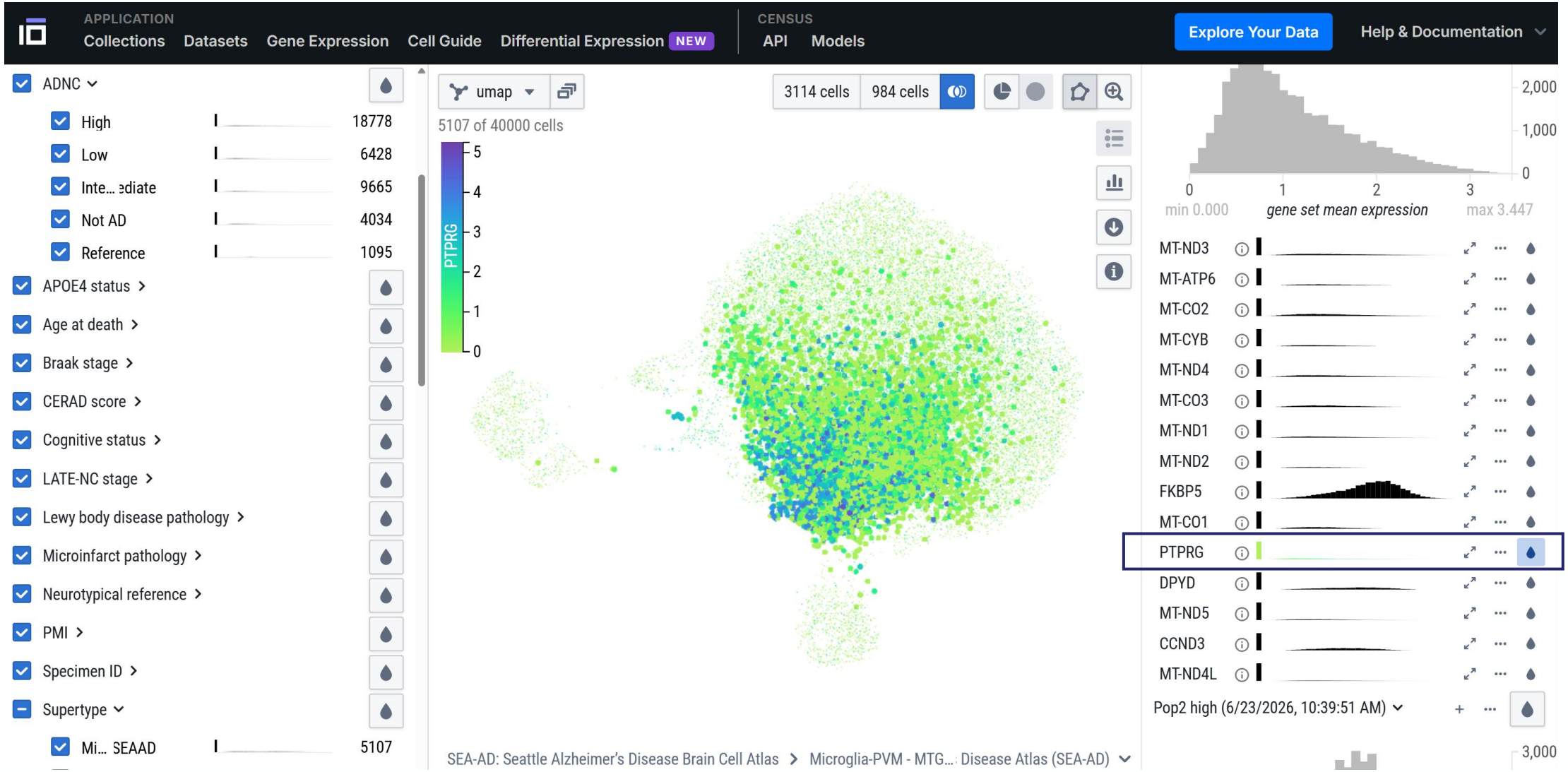
Term sets: neighborhood, subclass, and supertype define the SEA-AD CaH taxonomy. We can visualize the taxonomy as a sunburst diagram that shows the single inheritance hierarchy through a series of rings, that are sliced for each annotation term. Each ring corresponds to a level in the hierarchy. We have ordered the rings so that the class level. Rings are divided based on their hierarchical relationship to the parent slice.

In [46]:
```
levels = ['Neighborhood', 'Subclass', 'Supertype']
df = {}

# Copy the term order of the parent into each of the level below it.
if term_with_counts.index.name != 'cluster_annotation_term_label':
    term_with_counts = term_with_counts.set_index('cluster_annotation_term_label')
```



CELLxGENE/ identify genes of interest



can find some of the same microglial genes associated with AD as complex analyses...

but need to note covariates

MTG gene trajectory viewer/ genes changing with AD pathology

SEA-AD
Gene Trajectories

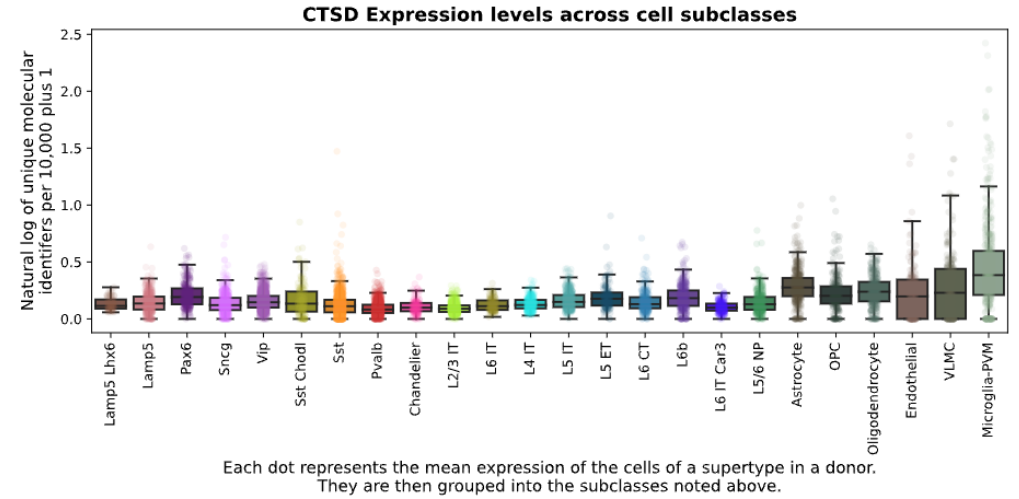
Please select an option and click open table to begin.

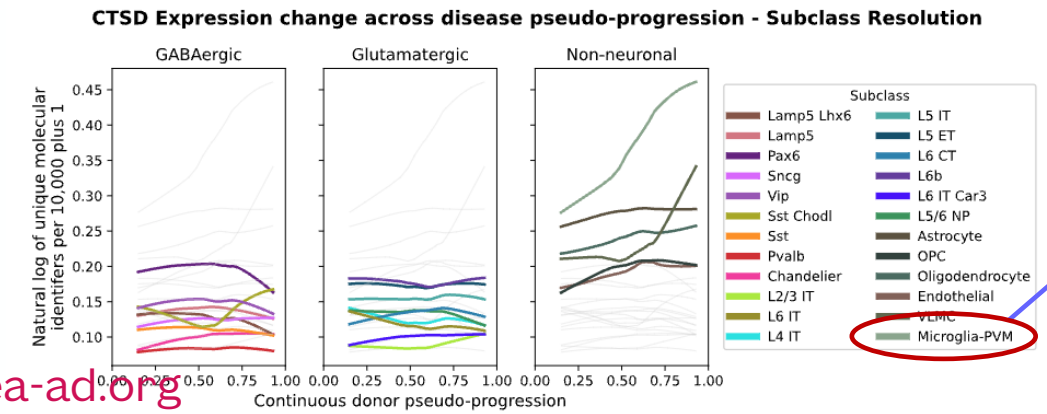
Taxonomy Level

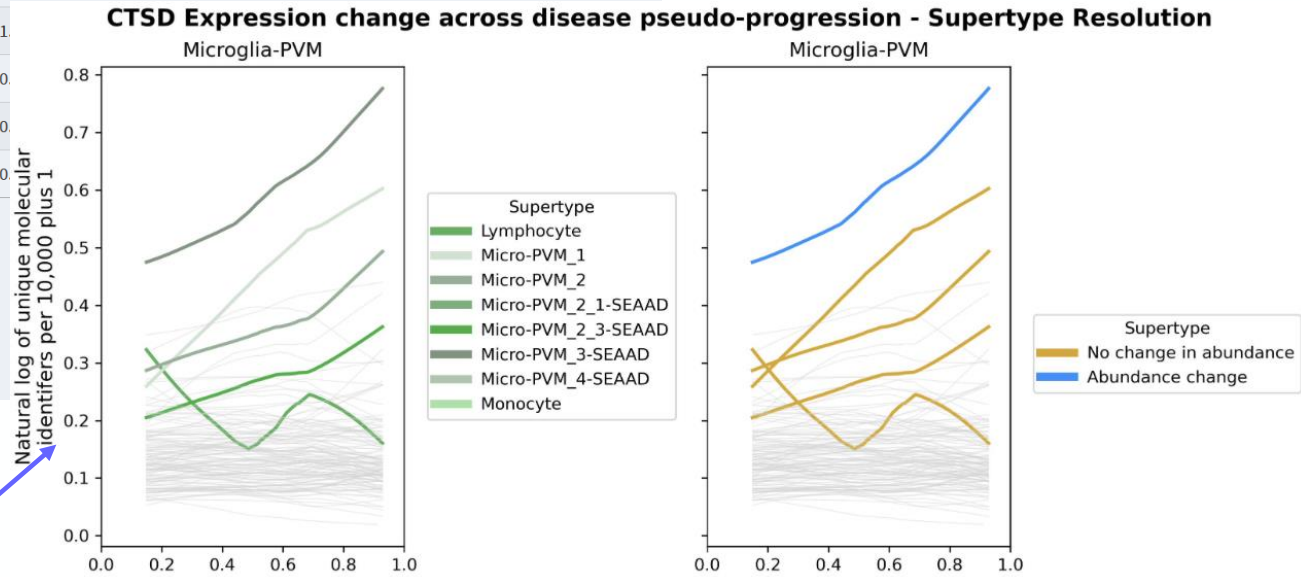
Subclass

Open beta coefficent table

Row.number	Gene	Taxonomy.Level	Population	all	early	late	Mean.expression	Comparative.Viewer	Pseudoprogression.Plot
All		All	All				All	All	All
5615742	CTSD	Subclass	Microglia-PVM	3.46	1.1	1.42	0.49	Comparative Viewer	Pseudoprogression Plot
						1.36	0.3	Comparative Viewer	Pseudoprogression Plot
						-0.73	0.29	Comparative Viewer	Pseudoprogression Plot
						0.92	0.24	Comparative Viewer	Pseudoprogression Plot
						0.38	0.23	Comparative Viewer	Pseudoprogression Plot
						0.14	0.23	Comparative Viewer	Pseudoprogression Plot

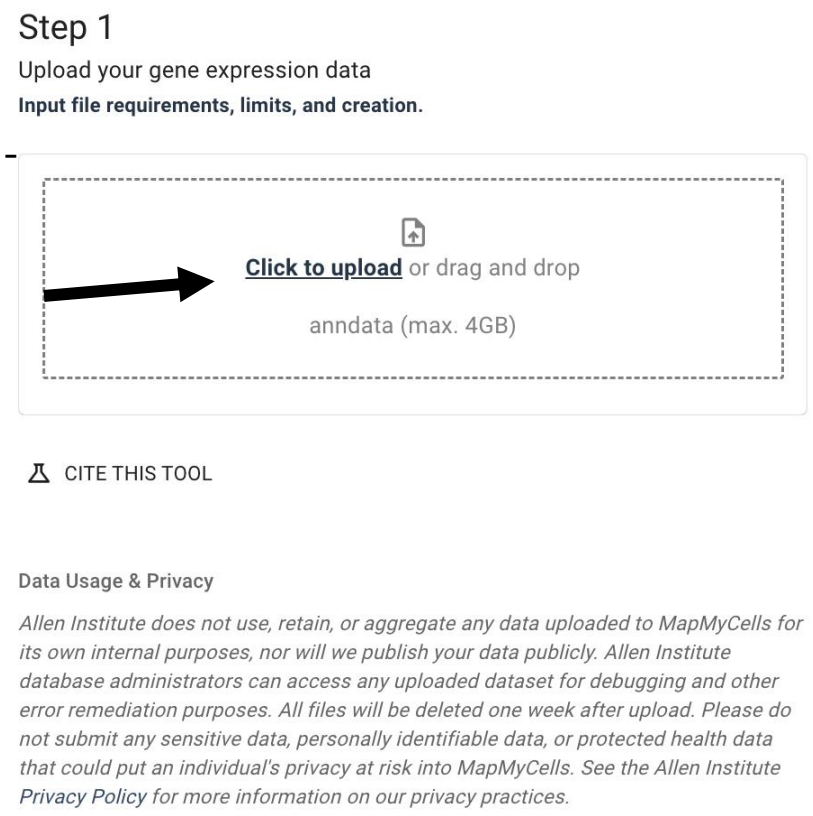


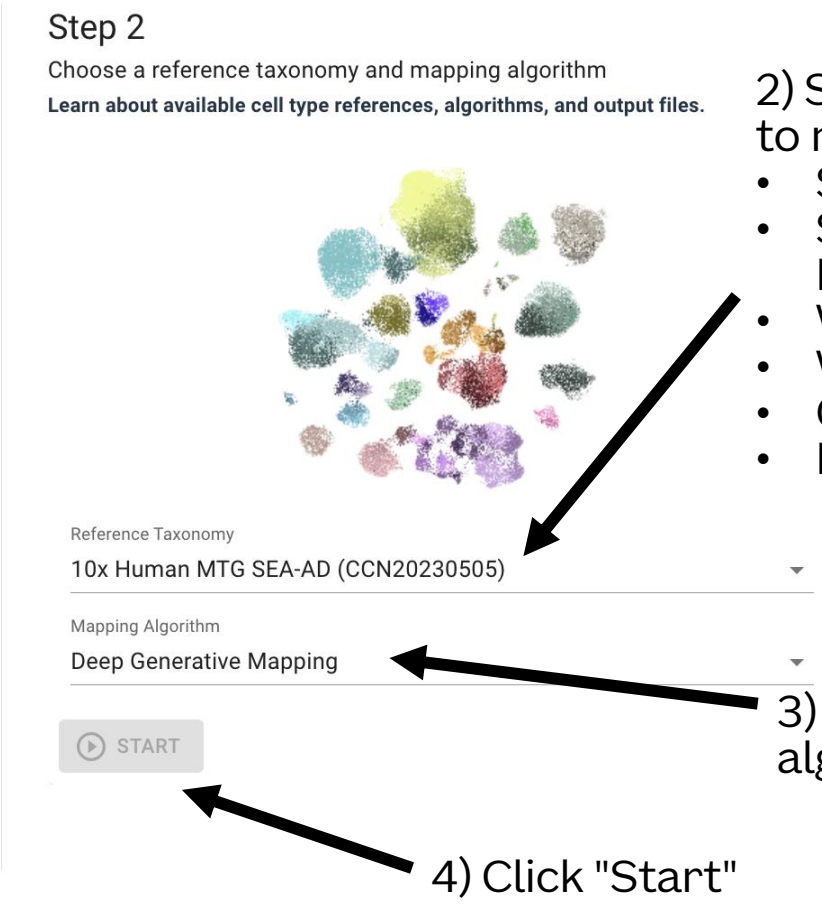




MapMyCells/ adding our cell type names to your data

1) Drag-and-drop input file here (csv or h5ad; any species)





2) Select taxonomy to which to map

- SEA-AD (human MTG)
- SEA-AD (multi-region; NEW)
- Whole Mouse Brain
- Whole Human Brain
- Cross-species BG
- Developing mouse VIS

3) Select mapping algorithm

4) Click "Start"

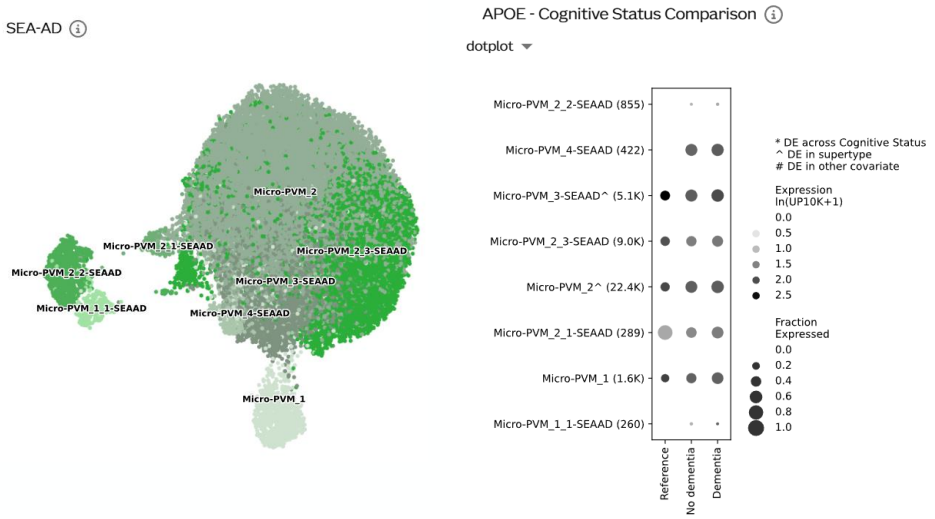
5) Wait 5 minutes and download results

brain-map.org/bkp/analyze/mapmycells

sea-ad/ other tools

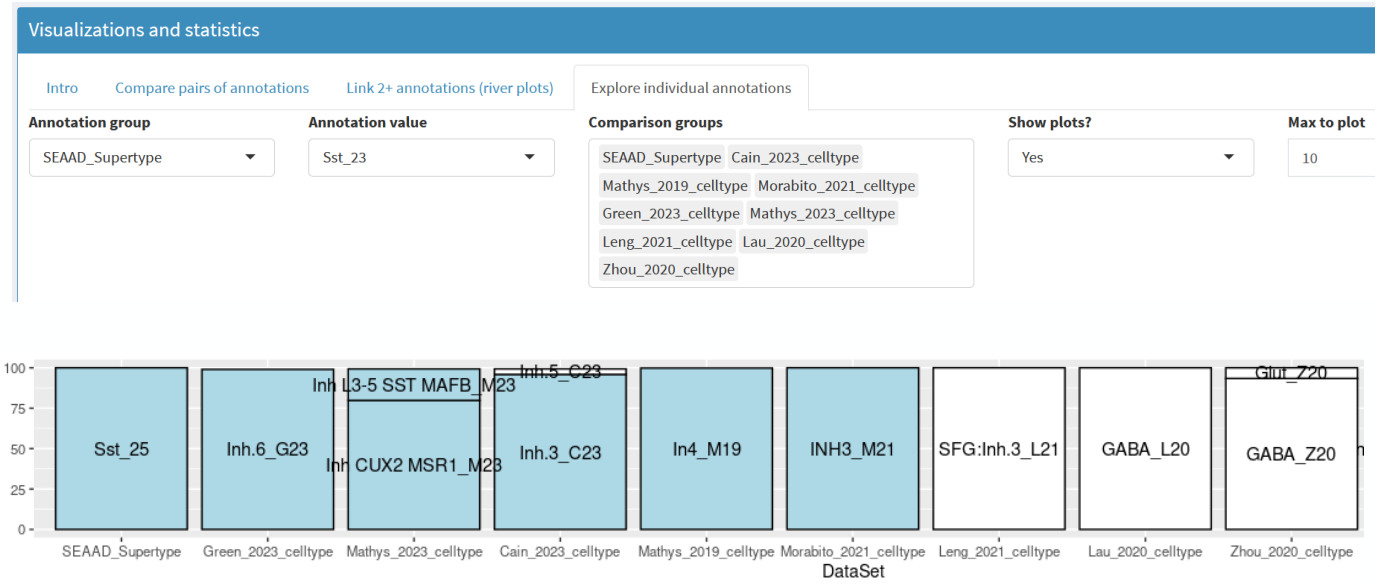
comparative viewer/

view gene expression in the context of AD metrics and other donor metadata



annotation comparison explorer/

connect cell type names and abundance changes between SEA-AD and multiple community studies of AD.



annotation comparison explorer (ACE)/

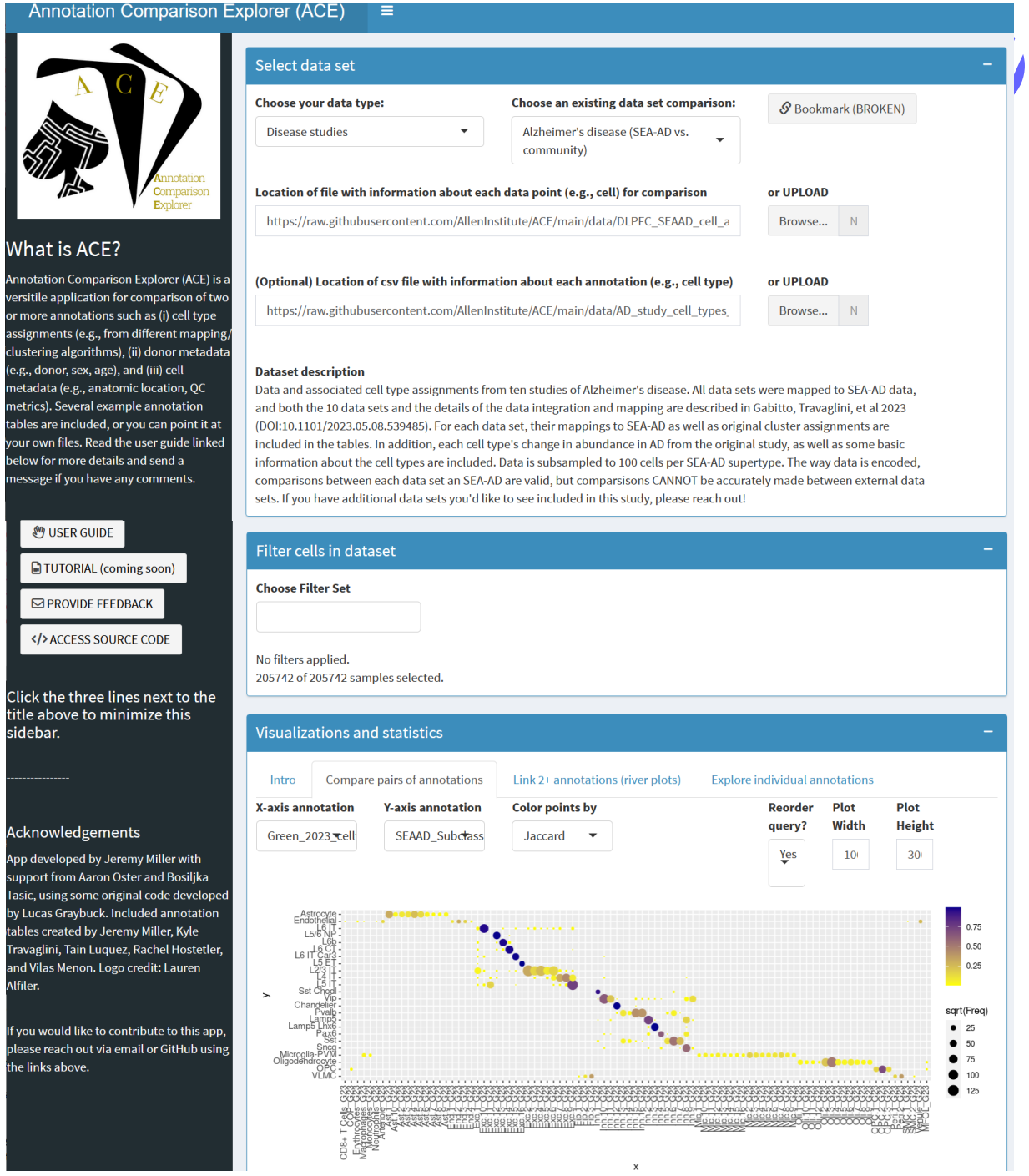
- ACE compares and visualizes different annotations (typically cell type assignments and/or other cell metadata) for a single set of data (typically cells)
- Includes a user guide and webinar for details on how to use the tool (or ask me!)

Annotation Comparison Explorer (ACE): connecting brain cell types across studies of health and Alzheimer's Disease

Jeremy A. Miller, Kyle J. Travaglini, Tain Luquez, Rachel E. Hostetler, Aaron Oster, Scott Daniel, Bosiljka Tasic, Vilas Menon

doi: https://doi.org/10.1101/2025.02.11.637559 Posted February 12, 2025.

sea-ad.shinyapps.io/ACEapp/



ACE/ components

ACE must be run in three sequential steps separated on the app:

1. Select a dataset for which annotations will be compared. These can be preselected or user provided.

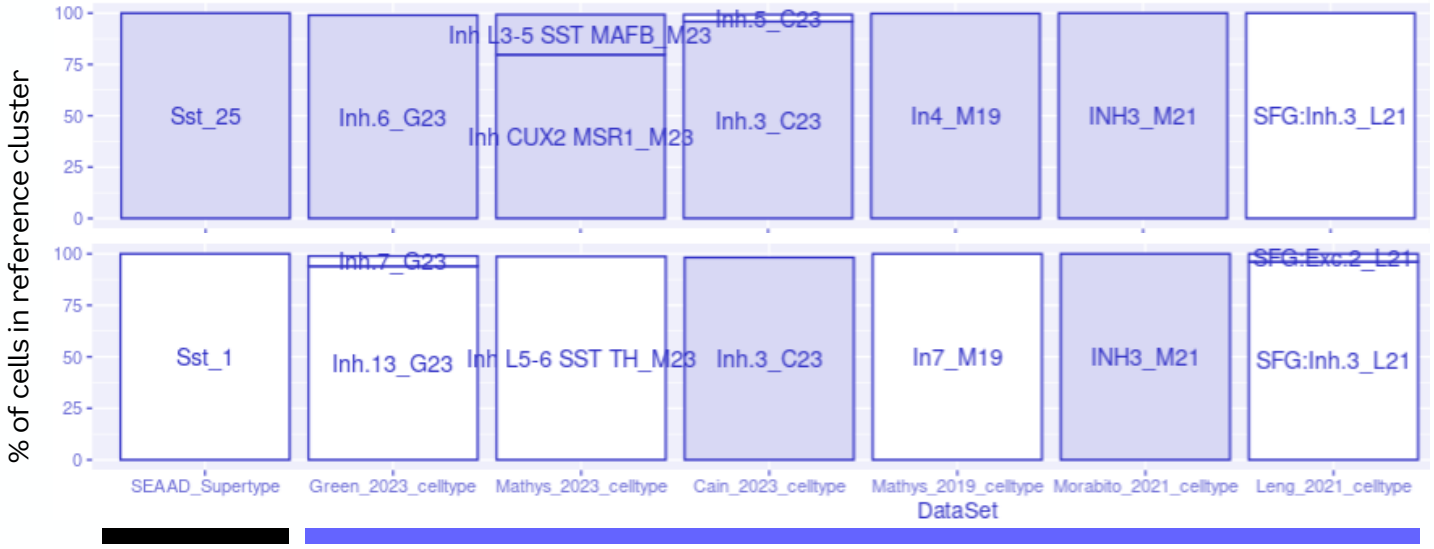
2. Filter cells in dataset to include only annotations of interest (e.g., all cells, neurons, microglia)

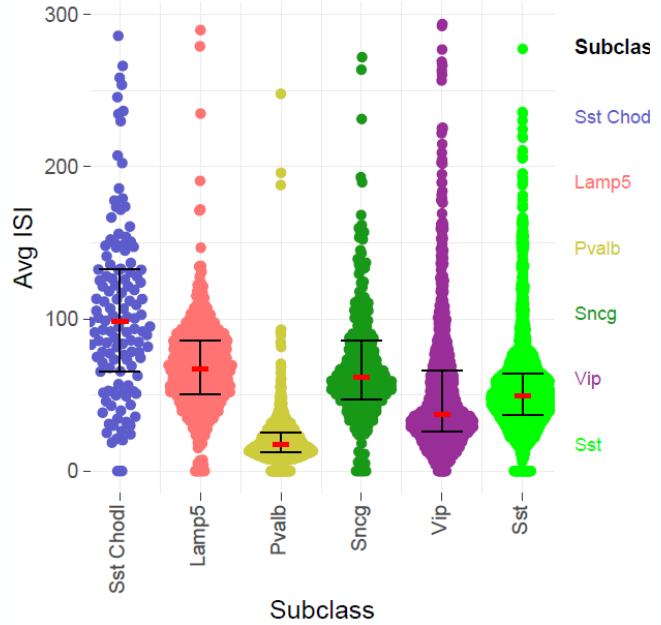
3. Visualizations and statistics for doing the actual comparisons

sea-ad.shinyapps.io/ACEapp/

ACE/ use cases

1. Comparing brain cell types in across 11 studies of Alzheimer's disease
2. Visualizing morpho-electric features of Patch-seq cells of different types
3. Translating between reference brain cell type taxonomies
4. Comparing user-provided clustering and MapMyCells mapping results



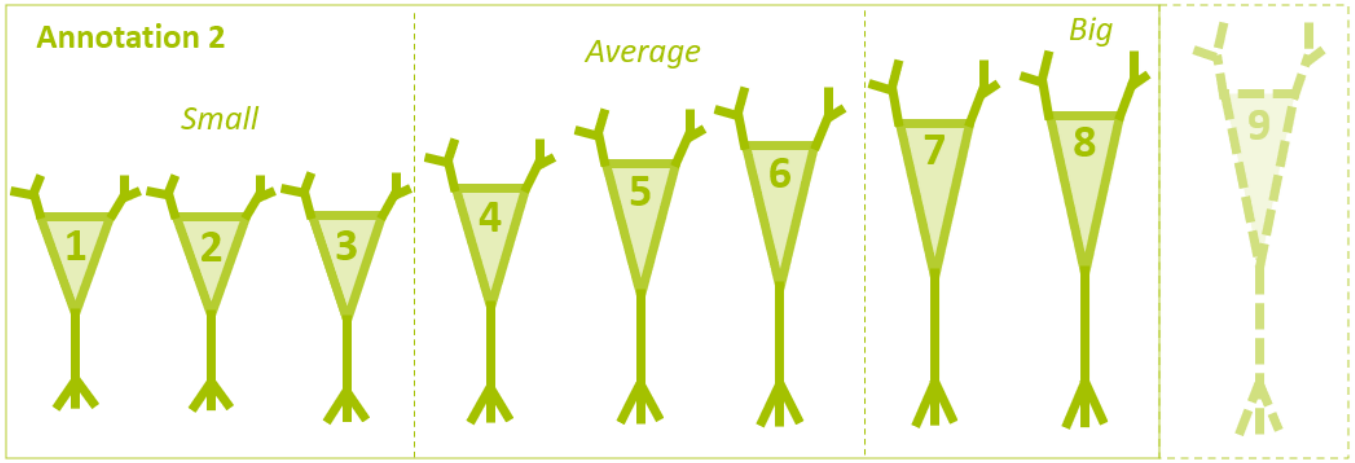


sea-ad.shinyapps.io/ACEapp/

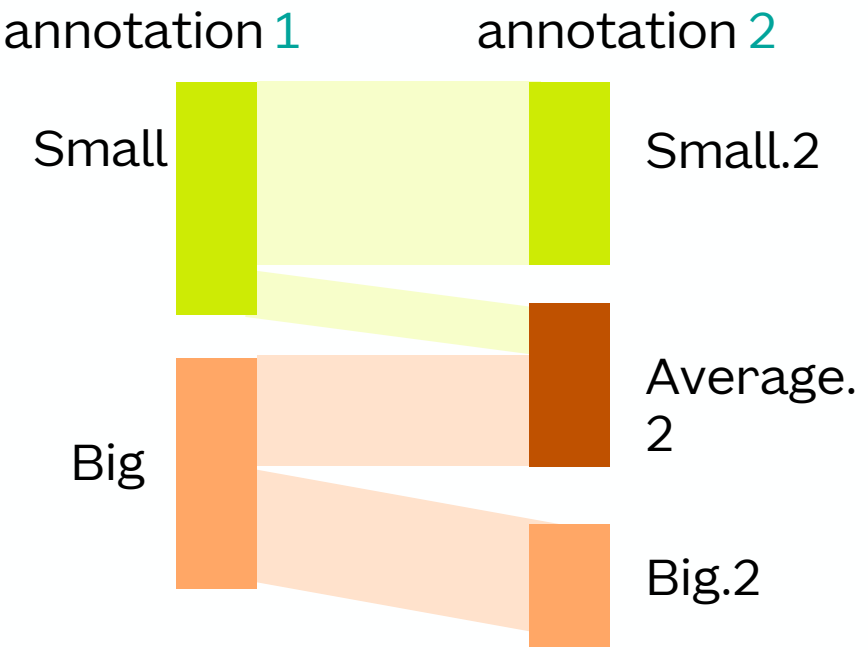
ACE/ comparing cell annotations/ example

I have some cells that I divided into groups (e.g., cell types!)

I added a new cell and now have regrouped them



How did the groups change?

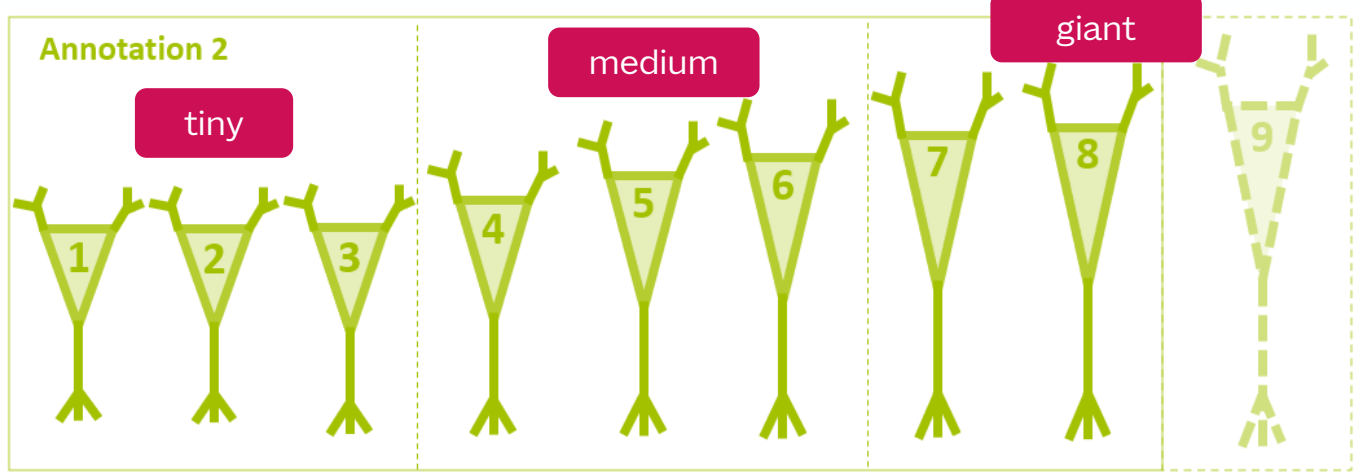


ACE/ comparing cell annotations/ example

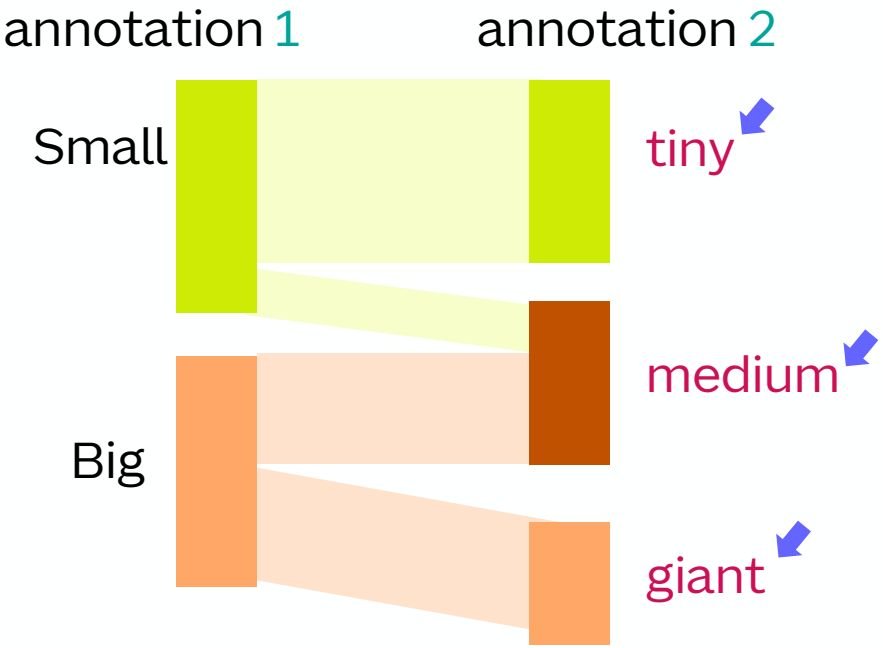
I have some cells that I divided into groups (e.g., cell types!)

ALSO: someone else did the analysis so the names have changed!

How did the groups change?



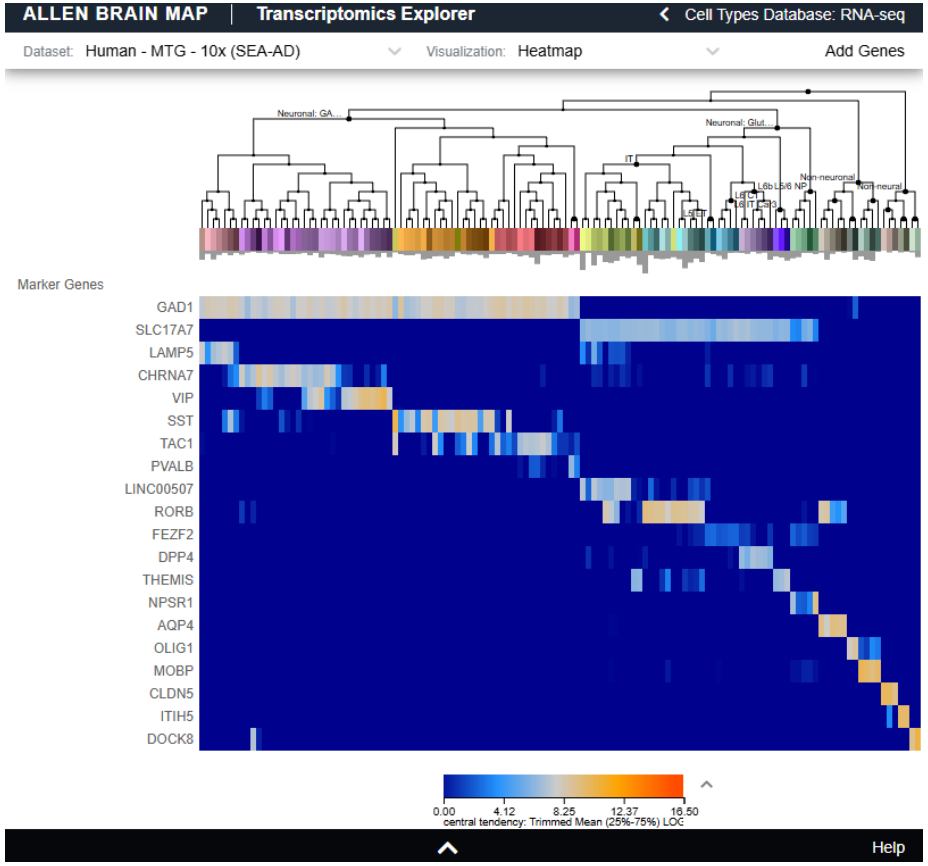
I added a new cell and now have regrouped them



other tools/

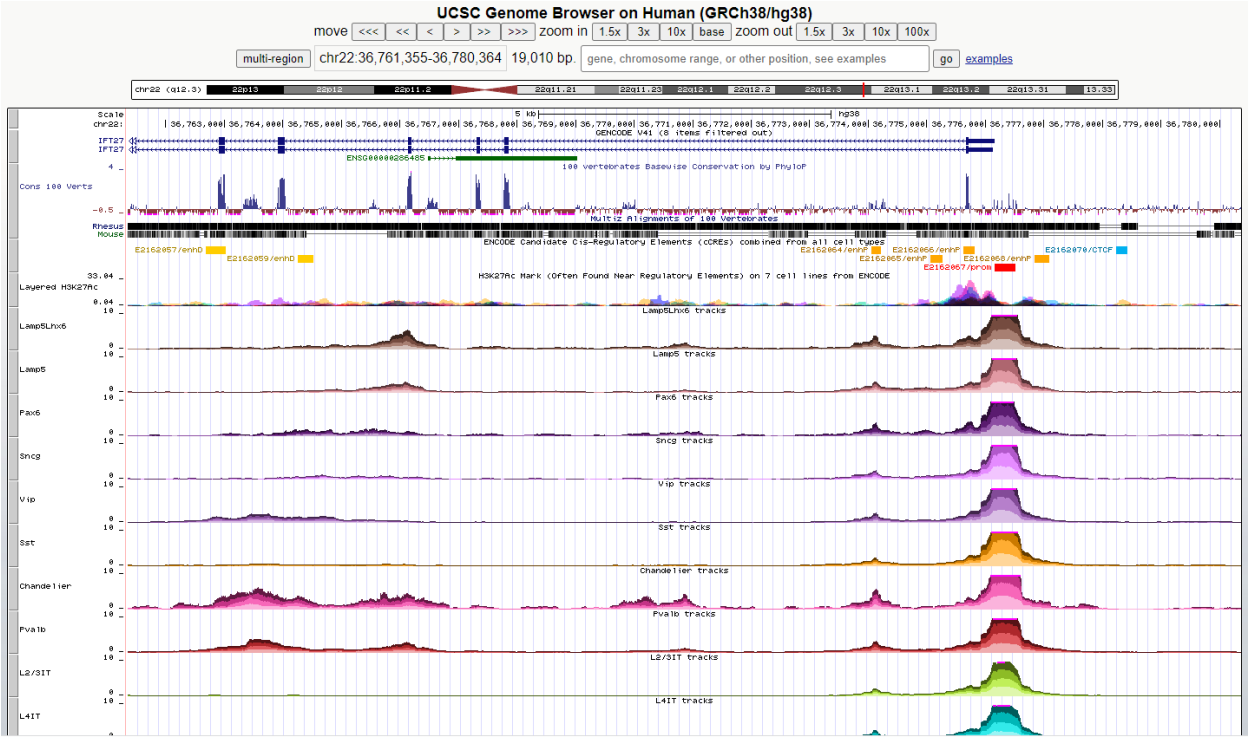
Transcriptomics explorer

Visualize cell type–specific expression of multiple genes in the young adult human MTG (reference)



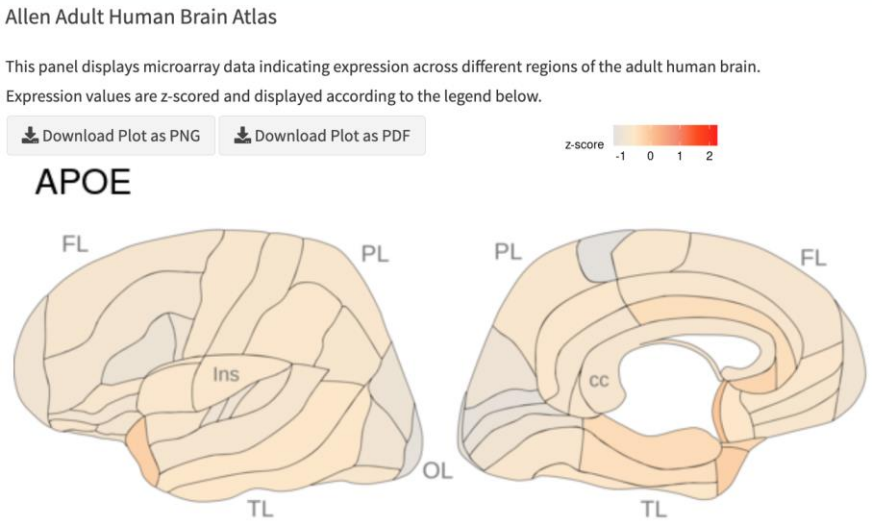
UCSC genome browser

Explore single cell ATAC-seq data in the context of AD pathology (ADNC) for known loci



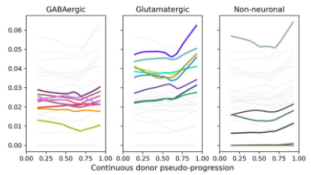
Gene Knowledge Cards ➔

Explore expression of individual genes and find similar genes across historic atlases and some single-cell transcriptomics data sets



tools in progress/ sea-ad sandbox

A collection of interactive tools and prototype applications for the Seattle Alzheimer's Disease Brain Cell Atlas (SEA-AD). These applications provide access to gene expression, cell type, trajectory, neuropathology, and annotation exploration resources.



MTG Trajectory Viewer

Identify and visualize cell type-specific gene expression changes across AD pseudo-progression in MTG.

Launch App | GitHub

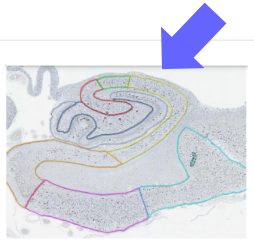
SEA-AD
Multiregion Gene Expression Viewer

SEA-AD Multiregion Viewer

Explore gene expression patterns in multi-region single cell data in the context of cell types, brain regions, and disease progression.

Launch App | GitHub

Related app: developing mouse VIS



Neuropathology Viewer

Explore neuropathology images and associated measurements from multiple brain region in SEA-AD donors.

Launch App | GitHub

A C E
Annotation Comparison Explorer

Annotation Comparison Explorer (ACE)

Compare cell type annotations between SEA-AD and ten community studies of AD or user data from MapMyCells.

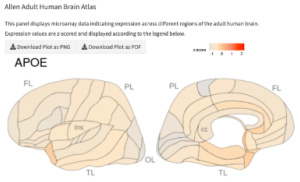
Launch App | GitHub

CHARGE

Cell Hierarchy Gene Explorer (CHARGE)

Identify genes of potential biological interest in multiple cell type taxonomies (including SEA-AD MTG).

Launch App | GitHub



Gene Knowledge Cards

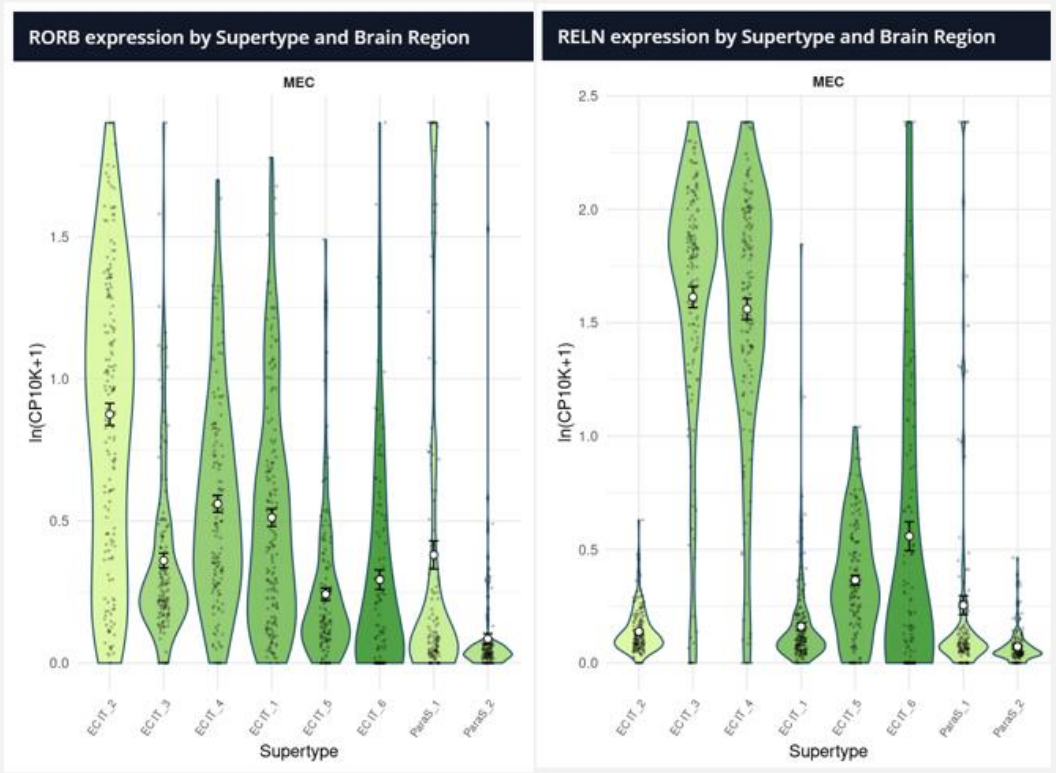
Explore patterns of gene expression in the brain across multiple Allen Brain Atlases (including SEA-AD).

Launch App | GitHub

alleninstitute.github.io/SEAAD_sandbox

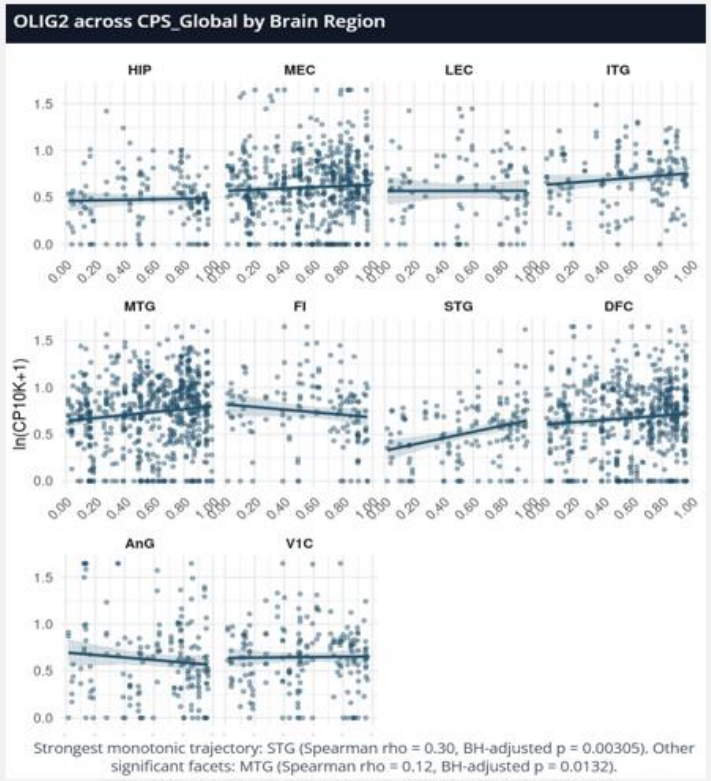
sea-ad/ sandbox/ multiregion viewer

Cell type selectivity in Entorhinal cortex



These genes were part of a panel used to distinguish EC IT_2 from EC IT_3 supertypes with RNAscope.

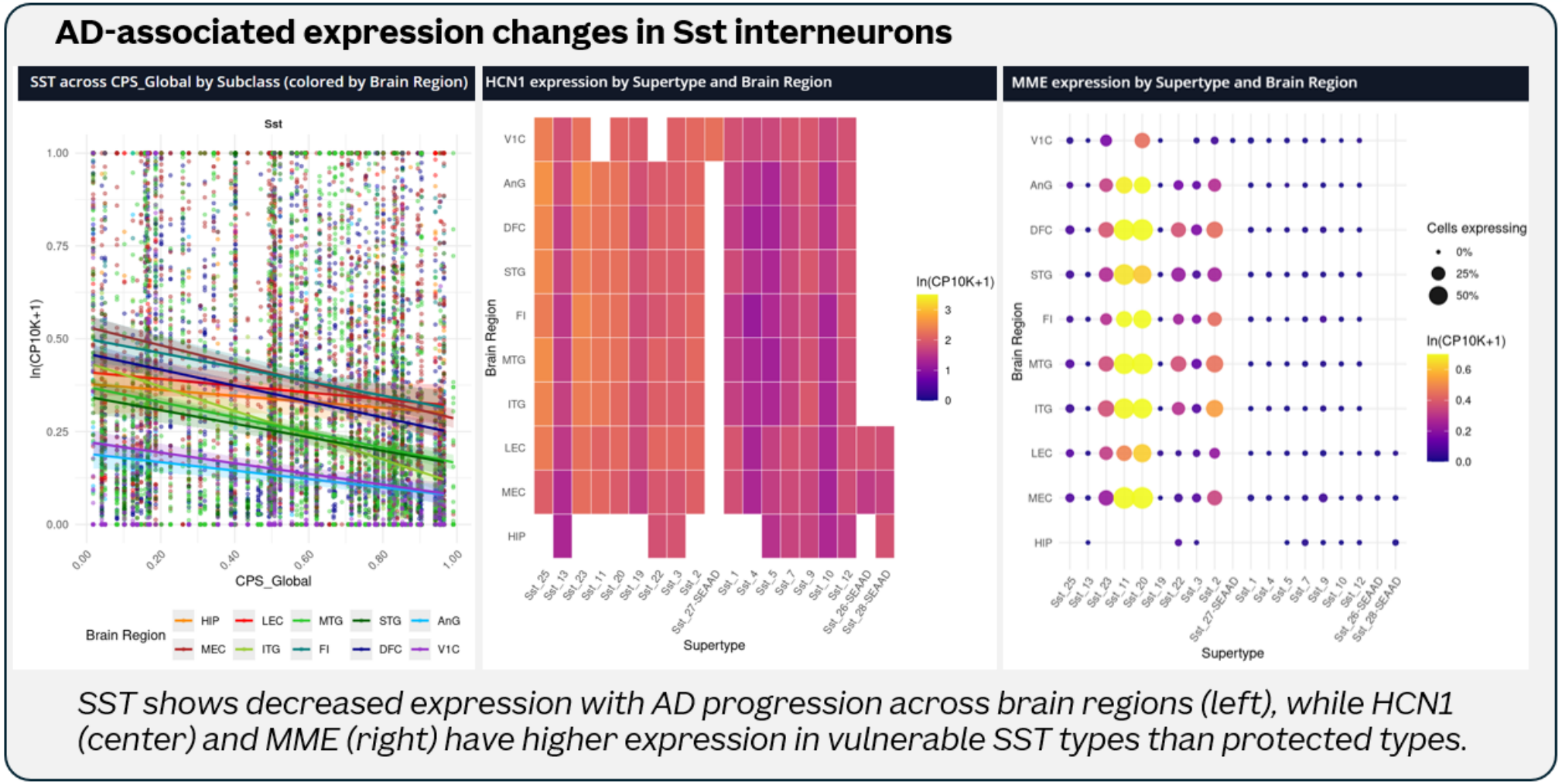
Region-specific changes in OPCs



While many AD-associated gene changes are conserved, this one is region-specific.

sea-ad.shinyapps.io/multiregion_gene_exploration

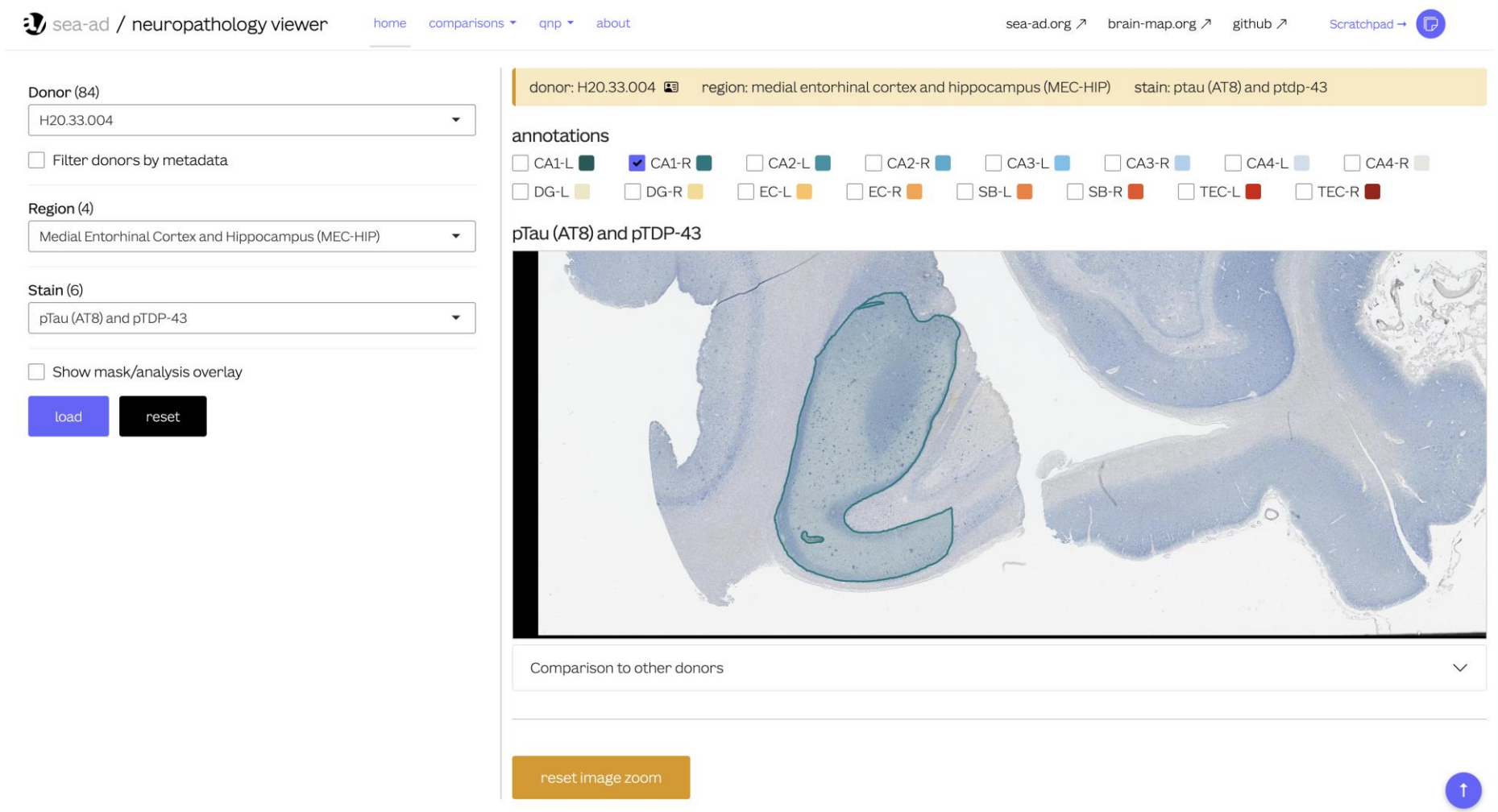
sea-ad/ sandbox/ multiregion viewer



SST shows decreased expression with AD progression across brain regions (left), while HCN1 (center) and MME (right) have higher expression in vulnerable SST types than protected types.

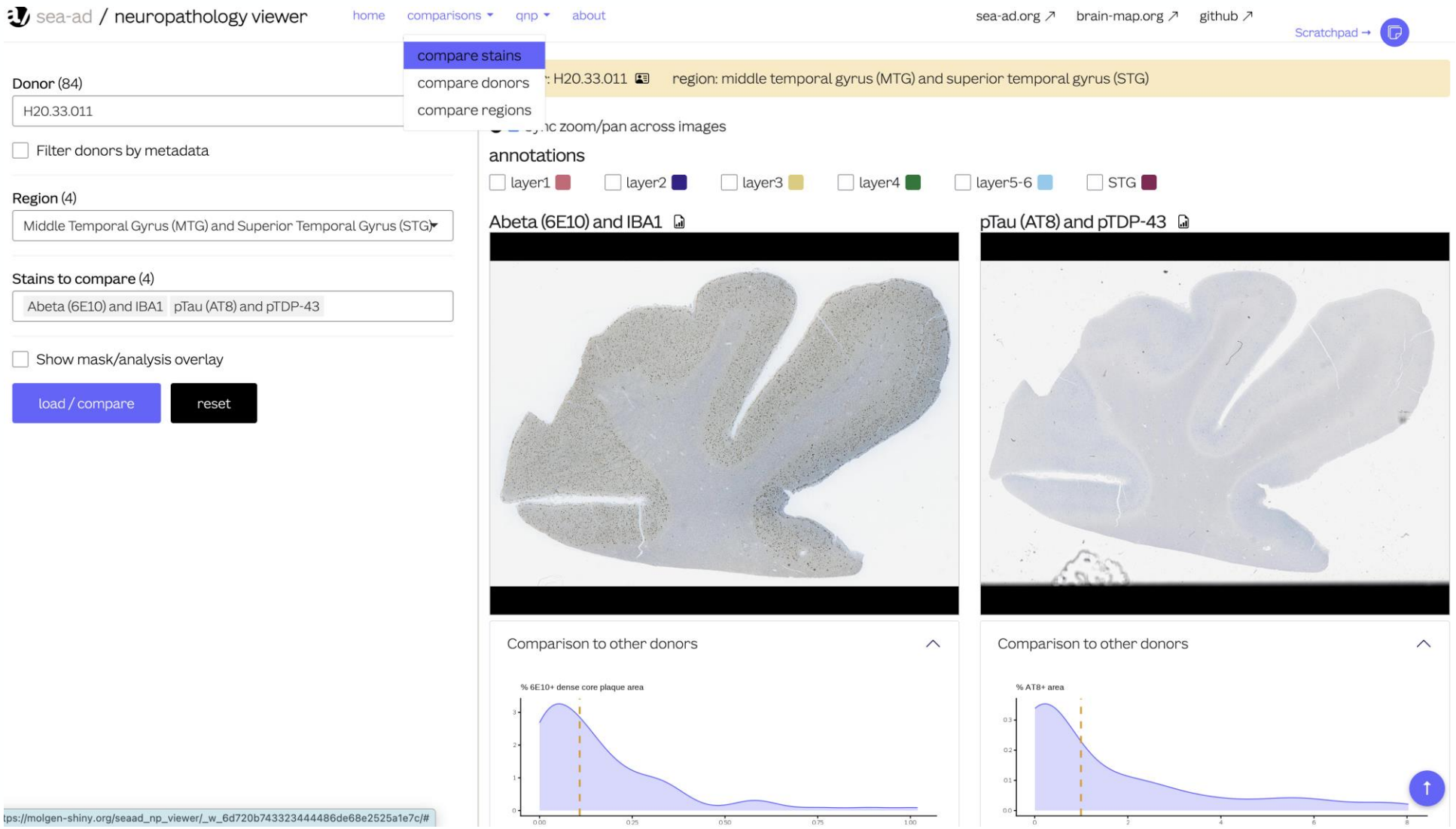
sea-ad.shinyapps.io/multiregion_gene_exploration

sea-ad/ sandbox/ neuropathology



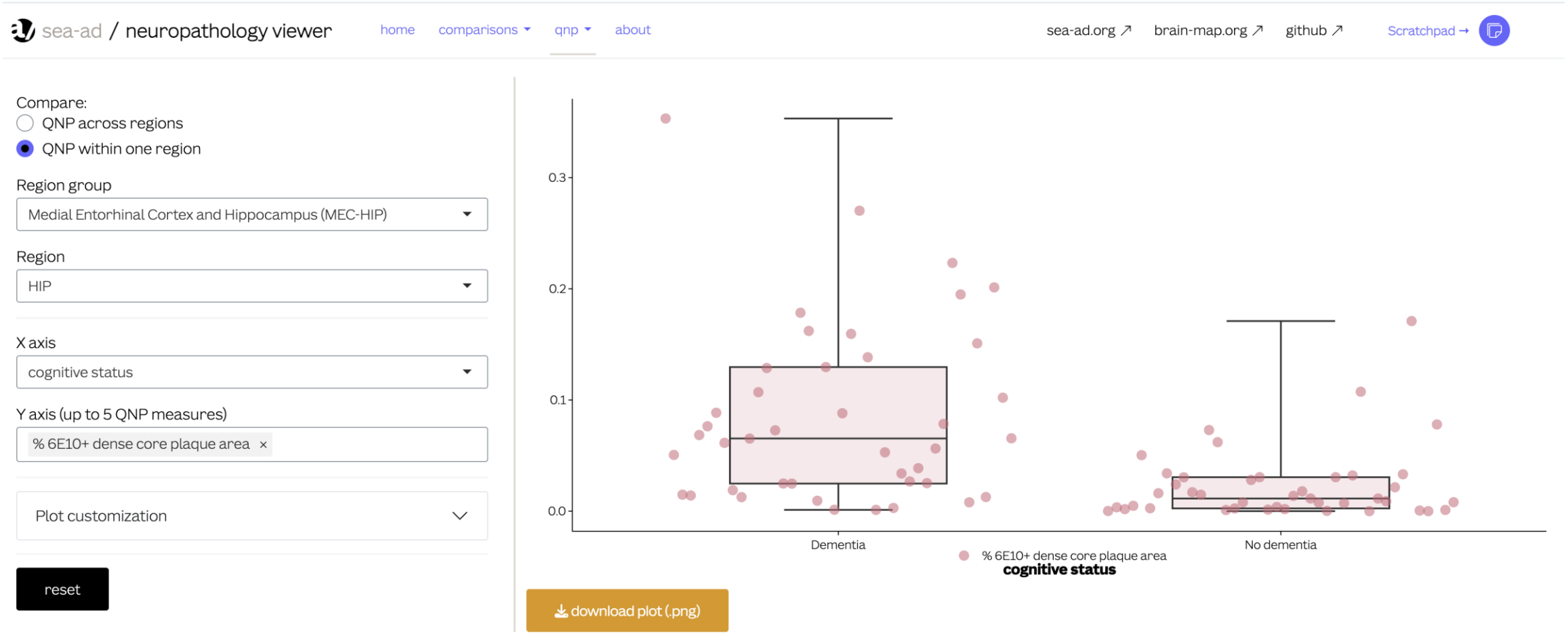
molgen-shiny.org/seaad_np_viewer

sea-ad/ sandbox/ neuropathology/ comparisons



molgen-shiny.org/seaad_np_viewer

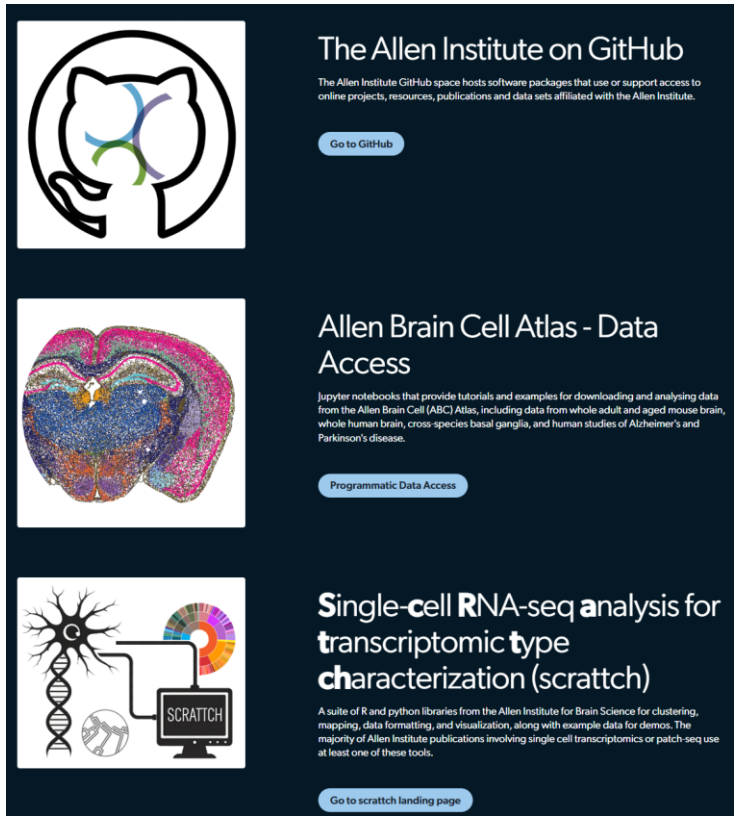
sea-ad/ sandbox/ neuropathology/ qnp plots



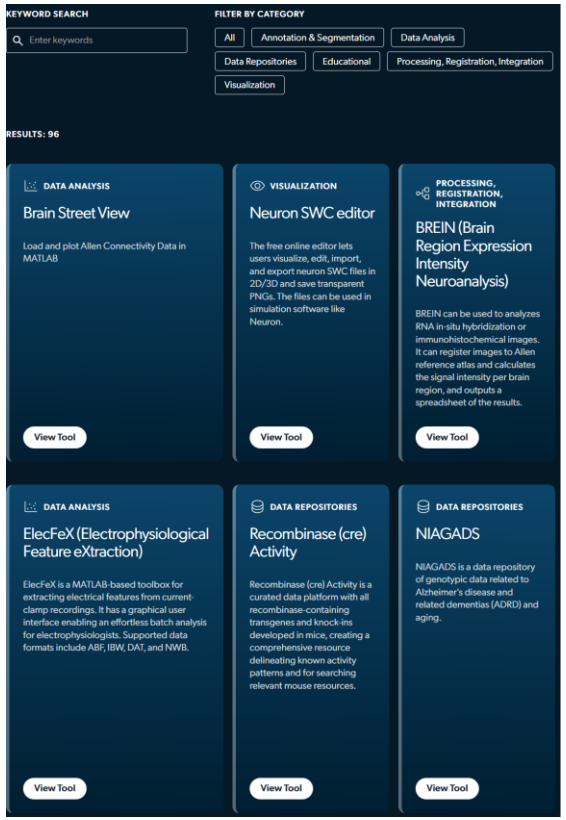
molgen-shiny.org/seaad_np_viewer

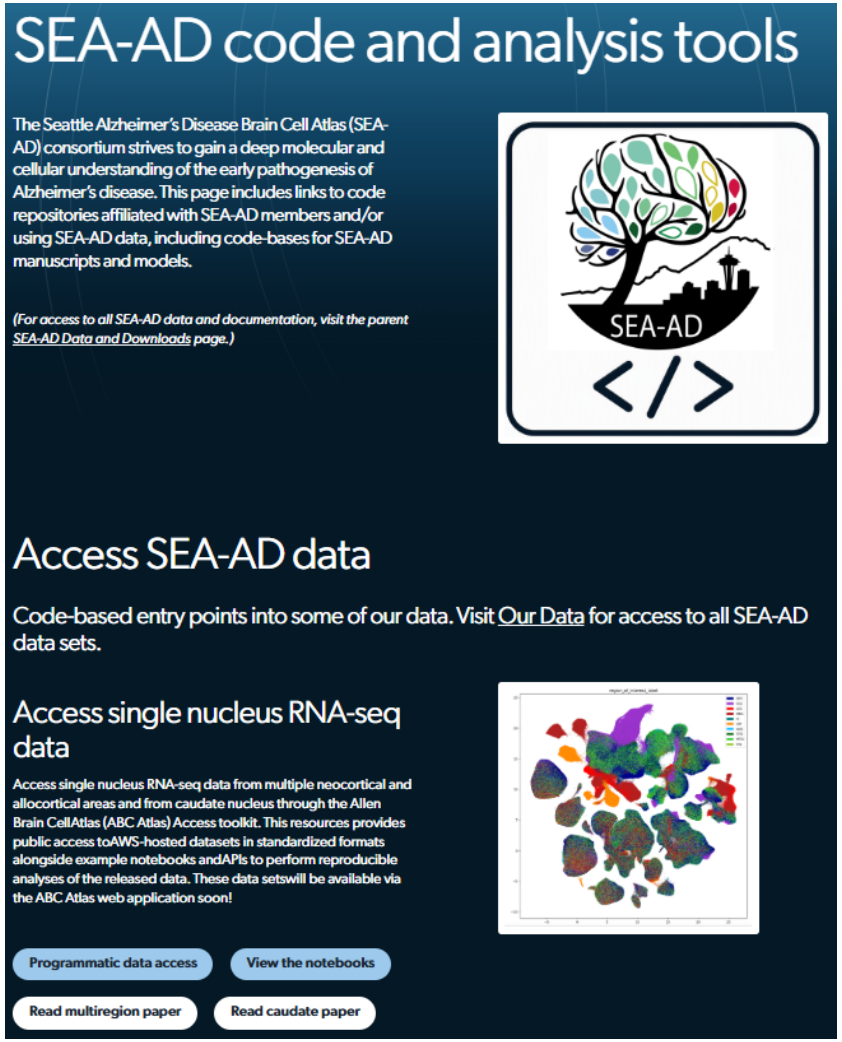
sea-ad/ computational tools & resources

computational analysis toolbox/



community tools/





agenda/afternoon

12:00-1:00 | lunch

1:00-2:15 | walkthrough of allen brain map tools (Rachel / Jeremy)

- walkthrough of allen brain map
- Brain Knowledge Platform: ABC Atlas, MapMyCells, Genetics Tools Atlas
- Additional tools in brief: Patch-seq, gene atlases, and anatomy

2:15-2:45 | SEA-AD tools for exploring changes in AD (Irika / Jeremy)

2:45-3:00 | break

3:00-3:30 | external use cases – Rachel

3:30-4:40 | user exploration of allen institute tools

4:40-5:00 | feedback survey (all – please fill out the survey!)

github.com/AllenInstitute/SPOCTT/tree/main/2026_Additional_Presentations

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github.com/AllenInstitute/SPOCTT/tree/main/2026_Additional_Presentations

demo & external use cases/

alleninstitute/

community use of BKP tools and data/

Title: Evolution of Cajal-Retzius Cells in Vertebrates from an Ancient Class of Tp73+ Neurons

Authors: Elias Gumnit[1], Giacomo Gattoni[1], Jamie Woych[1], Astrid Deryckere[1#], Prabesh Bhattarai[2,3], J Andrew Gillis[4], Caghan Kizil[2,3], Maria Antonietta Tosches[1*]

Next, we investigated whether the molecular and spatial heterogeneity of salamander *Tp73*+ neurons has counterparts in mammals. We isolated mouse *Tp73*+ neurons from the Allen Brain Cell (ABC) atlas (*34*) and compared them to the salamander *Tp73*+ neuron dataset. The

A unified analysis of atlas single-cell data

Hao Chen,[1,2] Nam D. Nguyen,[1] Matthew Ruffalo,[1] and Ziv Bar-Joseph[1,3]

To further test the ability of GIANT on integrating data from different modalities, we collected single-cell mouse brain data from the Allen Brain Cell Atlas (Yao et al. 2023), which contains 2,341,350 cells from scRNA-seq and 3,938,808 cells from MERFISH spatial transcriptomics (Chen et al. 2015). Both modali-

Morphine regulates astrocyte transcriptional dynamics in the ventral tegmental area by stimulation of glucocorticoid signaling

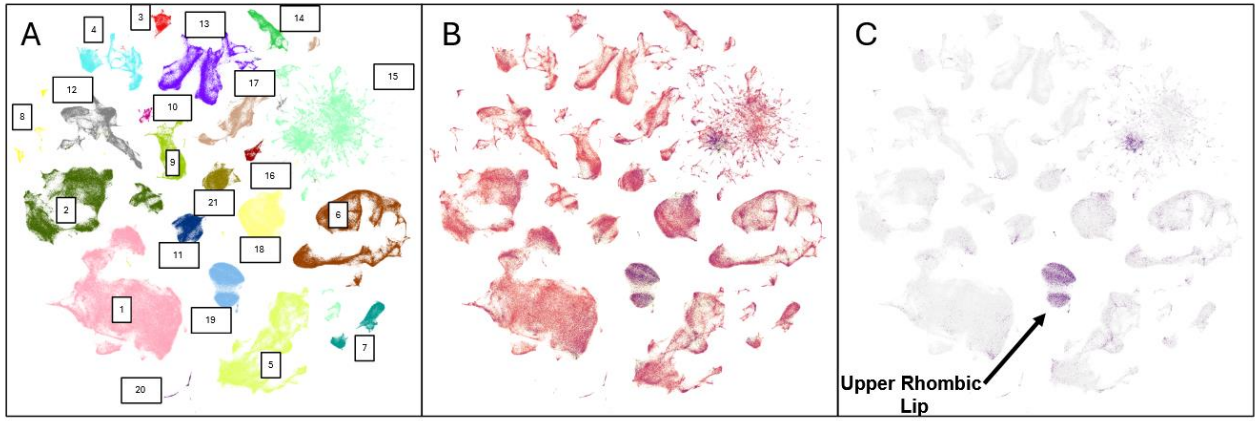
Jennifer J. Tuscher[1,3]*, Angela Cleere[1,3], Robert A. Phillips III[1,4], Catherine E. Newman[1], Guy Twa[1], Nathaniel J. Robinson[1], Lara Ianov[1], Robert E. Sorge[2], & Jeremy J. Day[1]*

clusters using lisi v1.0 in R[83]. To refine resolution of VTA neuronal subtypes and assign cells to classes, we followed the Allen Institute's Whole Mouse Brain taxonomy using the MapMyCells web interface[84] (RRID:SCR_024672). After removing additional low-

Review

Using the Allen Brain Cell Atlas of the Human Brain to Gain Insights into C-Terminal-Binding Protein 1 (CtBP1)'s Potential Function

Suhjin Lee [1] and Uthayashanker R. Ezekiel [2,*]



Use cases for ABC Atlas

“Are genes X and Y co-expressed in the same cell type?”
“Where is gene X expressed in the brain?”
“Is gene X expressed in a specific cell type?”
“Where is cell type X in the brain?”
“What is the spatial relationship between cell type X and cell type Y?”

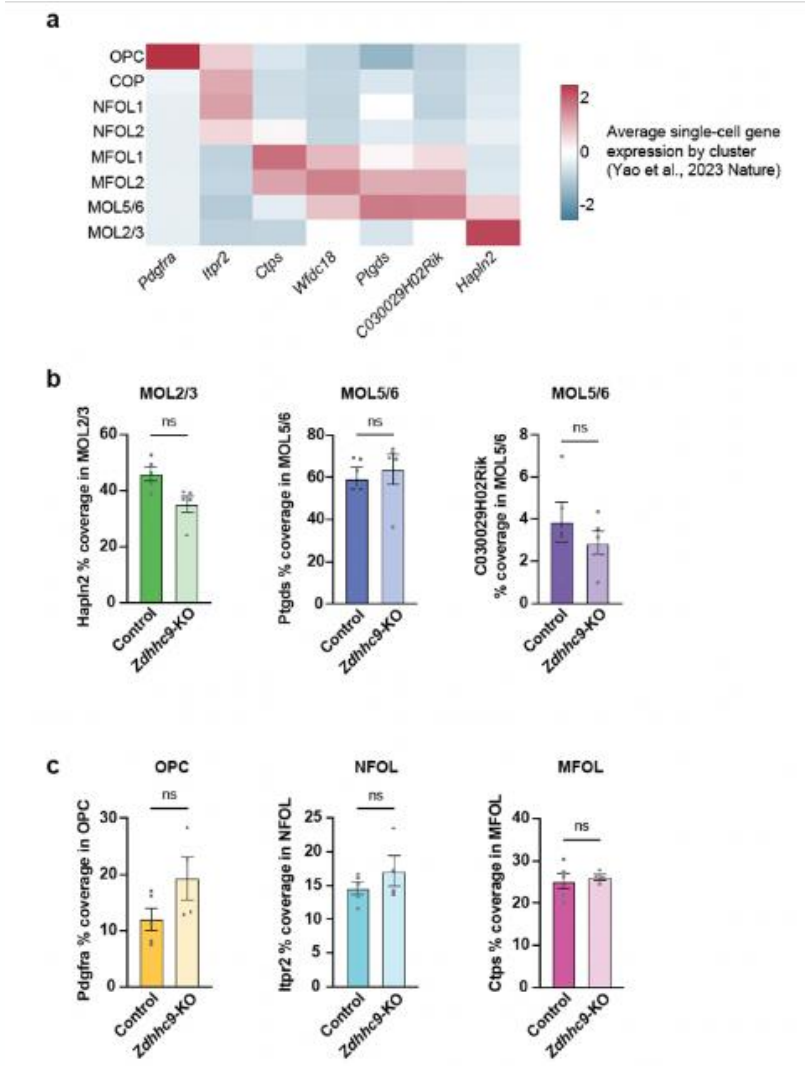
However,

- MERFISH data in ABC Atlas is based on cell segmentation; cell-to-cell or synaptic contacts are not shown.
  - “Does cell type X synapse onto cell type Y?” cannot be answered
- MERFISH data in ABC Atlas shows centroids of cells, not cell shape or size
  - “Is the soma of cell type X larger than the soma of cell type Y?” cannot be answered

ABC atlas data/ experimental design

Altered oligodendrocyte subtype proportions in *Zdhhc9*-KO mice

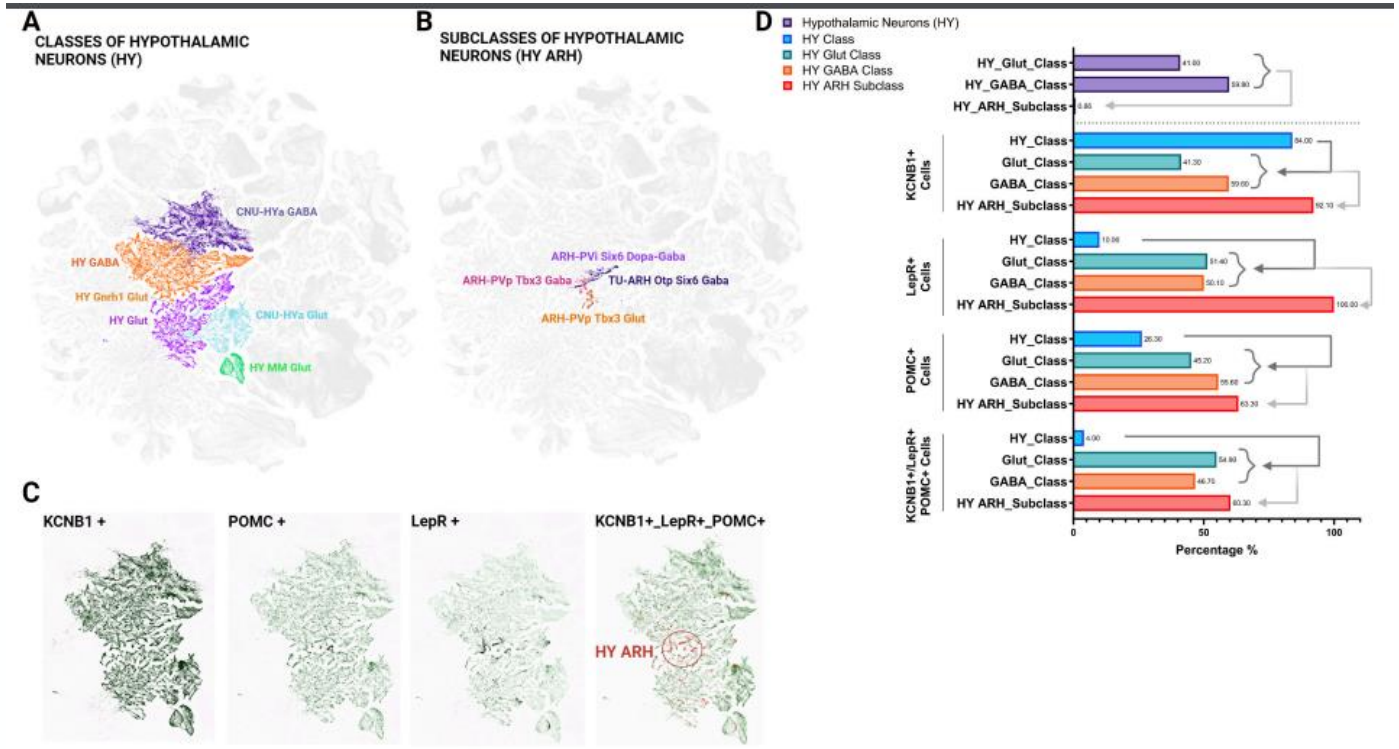
The changes in gene expression identified with RNAseq predict alterations in proportion of oligodendrocyte lineage cells in *Zdhhc9*-KO mice (Figure 1d). We therefore designed fluorescent *in situ* hybridization (FISH RNAscope) probes for the dominant MOL subtypes in the corpus callosum, guided by expression data from previously published scRNAseq and FISH studies(15, 28, 32)(Figure S4a). We chose marker genes with highly enriched expression in specific oligodendrocyte lineage cells, including *Hapln2* (MOL2/3 marker), *Ptgds* (MOL5/6 marker), and *C030029H02Rik* (MOL5/6 marker, Figure S4a). Importantly, within each cell type identified we



Hollman et al., 2024 (preprint)

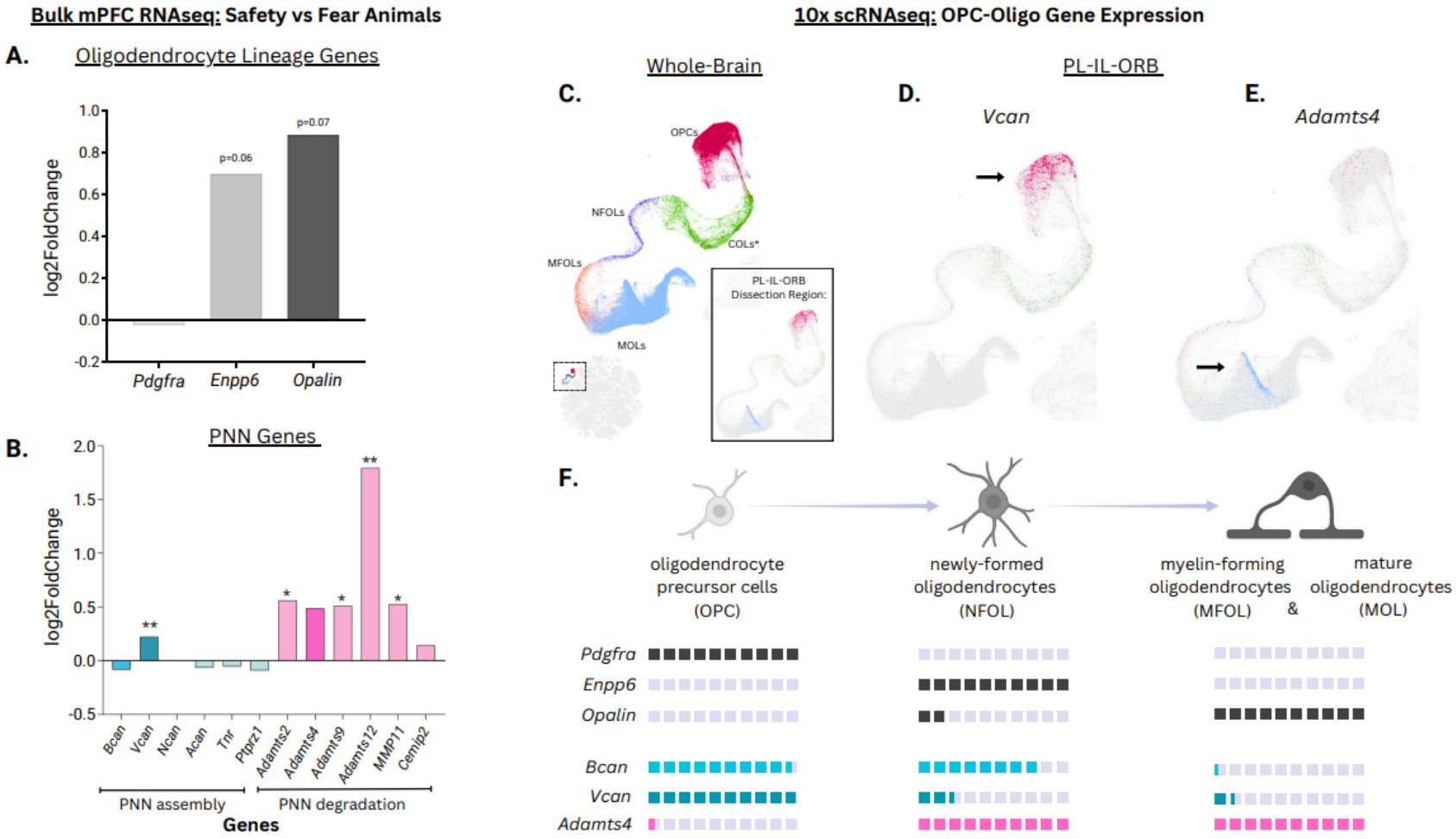
ABC atlas data/ experimental validation

These data are in agreement with the work of Yao and colleagues who constructed a mouse brain transcriptomic and cell type atlas.[26] Analysis of the atlas shows that KCNB1 mRNA is present in all hypothalamic neuron classes and is expressed in ~92% of ARH neurons (Figure S6). Likewise, POMC transcripts are maximally expressed in ARH neurons (~63%). Also, LepR transcripts are broadly expressed in ARH neurons where, in ~60% of the cases, they are colocalized with KCNB1 and POMC transcripts.



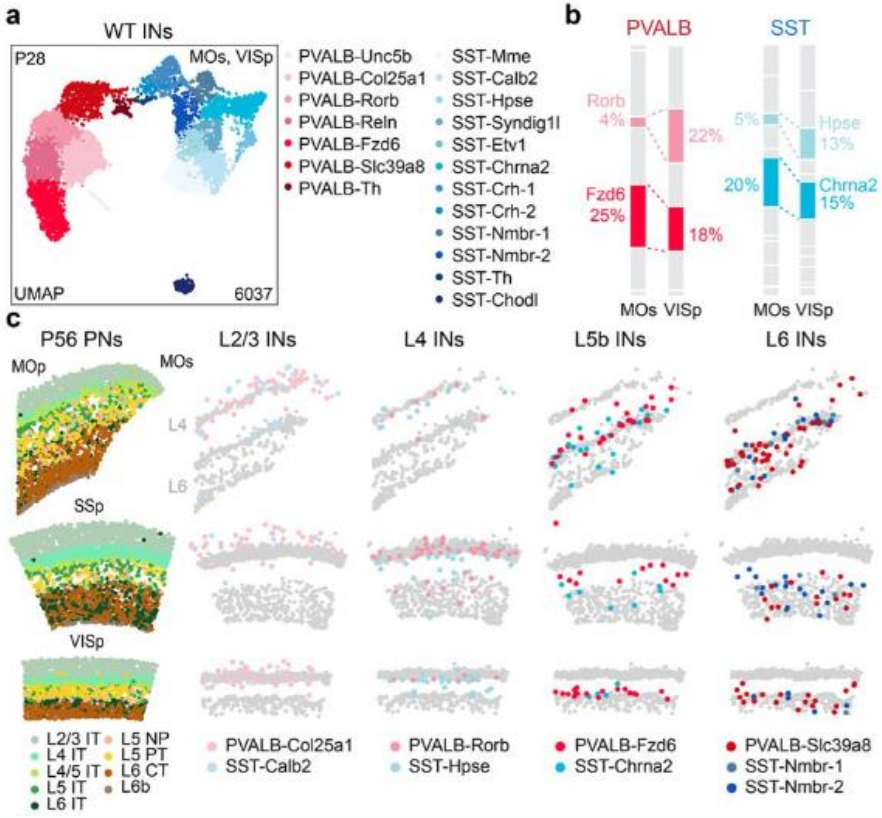
Forzisi-Kathera-Ibarra et al., 2024

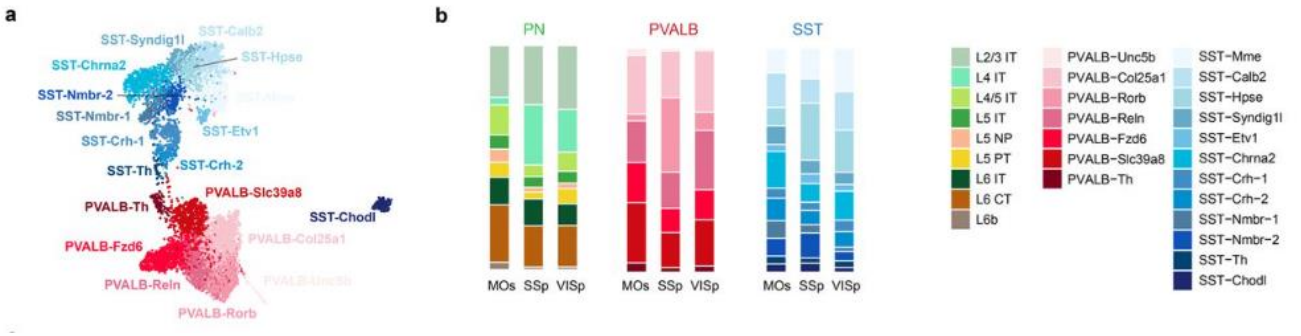
ABC atlas data/ bulk RNAseq comparison



Denholtz et al., 2025 (preprint)

ABC atlas data/ validation & analysis





Utilizing the published adult mouse whole-brain multiplexed error-robust fluorescence *in situ* hybridization (MERFISH) dataset[18], we examined the distribution patterns of these interneuron subtypes (Extended Data Fig. 2a). Each subtype showed a laminar specific distribution that is consistent across different cortical regions (Fig. 1c-d, Extended Data Fig. 3). In most cases, their spatial positioning correlates with the

Wu et al., 2024 (preprint)

ABC atlas data/ novel tool generation

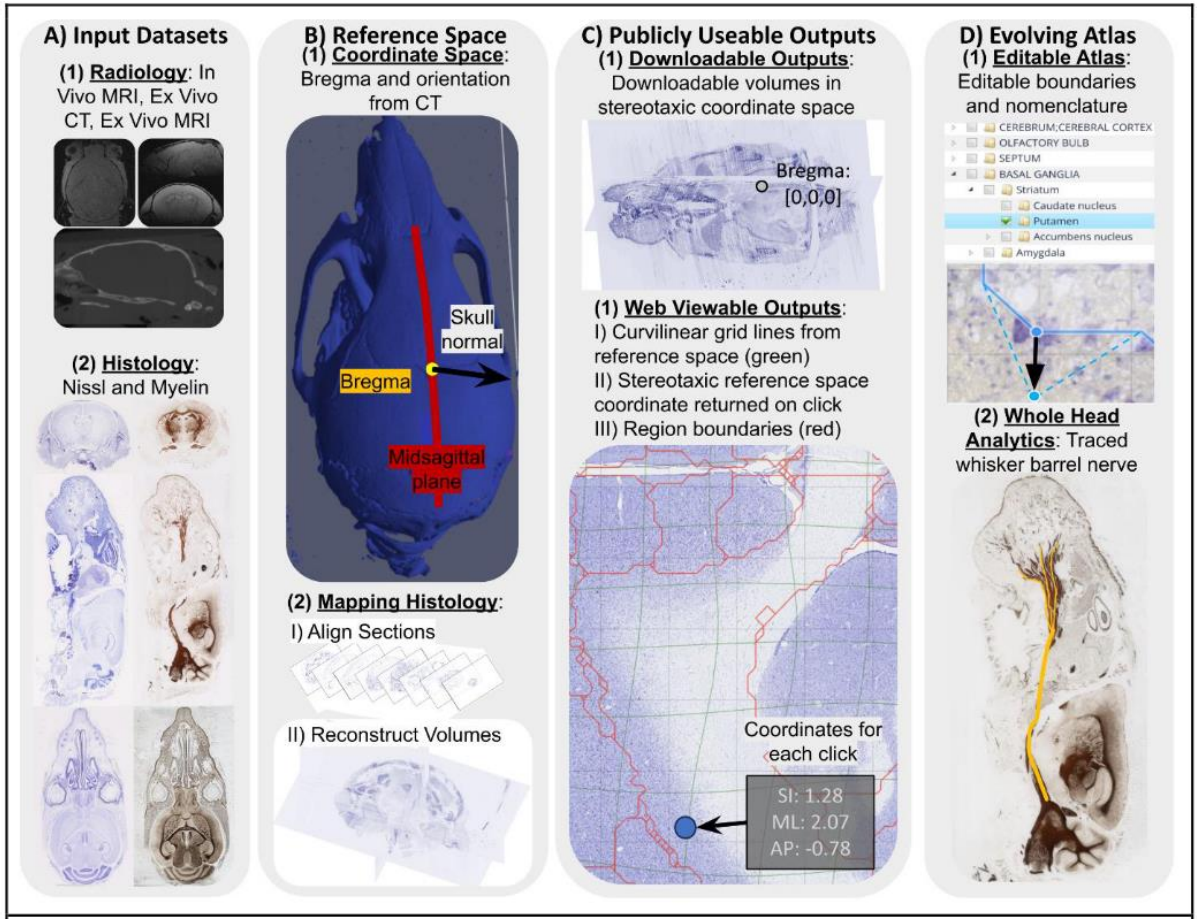


Fig. 1 | Summary of the pipeline for generating the next generation mouse common coordinate framework (CCFs) resource. Column 1: Whole-head *in vivo* and *ex vivo* MRI, CT

Tward et al., 2025 (preprint)

ABC atlas demo time

agenda/afternoon

12:00-1:00 | lunch

1:00-2:15 | walkthrough of allen brain map tools (Rachel / Jeremy)

- walkthrough of allen brain map
- Brain Knowledge Platform: ABC Atlas, MapMyCells, Genetics Tools Atlas
- Additional tools in brief: Patch-seq, gene atlases, and anatomy

2:15-2:45 | SEA-AD tools for exploring changes in AD (Irika / Jeremy)

2:45-3:00 | break

3:00-3:30 | external use cases – Rachel

3:30-4:40 | user exploration of allen institute tools

4:40-5:00 | feedback survey (all – please fill out the survey!)

github.com/AllenInstitute/SPOCTT/tree/main/2026_Additional_Presentations

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github.com/AllenInstitute/SPOCTT/tree/main/2026_Additional_Presentations

closing/session

allen institute/

need additional support/

Scientific support

brain-map.org/support

tutorials, **videos**, and other content aimed at making our **open science tools** usable

Educator support

alleninstitute.org/education

support for teaching students using the **Allen Brain Map**

Ask a scientist

community.brain-map.org

ask a **question** or submit **feedback** in the Community Forum

SPOCTT @alleninstitute.org

talk directly to an Allen Institute scientist

... or you can ask us any questions now!

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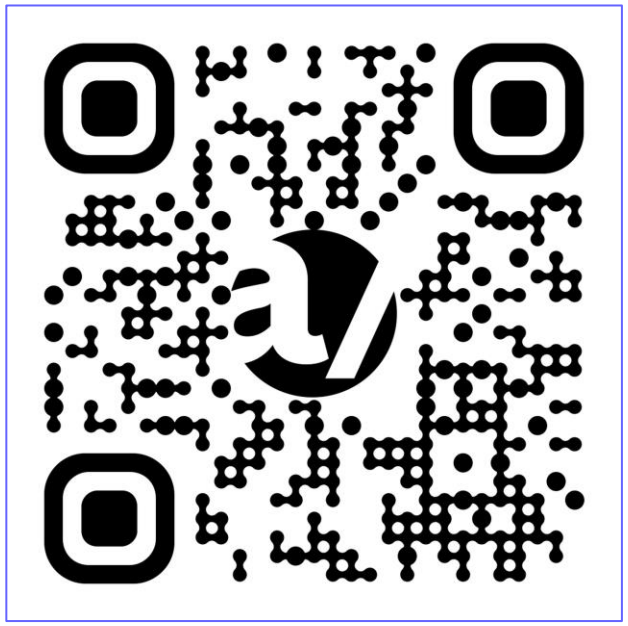

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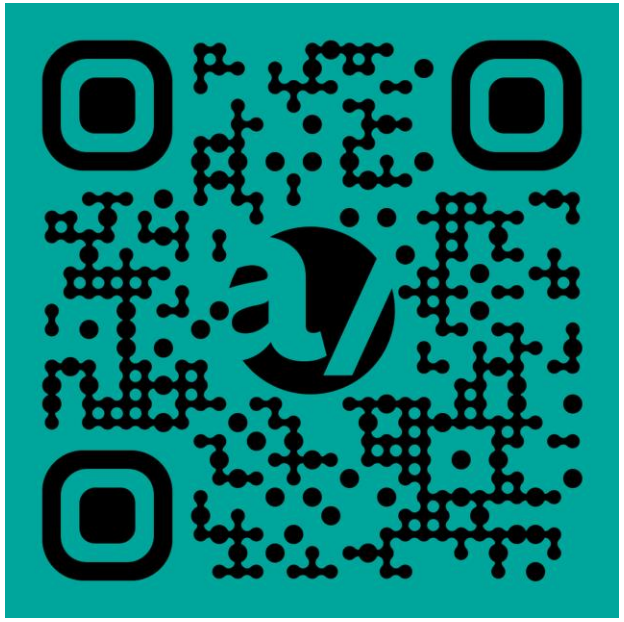
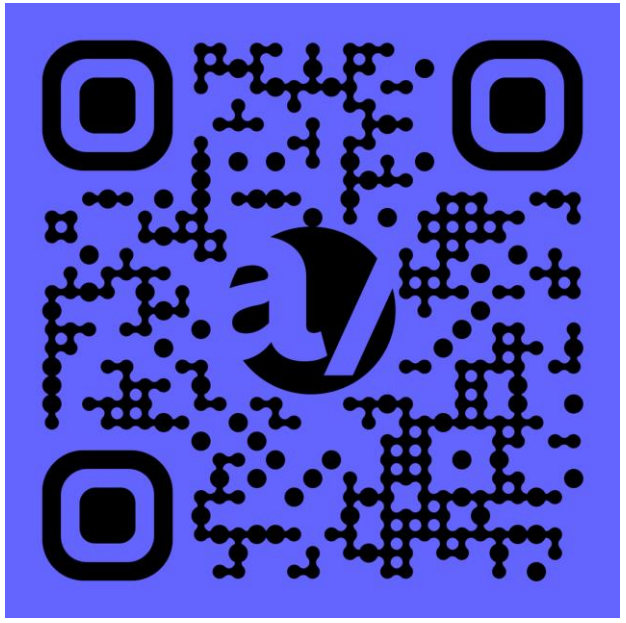
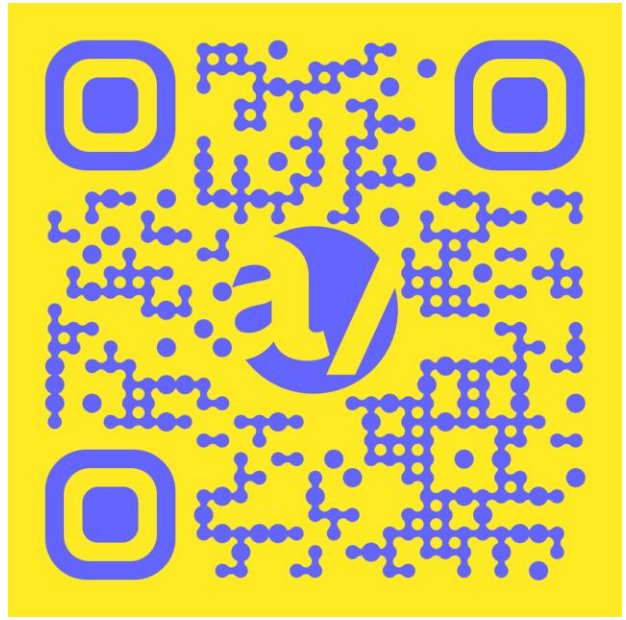


Read our protocols



Watch our YouTube videos



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work with us/

- we hire straight out of graduate school – no post-doc required!
  - this is typically a Scientist I
  - average “Scientist I” salary is about $80k-$100k USD
  - some **non-research** post-PhD roles (e.g., PM)
  - each position will be clear if it supports Visa or not - it is position dependent
- hiring is year-round, although sometimes more open spots during the start of the new year
- we often look for specific skill sets to fill open positions; the skill set depends on the position.

all careers are posted online
alleninstitute.org/careers

NOTE: We don’t have any undergraduate internships available for international students.

As of now, we also do not have any PhD-level internships.

Feedback survey

www.surveymonkey.com/r/LD69Y3P

thankyou

allen institute/

questions?

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